



API Reference

Amazon Elastic Container Service



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Amazon Elastic Container Service: API Reference

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Welcome

Amazon Elastic Container Service (Amazon ECS) is a highly scalable, fast, container management service. It makes it easy to run, stop, and manage Docker containers. You can host your cluster on a serverless infrastructure that's managed by Amazon ECS by launching your services or tasks on AWS Fargate. For more control, you can host your tasks on a cluster of Amazon Elastic Compute Cloud (Amazon EC2) or External (on-premises) instances that you manage.

Amazon ECS makes it easy to launch and stop container-based applications with simple API calls. This makes it easy to get the state of your cluster from a centralized service, and gives you access to many familiar Amazon EC2 features.

You can use Amazon ECS to schedule the placement of containers across your cluster based on your resource needs, isolation policies, and availability requirements. With Amazon ECS, you don't need to operate your own cluster management and configuration management systems. You also don't need to worry about scaling your management infrastructure.

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Actions by Amazon ECS resource

The following section lists the API actions by resource.

Resource actions

- [Account setting actions](#)
- [Cluster actions](#)
- [Cluster capacity provider actions](#)
- [Container actions](#)
- [Container agent actions](#)
- [Container instance actions](#)
- [Namespace actions](#)
- [Service actions](#)
- [Service deployment actions](#)
- [Service revision actions](#)
- [Task definition actions](#)
- [Task actions](#)
- [Task set actions](#)
- [Tag actions](#)

Account setting actions

The following API actions are available for account settings:

- [DeleteAccountSetting](#)
- [ListAccountSettings](#)
- [PutAccountSetting](#)
- [PutAccountSettingDefault](#)

Cluster actions

The following API actions are available for clusters:

- [CreateCluster](#)
- [DeleteCluster](#)
- [DescribeClusters](#)
- [ListClusters](#)
- [UpdateCluster](#)
- [UpdateClusterSettings](#)

Cluster capacity provider actions

The following APIs are available for cluster capacity providers:

- [CreateCapacityProvider](#)
- [DeleteCapacityProvider](#)
- [DescribeCapacityProviders](#)
- [PutClusterCapacityProviders](#)
- [UpdateCapacityProvider](#)

Container actions

The following APIs are available for containers:

- [ExecuteCommand](#)

Container agent actions

The following APIs are available for container agents:

- [UpdateContainerAgent](#)

Container instance actions

The following APIs are available for container instances:

- [DeregisterContainerInstance](#)

- [DescribeContainerInstances](#)
- [DeleteAttributes](#)
- [ListAttributes](#)
- [ListContainerInstances](#)
- [PutAttributes](#)
- [UpdateContainerInstancesState](#)

Namespace actions

Note

Namespaces are an AWS Cloud Map resource. Namespaces are necessary in Amazon ECS to use the ECS Service Connect or ECS service discovery features. Amazon ECS requests namespaces from AWS Cloud Map and then displays your namespaces in the AWS Management Console. Use the AWS Cloud Map console or API for additional configuration options.

The following APIs are available for namespaces:

- [CreateCluster](#)

Service actions

The following APIs are available for services:

- [CreateService](#)
- [DeleteService](#)
- [DescribeServices](#)
- [ListServices](#)
- [UpdateService](#)

Service deployment actions

The following APIs are available for service deployments:

- [DescribeServiceDeployments](#)
- [ListServiceDeployments](#)
- [StopServiceDeployment](#)

Service revision actions

The following APIs are available for service revisions:

- [DescribeServiceRevisions](#)

Task definition actions

The following APIs are available for task-definitions:

- [DeleteTaskDefinitions](#)
- [DeregisterTaskDefinition](#)
- [DescribeTaskDefinition](#)
- [ListTaskDefinitionFamilies](#)
- [ListTaskDefinitions](#)
- [RegisterTaskDefinition](#)

Task actions

The following APIs are available for tasks:

- [DescribeTasks](#)
- [GetTaskProtection](#)
- [ListTasks](#)
- [RunTask](#)
- [StartTask](#)

- [StopTask](#)
- [UpdateTaskProtection](#)

Task set actions

The following APIs are available for task sets:

- [CreateTaskSet](#)
- [DeleteTaskSet](#)
- [DescribeTaskSets](#)
- [UpdateServicePrimaryTaskSet](#)
- [UpdateTaskSet](#)

Tag actions

The following APIs are available for tags:

- [ListTagsForResource](#)
- [TagResource](#)
- [UntagResource](#)

Actions

The following actions are supported:

- [CreateCapacityProvider](#)
- [CreateCluster](#)
- [CreateService](#)
- [CreateTaskSet](#)
- [DeleteAccountSetting](#)
- [DeleteAttributes](#)
- [DeleteCapacityProvider](#)
- [DeleteCluster](#)
- [DeleteService](#)
- [DeleteTaskDefinitions](#)
- [DeleteTaskSet](#)
- [DeregisterContainerInstance](#)
- [DeregisterTaskDefinition](#)
- [DescribeCapacityProviders](#)
- [DescribeClusters](#)
- [DescribeContainerInstances](#)
- [DescribeServiceDeployments](#)
- [DescribeServiceRevisions](#)
- [DescribeServices](#)
- [DescribeTaskDefinition](#)
- [DescribeTasks](#)
- [DescribeTaskSets](#)
- [DiscoverPollEndpoint](#)
- [ExecuteCommand](#)
- [GetTaskProtection](#)
- [ListAccountSettings](#)
- [ListAttributes](#)

- [ListClusters](#)
- [ListContainerInstances](#)
- [ListServiceDeployments](#)
- [ListServices](#)
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- [ListTaskDefinitionFamilies](#)
- [ListTaskDefinitions](#)
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- [PutAccountSetting](#)
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- [PutAttributes](#)
- [PutClusterCapacityProviders](#)
- [RegisterContainerInstance](#)
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- [StopTask](#)
- [SubmitAttachmentStateChanges](#)
- [SubmitContainerStateChange](#)
- [SubmitTaskStateChange](#)
- [TagResource](#)
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- [UpdateCapacityProvider](#)
- [UpdateCluster](#)
- [UpdateClusterSettings](#)
- [UpdateContainerAgent](#)
- [UpdateContainerInstancesState](#)
- [UpdateService](#)

- [UpdateServicePrimaryTaskSet](#)
- [UpdateTaskProtection](#)
- [UpdateTaskSet](#)

CreateCapacityProvider

Creates a capacity provider. Capacity providers are associated with a cluster and are used in capacity provider strategies to facilitate cluster auto scaling. You can create capacity providers for Amazon ECS Managed Instances and EC2 instances. AWS Fargate has the predefined FARGATE and FARGATE_SPOT capacity providers.

Request Syntax

```
{
  "autoScalingGroupProvider": {
    "autoScalingGroupArn": "string",
    "managedDraining": "string",
    "managedScaling": {
      "instanceWarmupPeriod": number,
      "maximumScalingStepSize": number,
      "minimumScalingStepSize": number,
      "status": "string",
      "targetCapacity": number
    },
    "managedTerminationProtection": "string"
  },
  "cluster": "string",
  "managedInstancesProvider": {
    "infrastructureRoleArn": "string",
    "instanceLaunchTemplate": {
      "ec2InstanceProfileArn": "string",
      "instanceRequirements": {
        "acceleratorCount": {
          "max": number,
          "min": number
        },
        "acceleratorManufacturers": [ "string" ],
        "acceleratorNames": [ "string" ],
        "acceleratorTotalMemoryMiB": {
          "max": number,
          "min": number
        },
        "acceleratorTypes": [ "string" ],
        "allowedInstanceTypes": [ "string" ],
        "bareMetal": "string",
        "baselineEbsBandwidthMbps": {
```

```

        "max": number,
        "min": number
    },
    "burstablePerformance": "string",
    "cpuManufacturers": [ "string" ],
    "excludedInstanceTypes": [ "string" ],
    "instanceGenerations": [ "string" ],
    "localStorage": "string",
    "localStorageTypes": [ "string" ],
    "maxSpotPriceAsPercentageOfOptimalOnDemandPrice": number,
    "memoryGiBPerVCpu": {
        "max": number,
        "min": number
    },
    "memoryMiB": {
        "max": number,
        "min": number
    },
    "networkBandwidthGbps": {
        "max": number,
        "min": number
    },
    "networkInterfaceCount": {
        "max": number,
        "min": number
    },
    "onDemandMaxPricePercentageOverLowestPrice": number,
    "requireHibernateSupport": boolean,
    "spotMaxPricePercentageOverLowestPrice": number,
    "totalLocalStorageGB": {
        "max": number,
        "min": number
    },
    "vCpuCount": {
        "max": number,
        "min": number
    }
},
"monitoring": "string",
"networkConfiguration": {
    "securityGroups": [ "string" ],
    "subnets": [ "string" ]
},
"storageConfiguration": {

```

```
        "storageSizeGiB": number
      },
      "propagateTags": "string"
    },
    "name": "string",
    "tags": [
      {
        "key": "string",
        "value": "string"
      }
    ]
  }
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[autoScalingGroupProvider](#)

The details of the Auto Scaling group for the capacity provider.

Type: [AutoScalingGroupProvider](#) object

Required: No

[cluster](#)

The name of the cluster to associate with the capacity provider. When you create a capacity provider with Amazon ECS Managed Instances, it becomes available only within the specified cluster.

Type: String

Required: No

[managedInstancesProvider](#)

The configuration for the Amazon ECS Managed Instances provider. This configuration specifies how Amazon ECS manages Amazon EC2 instances on your behalf, including the infrastructure role, instance launch template, and tag propagation settings.

Type: [CreateManagedInstancesProviderConfiguration](#) object

Required: No

name

The name of the capacity provider. Up to 255 characters are allowed. They include letters (both upper and lowercase letters), numbers, underscores (_), and hyphens (-). The name can't be prefixed with "aws", "ecs", or "fargate".

Type: String

Required: Yes

tags

The metadata that you apply to the capacity provider to categorize and organize them more conveniently. Each tag consists of a key and an optional value. You define both of them.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws :`, `AWS :`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

Response Syntax

```
{  
  "capacityProvider": {
```

```

    "autoScalingGroupProvider": {
      "autoScalingGroupArn": "string",
      "managedDraining": "string",
      "managedScaling": {
        "instanceWarmupPeriod": number,
        "maximumScalingStepSize": number,
        "minimumScalingStepSize": number,
        "status": "string",
        "targetCapacity": number
      },
      "managedTerminationProtection": "string"
    },
    "capacityProviderArn": "string",
    "cluster": "string",
    "managedInstancesProvider": {
      "infrastructureRoleArn": "string",
      "instanceLaunchTemplate": {
        "ec2InstanceProfileArn": "string",
        "instanceRequirements": {
          "acceleratorCount": {
            "max": number,
            "min": number
          },
          "acceleratorManufacturers": [ "string" ],
          "acceleratorNames": [ "string" ],
          "acceleratorTotalMemoryMiB": {
            "max": number,
            "min": number
          },
          "acceleratorTypes": [ "string" ],
          "allowedInstanceTypes": [ "string" ],
          "bareMetal": "string",
          "baselineEbsBandwidthMbps": {
            "max": number,
            "min": number
          },
          "burstablePerformance": "string",
          "cpuManufacturers": [ "string" ],
          "excludedInstanceTypes": [ "string" ],
          "instanceGenerations": [ "string" ],
          "localStorage": "string",
          "localStorageTypes": [ "string" ],
          "maxSpotPriceAsPercentageOfOptimalOnDemandPrice": number,
          "memoryGiBPerVCpu": {

```

```

        "max": number,
        "min": number
    },
    "memoryMiB": {
        "max": number,
        "min": number
    },
    "networkBandwidthGbps": {
        "max": number,
        "min": number
    },
    "networkInterfaceCount": {
        "max": number,
        "min": number
    },
    "onDemandMaxPricePercentageOverLowestPrice": number,
    "requireHibernateSupport": boolean,
    "spotMaxPricePercentageOverLowestPrice": number,
    "totalLocalStorageGB": {
        "max": number,
        "min": number
    },
    "vCpuCount": {
        "max": number,
        "min": number
    }
},
"monitoring": "string",
"networkConfiguration": {
    "securityGroups": [ "string" ],
    "subnets": [ "string" ]
},
"storageConfiguration": {
    "storageSizeGiB": number
}
},
"propagateTags": "string"
},
"name": "string",
"status": "string",
"tags": [
    {
        "key": "string",
        "value": "string"
    }
]

```

```
    }  
  ],  
  "type": "string",  
  "updateStatus": "string",  
  "updateStatusReason": "string"  
}  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[capacityProvider](#)

The full description of the new capacity provider.

Type: [CapacityProvider](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

LimitExceededException

The limit for the resource was exceeded.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

UpdateInProgressException

There's already a current Amazon ECS container agent update in progress on the container instance that's specified. If the container agent becomes disconnected while it's in a transitional stage, such as PENDING or STAGING, the update process can get stuck in that state. However, when the agent reconnects, it resumes where it stopped previously.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)

- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

CreateCluster

Creates a new Amazon ECS cluster. By default, your account receives a default cluster when you launch your first container instance. However, you can create your own cluster with a unique name.

Note

When you call the [CreateCluster](#) API operation, Amazon ECS attempts to create the Amazon ECS service-linked role for your account. This is so that it can manage required resources in other AWS services on your behalf. However, if the user that makes the call doesn't have permissions to create the service-linked role, it isn't created. For more information, see [Using service-linked roles for Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{
  "capacityProviders": [ "string" ],
  "clusterName": "string",
  "configuration": {
    "executeCommandConfiguration": {
      "kmsKeyId": "string",
      "logConfiguration": {
        "cloudWatchEncryptionEnabled": boolean,
        "cloudWatchLogGroupName": "string",
        "s3BucketName": "string",
        "s3EncryptionEnabled": boolean,
        "s3KeyPrefix": "string"
      },
      "logging": "string"
    },
    "managedStorageConfiguration": {
      "fargateEphemeralStorageKmsKeyId": "string",
      "kmsKeyId": "string"
    }
  },
  "defaultCapacityProviderStrategy": [
    {
      "base": number,
      "capacityProvider": "string",
```

```
    "weight": number
  },
],
"serviceConnectDefaults": {
  "namespace": "string"
},
"settings": [
  {
    "name": "string",
    "value": "string"
  }
],
"tags": [
  {
    "key": "string",
    "value": "string"
  }
]
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

capacityProviders

The short name of one or more capacity providers to associate with the cluster. A capacity provider must be associated with a cluster before it can be included as part of the default capacity provider strategy of the cluster or used in a capacity provider strategy when calling the [CreateService](#) or [RunTask](#) actions.

If specifying a capacity provider that uses an Auto Scaling group, the capacity provider must be created but not associated with another cluster. New Auto Scaling group capacity providers can be created with the [CreateCapacityProvider](#) API operation.

To use a AWS Fargate capacity provider, specify either the FARGATE or FARGATE_SPOT capacity providers. The AWS Fargate capacity providers are available to all accounts and only need to be associated with a cluster to be used.

The [PutCapacityProvider](#) API operation is used to update the list of available capacity providers for a cluster after the cluster is created.

Type: Array of strings

Required: No

[clusterName](#)

The name of your cluster. If you don't specify a name for your cluster, you create a cluster that's named `default`. Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed.

Type: String

Required: No

[configuration](#)

The execute command configuration for the cluster.

Type: [ClusterConfiguration](#) object

Required: No

[defaultCapacityProviderStrategy](#)

The capacity provider strategy to set as the default for the cluster. After a default capacity provider strategy is set for a cluster, when you call the [CreateService](#) or [RunTask](#) APIs with no capacity provider strategy or launch type specified, the default capacity provider strategy for the cluster is used.

If a default capacity provider strategy isn't defined for a cluster when it was created, it can be defined later with the [PutClusterCapacityProviders](#) API operation.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: No

[serviceConnectDefaults](#)

Use this parameter to set a default Service Connect namespace. After you set a default Service Connect namespace, any new services with Service Connect turned on that are created in the cluster are added as client services in the namespace. This setting only applies to new services that set the `enabled` parameter to `true` in the `ServiceConnectConfiguration`. You can

set the namespace of each service individually in the `ServiceConnectConfiguration` to override this default parameter.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [ClusterServiceConnectDefaultsRequest](#) object

Required: No

[settings](#)

The setting to use when creating a cluster. This parameter is used to turn on CloudWatch Container Insights for a cluster. If this value is specified, it overrides the `containerInsights` value set with [PutAccountSetting](#) or [PutAccountSettingDefault](#).

Type: Array of [ClusterSetting](#) objects

Required: No

[tags](#)

The metadata that you apply to the cluster to help you categorize and organize them. Each tag consists of a key and an optional value. You define both.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: `+ - = . _ : / @`.
- Tag keys and values are case-sensitive.
- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

Response Syntax

```
{
  "cluster": {
    "activeServicesCount": number,
    "attachments": [
      {
        "details": [
          {
            "name": "string",
            "value": "string"
          }
        ],
        "id": "string",
        "status": "string",
        "type": "string"
      }
    ],
    "attachmentsStatus": "string",
    "capacityProviders": [ "string" ],
    "clusterArn": "string",
    "clusterName": "string",
    "configuration": {
      "executeCommandConfiguration": {
        "kmsKeyId": "string",
        "logConfiguration": {
          "cloudWatchEncryptionEnabled": boolean,
          "cloudWatchLogGroupName": "string",
          "s3BucketName": "string",
          "s3EncryptionEnabled": boolean,
          "s3KeyPrefix": "string"
        },
        "logging": "string"
      },
      "managedStorageConfiguration": {
        "fargateEphemeralStorageKmsKeyId": "string",
        "kmsKeyId": "string"
      }
    }
  }
}
```

```
    }
  },
  "defaultCapacityProviderStrategy": [
    {
      "base": number,
      "capacityProvider": "string",
      "weight": number
    }
  ],
  "pendingTasksCount": number,
  "registeredContainerInstancesCount": number,
  "runningTasksCount": number,
  "serviceConnectDefaults": {
    "namespace": "string"
  },
  "settings": [
    {
      "name": "string",
      "value": "string"
    }
  ],
  "statistics": [
    {
      "name": "string",
      "value": "string"
    }
  ],
  "status": "string",
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

cluster

The full description of your new cluster.

Type: [Cluster](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

NamespaceNotFoundException

The specified namespace wasn't found.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request creates a cluster called `My-cluster`.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 29
X-Amz-Target: AmazonEC2ContainerServiceV20141113.CreateCluster
X-Amz-Date: 20150429T163840Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "clusterName": "My-cluster"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 16:38:41 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 209
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "cluster": {
```

```
"activeServicesCount": 0,  
"clusterArn": "arn:aws:ecs:us-east-1:012345678910:cluster/My-cluster",  
"clusterName": "My-cluster",  
"pendingTasksCount": 0,  
"registeredContainerInstancesCount": 0,  
"runningTasksCount": 0,  
"status": "ACTIVE"  
}  
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

CreateService

Runs and maintains your desired number of tasks from a specified task definition. If the number of tasks running in a service drops below the `desiredCount`, Amazon ECS runs another copy of the task in the specified cluster. To update an existing service, use [UpdateService](#).

Note

On March 21, 2024, a change was made to resolve the task definition revision before authorization. When a task definition revision is not specified, authorization will occur using the latest revision of a task definition.

Note

Amazon Elastic Inference (EI) is no longer available to customers.

In addition to maintaining the desired count of tasks in your service, you can optionally run your service behind one or more load balancers. The load balancers distribute traffic across the tasks that are associated with the service. For more information, see [Service load balancing](#) in the *Amazon Elastic Container Service Developer Guide*.

You can attach Amazon EBS volumes to Amazon ECS tasks by configuring the volume when creating or updating a service. `volumeConfigurations` is only supported for REPLICA service and not DAEMON service. For more information, see [Amazon EBS volumes](#) in the *Amazon Elastic Container Service Developer Guide*.

Tasks for services that don't use a load balancer are considered healthy if they're in the RUNNING state. Tasks for services that use a load balancer are considered healthy if they're in the RUNNING state and are reported as healthy by the load balancer.

There are two service scheduler strategies available:

- **REPLICA** - The replica scheduling strategy places and maintains your desired number of tasks across your cluster. By default, the service scheduler spreads tasks across Availability Zones. You can use task placement strategies and constraints to customize task placement decisions. For more information, see [Service scheduler concepts](#) in the *Amazon Elastic Container Service Developer Guide*.

- **DAEMON** - The daemon scheduling strategy deploys exactly one task on each active container instance that meets all of the task placement constraints that you specify in your cluster. The service scheduler also evaluates the task placement constraints for running tasks. It also stops tasks that don't meet the placement constraints. When using this strategy, you don't need to specify a desired number of tasks, a task placement strategy, or use Service Auto Scaling policies. For more information, see [Amazon ECS services](#) in the *Amazon Elastic Container Service Developer Guide*.

The deployment controller is the mechanism that determines how tasks are deployed for your service. The valid options are:

- **ECS**

When you create a service which uses the ECS deployment controller, you can choose between the following deployment strategies (which you can set in the "strategy" field in "deploymentConfiguration"):

- **ROLLING**: When you create a service which uses the *rolling update* (ROLLING) deployment strategy, the Amazon ECS service scheduler replaces the currently running tasks with new tasks. The number of tasks that Amazon ECS adds or removes from the service during a rolling update is controlled by the service deployment configuration. For more information, see [Deploy Amazon ECS services by replacing tasks](#) in the *Amazon Elastic Container Service Developer Guide*.

Rolling update deployments are best suited for the following scenarios:

- **Gradual service updates**: You need to update your service incrementally without taking the entire service offline at once.
- **Limited resource requirements**: You want to avoid the additional resource costs of running two complete environments simultaneously (as required by blue/green deployments).
- **Acceptable deployment time**: Your application can tolerate a longer deployment process, as rolling updates replace tasks one by one.
- **No need for instant roll back**: Your service can tolerate a rollback process that takes minutes rather than seconds.
- **Simple deployment process**: You prefer a straightforward deployment approach without the complexity of managing multiple environments, target groups, and listeners.

- **No load balancer requirement:** Your service doesn't use or require a load balancer, Application Load Balancer, Network Load Balancer, or Service Connect (which are required for blue/green deployments).
- **Stateful applications:** Your application maintains state that makes it difficult to run two parallel environments.
- **Cost sensitivity:** You want to minimize deployment costs by not running duplicate environments during deployment.

Rolling updates are the default deployment strategy for services and provide a balance between deployment safety and resource efficiency for many common application scenarios.

- **BLUE_GREEN:** A *blue/green* deployment strategy (BLUE_GREEN) is a release methodology that reduces downtime and risk by running two identical production environments called blue and green. With Amazon ECS blue/green deployments, you can validate new service revisions before directing production traffic to them. This approach provides a safer way to deploy changes with the ability to quickly roll back if needed. For more information, see [Amazon ECS blue/green deployments](#) in the *Amazon Elastic Container Service Developer Guide*.

Amazon ECS blue/green deployments are best suited for the following scenarios:

- **Service validation:** When you need to validate new service revisions before directing production traffic to them
- **Zero downtime:** When your service requires zero-downtime deployments
- **Instant roll back:** When you need the ability to quickly roll back if issues are detected
- **Load balancer requirement:** When your service uses Application Load Balancer, Network Load Balancer, or Service Connect
- **LINEAR:** A *linear* deployment strategy (LINEAR) gradually shifts traffic from the current production environment to a new environment in equal percentage increments over a specified time period. With Amazon ECS linear deployments, you can control the pace of traffic shifting and validate new service revisions with increasing amounts of production traffic.

Linear deployments are best suited for the following scenarios:

- **Gradual validation:** When you want to gradually validate your new service version with increasing traffic
- **Performance monitoring:** When you need time to monitor metrics and performance during the deployment

- Risk minimization: When you want to minimize risk by exposing the new version to production traffic incrementally
- Load balancer requirement: When your service uses Application Load Balancer, Network Load Balancer, or Service Connect
- CANARY: A *canary* deployment strategy (CANARY) shifts a small percentage of traffic to the new service revision first, then shifts the remaining traffic all at once after a specified time period. This allows you to test the new version with a subset of users before full deployment.

Canary deployments are best suited for the following scenarios:

- Feature testing: When you want to test new features with a small subset of users before full rollout
- Production validation: When you need to validate performance and functionality with real production traffic
- Blast radius control: When you want to minimize blast radius if issues are discovered in the new version
- Load balancer requirement: When your service uses Application Load Balancer, Network Load Balancer, or Service Connect
- External

Use a third-party deployment controller.

- Blue/green deployment (powered by AWS CodeDeploy)

AWS CodeDeploy installs an updated version of the application as a new replacement task set and reroutes production traffic from the original application task set to the replacement task set. The original task set is terminated after a successful deployment. Use this deployment controller to verify a new deployment of a service before sending production traffic to it.

When creating a service that uses the EXTERNAL deployment controller, you can specify only parameters that aren't controlled at the task set level. The only required parameter is the service name. You control your services using the [CreateTaskSet](#). For more information, see [Amazon ECS deployment types](#) in the *Amazon Elastic Container Service Developer Guide*.

When the service scheduler launches new tasks, it determines task placement. For information about task placement and task placement strategies, see [Amazon ECS task placement](#) in the *Amazon Elastic Container Service Developer Guide*

Request Syntax

```
{
  "availabilityZoneRebalancing": "string",
  "capacityProviderStrategy": [
    {
      "base": number,
      "capacityProvider": "string",
      "weight": number
    }
  ],
  "clientToken": "string",
  "cluster": "string",
  "deploymentConfiguration": {
    "alarms": {
      "alarmNames": [ "string" ],
      "enable": boolean,
      "rollback": boolean
    },
    "bakeTimeInMinutes": number,
    "canaryConfiguration": {
      "canaryBakeTimeInMinutes": number,
      "canaryPercent": number
    },
    "deploymentCircuitBreaker": {
      "enable": boolean,
      "rollback": boolean
    },
    "lifecycleHooks": [
      {
        "hookDetails": JSON value,
        "hookTargetArn": "string",
        "lifecycleStages": [ "string" ],
        "roleArn": "string"
      }
    ],
    "linearConfiguration": {
      "stepBakeTimeInMinutes": number,
      "stepPercent": number
    },
    "maximumPercent": number,
    "minimumHealthyPercent": number,
    "strategy": "string"
  }
}
```

```
},
"deploymentController": {
  "type": "string"
},
"desiredCount": number,
"enableECSTags": boolean,
"enableExecuteCommand": boolean,
"healthCheckGracePeriodSeconds": number,
"launchType": "string",
"loadBalancers": [
  {
    "advancedConfiguration": {
      "alternateTargetGroupArn": "string",
      "productionListenerRule": "string",
      "roleArn": "string",
      "testListenerRule": "string"
    },
    "containerName": "string",
    "containerPort": number,
    "loadBalancerName": "string",
    "targetGroupArn": "string"
  }
],
"networkConfiguration": {
  "awsvpcConfiguration": {
    "assignPublicIp": "string",
    "securityGroups": [ "string" ],
    "subnets": [ "string" ]
  }
},
"placementConstraints": [
  {
    "expression": "string",
    "type": "string"
  }
],
"placementStrategy": [
  {
    "field": "string",
    "type": "string"
  }
],
"platformVersion": "string",
"propagateTags": "string",
```

```

"role": "string",
"schedulingStrategy": "string",
"serviceConnectConfiguration": {
  "accessLogConfiguration": {
    "format": "string",
    "includeQueryParameters": "string"
  },
  "enabled": boolean,
  "logConfiguration": {
    "logDriver": "string",
    "options": {
      "string" : "string"
    },
    "secretOptions": [
      {
        "name": "string",
        "valueFrom": "string"
      }
    ]
  },
  "namespace": "string",
  "services": [
    {
      "clientAliases": [
        {
          "dnsName": "string",
          "port": number,
          "testTrafficRules": {
            "header": {
              "name": "string",
              "value": {
                "exact": "string"
              }
            }
          }
        }
      ],
      "discoveryName": "string",
      "ingressPortOverride": number,
      "portName": "string",
      "timeout": {
        "idleTimeoutSeconds": number,
        "perRequestTimeoutSeconds": number
      }
    },

```

```

    "tls": {
      "issuerCertificateAuthority": {
        "awsPcaAuthorityArn": "string"
      },
      "kmsKey": "string",
      "roleArn": "string"
    }
  ]
},
"serviceName": "string",
"serviceRegistries": [
  {
    "containerName": "string",
    "containerPort": number,
    "port": number,
    "registryArn": "string"
  }
],
"tags": [
  {
    "key": "string",
    "value": "string"
  }
],
"taskDefinition": "string",
"volumeConfigurations": [
  {
    "managedEBSVolume": {
      "encrypted": boolean,
      "filesystemType": "string",
      "iops": number,
      "kmsKeyId": "string",
      "roleArn": "string",
      "sizeInGiB": number,
      "snapshotId": "string",
      "tagSpecifications": [
        {
          "propagateTags": "string",
          "resourceType": "string",
          "tags": [
            {
              "key": "string",
              "value": "string"
            }
          ]
        }
      ]
    }
  }
]

```

```

        }
      ]
    },
    "throughput": number,
    "volumeInitializationRate": number,
    "volumeType": "string"
  },
  "name": "string"
}
],
"vpcLatticeConfigurations": [
  {
    "portName": "string",
    "roleArn": "string",
    "targetGroupArn": "string"
  }
]
}

```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[availabilityZoneRebalancing](#)

Indicates whether to use Availability Zone rebalancing for the service.

For more information, see [Balancing an Amazon ECS service across Availability Zones](#) in the Amazon Elastic Container Service Developer Guide .

The default behavior of AvailabilityZoneRebalancing differs between create and update requests:

- For create service requests, when no value is specified for AvailabilityZoneRebalancing, Amazon ECS defaults the value to ENABLED.
- For update service requests, when no value is specified for AvailabilityZoneRebalancing, Amazon ECS defaults to the existing service's AvailabilityZoneRebalancing value. If the service never had an AvailabilityZoneRebalancing value set, Amazon ECS treats this as DISABLED.

Type: String

Valid Values: ENABLED | DISABLED

Required: No

capacityProviderStrategy

The capacity provider strategy to use for the service.

Note

If you want to use Amazon ECS Managed Instances, you must use the `capacityProviderStrategy` request parameter and omit the `launchType` request parameter.

If a `capacityProviderStrategy` is specified, the `launchType` parameter must be omitted. If no `capacityProviderStrategy` or `launchType` is specified, the `defaultCapacityProviderStrategy` for the cluster is used.

A capacity provider strategy can contain a maximum of 20 capacity providers.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: No

clientToken

An identifier that you provide to ensure the idempotency of the request. It must be unique and is case sensitive. Up to 36 ASCII characters in the range of 33-126 (inclusive) are allowed.

Type: String

Required: No

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that you run your service on. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

deploymentConfiguration

Optional deployment parameters that control how many tasks run during the deployment and the ordering of stopping and starting tasks.

Type: [DeploymentConfiguration](#) object

Required: No

deploymentController

The deployment controller to use for the service. If no deployment controller is specified, the default value of ECS is used.

Type: [DeploymentController](#) object

Required: No

desiredCount

The number of instantiations of the specified task definition to place and keep running in your service.

This is required if `schedulingStrategy` is `REPLICA` or isn't specified. If `schedulingStrategy` is `DAEMON` then this isn't required.

Type: Integer

Required: No

enableECSTags

Specifies whether to turn on Amazon ECS managed tags for the tasks within the service. For more information, see [Tagging your Amazon ECS resources](#) in the *Amazon Elastic Container Service Developer Guide*.

When you use Amazon ECS managed tags, you must set the `propagateTags` request parameter.

Type: Boolean

Required: No

enableExecuteCommand

Determines whether the execute command functionality is turned on for the service. If `true`, this enables execute command functionality on all containers in the service tasks.

Type: Boolean

Required: No

healthCheckGracePeriodSeconds

The period of time, in seconds, that the Amazon ECS service scheduler ignores unhealthy Elastic Load Balancing, VPC Lattice, and container health checks after a task has first started. If you do not specify a health check grace period value, the default value of 0 is used. If you do not use any of the health checks, then `healthCheckGracePeriodSeconds` is unused.

If your service has more running tasks than desired, unhealthy tasks in the grace period might be stopped to reach the desired count.

Type: Integer

Required: No

launchType

The infrastructure that you run your service on. For more information, see [Amazon ECS launch types](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

If you want to use Amazon ECS Managed Instances, you must use the `capacityProviderStrategy` request parameter and omit the `launchType` request parameter.

The FARGATE launch type runs your tasks on AWS Fargate On-Demand infrastructure.

Note

Fargate Spot infrastructure is available for use but a capacity provider strategy must be used. For more information, see [AWS Fargate capacity providers](#) in the *Amazon ECS Developer Guide*.

The EC2 launch type runs your tasks on Amazon EC2 instances registered to your cluster.

The EXTERNAL launch type runs your tasks on your on-premises server or virtual machine (VM) capacity registered to your cluster.

A service can use either a launch type or a capacity provider strategy. If a `launchType` is specified, the `capacityProviderStrategy` parameter must be omitted.

Type: String

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

loadBalancers

A load balancer object representing the load balancers to use with your service. For more information, see [Service load balancing](#) in the *Amazon Elastic Container Service Developer Guide*.

If the service uses the ECS deployment controller and using either an Application Load Balancer or Network Load Balancer, you must specify one or more target group ARNs to attach to the service. The service-linked role is required for services that use multiple target groups. For more information, see [Using service-linked roles for Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*.

If the service uses the CODE_DEPLOY deployment controller, the service is required to use either an Application Load Balancer or Network Load Balancer. When creating an AWS CodeDeploy deployment group, you specify two target groups (referred to as a `targetGroupPair`). During a deployment, AWS CodeDeploy determines which task set in your service has the status PRIMARY, and it associates one target group with it. Then, it also associates the other target group with the replacement task set. The load balancer can also have up to two listeners: a required listener for production traffic and an optional listener that you can use to perform validation tests with Lambda functions before routing production traffic to it.

If you use the CODE_DEPLOY deployment controller, these values can be changed when updating the service.

For Application Load Balancers and Network Load Balancers, this object must contain the load balancer target group ARN, the container name, and the container port to access from the load balancer. The container name must be as it appears in a container definition. The load balancer name parameter must be omitted. When a task from this service is placed on a container

instance, the container instance and port combination is registered as a target in the target group that's specified here.

For Classic Load Balancers, this object must contain the load balancer name, the container name, and the container port to access from the load balancer. The container name must be as it appears in a container definition. The target group ARN parameter must be omitted. When a task from this service is placed on a container instance, the container instance is registered with the load balancer that's specified here.

Services with tasks that use the `awsvpc` network mode (for example, those with the Fargate launch type) only support Application Load Balancers and Network Load Balancers. Classic Load Balancers aren't supported. Also, when you create any target groups for these services, you must choose `ip` as the target type, not `instance`. This is because tasks that use the `awsvpc` network mode are associated with an elastic network interface, not an Amazon EC2 instance.

Type: Array of [LoadBalancer](#) objects

Required: No

[networkConfiguration](#)

The network configuration for the service. This parameter is required for task definitions that use the `awsvpc` network mode to receive their own elastic network interface, and it isn't supported for other network modes. For more information, see [Task networking](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [NetworkConfiguration](#) object

Required: No

[placementConstraints](#)

An array of placement constraint objects to use for tasks in your service. You can specify a maximum of 10 constraints for each task. This limit includes constraints in the task definition and those specified at runtime.

Type: Array of [PlacementConstraint](#) objects

Required: No

[placementStrategy](#)

The placement strategy objects to use for tasks in your service. You can specify a maximum of 5 strategy rules for each service.

Type: Array of [PlacementStrategy](#) objects

Required: No

[platformVersion](#)

The platform version that your tasks in the service are running on. A platform version is specified only for tasks using the Fargate launch type. If one isn't specified, the LATEST platform version is used. For more information, see [AWS Fargate platform versions](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

[propagateTags](#)

Specifies whether to propagate the tags from the task definition to the task. If no value is specified, the tags aren't propagated. Tags can only be propagated to the task during task creation. To add tags to a task after task creation, use the [TagResource](#) API action.

You must set this to a value other than NONE when you use Cost Explorer. For more information, see [Amazon ECS usage reports](#) in the *Amazon Elastic Container Service Developer Guide*.

The default is NONE.

Type: String

Valid Values: TASK_DEFINITION | SERVICE | NONE

Required: No

[role](#)

The name or full Amazon Resource Name (ARN) of the IAM role that allows Amazon ECS to make calls to your load balancer on your behalf. This parameter is only permitted if you are using a load balancer with your service and your task definition doesn't use the awsvpc network mode. If you specify the `role` parameter, you must also specify a load balancer object with the `loadBalancers` parameter.

Important

If your account has already created the Amazon ECS service-linked role, that role is used for your service unless you specify a role here. The service-linked role is required if your

task definition uses the `awsvpc` network mode or if the service is configured to use service discovery, an external deployment controller, multiple target groups, or Elastic Inference accelerators in which case you don't specify a role here. For more information, see [Using service-linked roles for Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*.

If your specified role has a path other than `/`, then you must either specify the full role ARN (this is recommended) or prefix the role name with the path. For example, if a role with the name `bar` has a path of `/foo/` then you would specify `/foo/bar` as the role name. For more information, see [Friendly names and paths](#) in the *IAM User Guide*.

Type: String

Required: No

[schedulingStrategy](#)

The scheduling strategy to use for the service. For more information, see [Services](#).

There are two service scheduler strategies available:

- **REPLICA**—The replica scheduling strategy places and maintains the desired number of tasks across your cluster. By default, the service scheduler spreads tasks across Availability Zones. You can use task placement strategies and constraints to customize task placement decisions. This scheduler strategy is required if the service uses the `CODE_DEPLOY` or `EXTERNAL` deployment controller types.
- **DAEMON**—The daemon scheduling strategy deploys exactly one task on each active container instance that meets all of the task placement constraints that you specify in your cluster. The service scheduler also evaluates the task placement constraints for running tasks and will stop tasks that don't meet the placement constraints. When you're using this strategy, you don't need to specify a desired number of tasks, a task placement strategy, or use Service Auto Scaling policies.

Note

Tasks using the Fargate launch type or the `CODE_DEPLOY` or `EXTERNAL` deployment controller types don't support the `DAEMON` scheduling strategy.

Type: String

Valid Values: REPLICA | DAEMON

Required: No

serviceConnectConfiguration

The configuration for this service to discover and connect to services, and be discovered by, and connected from, other services within a namespace.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [ServiceConnectConfiguration](#) object

Required: No

serviceName

The name of your service. Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed. Service names must be unique within a cluster, but you can have similarly named services in multiple clusters within a Region or across multiple Regions.

Type: String

Required: Yes

serviceRegistries

The details of the service discovery registry to associate with this service. For more information, see [Service discovery](#).

Note

Each service may be associated with one service registry. Multiple service registries for each service isn't supported.

Type: Array of [ServiceRegistry](#) objects

Required: No

tags

The metadata that you apply to the service to help you categorize and organize them. Each tag consists of a key and an optional value, both of which you define. When a service is deleted, the tags are deleted as well.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

taskDefinition

The family and revision (`family:revision`) or full ARN of the task definition to run in your service. If a revision isn't specified, the latest ACTIVE revision is used.

A task definition must be specified if the service uses either the ECS or CODE_DEPLOY deployment controllers.

For more information about deployment types, see [Amazon ECS deployment types](#).

Type: String

Required: No

volumeConfigurations

The configuration for a volume specified in the task definition as a volume that is configured at launch time. Currently, the only supported volume type is an Amazon EBS volume.

Type: Array of [ServiceVolumeConfiguration](#) objects

Required: No

vpcLatticeConfigurations

The VPC Lattice configuration for the service being created.

Type: Array of [VpcLatticeConfiguration](#) objects

Required: No

Response Syntax

```
{
  "service": {
    "availabilityZoneRebalancing": "string",
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "clusterArn": "string",
    "createdAt": number,
    "createdBy": "string",
    "deploymentConfiguration": {
      "alarms": {
        "alarmNames": [ "string" ],
        "enable": boolean,
        "rollback": boolean
      },
      "bakeTimeInMinutes": number,
      "canaryConfiguration": {
        "canaryBakeTimeInMinutes": number,
        "canaryPercent": number
      },
      "deploymentCircuitBreaker": {
```

```

    "enable": boolean,
    "rollback": boolean
  },
  "lifecycleHooks": [
    {
      "hookDetails": JSON value,
      "hookTargetArn": "string",
      "lifecycleStages": [ "string " ],
      "roleArn": "string"
    }
  ],
  "linearConfiguration": {
    "stepBakeTimeInMinutes": number,
    "stepPercent": number
  },
  "maximumPercent": number,
  "minimumHealthyPercent": number,
  "strategy": "string"
},
"deploymentController": {
  "type": "string"
},
"deployments": [
  {
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "createdAt": number,
    "desiredCount": number,
    "failedTasks": number,
    "fargateEphemeralStorage": {
      "kmsKeyId": "string"
    },
    "id": "string",
    "launchType": "string",
    "networkConfiguration": {
      "awsvpcConfiguration": {
        "assignPublicIp": "string",
        "securityGroups": [ "string " ],
        "subnets": [ "string " ]
      }
    }
  }
]

```

```

    }
  },
  "pendingCount": number,
  "platformFamily": "string",
  "platformVersion": "string",
  "rolloutState": "string",
  "rolloutStateReason": "string",
  "runningCount": number,
  "serviceConnectConfiguration": {
    "accessLogConfiguration": {
      "format": "string",
      "includeQueryParameters": "string"
    },
    "enabled": boolean,
    "logConfiguration": {
      "logDriver": "string",
      "options": {
        "string" : "string"
      },
      "secretOptions": [
        {
          "name": "string",
          "valueFrom": "string"
        }
      ]
    },
    "namespace": "string",
    "services": [
      {
        "clientAliases": [
          {
            "dnsName": "string",
            "port": number,
            "testTrafficRules": {
              "header": {
                "name": "string",
                "value": {
                  "exact": "string"
                }
              }
            }
          }
        ]
      }
    ],
    "discoveryName": "string",

```

```

    "ingressPortOverride": number,
    "portName": "string",
    "timeout": {
        "idleTimeoutSeconds": number,
        "perRequestTimeoutSeconds": number
    },
    "tls": {
        "issuerCertificateAuthority": {
            "awsPcaAuthorityArn": "string"
        },
        "kmsKey": "string",
        "roleArn": "string"
    }
}
]
},
"serviceConnectResources": [
    {
        "discoveryArn": "string",
        "discoveryName": "string"
    }
],
"status": "string",
"taskDefinition": "string",
"updatedAt": number,
"volumeConfigurations": [
    {
        "managedEBSVolume": {
            "encrypted": boolean,
            "filesystemType": "string",
            "iops": number,
            "kmsKeyId": "string",
            "roleArn": "string",
            "sizeInGiB": number,
            "snapshotId": "string",
            "tagSpecifications": [
                {
                    "propagateTags": "string",
                    "resourceType": "string",
                    "tags": [
                        {
                            "key": "string",
                            "value": "string"
                        }
                    ]
                }
            ]
        }
    }
]
}

```

```

        ]
      }
    ],
    "throughput": number,
    "volumeInitializationRate": number,
    "volumeType": "string"
  },
  "name": "string"
}
],
"vpcLatticeConfigurations": [
  {
    "portName": "string",
    "roleArn": "string",
    "targetGroupArn": "string"
  }
]
}
],
"desiredCount": number,
"enableECSTags": boolean,
"enableExecuteCommand": boolean,
"events": [
  {
    "createdAt": number,
    "id": "string",
    "message": "string"
  }
],
"healthCheckGracePeriodSeconds": number,
"launchType": "string",
"loadBalancers": [
  {
    "advancedConfiguration": {
      "alternateTargetGroupArn": "string",
      "productionListenerRule": "string",
      "roleArn": "string",
      "testListenerRule": "string"
    },
    "containerName": "string",
    "containerPort": number,
    "loadBalancerName": "string",
    "targetGroupArn": "string"
  }
]
}

```

```
],
  "networkConfiguration": {
    "awsVpcConfiguration": {
      "assignPublicIp": "string",
      "securityGroups": [ "string" ],
      "subnets": [ "string" ]
    }
  },
  "pendingCount": number,
  "placementConstraints": [
    {
      "expression": "string",
      "type": "string"
    }
  ],
  "placementStrategy": [
    {
      "field": "string",
      "type": "string"
    }
  ],
  "platformFamily": "string",
  "platformVersion": "string",
  "propagateTags": "string",
  "roleArn": "string",
  "runningCount": number,
  "schedulingStrategy": "string",
  "serviceArn": "string",
  "serviceName": "string",
  "serviceRegistries": [
    {
      "containerName": "string",
      "containerPort": number,
      "port": number,
      "registryArn": "string"
    }
  ],
  "status": "string",
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ],
],
```

```

    "taskDefinition": "string",
    "taskSets": [
      {
        "capacityProviderStrategy": [
          {
            "base": number,
            "capacityProvider": "string",
            "weight": number
          }
        ],
        "clusterArn": "string",
        "computedDesiredCount": number,
        "createdAt": number,
        "externalId": "string",
        "fargateEphemeralStorage": {
          "kmsKeyId": "string"
        },
        "id": "string",
        "launchType": "string",
        "loadBalancers": [
          {
            "advancedConfiguration": {
              "alternateTargetGroupArn": "string",
              "productionListenerRule": "string",
              "roleArn": "string",
              "testListenerRule": "string"
            },
            "containerName": "string",
            "containerPort": number,
            "loadBalancerName": "string",
            "targetGroupArn": "string"
          }
        ],
        "networkConfiguration": {
          "awsvpcConfiguration": {
            "assignPublicIp": "string",
            "securityGroups": [ "string" ],
            "subnets": [ "string" ]
          }
        },
        "pendingCount": number,
        "platformFamily": "string",
        "platformVersion": "string",
        "runningCount": number,

```



```

    "scale": {
      "unit": "string",
      "value": number
    },
    "serviceArn": "string",
    "serviceRegistries": [
      {
        "containerName": "string",
        "containerPort": number,
        "port": number,
        "registryArn": "string"
      }
    ],
    "stabilityStatus": "string",
    "stabilityStatusAt": number,
    "startedBy": "string",
    "status": "string",
    "tags": [
      {
        "key": "string",
        "value": "string"
      }
    ],
    "taskDefinition": "string",
    "taskSetArn": "string",
    "updatedAt": number
  }
]
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

service

The full description of your service following the create call.

A service will return either a `capacityProviderStrategy` or `launchType` parameter, but not both, depending where one was specified when it was created.

If a service is using the ECS deployment controller, the `deploymentController` and `taskSets` parameters will not be returned.

if the service uses the `CODE_DEPLOY` deployment controller, the `deploymentController`, `taskSets` and `deployments` parameters will be returned, however the `deployments` parameter will be an empty list.

Type: [Service](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

NamespaceNotFoundException

The specified namespace wasn't found.

HTTP Status Code: 400

PlatformTaskDefinitionIncompatibilityException

The specified platform version doesn't satisfy the required capabilities of the task definition.

HTTP Status Code: 400

PlatformUnknownException

The specified platform version doesn't exist.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example 1

This example API request creates a service in your default Region called `ecs-simple-service`. The service uses the `ecs-demo` task definition and it maintains 10 instantiations of that task. It uses both the `FARGATE` and `FARGATE_SPOT` capacity providers.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 87
X-Amz-Target: AmazonEC2ContainerServiceV20141113.CreateService
X-Amz-Date: 20150429T170125Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "serviceName": "ecs-simple-service",
  "taskDefinition": "ecs-demo",
  "desiredCount": 10,
  "capacityProviderStrategy": [
    {
      "base": number,
      "capacityProvider": "FARGATE",
      "weight": 1
    }
  ],
  "capacityProviderStrategy": [
    {
      "base": number,
      "capacityProvider": "FARGATE_SPOT",
      "weight": 1
    }
  ]
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 17:01:27 GMT
Content-Type: application/x-amz-json-1.1
```

```
Content-Length: 636
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "service": {
    "clusterArn": "arn:aws:ecs:us-east-1:012345678910:cluster/default",
    "deploymentConfiguration": {
      "maximumPercent": 200,
      "minimumHealthyPercent": 100
    },
    "deployments": [
      {
        "createdAt": 1430326887.362,
        "desiredCount": 10,
        "id": "ecs-svc/9223370606527888445",
        "pendingCount": 0,
        "runningCount": 0,
        "status": "PRIMARY",
        "taskDefinition": "arn:aws:ecs:us-east-1:012345678910:task-definition/ecs-
demo:1",
        "updatedAt": 1430326887.362
      }
    ],
    "capacityProviderStrategy": [
      {
        "capacityProvider": "FARGATE",
        "weight": 1,
        "base": 0
      },
      {
        "capacityProvider": "FARGATE_SPOT",
        "weight": 1,
        "base": 0
      }
    ],
    "desiredCount": 10,
    "events": [],
    "loadBalancers": [],
    "pendingCount": 0,
    "runningCount": 0,
    "serviceArn": "arn:aws:ecs:us-east-1:012345678910:service/default/ecs-simple-
service",
    "serviceName": "ecs-simple-service",
```

```
    "status": "ACTIVE",
    "taskDefinition": "arn:aws:ecs:us-east-1:012345678910:task-definition/ecs-demo:1"
  }
}
```

Example 2

This example API request creates a service with multiple load balancer target groups.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
X-Amz-Target: AmazonEC2ContainerServiceV20141113.CreateService
Content-Type: application/x-amz-json-1.1
User-Agent: aws-cli/1.16.190 Python/3.6.1 Darwin/16.7.0 botocore/1.12.180
X-Amz-Date: 20190723T001203Z
Authorization: AUTHPARAMS
Content-Length: 453

{
  "serviceName": "ecs-multiplealb-service",
  "taskDefinition": "ecs-multiplealb-demo",
  "loadBalancers": [
    {
      "targetGroupArn": "arn:aws:elasticloadbalancing:us-east-1:012345678910:targetgroup/tg1/18ce32cc074018ed",
      "containerName": "simple-app",
      "containerPort": 80
    },
    {
      "targetGroupArn": "arn:aws:elasticloadbalancing:us-east-1:012345678910:targetgroup/tg2/737bead11d516e2a",
      "containerName": "simple-app",
      "containerPort": 8080
    }
  ],
  "desiredCount": 10
}
```

Sample Response

```
HTTP/1.1 200 OK
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f
Content-Type: application/x-amz-json-1.1
Content-Length: 1440
Date: Tue, 23 Jul 2019 00:12:03 GMT
Connection: keep-alive

{
  "service": {
    "serviceArn": "arn:aws:ecs:us-east-1:012345678910:service/default/ecs-
multiplealb-service",
    "serviceName": "ecs-multiplealb-service",
    "clusterArn": "arn:aws:ecs:us-east-1:012345678910:cluster/default",
    "loadBalancers": [
      {
        "targetGroupArn": "arn:aws:elasticloadbalancing:us-
east-1:012345678910:targetgroup/tg1/18ce32cc074018ed",
        "containerName": "simple-app",
        "containerPort": 80
      },
      {
        "targetGroupArn": "arn:aws:elasticloadbalancing:us-
east-1:012345678910:targetgroup/tg2/737bead11d516e2a",
        "containerName": "simple-app",
        "containerPort": 8080
      }
    ],
    "serviceRegistries": [],
    "status": "ACTIVE",
    "desiredCount": 10,
    "runningCount": 0,
    "pendingCount": 0,
    "launchType": "EC2",
    "taskDefinition": "arn:aws:ecs:us-east-1:012345678910:task-definition/ecs-
multiplealb-demo",
    "deploymentConfiguration": {
      "maximumPercent": 200,
      "minimumHealthyPercent": 100
    },
    "deployments": [
      {
        "id": "ecs-svc/9223370473014051517",
```

```

        "status": "PRIMARY",
        "taskDefinition": "arn:aws:ecs:us-east-1:012345678910:task-definition/
ecs-multiplealb-demo",
        "desiredCount": 10,
        "pendingCount": 0,
        "runningCount": 0,
        "createdAt": 1563840724.29,
        "updatedAt": 1563840724.29,
        "launchType": "EC2"
    }
],
    "roleArn": "arn:aws:iam::012345678910:role/aws-service-role/ecs.amazonaws.com/
AWSServiceRoleForECS",
    "events": [],
    "createdAt": 1563840724.29,
    "placementConstraints": [],
    "placementStrategy": [],
    "healthCheckGracePeriodSeconds": 0,
    "schedulingStrategy": "REPLICA",
    "enableECSTags": false,
    "propagateTags": "NONE"
}
}

```

Example 3

This example API request creates a service with a strategy that distributes tasks evenly across Availability Zones and then bin packs tasks based on memory within each Availability Zone.

Sample Request

```

POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
X-Amz-Target: AmazonEC2ContainerServiceV20141113.CreateService
Content-Type: application/x-amz-json-1.1
User-Agent: aws-cli/1.16.190 Python/3.6.1 Darwin/16.7.0 botocore/1.12.180
X-Amz-Date: 20190723T001203Z
Authorization: AUTHPARAMS
Content-Length: 453

{
    "serviceName": "example-placement1",

```



```
"taskDefinition": "windows-simple-iis",
"placementStrategy": [
  {
    "field": "attribute:ecs.availability-zone",
    "type": "spread"
  },
  {
    "field": "memory",
    "type": "binpack"
  }
]
"desiredCount": 3
}
```

Sample Response

```
HTTP/1.1 200 OK
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f
Content-Type: application/x-amz-json-1.1
Content-Length: 1440
Date: Tue, 23 Jul 2019 00:12:03 GMT
Connection: keep-alive

{
  "service": {
    "serviceArn": "arn:aws:ecs:us-east-1:123456789012:service/windows-ec2/default/example-placement1",
    "serviceName": "example-placement1",
    "clusterArn": "arn:aws:ecs:us-east-1:123456789012:cluster/windows-ec2",
    "loadBalancers": [],
    "serviceRegistries": [],
    "status": "ACTIVE",
    "desiredCount": 3,
    "runningCount": 0,
    "pendingCount": 0,
    "launchType": "EC2",
    "taskDefinition": "arn:aws:ecs:us-east-1:123456789012:task-definition/windows-simple-iis:2",
    "deploymentConfiguration": {
      "deploymentCircuitBreaker": {
        "enable": false,
        "rollback": false
      }
    },
  },
}
```

```

        "maximumPercent": 200,
        "minimumHealthyPercent": 100
    },
    "deployments": [
        {
            "id": "ecs-svc/409552086333EXAMPLE",
            "status": "PRIMARY",
            "taskDefinition": "arn:aws:ecs:us-east-1:123456789012:task-definition/
windows-simple-iis:2",
            "desiredCount": 3,
            "pendingCount": 0,
            "runningCount": 0,
            "failedTasks": 0,
            "createdAt": "2022-07-21T15:06:08.787000-04:00",
            "updatedAt": "2022-07-21T15:06:08.787000-04:00",
            "launchType": "EC2",
            "rolloutState": "IN_PROGRESS",
            "rolloutStateReason": "ECS deployment ecs-svc/409552086333EXAMPLE in
progress."
        }
    ],
    "events": [],
    "createdAt": "2022-07-21T15:06:08.787000-04:00",
    "placementConstraints": [],
    "placementStrategy": [
        {
            "type": "spread",
            "field": "attribute:ecs.availability-zone"
        },
        {
            "type": "binpack",
            "field": "MEMORY"
        }
    ],
    "schedulingStrategy": "REPLICA",
    "createdBy": "arn:aws:iam::123456789012:user/johndoe",
    "enableECSTags": false,
    "propagateTags": "NONE",
    "enableExecuteCommand": false
}

```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

CreateTaskSet

Create a task set in the specified cluster and service. This is used when a service uses the EXTERNAL deployment controller type. For more information, see [Amazon ECS deployment types](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

On March 21, 2024, a change was made to resolve the task definition revision before authorization. When a task definition revision is not specified, authorization will occur using the latest revision of a task definition.

For information about the maximum number of task sets and other quotas, see [Amazon ECS service quotas](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{
  "capacityProviderStrategy": [
    {
      "base": number,
      "capacityProvider": "string",
      "weight": number
    }
  ],
  "clientToken": "string",
  "cluster": "string",
  "externalId": "string",
  "launchType": "string",
  "loadBalancers": [
    {
      "advancedConfiguration": {
        "alternateTargetGroupArn": "string",
        "productionListenerRule": "string",
        "roleArn": "string",
        "testListenerRule": "string"
      },
      "containerName": "string",
      "containerPort": number,
      "loadBalancerName": "string",
```

```
    "targetGroupArn": "string"
  },
],
"networkConfiguration": {
  "awsvpcConfiguration": {
    "assignPublicIp": "string",
    "securityGroups": [ "string" ],
    "subnets": [ "string" ]
  }
},
"platformVersion": "string",
"scale": {
  "unit": "string",
  "value": number
},
"service": "string",
"serviceRegistries": [
  {
    "containerName": "string",
    "containerPort": number,
    "port": number,
    "registryArn": "string"
  }
],
"tags": [
  {
    "key": "string",
    "value": "string"
  }
],
"taskDefinition": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[capacityProviderStrategy](#)

The capacity provider strategy to use for the task set.

A capacity provider strategy consists of one or more capacity providers along with the base and weight to assign to them. A capacity provider must be associated with the cluster to be used in a capacity provider strategy. The [PutClusterCapacityProviders](#) API is used to associate a capacity provider with a cluster. Only capacity providers with an ACTIVE or UPDATING status can be used.

If a `capacityProviderStrategy` is specified, the `launchType` parameter must be omitted. If no `capacityProviderStrategy` or `launchType` is specified, the `defaultCapacityProviderStrategy` for the cluster is used.

If specifying a capacity provider that uses an Auto Scaling group, the capacity provider must already be created. New capacity providers can be created with the [CreateCapacityProvider](#) API operation.

To use a AWS Fargate capacity provider, specify either the FARGATE or FARGATE_SPOT capacity providers. The AWS Fargate capacity providers are available to all accounts and only need to be associated with a cluster to be used.

The [PutClusterCapacityProviders](#) API operation is used to update the list of available capacity providers for a cluster after the cluster is created.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: No

[clientToken](#)

An identifier that you provide to ensure the idempotency of the request. It must be unique and is case sensitive. Up to 36 ASCII characters in the range of 33-126 (inclusive) are allowed.

Type: String

Required: No

[cluster](#)

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the service to create the task set in.

Type: String

Required: Yes

externalId

An optional non-unique tag that identifies this task set in external systems. If the task set is associated with a service discovery registry, the tasks in this task set will have the ECS_TASK_SET_EXTERNAL_ID AWS Cloud Map attribute set to the provided value.

Type: String

Required: No

launchType

The launch type that new tasks in the task set uses. For more information, see [Amazon ECS launch types](#) in the *Amazon Elastic Container Service Developer Guide*.

If a launchType is specified, the capacityProviderStrategy parameter must be omitted.

Type: String

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

loadBalancers

A load balancer object representing the load balancer to use with the task set. The supported load balancer types are either an Application Load Balancer or a Network Load Balancer.

Type: Array of [LoadBalancer](#) objects

Required: No

networkConfiguration

An object representing the network configuration for a task set.

Type: [NetworkConfiguration](#) object

Required: No

platformVersion

The platform version that the tasks in the task set uses. A platform version is specified only for tasks using the Fargate launch type. If one isn't specified, the LATEST platform version is used.

Type: String

Required: No

scale

A floating-point percentage of the desired number of tasks to place and keep running in the task set.

Type: [Scale](#) object

Required: No

service

The short name or full Amazon Resource Name (ARN) of the service to create the task set in.

Type: String

Required: Yes

serviceRegistries

The details of the service discovery registries to assign to this task set. For more information, see [Service discovery](#).

Type: Array of [ServiceRegistry](#) objects

Required: No

tags

The metadata that you apply to the task set to help you categorize and organize them. Each tag consists of a key and an optional value. You define both. When a service is deleted, the tags are deleted.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.

- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

[taskDefinition](#)

The task definition for the tasks in the task set to use. If a revision isn't specified, the latest ACTIVE revision is used.

Type: String

Required: Yes

Response Syntax

```
{
  "taskSet": {
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "clusterArn": "string",
    "computedDesiredCount": number,
    "createdAt": number,
    "externalId": "string",
    "fargateEphemeralStorage": {
      "kmsKeyId": "string"
    },
    "id": "string",
    "launchType": "string",
    "loadBalancers": [
      {
        "advancedConfiguration": {
          "alternateTargetGroupArn": "string",
```

```

        "productionListenerRule": "string",
        "roleArn": "string",
        "testListenerRule": "string"
    },
    "containerName": "string",
    "containerPort": number,
    "loadBalancerName": "string",
    "targetGroupArn": "string"
}
],
"networkConfiguration": {
    "awsvpcConfiguration": {
        "assignPublicIp": "string",
        "securityGroups": [ "string" ],
        "subnets": [ "string" ]
    }
},
"pendingCount": number,
"platformFamily": "string",
"platformVersion": "string",
"runningCount": number,
"scale": {
    "unit": "string",
    "value": number
},
"serviceArn": "string",
"serviceRegistries": [
    {
        "containerName": "string",
        "containerPort": number,
        "port": number,
        "registryArn": "string"
    }
],
"stabilityStatus": "string",
"stabilityStatusAt": number,
"startedBy": "string",
"status": "string",
"tags": [
    {
        "key": "string",
        "value": "string"
    }
]
],

```

```
    "taskDefinition": "string",  
    "taskSetArn": "string",  
    "updatedAt": number  
  }  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

taskSet

Information about a set of Amazon ECS tasks in either an AWS CodeDeploy or an EXTERNAL deployment. A task set includes details such as the desired number of tasks, how many tasks are running, and whether the task set serves production traffic.

Type: [TaskSet](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

NamespaceNotFoundException

The specified namespace wasn't found.

HTTP Status Code: 400

PlatformTaskDefinitionIncompatibilityException

The specified platform version doesn't satisfy the required capabilities of the task definition.

HTTP Status Code: 400

PlatformUnknownException

The specified platform version doesn't exist.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotActiveException

The specified service isn't active. You can't update a service that's inactive. If you have previously deleted a service, you can re-create it with [CreateService](#).

HTTP Status Code: 400

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteAccountSetting

Disables an account setting for a specified user, role, or the root user for an account.

Request Syntax

```
{  
  "name": "string",  
  "principalArn": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

name

The resource name to disable the account setting for. If `serviceLongArnFormat` is specified, the ARN for your Amazon ECS services is affected. If `taskLongArnFormat` is specified, the ARN and resource ID for your Amazon ECS tasks is affected. If `containerInstanceLongArnFormat` is specified, the ARN and resource ID for your Amazon ECS container instances is affected. If `awsvpcTrunking` is specified, the ENI limit for your Amazon ECS container instances is affected.

Type: String

Valid Values: `serviceLongArnFormat` | `taskLongArnFormat` | `containerInstanceLongArnFormat` | `awsvpcTrunking` | `containerInsights` | `fargateFIPSMODE` | `tagResourceAuthorization` | `fargateTaskRetirementWaitPeriod` | `guardDutyActivate` | `defaultLogDriverMode`

Required: Yes

principalArn

The Amazon Resource Name (ARN) of the principal. It can be a user, role, or the root user. If you specify the root user, it disables the account setting for all users, roles, and the root user of

the account unless a user or role explicitly overrides these settings. If this field is omitted, the setting is changed only for the authenticated user.

In order to use this parameter, you must be the root user, or the principal.

Type: String

Required: No

Response Syntax

```
{
  "setting": {
    "name": "string",
    "principalArn": "string",
    "type": "string",
    "value": "string"
  }
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[setting](#)

The account setting for the specified principal ARN.

Type: [Setting](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteAttributes

Deletes one or more custom attributes from an Amazon ECS resource.

Request Syntax

```
{
  "attributes": [
    {
      "name": "string",
      "targetId": "string",
      "targetType": "string",
      "value": "string"
    }
  ],
  "cluster": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

attributes

The attributes to delete from your resource. You can specify up to 10 attributes for each request. For custom attributes, specify the attribute name and target ID, but don't specify the value. If you specify the target ID using the short form, you must also specify the target type.

Type: Array of [Attribute](#) objects

Required: Yes

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that contains the resource to delete attributes. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

Response Syntax

```
{
  "attributes": [
    {
      "name": "string",
      "targetId": "string",
      "targetType": "string",
      "value": "string"
    }
  ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[attributes](#)

A list of attribute objects that were successfully deleted from your resource.

Type: Array of [Attribute](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

TargetNotFoundException

The specified target wasn't found. You can view your available container instances with [ListContainerInstances](#). Amazon ECS container instances are cluster-specific and Region-specific.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example deletes an attribute with the name `stack` from a container instance.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 169
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DeleteAttributes
X-Amz-Date: 20161222T193851Z
User-Agent: aws-cli/1.11.30 Python/2.7.12 Darwin/16.3.0 botocore/1.4.87
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "cluster": "default",
  "attributes": [
    {
      "targetId": "arn:aws:ecs:us-west-2:123456789012:container-instance/1c3be8ed-
df30-47b4-8f1e-6e68ebd01f34",
```

```
    "name": "stack"
  }
]
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Thu, 22 Dec 2016 19:38:51 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 158
Connection: keep-alive
x-amzn-RequestId: 445193ca-c87e-11e6-86db-1bd3d9928caf

{
  "attributes": [
    {
      "name": "stack",
      "targetId": "arn:aws:ecs:us-west-2:123456789012:container-instance/1c3be8ed-
df30-47b4-8f1e-6e68ebd01f34",
      "value": "production"
    }
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)

- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteCapacityProvider

Deletes the specified capacity provider.

Note

The FARGATE and FARGATE_SPOT capacity providers are reserved and can't be deleted. You can disassociate them from a cluster using either [PutClusterCapacityProviders](#) or by deleting the cluster.

Prior to a capacity provider being deleted, the capacity provider must be removed from the capacity provider strategy from all services. The [UpdateService](#) API can be used to remove a capacity provider from a service's capacity provider strategy. When updating a service, the `forceNewDeployment` option can be used to ensure that any tasks using the Amazon EC2 instance capacity provided by the capacity provider are transitioned to use the capacity from the remaining capacity providers. Only capacity providers that aren't associated with a cluster can be deleted. To remove a capacity provider from a cluster, you can either use [PutClusterCapacityProviders](#) or delete the cluster.

Request Syntax

```
{
  "capacityProvider": "string",
  "cluster": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[capacityProvider](#)

The short name or full Amazon Resource Name (ARN) of the capacity provider to delete.

Type: String

Required: Yes

cluster

The name of the cluster that contains the capacity provider to delete. Managed instances capacity providers are cluster-scoped and can only be deleted from their associated cluster.

Type: String

Required: No

Response Syntax

```
{
  "capacityProvider": {
    "autoScalingGroupProvider": {
      "autoScalingGroupArn": "string",
      "managedDraining": "string",
      "managedScaling": {
        "instanceWarmupPeriod": number,
        "maximumScalingStepSize": number,
        "minimumScalingStepSize": number,
        "status": "string",
        "targetCapacity": number
      },
      "managedTerminationProtection": "string"
    },
    "capacityProviderArn": "string",
    "cluster": "string",
    "managedInstancesProvider": {
      "infrastructureRoleArn": "string",
      "instanceLaunchTemplate": {
        "ec2InstanceProfileArn": "string",
        "instanceRequirements": {
          "acceleratorCount": {
            "max": number,
            "min": number
          },
          "acceleratorManufacturers": [ "string" ],
          "acceleratorNames": [ "string" ],
          "acceleratorTotalMemoryMiB": {
            "max": number,
            "min": number
          },
          "acceleratorTypes": [ "string" ],
```

```

    "allowedInstanceTypes": [ "string" ],
    "bareMetal": "string",
    "baselineEbsBandwidthMbps": {
        "max": number,
        "min": number
    },
    "burstablePerformance": "string",
    "cpuManufacturers": [ "string" ],
    "excludedInstanceTypes": [ "string" ],
    "instanceGenerations": [ "string" ],
    "localStorage": "string",
    "localStorageTypes": [ "string" ],
    "maxSpotPriceAsPercentageOfOptimalOnDemandPrice": number,
    "memoryGiBPerVCpu": {
        "max": number,
        "min": number
    },
    "memoryMiB": {
        "max": number,
        "min": number
    },
    "networkBandwidthGbps": {
        "max": number,
        "min": number
    },
    "networkInterfaceCount": {
        "max": number,
        "min": number
    },
    "onDemandMaxPricePercentageOverLowestPrice": number,
    "requireHibernateSupport": boolean,
    "spotMaxPricePercentageOverLowestPrice": number,
    "totalLocalStorageGB": {
        "max": number,
        "min": number
    },
    "vCpuCount": {
        "max": number,
        "min": number
    }
},
"monitoring": "string",
"networkConfiguration": {
    "securityGroups": [ "string" ],

```



```
    "subnets": [ "string" ]
  },
  "storageConfiguration": {
    "storageSizeGiB": number
  }
},
"propagateTags": "string"
},
"name": "string",
"status": "string",
"tags": [
  {
    "key": "string",
    "value": "string"
  }
],
"type": "string",
"updateStatus": "string",
"updateStatusReason": "string"
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

capacityProvider

The details of the capacity provider.

Type: [CapacityProvider](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteCluster

Deletes the specified cluster. The cluster transitions to the INACTIVE state. Clusters with an INACTIVE status might remain discoverable in your account for a period of time. However, this behavior is subject to change in the future. We don't recommend that you rely on INACTIVE clusters persisting.

You must deregister all container instances from this cluster before you may delete it. You can list the container instances in a cluster with [ListContainerInstances](#) and deregister them with [DeregisterContainerInstance](#).

Request Syntax

```
{
  "cluster": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster to delete.

Type: String

Required: Yes

Response Syntax

```
{
  "cluster": {
    "activeServicesCount": number,
    "attachments": [
      {
        "details": [
          {
            "name": "string",
```

```

        "value": "string"
      }
    ],
    "id": "string",
    "status": "string",
    "type": "string"
  }
],
"attachmentsStatus": "string",
"capacityProviders": [ "string" ],
"clusterArn": "string",
"clusterName": "string",
"configuration": {
  "executeCommandConfiguration": {
    "kmsKeyId": "string",
    "logConfiguration": {
      "cloudWatchEncryptionEnabled": boolean,
      "cloudWatchLogGroupName": "string",
      "s3BucketName": "string",
      "s3EncryptionEnabled": boolean,
      "s3KeyPrefix": "string"
    },
    "logging": "string"
  },
  "managedStorageConfiguration": {
    "fargateEphemeralStorageKmsKeyId": "string",
    "kmsKeyId": "string"
  }
},
"defaultCapacityProviderStrategy": [
  {
    "base": number,
    "capacityProvider": "string",
    "weight": number
  }
],
"pendingTasksCount": number,
"registeredContainerInstancesCount": number,
"runningTasksCount": number,
"serviceConnectDefaults": {
  "namespace": "string"
},
"settings": [
  {

```

```
        "name": "string",
        "value": "string"
    },
    ],
    "statistics": [
        {
            "name": "string",
            "value": "string"
        }
    ],
    "status": "string",
    "tags": [
        {
            "key": "string",
            "value": "string"
        }
    ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

cluster

The full description of the deleted cluster.

Type: [Cluster](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterContainsCapacityProviderException

The cluster contains one or more capacity providers that prevent the requested operation. This exception occurs when you try to delete a cluster that still has active capacity providers, including Amazon ECS Managed Instances capacity providers. You must first delete all capacity providers from the cluster before you can delete the cluster itself.

HTTP Status Code: 400

ClusterContainsContainerInstancesException

You can't delete a cluster that has registered container instances. First, deregister the container instances before you can delete the cluster. For more information, see [DeregisterContainerInstance](#).

HTTP Status Code: 400

ClusterContainsServicesException

You can't delete a cluster that contains services. First, update the service to reduce its desired task count to 0, and then delete the service. For more information, see [UpdateService](#) and [DeleteService](#).

HTTP Status Code: 400

ClusterContainsTasksException

You can't delete a cluster that has active tasks.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UpdateInProgressException

There's already a current Amazon ECS container agent update in progress on the container instance that's specified. If the container agent becomes disconnected while it's in a transitional stage, such as PENDING or STAGING, the update process can get stuck in that state. However, when the agent reconnects, it resumes where it stopped previously.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request deletes the cluster called My-cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
```



```
Content-Length: 25
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DeleteCluster
X-Amz-Date: 20150429T170952Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "cluster": "My-cluster"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 17:09:54 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 211
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "cluster": {
    "activeServicesCount": 0,
    "clusterArn": "arn:aws:ecs:us-east-1:012345678910:cluster/My-cluster",
    "clusterName": "My-cluster",
    "pendingTasksCount": 0,
    "registeredContainerInstancesCount": 0,
    "runningTasksCount": 0,
    "status": "INACTIVE"
  }
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteService

Deletes a specified service within a cluster. You can delete a service if you have no running tasks in it and the desired task count is zero. If the service is actively maintaining tasks, you can't delete it, and you must update the service to a desired task count of zero. For more information, see [UpdateService](#).

Note

When you delete a service, if there are still running tasks that require cleanup, the service status moves from ACTIVE to DRAINING, and the service is no longer visible in the console or in the [ListServices](#) API operation. After all tasks have transitioned to either STOPPING or STOPPED status, the service status moves from DRAINING to INACTIVE. Services in the DRAINING or INACTIVE status can still be viewed with the [DescribeServices](#) API operation. However, in the future, INACTIVE services may be cleaned up and purged from Amazon ECS record keeping, and [DescribeServices](#) calls on those services return a `ServiceNotFoundException` error.

Important

If you attempt to create a new service with the same name as an existing service in either ACTIVE or DRAINING status, you receive an error.

Request Syntax

```
{
  "cluster": "string",
  "force": boolean,
  "service": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the service to delete. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

force

If `true`, allows you to delete a service even if it wasn't scaled down to zero tasks. It's only necessary to use this if the service uses the REPLICAS scheduling strategy.

Type: Boolean

Required: No

service

The name of the service to delete.

Type: String

Required: Yes

Response Syntax

```
{
  "service": {
    "availabilityZoneRebalancing": "string",
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "clusterArn": "string",
    "createdAt": number,
    "createdBy": "string",
    "deploymentConfiguration": {
      "alarms": {
        "alarmNames": [ "string" ],
        "enable": boolean,
```

```

    "rollback": boolean
  },
  "bakeTimeInMinutes": number,
  "canaryConfiguration": {
    "canaryBakeTimeInMinutes": number,
    "canaryPercent": number
  },
  "deploymentCircuitBreaker": {
    "enable": boolean,
    "rollback": boolean
  },
  "lifecycleHooks": [
    {
      "hookDetails": JSON value,
      "hookTargetArn": "string",
      "lifecycleStages": [ "string" ],
      "roleArn": "string"
    }
  ],
  "linearConfiguration": {
    "stepBakeTimeInMinutes": number,
    "stepPercent": number
  },
  "maximumPercent": number,
  "minimumHealthyPercent": number,
  "strategy": "string"
},
"deploymentController": {
  "type": "string"
},
"deployments": [
  {
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "createdAt": number,
    "desiredCount": number,
    "failedTasks": number,
    "fargateEphemeralStorage": {
      "kmsKeyId": "string"
    }
  }
]

```

```

    },
    "id": "string",
    "launchType": "string",
    "networkConfiguration": {
        "awsVpcConfiguration": {
            "assignPublicIp": "string",
            "securityGroups": [ "string" ],
            "subnets": [ "string" ]
        }
    },
    "pendingCount": number,
    "platformFamily": "string",
    "platformVersion": "string",
    "rolloutState": "string",
    "rolloutStateReason": "string",
    "runningCount": number,
    "serviceConnectConfiguration": {
        "accessLogConfiguration": {
            "format": "string",
            "includeQueryParameters": "string"
        },
        "enabled": boolean,
        "logConfiguration": {
            "logDriver": "string",
            "options": {
                "string" : "string"
            },
            "secretOptions": [
                {
                    "name": "string",
                    "valueFrom": "string"
                }
            ]
        },
        "namespace": "string",
        "services": [
            {
                "clientAliases": [
                    {
                        "dnsName": "string",
                        "port": number,
                        "testTrafficRules": {
                            "header": {
                                "name": "string",

```

```

        "value": {
            "exact": "string"
        }
    }
}
},
"discoveryName": "string",
"ingressPortOverride": number,
"portName": "string",
"timeout": {
    "idleTimeoutSeconds": number,
    "perRequestTimeoutSeconds": number
},
"tls": {
    "issuerCertificateAuthority": {
        "awsPcaAuthorityArn": "string"
    },
    "kmsKey": "string",
    "roleArn": "string"
}
}
],
},
"serviceConnectResources": [
    {
        "discoveryArn": "string",
        "discoveryName": "string"
    }
],
"status": "string",
"taskDefinition": "string",
"updatedAt": number,
"volumeConfigurations": [
    {
        "managedEBSVolume": {
            "encrypted": boolean,
            "filesystemType": "string",
            "iops": number,
            "kmsKeyId": "string",
            "roleArn": "string",
            "sizeInGiB": number,
            "snapshotId": "string",
            "tagSpecifications": [

```

```

        {
            "propagateTags": "string",
            "resourceType": "string",
            "tags": [
                {
                    "key": "string",
                    "value": "string"
                }
            ]
        }
    ],
    "throughput": number,
    "volumeInitializationRate": number,
    "volumeType": "string"
},
"name": "string"
}
],
"vpcLatticeConfigurations": [
    {
        "portName": "string",
        "roleArn": "string",
        "targetGroupArn": "string"
    }
]
}
],
"desiredCount": number,
"enableECSTags": boolean,
"enableExecuteCommand": boolean,
"events": [
    {
        "createdAt": number,
        "id": "string",
        "message": "string"
    }
],
"healthCheckGracePeriodSeconds": number,
"launchType": "string",
"loadBalancers": [
    {
        "advancedConfiguration": {
            "alternateTargetGroupArn": "string",
            "productionListenerRule": "string",

```



```

        "roleArn": "string",
        "testListenerRule": "string"
    },
    "containerName": "string",
    "containerPort": number,
    "loadBalancerName": "string",
    "targetGroupArn": "string"
}
],
"networkConfiguration": {
    "awsvpcConfiguration": {
        "assignPublicIp": "string",
        "securityGroups": [ "string" ],
        "subnets": [ "string" ]
    }
},
"pendingCount": number,
"placementConstraints": [
{
    "expression": "string",
    "type": "string"
}
],
"placementStrategy": [
{
    "field": "string",
    "type": "string"
}
],
"platformFamily": "string",
"platformVersion": "string",
"propagateTags": "string",
"roleArn": "string",
"runningCount": number,
"schedulingStrategy": "string",
"serviceArn": "string",
"serviceName": "string",
"serviceRegistries": [
{
    "containerName": "string",
    "containerPort": number,
    "port": number,
    "registryArn": "string"
}
]

```

```

],
"status": "string",
"tags": [
  {
    "key": "string",
    "value": "string"
  }
],
"taskDefinition": "string",
"taskSets": [
  {
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "clusterArn": "string",
    "computedDesiredCount": number,
    "createdAt": number,
    "externalId": "string",
    "fargateEphemeralStorage": {
      "kmsKeyId": "string"
    },
    "id": "string",
    "launchType": "string",
    "loadBalancers": [
      {
        "advancedConfiguration": {
          "alternateTargetGroupArn": "string",
          "productionListenerRule": "string",
          "roleArn": "string",
          "testListenerRule": "string"
        },
        "containerName": "string",
        "containerPort": number,
        "loadBalancerName": "string",
        "targetGroupArn": "string"
      }
    ],
    "networkConfiguration": {
      "awsvpcConfiguration": {
        "assignPublicIp": "string",

```

```

        "securityGroups": [ "string" ],
        "subnets": [ "string" ]
    },
    "pendingCount": number,
    "platformFamily": "string",
    "platformVersion": "string",
    "runningCount": number,
    "scale": {
        "unit": "string",
        "value": number
    },
    "serviceArn": "string",
    "serviceRegistries": [
        {
            "containerName": "string",
            "containerPort": number,
            "port": number,
            "registryArn": "string"
        }
    ],
    "stabilityStatus": "string",
    "stabilityStatusAt": number,
    "startedBy": "string",
    "status": "string",
    "tags": [
        {
            "key": "string",
            "value": "string"
        }
    ],
    "taskDefinition": "string",
    "taskSetArn": "string",
    "updatedAt": number
}
]
}
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

service

The full description of the deleted service.

Type: [Service](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example API request deletes the test service from the default cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 19
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DeleteService
X-Amz-Date: 20150429T172539Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "service": "test"
}
```

Sample Response

```
HTTP/1.1 200 OK
```

```
Server: Server
Date: Wed, 29 Apr 2015 17:25:40 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 13590
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "service": {
    "clusterArn": "arn:aws:ecs:us-east-1:012345678910:cluster/default",
    "deploymentConfiguration": {
      "maximumPercent": 200,
      "minimumHealthyPercent": 100
    },
    "deployments": [
      {
        "createdAt": 1430320735.285,
        "desiredCount": 0,
        "id": "ecs-svc/9223370606534040511",
        "pendingCount": 0,
        "runningCount": 0,
        "status": "PRIMARY",
        "taskDefinition": "arn:aws:ecs:us-east-1:012345678910:task-definition/sleep360:27",
        "updatedAt": 1430320735.285
      }
    ],
    "desiredCount": 0,
    "events": [],
    "loadBalancers": [],
    "pendingCount": 0,
    "runningCount": 0,
    "serviceArn": "arn:aws:ecs:us-east-1:012345678910:service/default/test",
    "serviceName": "test",
    "status": "DRAINING",
    "taskDefinition": "arn:aws:ecs:us-east-1:012345678910:task-definition/sleep360:27"
  }
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteTaskDefinitions

Deletes one or more task definitions.

You must deregister a task definition revision before you delete it. For more information, see [DeregisterTaskDefinition](#).

When you delete a task definition revision, it is immediately transitions from the INACTIVE to DELETE_IN_PROGRESS. Existing tasks and services that reference a DELETE_IN_PROGRESS task definition revision continue to run without disruption. Existing services that reference a DELETE_IN_PROGRESS task definition revision can still scale up or down by modifying the service's desired count.

You can't use a DELETE_IN_PROGRESS task definition revision to run new tasks or create new services. You also can't update an existing service to reference a DELETE_IN_PROGRESS task definition revision.

A task definition revision will stay in DELETE_IN_PROGRESS status until all the associated tasks and services have been terminated.

When you delete all INACTIVE task definition revisions, the task definition name is not displayed in the console and not returned in the API. If a task definition revisions are in the DELETE_IN_PROGRESS state, the task definition name is displayed in the console and returned in the API. The task definition name is retained by Amazon ECS and the revision is incremented the next time you create a task definition with that name.

Request Syntax

```
{
  "taskDefinitions": [ "string" ]
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

taskDefinitions

The family and revision (family:revision) or full Amazon Resource Name (ARN) of the task definition to delete. You must specify a revision.

You can specify up to 10 task definitions as a comma separated list.

Type: Array of strings

Required: Yes

Response Syntax

```
{
  "failures": [
    {
      "arn": "string",
      "detail": "string",
      "reason": "string"
    }
  ],
  "taskDefinitions": [
    {
      "compatibilities": [ "string" ],
      "containerDefinitions": [
        {
          "command": [ "string" ],
          "cpu": number,
          "credentialSpecs": [ "string" ],
          "dependsOn": [
            {
              "condition": "string",
              "containerName": "string"
            }
          ],
          "disableNetworking": boolean,
          "dnsSearchDomains": [ "string" ],
          "dnsServers": [ "string" ],
          "dockerLabels": {
            "string" : "string"
          },
          "dockerSecurityOptions": [ "string" ],
          "entryPoint": [ "string" ],
```

```
"environment": [  
  {  
    "name": "string",  
    "value": "string"  
  }  
],  
"environmentFiles": [  
  {  
    "type": "string",  
    "value": "string"  
  }  
],  
"essential": boolean,  
"extraHosts": [  
  {  
    "hostname": "string",  
    "ipAddress": "string"  
  }  
],  
"firelensConfiguration": {  
  "options": {  
    "string": "string"  
  },  
  "type": "string"  
},  
"healthCheck": {  
  "command": [ "string" ],  
  "interval": number,  
  "retries": number,  
  "startPeriod": number,  
  "timeout": number  
},  
"hostname": "string",  
"image": "string",  
"interactive": boolean,  
"links": [ "string" ],  
"linuxParameters": {  
  "capabilities": {  
    "add": [ "string" ],  
    "drop": [ "string" ]  
  },  
  "devices": [  
    {  
      "containerPath": "string",
```

```

        "hostPath": "string",
        "permissions": [ "string" ]
    }
],
"initProcessEnabled": boolean,
"maxSwap": number,
"sharedMemorySize": number,
"swappiness": number,
"tmpfs": [
    {
        "containerPath": "string",
        "mountOptions": [ "string" ],
        "size": number
    }
]
},
"logConfiguration": {
    "logDriver": "string",
    "options": {
        "string" : "string"
    },
    "secretOptions": [
        {
            "name": "string",
            "valueFrom": "string"
        }
    ]
},
"memory": number,
"memoryReservation": number,
"mountPoints": [
    {
        "containerPath": "string",
        "readOnly": boolean,
        "sourceVolume": "string"
    }
],
"name": "string",
"portMappings": [
    {
        "appProtocol": "string",
        "containerPort": number,
        "containerPortRange": "string",
        "hostPort": number,

```

```

        "name": "string",
        "protocol": "string"
    }
],
"privileged": boolean,
"pseudoTerminal": boolean,
"readonlyRootFilesystem": boolean,
"repositoryCredentials": {
    "credentialsParameter": "string"
},
"resourceRequirements": [
    {
        "type": "string",
        "value": "string"
    }
],
"restartPolicy": {
    "enabled": boolean,
    "ignoredExitCodes": [ number ],
    "restartAttemptPeriod": number
},
"secrets": [
    {
        "name": "string",
        "valueFrom": "string"
    }
],
"startTimeout": number,
"stopTimeout": number,
"systemControls": [
    {
        "namespace": "string",
        "value": "string"
    }
],
"ulimits": [
    {
        "hardLimit": number,
        "name": "string",
        "softLimit": number
    }
],
"user": "string",
"versionConsistency": "string",

```

```

        "volumesFrom": [
            {
                "readOnly": boolean,
                "sourceContainer": "string"
            }
        ],
        "workingDirectory": "string"
    }
],
"cpu": "string",
"deregisteredAt": number,
"enableFaultInjection": boolean,
"ephemeralStorage": {
    "sizeInGiB": number
},
"executionRoleArn": "string",
"family": "string",
"inferenceAccelerators": [
    {
        "deviceName": "string",
        "deviceType": "string"
    }
],
"ipcMode": "string",
"memory": "string",
"networkMode": "string",
"pidMode": "string",
"placementConstraints": [
    {
        "expression": "string",
        "type": "string"
    }
],
"proxyConfiguration": {
    "containerName": "string",
    "properties": [
        {
            "name": "string",
            "value": "string"
        }
    ],
    "type": "string"
},
"registeredAt": number,

```

```

    "registeredBy": "string",
    "requiresAttributes": [
      {
        "name": "string",
        "targetId": "string",
        "targetType": "string",
        "value": "string"
      }
    ],
    "requiresCompatibilities": [ "string" ],
    "revision": number,
    "runtimePlatform": {
      "cpuArchitecture": "string",
      "operatingSystemFamily": "string"
    },
    "status": "string",
    "taskDefinitionArn": "string",
    "taskRoleArn": "string",
    "volumes": [
      {
        "configuredAtLaunch": boolean,
        "dockerVolumeConfiguration": {
          "autoprovision": boolean,
          "driver": "string",
          "driverOpts": {
            "string" : "string"
          },
          "labels": {
            "string" : "string"
          },
          "scope": "string"
        },
        "efsVolumeConfiguration": {
          "authorizationConfig": {
            "accessPointId": "string",
            "iam": "string"
          },
          "fileSystemId": "string",
          "rootDirectory": "string",
          "transitEncryption": "string",
          "transitEncryptionPort": number
        },
        "fsxWindowsFileServerVolumeConfiguration": {
          "authorizationConfig": {

```

```
        "credentialsParameter": "string",
        "domain": "string"
    },
    "fileSystemId": "string",
    "rootDirectory": "string"
},
"host": {
    "sourcePath": "string"
},
"name": "string"
}
]
}
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

taskDefinitions

The list of deleted task definitions.

Type: Array of [TaskDefinition](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request deletes the task definition named `Example-task-definition:1`.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 25
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DeleteTaskDefinitions
X-Amz-Date: 20150429T170952Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "taskDefinitions": [
    "Example-task-definition:1"
  ]
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 7 Dec 2022 17:09:54 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 211
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "failures": [],
  "taskDefinitions": [
    {
      "containerDefinitions": [
        {
          "command": [
            "apt-get update; apt-get install stress; while true; do stress --cpu"
            "$(( RANDOM % 4 )) -t $(( RANDOM % 10 )); done"
          ],
          "cpu": 50,
          "entryPoint": [
            "bash",
            "-c"
          ],
          "environment": [],
```

```
        "essential": true,
        "image": "public.ecr.aws/docker/library/ubuntu:latest",
        "memory": 100,
        "mountPoints": [],
        "name": "wave",
        "portMappings": [],
        "volumesFrom": []
    }
],
"family": "cpu-wave",
"revision": 1,
"status": "DELETE_IN_PROGRESS",
"taskDefinitionArn": "arn:aws:ecs:us-east-1:012345678910:task-definition/Example-
task-definition:1",
"volumes": []
}
]
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeleteTaskSet

Deletes a specified task set within a service. This is used when a service uses the EXTERNAL deployment controller type. For more information, see [Amazon ECS deployment types](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{  
  "cluster": "string",  
  "force": boolean,  
  "service": "string",  
  "taskSet": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the service that the task set found in to delete.

Type: String

Required: Yes

force

If `true`, you can delete a task set even if it hasn't been scaled down to zero.

Type: Boolean

Required: No

service

The short name or full Amazon Resource Name (ARN) of the service that hosts the task set to delete.

Type: String

Required: Yes

taskSet

The task set ID or full Amazon Resource Name (ARN) of the task set to delete.

Type: String

Required: Yes

Response Syntax

```
{
  "taskSet": {
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "clusterArn": "string",
    "computedDesiredCount": number,
    "createdAt": number,
    "externalId": "string",
    "fargateEphemeralStorage": {
      "kmsKeyId": "string"
    },
    "id": "string",
    "launchType": "string",
    "loadBalancers": [
      {
        "advancedConfiguration": {
          "alternateTargetGroupArn": "string",
          "productionListenerRule": "string",
          "roleArn": "string",
          "testListenerRule": "string"
        },
        "containerName": "string",
        "containerPort": number,
        "loadBalancerName": "string",
        "targetGroupArn": "string"
      }
    ]
  }
}
```

```
],
  "networkConfiguration": {
    "awsVpcConfiguration": {
      "assignPublicIp": "string",
      "securityGroups": [ "string" ],
      "subnets": [ "string" ]
    }
  },
  "pendingCount": number,
  "platformFamily": "string",
  "platformVersion": "string",
  "runningCount": number,
  "scale": {
    "unit": "string",
    "value": number
  },
  "serviceArn": "string",
  "serviceRegistries": [
    {
      "containerName": "string",
      "containerPort": number,
      "port": number,
      "registryArn": "string"
    }
  ],
  "stabilityStatus": "string",
  "stabilityStatusAt": number,
  "startedBy": "string",
  "status": "string",
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ],
  "taskDefinition": "string",
  "taskSetArn": "string",
  "updatedAt": number
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

taskSet

Details about the task set.

Type: [TaskSet](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotActiveException

The specified service isn't active. You can't update a service that's inactive. If you have previously deleted a service, you can re-create it with [CreateService](#).

HTTP Status Code: 400

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

TaskSetNotFoundException

The specified task set wasn't found. You can view your available task sets with [DescribeTaskSets](#). Task sets are specific to each cluster, service and Region.

HTTP Status Code: 400

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeregisterContainerInstance

Deregisters an Amazon ECS container instance from the specified cluster. This instance is no longer available to run tasks.

If you intend to use the container instance for some other purpose after deregistration, we recommend that you stop all of the tasks running on the container instance before deregistration. That prevents any orphaned tasks from consuming resources.

Deregistering a container instance removes the instance from a cluster, but it doesn't terminate the EC2 instance. If you are finished using the instance, be sure to terminate it in the Amazon EC2 console to stop billing.

Note

If you terminate a running container instance, Amazon ECS automatically deregisters the instance from your cluster (stopped container instances or instances with disconnected agents aren't automatically deregistered when terminated).

Request Syntax

```
{  
  "cluster": "string",  
  "containerInstance": "string",  
  "force": boolean  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the container instance to deregister. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

containerInstance

The container instance ID or full ARN of the container instance to deregister. For more information about the ARN format, see [Amazon Resource Name \(ARN\)](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: Yes

force

Forces the container instance to be deregistered. If you have tasks running on the container instance when you deregister it with the `force` option, these tasks remain running until you terminate the instance or the tasks stop through some other means, but they're orphaned (no longer monitored or accounted for by Amazon ECS). If an orphaned task on your container instance is part of an Amazon ECS service, then the service scheduler starts another copy of that task, on a different container instance if possible.

Any containers in orphaned service tasks that are registered with a Classic Load Balancer or an Application Load Balancer target group are deregistered. They begin connection draining according to the settings on the load balancer or target group.

Type: Boolean

Required: No

Response Syntax

```
{
  "containerInstance": {
    "agentConnected": boolean,
    "agentUpdateStatus": "string",
    "attachments": [
      {
        "details": [
          {
            "name": "string",
            "value": "string"
          }
        ]
      }
    ],
  },
}
```

```

        "id": "string",
        "status": "string",
        "type": "string"
    }
],
"attributes": [
    {
        "name": "string",
        "targetId": "string",
        "targetType": "string",
        "value": "string"
    }
],
"capacityProviderName": "string",
"containerInstanceArn": "string",
"ec2InstanceId": "string",
"healthStatus": {
    "details": [
        {
            "lastStatusChange": number,
            "lastUpdated": number,
            "status": "string",
            "type": "string"
        }
    ],
    "overallStatus": "string"
},
"pendingTasksCount": number,
"registeredAt": number,
"registeredResources": [
    {
        "doubleValue": number,
        "integerValue": number,
        "longValue": number,
        "name": "string",
        "stringSetValue": [ "string" ],
        "type": "string"
    }
],
"remainingResources": [
    {
        "doubleValue": number,
        "integerValue": number,
        "longValue": number,

```

```
        "name": "string",
        "stringValue": [ "string" ],
        "type": "string"
    }
],
"runningTasksCount": number,
"status": "string",
"statusReason": "string",
"tags": [
    {
        "key": "string",
        "value": "string"
    }
],
"version": number,
"versionInfo": {
    "agentHash": "string",
    "agentVersion": "string",
    "dockerVersion": "string"
}
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

containerInstance

The container instance that was deregistered.

Type: [ContainerInstance](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS,

these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request deregisters a container instance with the ID `f4292606-fbed-4b53-833b-92cad7c687c2` in the default cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 61
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DeregisterContainerInstance
X-Amz-Date: 20151001T191224Z
User-Agent: aws-cli/1.8.7 Python/2.7.9 Darwin/14.5.0
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "containerInstance": "c9c9a6f2-8766-464b-8805-9c57b9368fb0"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Thu, 01 Oct 2015 19:12:25 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 1613
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "containerInstance": {
    "agentConnected": true,
    "attributes": [
      {
        "name": "com.amazonaws.ecs.capability.privileged-container"
      },
      {
        "name": "com.amazonaws.ecs.capability.docker-remote-api.1.17"
      }
    ]
  }
}
```

```
    },
    {
      "name": "com.amazonaws.ecs.capability.docker-remote-api.1.18"
    },
    {
      "name": "com.amazonaws.ecs.capability.docker-remote-api.1.19"
    },
    {
      "name": "com.amazonaws.ecs.capability.logging-driver.json-file"
    },
    {
      "name": "com.amazonaws.ecs.capability.logging-driver.syslog"
    }
  ],
  "containerInstanceArn": "arn:aws:ecs:us-west-2:012345678910:container-instance/default/c9c9a6f2-8766-464b-8805-9c57b9368fb0",
  "ec2InstanceId": "i-0c3826c9",
  "pendingTasksCount": 0,
  "registeredResources": [
    {
      "doubleValue": 0,
      "integerValue": 1024,
      "longValue": 0,
      "name": "CPU",
      "type": "INTEGER"
    },
    {
      "doubleValue": 0,
      "integerValue": 995,
      "longValue": 0,
      "name": "MEMORY",
      "type": "INTEGER"
    },
    {
      "doubleValue": 0,
      "integerValue": 0,
      "longValue": 0,
      "name": "PORTS",
      "stringSetValue": [
        "22",
        "2376",
        "2375",
        "51678"
      ]
    }
  ],
```

```
    "type": "STRINGSET"
  },
  {
    "doubleValue": 0,
    "integerValue": 0,
    "longValue": 0,
    "name": "PORTS_UDP",
    "stringSetValue": [],
    "type": "STRINGSET"
  }
],
"remainingResources": [
  {
    "doubleValue": 0,
    "integerValue": 1024,
    "longValue": 0,
    "name": "CPU",
    "type": "INTEGER"
  },
  {
    "doubleValue": 0,
    "integerValue": 995,
    "longValue": 0,
    "name": "MEMORY",
    "type": "INTEGER"
  },
  {
    "doubleValue": 0,
    "integerValue": 0,
    "longValue": 0,
    "name": "PORTS",
    "stringSetValue": [
      "22",
      "2376",
      "2375",
      "51678"
    ],
    "type": "STRINGSET"
  },
  {
    "doubleValue": 0,
    "integerValue": 0,
    "longValue": 0,
    "name": "PORTS_UDP",
```



```
        "stringValue": [],
        "type": "STRINGSET"
    }
],
"runningTasksCount": 0,
"status": "INACTIVE",
"versionInfo": {
    "agentHash": "b197edd",
    "agentVersion": "1.5.0",
    "dockerVersion": "DockerVersion: 1.7.1"
}
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DeregisterTaskDefinition

Deregisters the specified task definition by family and revision. Upon deregistration, the task definition is marked as INACTIVE. Existing tasks and services that reference an INACTIVE task definition continue to run without disruption. Existing services that reference an INACTIVE task definition can still scale up or down by modifying the service's desired count. If you want to delete a task definition revision, you must first deregister the task definition revision.

You can't use an INACTIVE task definition to run new tasks or create new services, and you can't update an existing service to reference an INACTIVE task definition. However, there may be up to a 10-minute window following deregistration where these restrictions have not yet taken effect.

Note

At this time, INACTIVE task definitions remain discoverable in your account indefinitely. However, this behavior is subject to change in the future. We don't recommend that you rely on INACTIVE task definitions persisting beyond the lifecycle of any associated tasks and services.

You must deregister a task definition revision before you delete it. For more information, see [DeleteTaskDefinitions](#).

Request Syntax

```
{
  "taskDefinition": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[taskDefinition](#)

The family and revision (family:revision) or full Amazon Resource Name (ARN) of the task definition to deregister. You must specify a revision.

Type: String

Required: Yes

Response Syntax

```
{
  "taskDefinition": {
    "compatibilities": [ "string" ],
    "containerDefinitions": [
      {
        "command": [ "string" ],
        "cpu": number,
        "credentialSpecs": [ "string" ],
        "dependsOn": [
          {
            "condition": "string",
            "containerName": "string"
          }
        ],
        "disableNetworking": boolean,
        "dnsSearchDomains": [ "string" ],
        "dnsServers": [ "string" ],
        "dockerLabels": {
          "string" : "string"
        },
        "dockerSecurityOptions": [ "string" ],
        "entryPoint": [ "string" ],
        "environment": [
          {
            "name": "string",
            "value": "string"
          }
        ],
        "environmentFiles": [
          {
            "type": "string",
            "value": "string"
          }
        ],
        "essential": boolean,
        "extraHosts": [
          {
```

```

        "hostname": "string",
        "ipAddress": "string"
    }
],
"firelensConfiguration": {
    "options": {
        "string" : "string"
    },
    "type": "string"
},
"healthCheck": {
    "command": [ "string" ],
    "interval": number,
    "retries": number,
    "startPeriod": number,
    "timeout": number
},
"hostname": "string",
"image": "string",
"interactive": boolean,
"links": [ "string" ],
"linuxParameters": {
    "capabilities": {
        "add": [ "string" ],
        "drop": [ "string" ]
    },
    "devices": [
        {
            "containerPath": "string",
            "hostPath": "string",
            "permissions": [ "string" ]
        }
    ],
    "initProcessEnabled": boolean,
    "maxSwap": number,
    "sharedMemorySize": number,
    "swappiness": number,
    "tmpfs": [
        {
            "containerPath": "string",
            "mountOptions": [ "string" ],
            "size": number
        }
    ]
}
]
```

```

    },
    "logConfiguration": {
      "logDriver": "string",
      "options": {
        "string" : "string"
      },
      "secretOptions": [
        {
          "name": "string",
          "valueFrom": "string"
        }
      ]
    },
    "memory": number,
    "memoryReservation": number,
    "mountPoints": [
      {
        "containerPath": "string",
        "readOnly": boolean,
        "sourceVolume": "string"
      }
    ],
    "name": "string",
    "portMappings": [
      {
        "appProtocol": "string",
        "containerPort": number,
        "containerPortRange": "string",
        "hostPort": number,
        "name": "string",
        "protocol": "string"
      }
    ],
    "privileged": boolean,
    "pseudoTerminal": boolean,
    "readonlyRootFilesystem": boolean,
    "repositoryCredentials": {
      "credentialsParameter": "string"
    },
    "resourceRequirements": [
      {
        "type": "string",
        "value": "string"
      }
    ]
  }

```

```

    ],
    "restartPolicy": {
      "enabled": boolean,
      "ignoredExitCodes": [ number ],
      "restartAttemptPeriod": number
    },
    "secrets": [
      {
        "name": "string",
        "valueFrom": "string"
      }
    ],
    "startTimeout": number,
    "stopTimeout": number,
    "systemControls": [
      {
        "namespace": "string",
        "value": "string"
      }
    ],
    "ulimits": [
      {
        "hardLimit": number,
        "name": "string",
        "softLimit": number
      }
    ],
    "user": "string",
    "versionConsistency": "string",
    "volumesFrom": [
      {
        "readOnly": boolean,
        "sourceContainer": "string"
      }
    ],
    "workingDirectory": "string"
  }
],
"cpu": "string",
"deregisteredAt": number,
"enableFaultInjection": boolean,
"ephemeralStorage": {
  "sizeInGiB": number
},

```

```
"executionRoleArn": "string",
"family": "string",
"inferenceAccelerators": [
  {
    "deviceName": "string",
    "deviceType": "string"
  }
],
"ipcMode": "string",
"memory": "string",
"networkMode": "string",
"pidMode": "string",
"placementConstraints": [
  {
    "expression": "string",
    "type": "string"
  }
],
"proxyConfiguration": {
  "containerName": "string",
  "properties": [
    {
      "name": "string",
      "value": "string"
    }
  ],
  "type": "string"
},
"registeredAt": number,
"registeredBy": "string",
"requiresAttributes": [
  {
    "name": "string",
    "targetId": "string",
    "targetType": "string",
    "value": "string"
  }
],
"requiresCompatibilities": [ "string" ],
"revision": number,
"runtimePlatform": {
  "cpuArchitecture": "string",
  "operatingSystemFamily": "string"
},
```

```

    "status": "string",
    "taskDefinitionArn": "string",
    "taskRoleArn": "string",
    "volumes": [
      {
        "configuredAtLaunch": boolean,
        "dockerVolumeConfiguration": {
          "autoprovision": boolean,
          "driver": "string",
          "driverOpts": {
            "string" : "string"
          },
          "labels": {
            "string" : "string"
          },
          "scope": "string"
        },
        "efsVolumeConfiguration": {
          "authorizationConfig": {
            "accessPointId": "string",
            "iam": "string"
          },
          "fileSystemId": "string",
          "rootDirectory": "string",
          "transitEncryption": "string",
          "transitEncryptionPort": number
        },
        "fsxWindowsFileServerVolumeConfiguration": {
          "authorizationConfig": {
            "credentialsParameter": "string",
            "domain": "string"
          },
          "fileSystemId": "string",
          "rootDirectory": "string"
        },
        "host": {
          "sourcePath": "string"
        },
        "name": "string"
      }
    ]
  }
}

```


Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

taskDefinition

The full description of the deregistered task.

Type: [TaskDefinition](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

The following example request deregisters the first revision of the cpu-wave task definition family (cpu-wave:1). In the resulting output, the task definition status becomes INACTIVE.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 35
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DeregisterTaskDefinition
X-Amz-Date: 20150429T184806Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "taskDefinition": "cpu-wave:1"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Fri, 12 Jun 2015 23:07:39 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 491
Connection: keep-alive
```

```
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "taskDefinition": {
    "containerDefinitions": [
      {
        "command": [
          "apt-get update; apt-get install stress; while true; do stress --cpu"
          "$(( RANDOM % 4 )) -t $(( RANDOM % 10 )); done"
        ],
        "cpu": 50,
        "entryPoint": [
          "bash",
          "-c"
        ],
        "environment": [],
        "essential": true,
        "image": "public.ecr.aws/docker/library/ubuntu:latest",
        "memory": 100,
        "mountPoints": [],
        "name": "wave",
        "portMappings": [],
        "volumesFrom": []
      }
    ],
    "family": "cpu-wave",
    "revision": 1,
    "status": "INACTIVE",
    "taskDefinitionArn": "arn:aws:ecs:us-west-2:012345678910:task-definition/cpu-wave:1",
    "volumes": []
  }
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)

- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeCapacityProviders

Describes one or more of your capacity providers.

Request Syntax

```
{  
  "capacityProviders": [ "string" ],  
  "cluster": "string",  
  "include": [ "string" ],  
  "maxResults": number,  
  "nextToken": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

capacityProviders

The short name or full Amazon Resource Name (ARN) of one or more capacity providers. Up to 100 capacity providers can be described in an action.

Type: Array of strings

Required: No

cluster

The name of the cluster to describe capacity providers for. When specified, only capacity providers associated with this cluster are returned, including Amazon ECS Managed Instances capacity providers.

Type: String

Required: No

include

Specifies whether or not you want to see the resource tags for the capacity provider. If TAGS is specified, the tags are included in the response. If this field is omitted, tags aren't included in the response.

Type: Array of strings

Valid Values: TAGS

Required: No

maxResults

The maximum number of account setting results returned by `DescribeCapacityProviders` in paginated output. When this parameter is used, `DescribeCapacityProviders` only returns `maxResults` results in a single page along with a `nextToken` response element. The remaining results of the initial request can be seen by sending another `DescribeCapacityProviders` request with the returned `nextToken` value. This value can be between 1 and 10. If this parameter is not used, then `DescribeCapacityProviders` returns up to 10 results and a `nextToken` value if applicable.

Type: Integer

Required: No

nextToken

The `nextToken` value returned from a previous paginated `DescribeCapacityProviders` request where `maxResults` was used and the results exceeded the value of that parameter. Pagination continues from the end of the previous results that returned the `nextToken` value.

Note

This token should be treated as an opaque identifier that is only used to retrieve the next items in a list and not for other programmatic purposes.

Type: String

Required: No

Response Syntax

```
{
  "capacityProviders": [
    {
      "autoScalingGroupProvider": {
```

```

    "autoScalingGroupArn": "string",
    "managedDraining": "string",
    "managedScaling": {
        "instanceWarmupPeriod": number,
        "maximumScalingStepSize": number,
        "minimumScalingStepSize": number,
        "status": "string",
        "targetCapacity": number
    },
    "managedTerminationProtection": "string"
},
"capacityProviderArn": "string",
"cluster": "string",
"managedInstancesProvider": {
    "infrastructureRoleArn": "string",
    "instanceLaunchTemplate": {
        "ec2InstanceProfileArn": "string",
        "instanceRequirements": {
            "acceleratorCount": {
                "max": number,
                "min": number
            },
            "acceleratorManufacturers": [ "string" ],
            "acceleratorNames": [ "string" ],
            "acceleratorTotalMemoryMiB": {
                "max": number,
                "min": number
            },
            "acceleratorTypes": [ "string" ],
            "allowedInstanceTypes": [ "string" ],
            "bareMetal": "string",
            "baselineEbsBandwidthMbps": {
                "max": number,
                "min": number
            },
            "burstablePerformance": "string",
            "cpuManufacturers": [ "string" ],
            "excludedInstanceTypes": [ "string" ],
            "instanceGenerations": [ "string" ],
            "localStorage": "string",
            "localStorageTypes": [ "string" ],
            "maxSpotPriceAsPercentageOfOptimalOnDemandPrice": number,
            "memoryGiBPerVCpu": {
                "max": number,

```

```

        "min": number
    },
    "memoryMiB": {
        "max": number,
        "min": number
    },
    "networkBandwidthGbps": {
        "max": number,
        "min": number
    },
    "networkInterfaceCount": {
        "max": number,
        "min": number
    },
    "onDemandMaxPricePercentageOverLowestPrice": number,
    "requireHibernateSupport": boolean,
    "spotMaxPricePercentageOverLowestPrice": number,
    "totalLocalStorageGB": {
        "max": number,
        "min": number
    },
    "vCpuCount": {
        "max": number,
        "min": number
    }
},
"monitoring": "string",
"networkConfiguration": {
    "securityGroups": [ "string" ],
    "subnets": [ "string" ]
},
"storageConfiguration": {
    "storageSizeGiB": number
}
},
"propagateTags": "string"
},
"name": "string",
"status": "string",
"tags": [
    {
        "key": "string",
        "value": "string"
    }
]

```



```
    ],
    "type": "string",
    "updateStatus": "string",
    "updateStatusReason": "string"
  }
],
"failures": [
  {
    "arn": "string",
    "detail": "string",
    "reason": "string"
  }
],
"nextToken": "string"
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

capacityProviders

The list of capacity providers.

Type: Array of [CapacityProvider](#) objects

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

nextToken

The nextToken value to include in a future DescribeCapacityProviders request. When the results of a DescribeCapacityProviders request exceed maxResults, this value can be used to retrieve the next page of results. This value is null when there are no more results to return.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeClusters

Describes one or more of your clusters.

For AWS CLI examples, see [describe-clusters.rst](#) on GitHub.

Request Syntax

```
{  
  "clusters": [ "string" ],  
  "include": [ "string" ]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[clusters](#)

A list of up to 100 cluster names or full cluster Amazon Resource Name (ARN) entries. If you do not specify a cluster, the default cluster is assumed.

Type: Array of strings

Required: No

[include](#)

Determines whether to include additional information about the clusters in the response. If this field is omitted, this information isn't included.

If ATTACHMENTS is specified, the attachments for the container instances or tasks within the cluster are included, for example the capacity providers.

If SETTINGS is specified, the settings for the cluster are included.

If CONFIGURATIONS is specified, the configuration for the cluster is included.

If STATISTICS is specified, the task and service count is included, separated by launch type.

If TAGS is specified, the metadata tags associated with the cluster are included.

Type: Array of strings

Valid Values: ATTACHMENTS | CONFIGURATIONS | SETTINGS | STATISTICS | TAGS

Required: No

Response Syntax

```
{
  "clusters": [
    {
      "activeServicesCount": number,
      "attachments": [
        {
          "details": [
            {
              "name": "string",
              "value": "string"
            }
          ],
          "id": "string",
          "status": "string",
          "type": "string"
        }
      ],
      "attachmentsStatus": "string",
      "capacityProviders": [ "string" ],
      "clusterArn": "string",
      "clusterName": "string",
      "configuration": {
        "executeCommandConfiguration": {
          "kmsKeyId": "string",
          "logConfiguration": {
            "cloudWatchEncryptionEnabled": boolean,
            "cloudWatchLogGroupName": "string",
            "s3BucketName": "string",
            "s3EncryptionEnabled": boolean,
            "s3KeyPrefix": "string"
          },
          "logging": "string"
        },
        "managedStorageConfiguration": {
          "fargateEphemeralStorageKmsKeyId": "string",
```

```

        "kmsKeyId": "string"
    },
    "defaultCapacityProviderStrategy": [
        {
            "base": number,
            "capacityProvider": "string",
            "weight": number
        }
    ],
    "pendingTasksCount": number,
    "registeredContainerInstancesCount": number,
    "runningTasksCount": number,
    "serviceConnectDefaults": {
        "namespace": "string"
    },
    "settings": [
        {
            "name": "string",
            "value": "string"
        }
    ],
    "statistics": [
        {
            "name": "string",
            "value": "string"
        }
    ],
    "status": "string",
    "tags": [
        {
            "key": "string",
            "value": "string"
        }
    ]
}
],
"failures": [
    {
        "arn": "string",
        "detail": "string",
        "reason": "string"
    }
]

```

```
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

clusters

The list of clusters.

Type: Array of [Cluster](#) objects

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request provides descriptive information about the default cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 25
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DescribeClusters
X-Amz-Date: 20150429T185014Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "clusters": [
    "default"
  ]
}
```


Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 18:50:14 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 220
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "clusters": [
    {
      "activeServicesCount": 1,
      "clusterArn": "arn:aws:ecs:us-east-1:012345678910:cluster/default",
      "clusterName": "default",
      "pendingTasksCount": 0,
      "registeredContainerInstancesCount": 0,
      "runningTasksCount": 0,
      "status": "ACTIVE"
    }
  ],
  "failures": []
}
```

Example

This example request provides descriptive information about a cluster that has an associated capacity provider with managed scaling turned on. The capacity provider and the scaling policy that Amazon ECS creates on your behalf are included as attachments in the response.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 25
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DescribeClusters
X-Amz-Date: 20220429T185014Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
```

```
"clusters": [  
  "cluster-with-capacity-provider",  
  "include": ["ATTACHMENTS"]  
]  
}
```

Sample Response

```
HTTP/1.1 200 OK  
Server: Server  
Date: Wed, 29 Apr 2022 18:50:14 GMT  
Content-Type: application/x-amz-json-1.1  
Content-Length: 220  
Connection: keep-alive  
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f  
  
{  
  "clusters": [  
    {  
      "clusterArn": "arn:aws:ecs:us-west-2:012345678910:cluster/cluster-with-capacity-provider",  
      "clusterName": "cluster-with-capacity-provider",  
      "status": "ACTIVE",  
      "registeredContainerInstancesCount": 0,  
      "runningTasksCount": 0,  
      "pendingTasksCount": 0,  
      "activeServicesCount": 0,  
      "statistics": [],  
      "tags": [],  
      "settings": [],  
      "capacityProviders": [  
        "FARGATE",  
        "FARGATE_SPOT",  
        "Infra-ECS-Cluster-45ff394c-615f-4496-a336-8019a3b38d7a-EC2CapacityProvider-EXAMPLE"  
      ],  
      "defaultCapacityProviderStrategy": [],  
      "attachments": [  
        {  
          "id": "bf5ecfe0-f813-4356-b862-1dbe6025c57f",  
          "type": "as_policy",  
          "status": "CREATED",  
          "details": [  

```

```
        {
            "name": "capacityProviderName",
            "value": "Infra-ECS-Cluster-45ff394c-615f-4496-
a336-8019a3b38d7a-EC2CapacityProvider-EXAMPLE"
        },
        {
            "name": "scalingPolicyName",
            "value": "ECSManagedAutoScalingPolicy-55c67464-7691-4310-
a725-EXAMPLE"
        }
    ],
    "attachmentsStatus": "UPDATE_COMPLETE"
},
],
"failures": []
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeContainerInstances

Describes one or more container instances. Returns metadata about each container instance requested.

Request Syntax

```
{  
  "cluster": "string",  
  "containerInstances": [ "string" ],  
  "include": [ "string" ]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the container instances to describe. If you do not specify a cluster, the default cluster is assumed. This parameter is required if the container instance or container instances you are describing were launched in any cluster other than the default cluster.

Type: String

Required: No

containerInstances

A list of up to 100 container instance IDs or full Amazon Resource Name (ARN) entries.

Type: Array of strings

Required: Yes

include

Specifies whether you want to see the resource tags for the container instance. If TAGS is specified, the tags are included in the response. If CONTAINER_INSTANCE_HEALTH is specified,

the container instance health is included in the response. If this field is omitted, tags and container instance health status aren't included in the response.

Type: Array of strings

Valid Values: TAGS | CONTAINER_INSTANCE_HEALTH

Required: No

Response Syntax

```
{
  "containerInstances": [
    {
      "agentConnected": boolean,
      "agentUpdateStatus": "string",
      "attachments": [
        {
          "details": [
            {
              "name": "string",
              "value": "string"
            }
          ],
          "id": "string",
          "status": "string",
          "type": "string"
        }
      ],
      "attributes": [
        {
          "name": "string",
          "targetId": "string",
          "targetType": "string",
          "value": "string"
        }
      ],
      "capacityProviderName": "string",
      "containerInstanceArn": "string",
      "ec2InstanceId": "string",
      "healthStatus": {
        "details": [
          {
```

```

        "lastStatusChange": number,
        "lastUpdated": number,
        "status": "string",
        "type": "string"
    }
],
    "overallStatus": "string"
},
    "pendingTasksCount": number,
    "registeredAt": number,
    "registeredResources": [
        {
            "doubleValue": number,
            "integerValue": number,
            "longValue": number,
            "name": "string",
            "stringSetValue": [ "string" ],
            "type": "string"
        }
    ],
    "remainingResources": [
        {
            "doubleValue": number,
            "integerValue": number,
            "longValue": number,
            "name": "string",
            "stringSetValue": [ "string" ],
            "type": "string"
        }
    ],
    "runningTasksCount": number,
    "status": "string",
    "statusReason": "string",
    "tags": [
        {
            "key": "string",
            "value": "string"
        }
    ],
    "version": number,
    "versionInfo": {
        "agentHash": "string",
        "agentVersion": "string",
        "dockerVersion": "string"
    }
}

```

```
    }
  }
],
"failures": [
  {
    "arn": "string",
    "detail": "string",
    "reason": "string"
  }
]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

containerInstances

The list of container instances.

Type: Array of [ContainerInstance](#) objects

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request provides descriptive information about a container instance with an ID of f9cc75bb-0c94-46b9-bf6d-49d320bc1551 in the default cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 64
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DescribeContainerInstances
X-Amz-Date: 20160520T171518Z
User-Agent: aws-cli/1.10.30 Python/2.7.11 Darwin/15.4.0 botocore/1.4.17
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
```

```
{
  "containerInstances": [
    "f9cc75bb-0c94-46b9-bf6d-49d320bc1551"
  ]
}
```

Sample Response

```
{
  "containerInstances": [
    {
      "agentConnected": true,
      "attributes": [
        {
          "name": "com.amazonaws.ecs.capability.privileged-container"
        },
        {
          "name": "com.amazonaws.ecs.capability.docker-remote-api.1.17"
        },
        {
          "name": "com.amazonaws.ecs.capability.docker-remote-api.1.18"
        },
        {
          "name": "com.amazonaws.ecs.capability.docker-remote-api.1.19"
        },
        {
          "name": "com.amazonaws.ecs.capability.docker-remote-api.1.20"
        },
        {
          "name": "com.amazonaws.ecs.capability.docker-remote-api.1.21"
        }
      ]
    }
  ]
}
```

```
{
  "name": "com.amazonaws.ecs.capability.logging-driver.json-file"
},
{
  "name": "com.amazonaws.ecs.capability.logging-driver.syslog"
},
{
  "name": "com.amazonaws.ecs.capability.logging-driver.awslogs"
},
{
  "name": "com.amazonaws.ecs.capability.ecr-auth"
}
],
"containerInstanceArn": "arn:aws:ecs:us-west-2:012345678910:container-
instance/default/f9cc75bb-0c94-46b9-bf6d-49d320bc1551",
"ec2InstanceId": "i-042f39dc",
"pendingTasksCount": 0,
"registeredResources": [
  {
    "doubleValue": 0,
    "integerValue": 1024,
    "longValue": 0,
    "name": "CPU",
    "type": "INTEGER"
  },
  {
    "doubleValue": 0,
    "integerValue": 995,
    "longValue": 0,
    "name": "MEMORY",
    "type": "INTEGER"
  },
  {
    "doubleValue": 0,
    "integerValue": 0,
    "longValue": 0,
    "name": "PORTS",
    "stringSetValue": [
      "22",
      "2376",
      "2375",
      "51678"
    ],
    "type": "STRINGSET"
  }
]
```

```
    },
    {
      "doubleValue": 0,
      "integerValue": 0,
      "longValue": 0,
      "name": "PORTS_UDP",
      "stringSetValue": [],
      "type": "STRINGSET"
    }
  ],
  "remainingResources": [
    {
      "doubleValue": 0,
      "integerValue": 1024,
      "longValue": 0,
      "name": "CPU",
      "type": "INTEGER"
    },
    {
      "doubleValue": 0,
      "integerValue": 995,
      "longValue": 0,
      "name": "MEMORY",
      "type": "INTEGER"
    },
    {
      "doubleValue": 0,
      "integerValue": 0,
      "longValue": 0,
      "name": "PORTS",
      "stringSetValue": [
        "22",
        "2376",
        "2375",
        "51678"
      ],
      "type": "STRINGSET"
    },
    {
      "doubleValue": 0,
      "integerValue": 0,
      "longValue": 0,
      "name": "PORTS_UDP",
      "stringSetValue": [],

```

```
        "type": "STRINGSET"
      }
    ],
    "runningTasksCount": 0,
    "status": "ACTIVE",
    "version": 850,
    "versionInfo": {
      "agentHash": "0931217",
      "agentVersion": "1.9.0",
      "dockerVersion": "DockerVersion: 1.9.1"
    }
  },
  "failures": []
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeServiceDeployments

Describes one or more of your service deployments.

A service deployment happens when you release a software update for the service. For more information, see [View service history using Amazon ECS service deployments](#).

Request Syntax

```
{  
  "serviceDeploymentArns": [ "string" ]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[serviceDeploymentArns](#)

The ARN of the service deployment.

You can specify a maximum of 20 ARNs.

Type: Array of strings

Required: Yes

Response Syntax

```
{  
  "failures": [  
    {  
      "arn": "string",  
      "detail": "string",  
      "reason": "string"  
    }  
  ],  
  "serviceDeployments": [  
    {  
      "alarms": {  
        "alarms": {  
          "alarms": {  
            "alarms": {  
              "alarms": {  
                "alarms": {  
                  "alarms": {  
                    "alarms": {  
                      "alarms": {  
                        "alarms": {  
                          "alarms": {  
                        ...  
                      }  
                    }  
                  }  
                }  
              }  
            }  
          }  
        }  
      }  
    }  
  ]  
}
```

```

    "alarmNames": [ "string" ],
    "status": "string",
    "triggeredAlarmNames": [ "string" ]
  },
  "clusterArn": "string",
  "createdAt": number,
  "deploymentCircuitBreaker": {
    "failureCount": number,
    "status": "string",
    "threshold": number
  },
  "deploymentConfiguration": {
    "alarms": {
      "alarmNames": [ "string" ],
      "enable": boolean,
      "rollback": boolean
    },
    "bakeTimeInMinutes": number,
    "canaryConfiguration": {
      "canaryBakeTimeInMinutes": number,
      "canaryPercent": number
    },
    "deploymentCircuitBreaker": {
      "enable": boolean,
      "rollback": boolean
    },
    "lifecycleHooks": [
      {
        "hookDetails": JSON value,
        "hookTargetArn": "string",
        "lifecycleStages": [ "string" ],
        "roleArn": "string"
      }
    ],
    "linearConfiguration": {
      "stepBakeTimeInMinutes": number,
      "stepPercent": number
    },
    "maximumPercent": number,
    "minimumHealthyPercent": number,
    "strategy": "string"
  },
  "finishedAt": number,
  "lifecycleStage": "string",

```

```

    "rollback": {
      "reason": "string",
      "serviceRevisionArn": "string",
      "startedAt": number
    },
    "serviceArn": "string",
    "serviceDeploymentArn": "string",
    "sourceServiceRevisions": [
      {
        "arn": "string",
        "pendingTaskCount": number,
        "requestedProductionTrafficWeight": number,
        "requestedTaskCount": number,
        "requestedTestTrafficWeight": number,
        "runningTaskCount": number
      }
    ],
    "startedAt": number,
    "status": "string",
    "statusReason": "string",
    "stoppedAt": number,
    "targetServiceRevision": {
      "arn": "string",
      "pendingTaskCount": number,
      "requestedProductionTrafficWeight": number,
      "requestedTaskCount": number,
      "requestedTestTrafficWeight": number,
      "runningTaskCount": number
    },
    "updatedAt": number
  }
]
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

failures

Any failures associated with the call.

If you describe a deployment with a service revision created before October 25, 2024, the call fails. The failure includes the service revision ARN and the reason set to MISSING.

Type: Array of [Failure](#) objects

[serviceDeployments](#)

The list of service deployments described.

Type: Array of [ServiceDeployment](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

Examples

Example

This example request provides a full description of the service deployments for the sd-example service on the example cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 55
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DescribeServiceDeployments
X-Amz-Date: 20150528T163859Z
User-Agent: aws-cli/2.17 Python/3.11.9 Darwin/14.3.0
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
```

```
"serviceDeploymentArn": "arn:aws:ecs:us-west-2:123456789012:service-deployment/
example/sd-example/NCWGC2ZR-taawPAYrIaU5"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Tue, 10 Sep 2024 18:50:14 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 220
Connection: keep-alive
RequestId: 360c5551-123e-4e74-9914-7582d3a28807
{
  "serviceDeployments": [
    {
      "serviceDeploymentArn": "arn:aws:ecs:us-west-2:123456789012:service-
deployment/example/sd-example/NCWGC2ZR-taawPAYrIaU5",
      "serviceArn": "arn:aws:ecs:us-west-2:123456789012:service/example/sd-
example",
      "clusterArn": "arn:aws:ecs:us-west-2:123456789012:cluster/example",
      "updatedAt": "2024-09-10T16:49:35.572000+00:00",
      "targetServiceRevision": {
        "arn": "arn:aws:ecs:us-west-2:123456789012:service-revision/example/sd-
example/4980306466373577095",
        "requestedTaskCount": 0,
        "runningTaskCount": 0,
        "pendingTaskCount": 0
      },
      "status": "PENDING",
      "deploymentConfiguration": {
        "deploymentCircuitBreaker": {
          "enable": false,
          "rollback": false
        },
        "maximumPercent": 200,
        "minimumHealthyPercent": 100
      }
    }
  ],
  "failures": []
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeServiceRevisions

Describes one or more service revisions.

A service revision is a version of the service that includes the values for the Amazon ECS resources (for example, task definition) and the environment resources (for example, load balancers, subnets, and security groups). For more information, see [Amazon ECS service revisions](#).

You can't describe a service revision that was created before October 25, 2024.

Request Syntax

```
{  
  "serviceRevisionArns": [ "string" ]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[serviceRevisionArns](#)

The ARN of the service revision.

You can specify a maximum of 20 ARNs.

You can call [ListServiceDeployments](#) to get the ARNs.

Type: Array of strings

Required: Yes

Response Syntax

```
{  
  "failures": [  
    {  
      "arn": "string",  
      "detail": "string",  
      "reason": "string"  
    }  
  ]  
}
```

```

    }
  ],
  "serviceRevisions": [
    {
      "capacityProviderStrategy": [
        {
          "base": number,
          "capacityProvider": "string",
          "weight": number
        }
      ],
      "clusterArn": "string",
      "containerImages": [
        {
          "containerName": "string",
          "image": "string",
          "imageDigest": "string"
        }
      ],
      "createdAt": number,
      "fargateEphemeralStorage": {
        "kmsKeyId": "string"
      },
      "guardDutyEnabled": boolean,
      "launchType": "string",
      "loadBalancers": [
        {
          "advancedConfiguration": {
            "alternateTargetGroupArn": "string",
            "productionListenerRule": "string",
            "roleArn": "string",
            "testListenerRule": "string"
          },
          "containerName": "string",
          "containerPort": number,
          "loadBalancerName": "string",
          "targetGroupArn": "string"
        }
      ],
      "networkConfiguration": {
        "awsvpcConfiguration": {
          "assignPublicIp": "string",
          "securityGroups": [ "string " ],
          "subnets": [ "string " ]
        }
      }
    }
  ]
}

```

```

    }
  },
  "platformFamily": "string",
  "platformVersion": "string",
  "resolvedConfiguration": {
    "loadBalancers": [
      {
        "productionListenerRule": "string",
        "targetGroupArn": "string"
      }
    ]
  },
  "serviceArn": "string",
  "serviceConnectConfiguration": {
    "accessLogConfiguration": {
      "format": "string",
      "includeQueryParameters": "string"
    },
    "enabled": boolean,
    "logConfiguration": {
      "logDriver": "string",
      "options": {
        "string" : "string"
      },
      "secretOptions": [
        {
          "name": "string",
          "valueFrom": "string"
        }
      ]
    },
    "namespace": "string",
    "services": [
      {
        "clientAliases": [
          {
            "dnsName": "string",
            "port": number,
            "testTrafficRules": {
              "header": {
                "name": "string",
                "value": {
                  "exact": "string"
                }
              }
            }
          }
        ]
      }
    ]
  }
}

```

```

        }
    }
}
],
"discoveryName": "string",
"ingressPortOverride": number,
"portName": "string",
"timeout": {
    "idleTimeoutSeconds": number,
    "perRequestTimeoutSeconds": number
},
"tls": {
    "issuerCertificateAuthority": {
        "awsPcaAuthorityArn": "string"
    },
    "kmsKey": "string",
    "roleArn": "string"
}
}
],
},
"serviceRegistries": [
    {
        "containerName": "string",
        "containerPort": number,
        "port": number,
        "registryArn": "string"
    }
],
"serviceRevisionArn": "string",
"taskDefinition": "string",
"volumeConfigurations": [
    {
        "managedEBSVolume": {
            "encrypted": boolean,
            "filesystemType": "string",
            "iops": number,
            "kmsKeyId": "string",
            "roleArn": "string",
            "sizeInGiB": number,
            "snapshotId": "string",
            "tagSpecifications": [
                {
                    "propagateTags": "string",

```

```

        "resourceType": "string",
        "tags": [
            {
                "key": "string",
                "value": "string"
            }
        ]
    },
    "throughput": number,
    "volumeInitializationRate": number,
    "volumeType": "string"
},
"name": "string"
}
],
"vpcLatticeConfigurations": [
    {
        "portName": "string",
        "roleArn": "string",
        "targetGroupArn": "string"
    }
]
}
]
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

serviceRevisions

The list of service revisions described.

Type: Array of [ServiceRevision](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

Examples

Example

This example request provides a full description of the service revision with the ARN `arn:aws:ecs:us-west-2:123456789012:service-revision/testc/sd1/4980306466373577095`.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 55
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DescribeServiceRevisions
X-Amz-Date: 20150528T163859Z
User-Agent: aws-cli/2.17 Python/3.11.9 Darwin/14.3.0
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "serviceRevisionArn": "arn:aws:ecs:us-west-2:123456789012:service-revision/testc/sd1/4980306466373577095"
}
```

Sample Response

```
HTTP/1.1 200 OK
```

```

Server: Server
Date: Tue, 10 Sep 2024 18:50:14 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 220
Connection: keep-alive
RequestId: 360c5551-123e-4e74-9914-7582d3a28807
{
  "serviceRevisions": [
    {
      "serviceRevisionArn": "arn:aws:ecs:us-west-2:123456789012:service-revision/
example/sd-example/4980306466373577095",
      "serviceArn": "arn:aws:ecs:us-west-2:123456789012:service/example/sd-
example",
      "clusterArn": "arn:aws:ecs:us-west-2:123456789012:cluster/example",
      "taskDefinition": "arn:aws:ecs:us-west-2:123456789012:task-definition/
large-ngingx:1",
      "launchType": "FARGATE",
      "platformVersion": "1.4.0",
      "platformFamily": "DockerLinux",
      "networkConfiguration": {
        "awsvpcConfiguration": {
          "subnets": [
            "subnet-0a4040e73895f04e1"
          ],
          "securityGroups": [
            "sg-09605d60a6bc1b296"
          ],
          "assignPublicIp": "ENABLED"
        }
      },
      "createdAt": "2024-09-10T16:49:26.388000+00:00"
    }
  ],
  "failures": []
}

```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeServices

Describes the specified services running in your cluster.

Request Syntax

```
{  
  "cluster": "string",  
  "include": [ "string" ],  
  "services": [ "string" ]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) the cluster that hosts the service to describe. If you do not specify a cluster, the default cluster is assumed. This parameter is required if the service or services you are describing were launched in any cluster other than the default cluster.

Type: String

Required: No

include

Determines whether you want to see the resource tags for the service. If TAGS is specified, the tags are included in the response. If this field is omitted, tags aren't included in the response.

Type: Array of strings

Valid Values: TAGS

Required: No

services

A list of services to describe. You may specify up to 10 services to describe in a single operation.

Type: Array of strings

Required: Yes

Response Syntax

```
{
  "failures": [
    {
      "arn": "string",
      "detail": "string",
      "reason": "string"
    }
  ],
  "services": [
    {
      "availabilityZoneRebalancing": "string",
      "capacityProviderStrategy": [
        {
          "base": number,
          "capacityProvider": "string",
          "weight": number
        }
      ],
      "clusterArn": "string",
      "createdAt": number,
      "createdBy": "string",
      "deploymentConfiguration": {
        "alarms": {
          "alarmNames": [ "string" ],
          "enable": boolean,
          "rollback": boolean
        },
        "bakeTimeInMinutes": number,
        "canaryConfiguration": {
          "canaryBakeTimeInMinutes": number,
          "canaryPercent": number
        },
        "deploymentCircuitBreaker": {
          "enable": boolean,
          "rollback": boolean
        },
        "lifecycleHooks": [
```

```

    {
      "hookDetails": JSON value,
      "hookTargetArn": "string",
      "lifecycleStages": [ "string" ],
      "roleArn": "string"
    }
  ],
  "linearConfiguration": {
    "stepBakeTimeInMinutes": number,
    "stepPercent": number
  },
  "maximumPercent": number,
  "minimumHealthyPercent": number,
  "strategy": "string"
},
"deploymentController": {
  "type": "string"
},
"deployments": [
  {
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "createdAt": number,
    "desiredCount": number,
    "failedTasks": number,
    "fargateEphemeralStorage": {
      "kmsKeyId": "string"
    },
    "id": "string",
    "launchType": "string",
    "networkConfiguration": {
      "awsvpcConfiguration": {
        "assignPublicIp": "string",
        "securityGroups": [ "string" ],
        "subnets": [ "string" ]
      }
    },
    "pendingCount": number,
    "platformFamily": "string",

```

```

    "platformVersion": "string",
    "rolloutState": "string",
    "rolloutStateReason": "string",
    "runningCount": number,
    "serviceConnectConfiguration": {
      "accessLogConfiguration": {
        "format": "string",
        "includeQueryParameters": "string"
      },
      "enabled": boolean,
      "logConfiguration": {
        "logDriver": "string",
        "options": {
          "string" : "string"
        },
        "secretOptions": [
          {
            "name": "string",
            "valueFrom": "string"
          }
        ]
      },
      "namespace": "string",
      "services": [
        {
          "clientAliases": [
            {
              "dnsName": "string",
              "port": number,
              "testTrafficRules": {
                "header": {
                  "name": "string",
                  "value": {
                    "exact": "string"
                  }
                }
              }
            }
          ],
          "discoveryName": "string",
          "ingressPortOverride": number,
          "portName": "string",
          "timeout": {
            "idleTimeoutSeconds": number,

```



```

        "perRequestTimeoutSeconds": number
    },
    "tls": {
        "issuerCertificateAuthority": {
            "awsPcaAuthorityArn": "string"
        },
        "kmsKey": "string",
        "roleArn": "string"
    }
}
]
},
"serviceConnectResources": [
    {
        "discoveryArn": "string",
        "discoveryName": "string"
    }
],
"status": "string",
"taskDefinition": "string",
"updatedAt": number,
"volumeConfigurations": [
    {
        "managedEBSVolume": {
            "encrypted": boolean,
            "filesystemType": "string",
            "iops": number,
            "kmsKeyId": "string",
            "roleArn": "string",
            "sizeInGiB": number,
            "snapshotId": "string",
            "tagSpecifications": [
                {
                    "propagateTags": "string",
                    "resourceType": "string",
                    "tags": [
                        {
                            "key": "string",
                            "value": "string"
                        }
                    ]
                }
            ]
        }
    }
],
"throughput": number,

```

```

        "volumeInitializationRate": number,
        "volumeType": "string"
    },
    "name": "string"
}
],
"vpcLatticeConfigurations": [
    {
        "portName": "string",
        "roleArn": "string",
        "targetGroupArn": "string"
    }
]
},
"desiredCount": number,
"enableECSTags": boolean,
"enableExecuteCommand": boolean,
"events": [
    {
        "createdAt": number,
        "id": "string",
        "message": "string"
    }
],
"healthCheckGracePeriodSeconds": number,
"launchType": "string",
"loadBalancers": [
    {
        "advancedConfiguration": {
            "alternateTargetGroupArn": "string",
            "productionListenerRule": "string",
            "roleArn": "string",
            "testListenerRule": "string"
        },
        "containerName": "string",
        "containerPort": number,
        "loadBalancerName": "string",
        "targetGroupArn": "string"
    }
],
"networkConfiguration": {
    "awsVpcConfiguration": {
        "assignPublicIp": "string",

```

```

        "securityGroups": [ "string" ],
        "subnets": [ "string" ]
    }
},
"pendingCount": number,
"placementConstraints": [
    {
        "expression": "string",
        "type": "string"
    }
],
"placementStrategy": [
    {
        "field": "string",
        "type": "string"
    }
],
"platformFamily": "string",
"platformVersion": "string",
"propagateTags": "string",
"roleArn": "string",
"runningCount": number,
"schedulingStrategy": "string",
"serviceArn": "string",
"serviceName": "string",
"serviceRegistries": [
    {
        "containerName": "string",
        "containerPort": number,
        "port": number,
        "registryArn": "string"
    }
],
"status": "string",
"tags": [
    {
        "key": "string",
        "value": "string"
    }
],
"taskDefinition": "string",
"taskSets": [
    {
        "capacityProviderStrategy": [

```

```

    {
      "base": number,
      "capacityProvider": "string",
      "weight": number
    }
  ],
  "clusterArn": "string",
  "computedDesiredCount": number,
  "createdAt": number,
  "externalId": "string",
  "fargateEphemeralStorage": {
    "kmsKeyId": "string"
  },
  "id": "string",
  "launchType": "string",
  "loadBalancers": [
    {
      "advancedConfiguration": {
        "alternateTargetGroupArn": "string",
        "productionListenerRule": "string",
        "roleArn": "string",
        "testListenerRule": "string"
      },
      "containerName": "string",
      "containerPort": number,
      "loadBalancerName": "string",
      "targetGroupArn": "string"
    }
  ],
  "networkConfiguration": {
    "awsvpcConfiguration": {
      "assignPublicIp": "string",
      "securityGroups": [ "string " ],
      "subnets": [ "string " ]
    }
  },
  "pendingCount": number,
  "platformFamily": "string",
  "platformVersion": "string",
  "runningCount": number,
  "scale": {
    "unit": "string",
    "value": number
  },

```

```

    "serviceArn": "string",
    "serviceRegistries": [
      {
        "containerName": "string",
        "containerPort": number,
        "port": number,
        "registryArn": "string"
      }
    ],
    "stabilityStatus": "string",
    "stabilityStatusAt": number,
    "startedBy": "string",
    "status": "string",
    "tags": [
      {
        "key": "string",
        "value": "string"
      }
    ],
    "taskDefinition": "string",
    "taskSetArn": "string",
    "updatedAt": number
  }
]
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

services

The list of services described.

Type: Array of [Service](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request provides a full description of the test-service service in the telemetry cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 55
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DescribeServices
X-Amz-Date: 20150528T163859Z
User-Agent: aws-cli/1.7.30 Python/2.7.9 Darwin/14.3.0
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "services": [
    "test-service"
  ],
  "cluster": "telemetry"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 19:02:59 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 2449
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "failures": [],
  "services": [
```

```
{
  "clusterArn": "arn:aws:ecs:us-west-2:012345678910:cluster/telemetry",
  "deploymentConfiguration": {
    "maximumPercent": 200,
    "minimumHealthyPercent": 100
  },
  "deployments": [
    {
      "createdAt": 1432829320.611,
      "desiredCount": 4,
      "id": "ecs-svc/9223370604025455196",
      "pendingCount": 0,
      "runningCount": 4,
      "status": "PRIMARY",
      "taskDefinition": "arn:aws:ecs:us-west-2:012345678910:task-definition/hpcc-
t2-medium:1",
      "updatedAt": 1432829320.611
    }
  ],
  "desiredCount": 4,
  "events": [],
  "loadBalancers": [],
  "pendingCount": 0,
  "runningCount": 4,
  "serviceArn": "arn:aws:ecs:us-west-2:012345678910:service/default/test-service",
  "serviceName": "test-service",
  "status": "ACTIVE",
  "taskDefinition": "arn:aws:ecs:us-west-2:012345678910:task-definition/hpcc-t2-
medium:1"
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeTaskDefinition

Describes a task definition. You can specify a family and revision to find information about a specific task definition, or you can simply specify the family to find the latest ACTIVE revision in that family.

Note

You can only describe INACTIVE task definitions while an active task or service references them.

Request Syntax

```
{  
  "include": [ "string" ],  
  "taskDefinition": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

include

Determines whether to see the resource tags for the task definition. If TAGS is specified, the tags are included in the response. If this field is omitted, tags aren't included in the response.

Type: Array of strings

Valid Values: TAGS

Required: No

taskDefinition

The family for the latest ACTIVE revision, family and revision (family:revision) for a specific revision in the family, or full Amazon Resource Name (ARN) of the task definition to describe.

Type: String

Required: Yes

Response Syntax

```
{
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ],
  "taskDefinition": {
    "compatibilities": [ "string" ],
    "containerDefinitions": [
      {
        "command": [ "string" ],
        "cpu": number,
        "credentialSpecs": [ "string" ],
        "dependsOn": [
          {
            "condition": "string",
            "containerName": "string"
          }
        ],
        "disableNetworking": boolean,
        "dnsSearchDomains": [ "string" ],
        "dnsServers": [ "string" ],
        "dockerLabels": {
          "string" : "string"
        },
        "dockerSecurityOptions": [ "string" ],
        "entryPoint": [ "string" ],
        "environment": [
          {
            "name": "string",
            "value": "string"
          }
        ],
        "environmentFiles": [
          {
            "type": "string",
```

```

        "value": "string"
    }
],
"essential": boolean,
"extraHosts": [
    {
        "hostname": "string",
        "ipAddress": "string"
    }
],
"firelensConfiguration": {
    "options": {
        "string" : "string"
    },
    "type": "string"
},
"healthCheck": {
    "command": [ "string" ],
    "interval": number,
    "retries": number,
    "startPeriod": number,
    "timeout": number
},
"hostname": "string",
"image": "string",
"interactive": boolean,
"links": [ "string" ],
"linuxParameters": {
    "capabilities": {
        "add": [ "string" ],
        "drop": [ "string" ]
    },
    "devices": [
        {
            "containerPath": "string",
            "hostPath": "string",
            "permissions": [ "string" ]
        }
    ],
    "initProcessEnabled": boolean,
    "maxSwap": number,
    "sharedMemorySize": number,
    "swappiness": number,
    "tmpfs": [

```

```

        {
            "containerPath": "string",
            "mountOptions": [ "string" ],
            "size": number
        }
    ],
    "logConfiguration": {
        "logDriver": "string",
        "options": {
            "string" : "string"
        },
        "secretOptions": [
            {
                "name": "string",
                "valueFrom": "string"
            }
        ]
    },
    "memory": number,
    "memoryReservation": number,
    "mountPoints": [
        {
            "containerPath": "string",
            "readOnly": boolean,
            "sourceVolume": "string"
        }
    ],
    "name": "string",
    "portMappings": [
        {
            "appProtocol": "string",
            "containerPort": number,
            "containerPortRange": "string",
            "hostPort": number,
            "name": "string",
            "protocol": "string"
        }
    ],
    "privileged": boolean,
    "pseudoTerminal": boolean,
    "readonlyRootFilesystem": boolean,
    "repositoryCredentials": {
        "credentialsParameter": "string"
    }

```

```

    },
    "resourceRequirements": [
      {
        "type": "string",
        "value": "string"
      }
    ],
    "restartPolicy": {
      "enabled": boolean,
      "ignoredExitCodes": [ number ],
      "restartAttemptPeriod": number
    },
    "secrets": [
      {
        "name": "string",
        "valueFrom": "string"
      }
    ],
    "startTimeout": number,
    "stopTimeout": number,
    "systemControls": [
      {
        "namespace": "string",
        "value": "string"
      }
    ],
    "ulimits": [
      {
        "hardLimit": number,
        "name": "string",
        "softLimit": number
      }
    ],
    "user": "string",
    "versionConsistency": "string",
    "volumesFrom": [
      {
        "readOnly": boolean,
        "sourceContainer": "string"
      }
    ],
    "workingDirectory": "string"
  }
],

```

```
"cpu": "string",
"deregisteredAt": number,
"enableFaultInjection": boolean,
"ephemeralStorage": {
  "sizeInGiB": number
},
"executionRoleArn": "string",
"family": "string",
"inferenceAccelerators": [
  {
    "deviceName": "string",
    "deviceType": "string"
  }
],
"ipcMode": "string",
"memory": "string",
"networkMode": "string",
"pidMode": "string",
"placementConstraints": [
  {
    "expression": "string",
    "type": "string"
  }
],
"proxyConfiguration": {
  "containerName": "string",
  "properties": [
    {
      "name": "string",
      "value": "string"
    }
  ],
  "type": "string"
},
"registeredAt": number,
"registeredBy": "string",
"requiresAttributes": [
  {
    "name": "string",
    "targetId": "string",
    "targetType": "string",
    "value": "string"
  }
],
```

```

    "requiresCompatibilities": [ "string" ],
    "revision": number,
    "runtimePlatform": {
        "cpuArchitecture": "string",
        "operatingSystemFamily": "string"
    },
    "status": "string",
    "taskDefinitionArn": "string",
    "taskRoleArn": "string",
    "volumes": [
        {
            "configuredAtLaunch": boolean,
            "dockerVolumeConfiguration": {
                "autoprovision": boolean,
                "driver": "string",
                "driverOpts": {
                    "string" : "string"
                },
                "labels": {
                    "string" : "string"
                },
                "scope": "string"
            },
            "efsVolumeConfiguration": {
                "authorizationConfig": {
                    "accessPointId": "string",
                    "iam": "string"
                },
                "fileSystemId": "string",
                "rootDirectory": "string",
                "transitEncryption": "string",
                "transitEncryptionPort": number
            },
            "fsxWindowsFileServerVolumeConfiguration": {
                "authorizationConfig": {
                    "credentialsParameter": "string",
                    "domain": "string"
                },
                "fileSystemId": "string",
                "rootDirectory": "string"
            },
            "host": {
                "sourcePath": "string"
            }
        },

```



```
    "name": "string"  
  }  
]  
}  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[tags](#)

The metadata that's applied to the task definition to help you categorize and organize them. Each tag consists of a key and an optional value. You define both.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

[taskDefinition](#)

The full task definition description.

Type: [TaskDefinition](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request provides descriptive information about the 10th revision of a task definition in the `hello_world` family.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 36
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DescribeTaskDefinition
X-Amz-Date: 20150429T190902Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "taskDefinition": "hello_world:10"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 19:09:03 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 574
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "taskDefinition": {
    "containerDefinitions": [
      {
        "cpu": 10,
        "environment": [],
        "essential": true,
        "image": "public.ecr.aws/docker/library/wordpress:latest",
        "links": [
          "mysql"
        ],
        "memory": 500,
        "mountPoints": [],
```

```
    "name": "wordpress",
    "portMappings": [
      {
        "containerPort": 80,
        "hostPort": 80
      }
    ],
    "volumesFrom": [],
  },
  {
    "cpu": 10,
    "environment": [
      {
        "name": "MYSQL_ROOT_PASSWORD",
        "value": "password"
      }
    ],
    "essential": true,
    "image": "public.ecr.aws/docker/library/mysql:latest",
    "memory": 500,
    "mountPoints": [],
    "name": "mysql",
    "portMappings": [],
    "volumesFrom": []
  }
],
"family": "hello_world",
"revision": 10,
"taskDefinitionArn": "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:10",
"volumes": []
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)

- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeTasks

Describes a specified task or tasks.

Currently, stopped tasks appear in the returned results for at least one hour.

If you have tasks with tags, and then delete the cluster, the tagged tasks are returned in the response. If you create a new cluster with the same name as the deleted cluster, the tagged tasks are not included in the response.

Request Syntax

```
{  
  "cluster": "string",  
  "include": [ "string" ],  
  "tasks": [ "string" ]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the task or tasks to describe. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

include

Specifies whether you want to see the resource tags for the task. If TAGS is specified, the tags are included in the response. If this field is omitted, tags aren't included in the response.

Type: Array of strings

Valid Values: TAGS

Required: No

tasks

A list of up to 100 task IDs or full ARN entries.

Type: Array of strings

Required: Yes

Response Syntax

```
{
  "failures": [
    {
      "arn": "string",
      "detail": "string",
      "reason": "string"
    }
  ],
  "tasks": [
    {
      "attachments": [
        {
          "details": [
            {
              "name": "string",
              "value": "string"
            }
          ],
          "id": "string",
          "status": "string",
          "type": "string"
        }
      ],
      "attributes": [
        {
          "name": "string",
          "targetId": "string",
          "targetType": "string",
          "value": "string"
        }
      ],
      "availabilityZone": "string",
      "capacityProviderName": "string",
```

```

    "clusterArn": "string",
    "connectivity": "string",
    "connectivityAt": number,
    "containerInstanceArn": "string",
    "containers": [
      {
        "containerArn": "string",
        "cpu": "string",
        "exitCode": number,
        "gpuIds": [ "string "],
        "healthStatus": "string",
        "image": "string",
        "imageDigest": "string",
        "lastStatus": "string",
        "managedAgents": [
          {
            "lastStartedAt": number,
            "lastStatus": "string",
            "name": "string",
            "reason": "string"
          }
        ],
        "memory": "string",
        "memoryReservation": "string",
        "name": "string",
        "networkBindings": [
          {
            "bindIP": "string",
            "containerPort": number,
            "containerPortRange": "string",
            "hostPort": number,
            "hostPortRange": "string",
            "protocol": "string"
          }
        ],
        "networkInterfaces": [
          {
            "attachmentId": "string",
            "ipv6Address": "string",
            "privateIpv4Address": "string"
          }
        ],
        "reason": "string",
        "runtimeId": "string",

```



```

        "taskArn": "string"
    }
],
"cpu": "string",
"createdAt": number,
"desiredStatus": "string",
"enableExecuteCommand": boolean,
"ephemeralStorage": {
    "sizeInGiB": number
},
"executionStoppedAt": number,
"fargateEphemeralStorage": {
    "kmsKeyId": "string",
    "sizeInGiB": number
},
"group": "string",
"healthStatus": "string",
"inferenceAccelerators": [
    {
        "deviceName": "string",
        "deviceType": "string"
    }
],
"lastStatus": "string",
"launchType": "string",
"memory": "string",
"overrides": {
    "containerOverrides": [
        {
            "command": [ "string" ],
            "cpu": number,
            "environment": [
                {
                    "name": "string",
                    "value": "string"
                }
            ],
            "environmentFiles": [
                {
                    "type": "string",
                    "value": "string"
                }
            ],
            "memory": number,

```

```

        "memoryReservation": number,
        "name": "string",
        "resourceRequirements": [
            {
                "type": "string",
                "value": "string"
            }
        ]
    },
    "cpu": "string",
    "ephemeralStorage": {
        "sizeInGiB": number
    },
    "executionRoleArn": "string",
    "inferenceAcceleratorOverrides": [
        {
            "deviceName": "string",
            "deviceType": "string"
        }
    ],
    "memory": "string",
    "taskRoleArn": "string"
},
"platformFamily": "string",
"platformVersion": "string",
"pullStartedAt": number,
"pullStoppedAt": number,
"startedAt": number,
"startedBy": "string",
"stopCode": "string",
"stoppedAt": number,
"stoppedReason": "string",
"stoppingAt": number,
"tags": [
    {
        "key": "string",
        "value": "string"
    }
],
"taskArn": "string",
"taskDefinitionArn": "string",
"version": number
}

```

```
]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[failures](#)

Any failures associated with the call.

Type: Array of [Failure](#) objects

[tasks](#)

The list of tasks.

Type: Array of [Task](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request provides descriptive information about a task with an ARN of `arn:aws:ecs:us-east-1:123456789012:cluster/default` in the default cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 51
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DescribeTasks
X-Amz-Date: 20250212T181908Z
```

```
User-Agent: aws-cli/2.2.26 Python/2.7.12 Darwin/16.1.0 boto3/1.4.66
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
```

```
{
  "tasks": [
    "arn:aws:ecs:us-east-1:1234567890:cluster/default"
  ]
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Mon, 21 Nov 2016 21:49:16 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 1238
Connection: keep-alive

x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f
```

```
{
  "tasks": [
    {
      "attachments": [
        {
          "id": "39888c47-dd47-482b-b6b5-1ea60b6633e7",
          "type": "ElasticNetworkInterface",
          "status": "ATTACHED",
          "details": [
            {
              "name": "subnetId",
              "value": "subnet-0db5010045995c2d5"
            },
            {
              "name": "networkInterfaceId",
              "value": "eni-0800851957d7918fa"
            },
            {
              "name": "macAddress",
              "value": "0a:ff:cc:73:2e:b7"
            }
          ]
        }
      ]
    }
  ]
}
```

```

        "name": "privateDnsName",
        "value": "ip-10-0-0-250.ec2.internal"
    },
    {
        "name": "privateIPv4Address",
        "value": "10.0.0.250"
    }
]
},
"attributes": [
    {
        "name": "ecs.cpu-architecture",
        "value": "x86_64"
    }
],
"availabilityZone": "us-east-1a",
"capacityProviderName": "FARGATE",
"clusterArn": "arn:aws:ecs:us-east-1:123456789012:cluster/default",
"connectivity": "CONNECTED",
"connectivityAt": "2025-02-03T16:01:58.863000-05:00",
"containers": [
    {
        "containerArn": "arn:aws:ecs:us-east-1:123456789012:container/
default/154bdb1df5b446b280fe900d12c452a1/417f6954-30e9-4e60-84e5-bad3743617ab",
        "taskArn": "arn:aws:ecs:us-east-1:123456789012:task/
default/154bdb1df5b446b280fe900d12c452a1",
        "name": "aws-otel-collector",
        "image": "public.ecr.aws/aws-observability/aws-otel-
collector:v0.32.0",
        "imageDigest":
            "sha256:7a1b3560655071bcacd66902c20ebe9a69470d5691fe3bd36baace7c2f3c4640",
        "runtimeId": "154bdb1df5b446b280fe900d12c452a1-4027662657",
        "lastStatus": "RUNNING",
        "networkBindings": [],
        "networkInterfaces": [
            {
                "attachmentId": "39888c47-dd47-482b-b6b5-1ea60b6633e7",
                "privateIpv4Address": "10.0.0.250"
            }
        ],
        "healthStatus": "UNKNOWN",
        "cpu": "0"
    },

```

```

        {
            "containerArn": "arn:aws:ecs:us-east-1:123456789012:container/
default/154bdb1df5b446b280fe900d12c452a1/52c6b197-710b-4860-9d52-262a56550b6a",
            "taskArn": "arn:aws:ecs:us-east-1:123456789012:task/
default/154bdb1df5b446b280fe900d12c452a1",
            "name": "web",
            "image": "nginx",
            "imageDigest":
"sha256:0a399eb16751829e1af26fea27b20c3ec28d7ab1fb72182879dcae1cca21206a",
            "runtimeId": "154bdb1df5b446b280fe900d12c452a1-0265927825",
            "lastStatus": "RUNNING",
            "networkBindings": [],
            "networkInterfaces": [
                {
                    "attachmentId": "39888c47-dd47-482b-b6b5-1ea60b6633e7",
                    "privateIpv4Address": "10.0.0.250"
                }
            ],
            "healthStatus": "HEALTHY",
            "cpu": "99",
            "memory": "100"
        }
    ],
    "cpu": "256",
    "createdAt": "2025-02-03T16:01:55.201000-05:00",
    "desiredStatus": "RUNNING",
    "enableExecuteCommand": false,
    "group": "family:webserver",
    "healthStatus": "HEALTHY",
    "lastStatus": "RUNNING",
    "launchType": "FARGATE",
    "memory": "512",
    "overrides": {
        "containerOverrides": [
            {
                "name": "aws-otel-collector"
            },
            {
                "name": "web"
            }
        ],
        "inferenceAcceleratorOverrides": []
    },
    "platformVersion": "1.4.0",

```

```
    "platformFamily": "Linux",
    "pullStartedAt": "2025-02-03T16:02:10.727000-05:00",
    "pullStoppedAt": "2025-02-03T16:02:17.868000-05:00",
    "startedAt": "2025-02-03T16:02:17.978000-05:00",
    "tags": [],
    "taskArn": "arn:aws:ecs:us-east-1:123456789012:task/
default/154bdb1df5b446b280fe900d12c452a1",
    "taskDefinitionArn": "arn:aws:ecs:us-east-1:123456789012:task-definition/
webserver:5",
    "version": 4,
    "ephemeralStorage": {
        "sizeInGiB": 20
    },
    "fargateEphemeralStorage": {
        "sizeInGiB": 20
    }
},
"failures": []
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DescribeTaskSets

Describes the task sets in the specified cluster and service. This is used when a service uses the EXTERNAL deployment controller type. For more information, see [Amazon ECS Deployment Types](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{  
  "cluster": "string",  
  "include": [ "string" ],  
  "service": "string",  
  "taskSets": [ "string" ]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the service that the task sets exist in.

Type: String

Required: Yes

include

Specifies whether to see the resource tags for the task set. If TAGS is specified, the tags are included in the response. If this field is omitted, tags aren't included in the response.

Type: Array of strings

Valid Values: TAGS

Required: No

service

The short name or full Amazon Resource Name (ARN) of the service that the task sets exist in.

Type: String

Required: Yes

taskSets

The ID or full Amazon Resource Name (ARN) of task sets to describe.

Type: Array of strings

Required: No

Response Syntax

```
{
  "failures": [
    {
      "arn": "string",
      "detail": "string",
      "reason": "string"
    }
  ],
  "taskSets": [
    {
      "capacityProviderStrategy": [
        {
          "base": number,
          "capacityProvider": "string",
          "weight": number
        }
      ],
      "clusterArn": "string",
      "computedDesiredCount": number,
      "createdAt": number,
      "externalId": "string",
      "fargateEphemeralStorage": {
        "kmsKeyId": "string"
      },
      "id": "string",
      "launchType": "string",
      "loadBalancers": [
        {
          "advancedConfiguration": {
            "alternateTargetGroupArn": "string",
```

```

        "productionListenerRule": "string",
        "roleArn": "string",
        "testListenerRule": "string"
    },
    "containerName": "string",
    "containerPort": number,
    "loadBalancerName": "string",
    "targetGroupArn": "string"
}
],
"networkConfiguration": {
    "awsvpcConfiguration": {
        "assignPublicIp": "string",
        "securityGroups": [ "string" ],
        "subnets": [ "string" ]
    }
},
"pendingCount": number,
"platformFamily": "string",
"platformVersion": "string",
"runningCount": number,
"scale": {
    "unit": "string",
    "value": number
},
"serviceArn": "string",
"serviceRegistries": [
    {
        "containerName": "string",
        "containerPort": number,
        "port": number,
        "registryArn": "string"
    }
],
"stabilityStatus": "string",
"stabilityStatusAt": number,
"startedBy": "string",
"status": "string",
"tags": [
    {
        "key": "string",
        "value": "string"
    }
],

```

```
    "taskDefinition": "string",  
    "taskSetArn": "string",  
    "updatedAt": number  
  }  
]  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

taskSets

The list of task sets described.

Type: Array of [TaskSet](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotActiveException

The specified service isn't active. You can't update a service that's inactive. If you have previously deleted a service, you can re-create it with [CreateService](#).

HTTP Status Code: 400

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

DiscoverPollEndpoint

Note

This action is only used by the Amazon ECS agent, and it is not intended for use outside of the agent.

Returns an endpoint for the Amazon ECS agent to poll for updates.

Request Syntax

```
{  
  "cluster": "string",  
  "containerInstance": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that the container instance belongs to.

Type: String

Required: No

containerInstance

The container instance ID or full ARN of the container instance. For more information about the ARN format, see [Amazon Resource Name \(ARN\)](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: No

Response Syntax

```
{  
  "endpoint": "string",  
  "serviceConnectEndpoint": "string",  
  "telemetryEndpoint": "string"  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

endpoint

The endpoint for the Amazon ECS agent to poll.

Type: String

serviceConnectEndpoint

The endpoint for the Amazon ECS agent to poll for Service Connect configuration. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

telemetryEndpoint

The telemetry endpoint for the Amazon ECS agent.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ExecuteCommand

Runs a command remotely on a container within a task.

If you use a condition key in your IAM policy to refine the conditions for the policy statement, for example limit the actions to a specific cluster, you receive an `AccessDeniedException` when there is a mismatch between the condition key value and the corresponding parameter value.

For information about required permissions and considerations, see [Using Amazon ECS Exec for debugging](#) in the *Amazon ECS Developer Guide*.

Request Syntax

```
{  
  "cluster": "string",  
  "command": "string",  
  "container": "string",  
  "interactive": boolean,  
  "task": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[cluster](#)

The Amazon Resource Name (ARN) or short name of the cluster the task is running in. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

[command](#)

The command to run on the container.

Type: String

Required: Yes

container

The name of the container to execute the command on. A container name only needs to be specified for tasks containing multiple containers.

Type: String

Required: No

interactive

Use this flag to run your command in interactive mode.

Type: Boolean

Required: Yes

task

The Amazon Resource Name (ARN) or ID of the task the container is part of.

Type: String

Required: Yes

Response Syntax

```
{
  "clusterArn": "string",
  "containerArn": "string",
  "containerName": "string",
  "interactive": boolean,
  "session": {
    "sessionId": "string",
    "streamUrl": "string",
    "tokenValue": "string"
  },
  "taskArn": "string"
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

clusterArn

The Amazon Resource Name (ARN) of the cluster.

Type: String

containerArn

The Amazon Resource Name (ARN) of the container.

Type: String

containerName

The name of the container.

Type: String

interactive

Determines whether the execute command session is running in interactive mode. Amazon ECS only supports initiating interactive sessions, so you must specify `true` for this value.

Type: Boolean

session

The details of the SSM session that was created for this instance of `execute-command`.

Type: [Session](#) object

taskArn

The Amazon Resource Name (ARN) of the task.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

TargetNotConnectedException

The execute command cannot run. This error can be caused by any of the following configuration issues:

- Incorrect IAM permissions
- The SSM agent is not installed or is not running

- There is an interface Amazon VPC endpoint for Amazon ECS, but there is not one for Systems Manager Session Manager

For information about how to troubleshoot the issues, see [Troubleshooting issues with ECS Exec](#) in the *Amazon Elastic Container Service Developer Guide*.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

GetTaskProtection

Retrieves the protection status of tasks in an Amazon ECS service.

Request Syntax

```
{  
  "cluster": "string",  
  "tasks": [ "string" ]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the service that the task sets exist in.

Type: String

Required: Yes

tasks

A list of up to 100 task IDs or full ARN entries.

Type: Array of strings

Required: No

Response Syntax

```
{  
  "failures": [  
    {  
      "arn": "string",  
      "detail": "string",  
    }  
  ]  
}
```

```
    "reason": "string"
  },
  "protectedTasks": [
    {
      "expirationDate": number,
      "protectionEnabled": boolean,
      "taskArn": "string"
    }
  ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

protectedTasks

A list of tasks with the following information.

- **taskArn**: The task ARN.
- **protectionEnabled**: The protection status of the task. If scale-in protection is turned on for a task, the value is true. Otherwise, it is false.
- **expirationDate**: The epoch time when protection for the task will expire.

Type: Array of [ProtectedTask](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ResourceNotFoundException

The specified resource wasn't found.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request gets the protection status for a task.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length:81
X-Amz-Target: AmazonEC2ContainerServiceV20141113.GetTaskProtection
X-Amz-Date: 20221102T190406Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "cluster": "test-task-protection",
  "tasks": [
    "b8b1cf532d0e46ba8d44a40d1de16772"
  ]
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 02 Nov 2022 19:04:06 GMT
Content-Type: application/x-amz-json-1.1
```

```
Content-Length:177
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "protectedTasks": [
    {
      "taskArn": "arn:aws:ecs:us-west-2:012345678910:task/
b8b1cf532d0e46ba8d44a40d1de16772",
      "protectionEnabled": true,
      "expirationDate": 1667416437.0
    }
  ],
  "failures": []
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListAccountSettings

Lists the account settings for a specified principal.

Request Syntax

```
{  
  "effectiveSettings": boolean,  
  "maxResults": number,  
  "name": "string",  
  "nextToken": "string",  
  "principalArn": "string",  
  "value": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[effectiveSettings](#)

Determines whether to return the effective settings. If true, the account settings for the root user or the default setting for the principalArn are returned. If false, the account settings for the principalArn are returned if they're set. Otherwise, no account settings are returned.

Type: Boolean

Required: No

[maxResults](#)

The maximum number of account setting results returned by ListAccountSettings in paginated output. When this parameter is used, ListAccountSettings only returns maxResults results in a single page along with a nextToken response element. The remaining results of the initial request can be seen by sending another ListAccountSettings request with the returned nextToken value. This value can be between 1 and 10. If this parameter isn't used, then ListAccountSettings returns up to 10 results and a nextToken value if applicable.

Type: Integer

Required: No

name

The name of the account setting you want to list the settings for.

Type: String

Valid Values: `serviceLongArnFormat` | `taskLongArnFormat` | `containerInstanceLongArnFormat` | `awsvpcTrunking` | `containerInsights` | `fargateFIPSMODE` | `tagResourceAuthorization` | `fargateTaskRetirementWaitPeriod` | `guardDutyActivate` | `defaultLogDriverMode`

Required: No

nextToken

The `nextToken` value returned from a `ListAccountSettings` request indicating that more results are available to fulfill the request and further calls will be needed. If `maxResults` was provided, it's possible the number of results to be fewer than `maxResults`.

Note

This token should be treated as an opaque identifier that is only used to retrieve the next items in a list and not for other programmatic purposes.

Type: String

Required: No

principalArn

The ARN of the principal, which can be a user, role, or the root user. If this field is omitted, the account settings are listed only for the authenticated user.

In order to use this parameter, you must be the root user, or the principal.

Note

Federated users assume the account setting of the root user and can't have explicit account settings set for them.

Type: String

Required: No

value

The value of the account settings to filter results with. You must also specify an account setting name to use this parameter.

Type: String

Required: No

Response Syntax

```
{
  "nextToken": "string",
  "settings": [
    {
      "name": "string",
      "principalArn": "string",
      "type": "string",
      "value": "string"
    }
  ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

nextToken

The nextToken value to include in a future ListAccountSettings request. When the results of a ListAccountSettings request exceed maxResults, this value can be used to retrieve the next page of results. This value is null when there are no more results to return.

Type: String

settings

The account settings for the resource.

Type: Array of [Setting](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListAttributes

Lists the attributes for Amazon ECS resources within a specified target type and cluster. When you specify a target type and cluster, `ListAttributes` returns a list of attribute objects, one for each attribute on each resource. You can filter the list of results to a single attribute name to only return results that have that name. You can also filter the results by attribute name and value. You can do this, for example, to see which container instances in a cluster are running a Linux AMI (`ecs.os-type=linux`).

Request Syntax

```
{
  "attributeName": "string",
  "attributeValue": "string",
  "cluster": "string",
  "maxResults": number,
  "nextToken": "string",
  "targetType": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

attributeName

The name of the attribute to filter the results with.

Type: String

Required: No

attributeValue

The value of the attribute to filter results with. You must also specify an attribute name to use this parameter.

Type: String

Required: No

cluster

The short name or full Amazon Resource Name (ARN) of the cluster to list attributes. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

maxResults

The maximum number of cluster results that `ListAttributes` returned in paginated output. When this parameter is used, `ListAttributes` only returns `maxResults` results in a single page along with a `nextToken` response element. The remaining results of the initial request can be seen by sending another `ListAttributes` request with the returned `nextToken` value. This value can be between 1 and 100. If this parameter isn't used, then `ListAttributes` returns up to 100 results and a `nextToken` value if applicable.

Type: Integer

Required: No

nextToken

The `nextToken` value returned from a `ListAttributes` request indicating that more results are available to fulfill the request and further calls are needed. If `maxResults` was provided, it's possible the number of results to be fewer than `maxResults`.

Note

This token should be treated as an opaque identifier that is only used to retrieve the next items in a list and not for other programmatic purposes.

Type: String

Required: No

targetType

The type of the target to list attributes with.

Type: String

Valid Values: container-instance

Required: Yes

Response Syntax

```
{
  "attributes": [
    {
      "name": "string",
      "targetId": "string",
      "targetType": "string",
      "value": "string"
    }
  ],
  "nextToken": "string"
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[attributes](#)

A list of attribute objects that meet the criteria of the request.

Type: Array of [Attribute](#) objects

[nextToken](#)

The nextToken value to include in a future ListAttributes request. When the results of a ListAttributes request exceed maxResults, this value can be used to retrieve the next page of results. This value is null when there are no more results to return.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example lists the attributes for container instances that have the `stack=production` attribute in the default cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 122
X-Amz-Target: AmazonEC2ContainerServiceV20141113.ListAttributes
X-Amz-Date: 20161222T181559Z
User-Agent: aws-cli/1.11.30 Python/2.7.12 Darwin/16.3.0 botocore/1.4.87
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
```

```
{
  "cluster": "default",
  "attributeName": "stack",
  "attributeValue": "production",
  "targetType": "container-instance"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Thu, 22 Dec 2016 18:16:00 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 158
Connection: keep-alive
x-amzn-RequestId: b0eb3407-c872-11e6-a3b0-295902c79de2

{
  "attributes": [
    {
      "name": "stack",
      "targetId": "arn:aws:ecs:us-west-2:123456789012:container-instance/1c3be8ed-
df30-47b4-8f1e-6e68ebd01f34",
      "value": "production"
    }
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)

- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListClusters

Returns a list of existing clusters.

Request Syntax

```
{  
  "maxResults": number,  
  "nextToken": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

maxResults

The maximum number of cluster results that `ListClusters` returned in paginated output. When this parameter is used, `ListClusters` only returns `maxResults` results in a single page along with a `nextToken` response element. The remaining results of the initial request can be seen by sending another `ListClusters` request with the returned `nextToken` value. This value can be between 1 and 100. If this parameter isn't used, then `ListClusters` returns up to 100 results and a `nextToken` value if applicable.

Type: Integer

Required: No

nextToken

The `nextToken` value returned from a `ListClusters` request indicating that more results are available to fulfill the request and further calls are needed. If `maxResults` was provided, it's possible the number of results to be fewer than `maxResults`.

Note

This token should be treated as an opaque identifier that is only used to retrieve the next items in a list and not for other programmatic purposes.

Type: String

Required: No

Response Syntax

```
{  
  "clusterArns": [ "string" ],  
  "nextToken": "string"  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[clusterArns](#)

The list of full Amazon Resource Name (ARN) entries for each cluster that's associated with your account.

Type: Array of strings

[nextToken](#)

The nextToken value to include in a future ListClusters request. When the results of a ListClusters request exceed maxResults, this value can be used to retrieve the next page of results. This value is null when there are no more results to return.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request lists the clusters for your account.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
```

```
Accept-Encoding: identity
Content-Length: 2
X-Amz-Target: AmazonEC2ContainerServiceV20141113.ListClusters
X-Amz-Date: 20150429T170621Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
```

```
{}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 17:06:21 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 126
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "clusterArns": [
    "arn:aws:ecs:us-east-1:012345678910:cluster/My-cluster",
    "arn:aws:ecs:us-east-1:012345678910:cluster/default"
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)

- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListContainerInstances

Returns a list of container instances in a specified cluster. You can filter the results of a `ListContainerInstances` operation with cluster query language statements inside the `filter` parameter. For more information, see [Cluster Query Language](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{
  "cluster": "string",
  "filter": "string",
  "maxResults": number,
  "nextToken": "string",
  "status": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the container instances to list. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

filter

You can filter the results of a `ListContainerInstances` operation with cluster query language statements. For more information, see [Cluster Query Language](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

maxResults

The maximum number of container instance results that `ListContainerInstances` returned in paginated output. When this parameter is used, `ListContainerInstances` only returns `maxResults` results in a single page along with a `nextToken` response element. The remaining results of the initial request can be seen by sending another `ListContainerInstances` request with the returned `nextToken` value. This value can be between 1 and 100. If this parameter isn't used, then `ListContainerInstances` returns up to 100 results and a `nextToken` value if applicable.

Type: Integer

Required: No

nextToken

The `nextToken` value returned from a `ListContainerInstances` request indicating that more results are available to fulfill the request and further calls are needed. If `maxResults` was provided, it's possible the number of results to be fewer than `maxResults`.

Note

This token should be treated as an opaque identifier that is only used to retrieve the next items in a list and not for other programmatic purposes.

Type: String

Required: No

status

Filters the container instances by status. For example, if you specify the `DRAINING` status, the results include only container instances that have been set to `DRAINING` using [UpdateContainerInstancesState](#). If you don't specify this parameter, the default is to include container instances set to all states other than `INACTIVE`.

Type: String

Valid Values: `ACTIVE` | `DRAINING` | `REGISTERING` | `DEREGISTERING` | `REGISTRATION_FAILED`

Required: No

Response Syntax

```
{  
  "containerInstanceArns": [ "string" ],  
  "nextToken": "string"  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[containerInstanceArns](#)

The list of container instances with full ARN entries for each container instance associated with the specified cluster.

Type: Array of strings

[nextToken](#)

The nextToken value to include in a future ListContainerInstances request. When the results of a ListContainerInstances request exceed maxResults, this value can be used to retrieve the next page of results. This value is null when there are no more results to return.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request lists the container instances in the "example" cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 2
X-Amz-Target: AmazonEC2ContainerServiceV20141113.ListContainerInstances
X-Amz-Date: 20150429T175306Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{"containerInstanceArns":["arn:aws:ecs:us-east-1:012345678910:container-
instance/example/1eb22c8ab33149b397dc769f68cc1319","arn:aws:ecs:us-
east-1:012345678910:container-instance/example/5cf7e311a2b74d3882650353cf3b2214"]}]}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 17:53:06 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 492
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "containerInstanceArns": [
    "arn:aws:ecs:us-east-1:012345678910:container-instance/
example/1eb22c8ab33149b397dc769f68cc1319",
    "arn:aws:ecs:us-east-1:012345678910:container-instance/
example/5cf7e311a2b74d3882650353cf3b2214"
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)

- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListServiceDeployments

This operation lists all the service deployments that meet the specified filter criteria.

A service deployment happens when you release a software update for the service. You route traffic from the running service revisions to the new service revision and control the number of running tasks.

This API returns the values that you use for the request parameters in [DescribeServiceRevisions](#).

Request Syntax

```
{
  "cluster": "string",
  "createdAt": {
    "after": number,
    "before": number
  },
  "maxResults": number,
  "nextToken": "string",
  "service": "string",
  "status": [ "string" ]
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The cluster that hosts the service. This can either be the cluster name or ARN. Starting April 15, 2023, Amazon Web Services will not onboard new customers to Amazon Elastic Inference (EI), and will help current customers migrate their workloads to options that offer better price and performance. If you don't specify a cluster, default is used.

Type: String

Required: No

createdAt

An optional filter you can use to narrow the results by the service creation date. If you do not specify a value, the result includes all services created before the current time. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: [CreatedAt](#) object

Required: No

maxResults

The maximum number of service deployment results that `ListServiceDeployments` returned in paginated output. When this parameter is used, `ListServiceDeployments` only returns `maxResults` results in a single page along with a `nextToken` response element. The remaining results of the initial request can be seen by sending another `ListServiceDeployments` request with the returned `nextToken` value. This value can be between 1 and 100. If this parameter isn't used, then `ListServiceDeployments` returns up to 20 results and a `nextToken` value if applicable.

Type: Integer

Required: No

nextToken

The `nextToken` value returned from a `ListServiceDeployments` request indicating that more results are available to fulfill the request and further calls are needed. If you provided `maxResults`, it's possible the number of results is fewer than `maxResults`.

Type: String

Required: No

service

The ARN or name of the service

Type: String

Required: Yes

status

An optional filter you can use to narrow the results. If you do not specify a status, then all status values are included in the result.

Type: Array of strings

Valid Values: PENDING | SUCCESSFUL | STOPPED | STOP_REQUESTED | IN_PROGRESS
| ROLLBACK_REQUESTED | ROLLBACK_IN_PROGRESS | ROLLBACK_SUCCESSFUL |
ROLLBACK_FAILED

Required: No

Response Syntax

```
{
  "nextToken": "string",
  "serviceDeployments": [
    {
      "clusterArn": "string",
      "createdAt": number,
      "finishedAt": number,
      "serviceArn": "string",
      "serviceDeploymentArn": "string",
      "startedAt": number,
      "status": "string",
      "statusReason": "string",
      "targetServiceRevisionArn": "string"
    }
  ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

nextToken

The nextToken value to include in a future ListServiceDeployments request. When the results of a ListServiceDeployments request exceed maxResults, this value can be used to retrieve the next page of results. This value is null when there are no more results to return.

Type: String

serviceDeployments

An overview of the service deployment, including the following properties:

- The ARN of the service deployment.
- The ARN of the service being deployed.
- The ARN of the cluster that hosts the service in the service deployment.
- The time that the service deployment started.
- The time that the service deployment completed.
- The service deployment status.
- Information about why the service deployment is in the current state.
- The ARN of the service revision that is being deployed.

Type: Array of [ServiceDeploymentBrief](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

Examples

Example

This example request lists the service deployments for the sd-example service on the example cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 55
X-Amz-Target: AmazonEC2ContainerServiceV20141113.DescribeServiceDeployments
X-Amz-Date: 20150528T163859Z
User-Agent: aws-cli/2.17 Python/3.11.9 Darwin/14.3.0
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
{
```

```
"cluster": "example",
"service": "sd-example"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Tue, 10 Sep 2024 18:50:14 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 220
Connection: keep-alive
RequestId: 360c5551-123e-4e74-9914-7582d3a28807
{
  "serviceDeployments": [
    {
      "serviceDeploymentArn": "arn:aws:ecs:us-west-2:123456789012:service-
deployment/example/sd-example/NCWGC2ZR-taawPAYrIaU5",
      "serviceArn": "arn:aws:ecs:us-west-2:123456789012:service/example/sd-
example",
      "clusterArn": "arn:aws:ecs:us-west-2:123456789012:cluster/example",
      "targetServiceRevisionArn": "arn:aws:ecs:us-west-2:123456789012:service-
revision/example/sd-example/4980306466373577095",
      "status": "SUCCESSFUL"
    }
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)

- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListServices

Returns a list of services. You can filter the results by cluster, launch type, and scheduling strategy.

Request Syntax

```
{  
  "cluster": "string",  
  "launchType": "string",  
  "maxResults": number,  
  "nextToken": "string",  
  "schedulingStrategy": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster to use when filtering the ListServices results. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

launchType

The launch type to use when filtering the ListServices results.

Type: String

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

maxResults

The maximum number of service results that ListServices returned in paginated output. When this parameter is used, ListServices only returns maxResults results in a single page

along with a `nextToken` response element. The remaining results of the initial request can be seen by sending another `ListServices` request with the returned `nextToken` value. This value can be between 1 and 100. If this parameter isn't used, then `ListServices` returns up to 10 results and a `nextToken` value if applicable.

Type: Integer

Required: No

`nextToken`

The `nextToken` value returned from a `ListServices` request indicating that more results are available to fulfill the request and further calls will be needed. If `maxResults` was provided, it is possible the number of results to be fewer than `maxResults`.

Note

This token should be treated as an opaque identifier that is only used to retrieve the next items in a list and not for other programmatic purposes.

Type: String

Required: No

`schedulingStrategy`

The scheduling strategy to use when filtering the `ListServices` results.

Type: String

Valid Values: `REPLICA` | `DAEMON`

Required: No

Response Syntax

```
{
  "nextToken": "string",
  "serviceArns": [ "string" ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

nextToken

The `nextToken` value to include in a future `ListServices` request. When the results of a `ListServices` request exceed `maxResults`, this value can be used to retrieve the next page of results. This value is `null` when there are no more results to return.

Type: String

serviceArns

The list of full ARN entries for each service that's associated with the specified cluster.

Type: Array of strings

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request lists the services in the default cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 2
X-Amz-Target: AmazonEC2ContainerServiceV20141113.ListServices
X-Amz-Date: 20150429T191342Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
```

```
{}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 19:13:42 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 138
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "serviceArns": [
    "arn:aws:ecs:us-east-1:012345678910:service/hello_world",
    "arn:aws:ecs:us-east-1:012345678910:service/ecs-simple-service"
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListServicesByNamespace

This operation lists all of the services that are associated with a AWS Cloud Map namespace. This list might include services in different clusters. In contrast, `ListServices` can only list services in one cluster at a time. If you need to filter the list of services in a single cluster by various parameters, use `ListServices`. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{  
  "maxResults": number,  
  "namespace": "string",  
  "nextToken": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[maxResults](#)

The maximum number of service results that `ListServicesByNamespace` returns in paginated output. When this parameter is used, `ListServicesByNamespace` only returns `maxResults` results in a single page along with a `nextToken` response element. The remaining results of the initial request can be seen by sending another `ListServicesByNamespace` request with the returned `nextToken` value. This value can be between 1 and 100. If this parameter isn't used, then `ListServicesByNamespace` returns up to 10 results and a `nextToken` value if applicable.

Type: Integer

Required: No

[namespace](#)

The namespace name or full Amazon Resource Name (ARN) of the AWS Cloud Map namespace to list the services in.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: Yes

[nextToken](#)

The `nextToken` value that's returned from a `ListServicesByNamespace` request. It indicates that more results are available to fulfill the request and further calls are needed. If `maxResults` is returned, it is possible the number of results is less than `maxResults`.

Type: String

Required: No

Response Syntax

```
{
  "nextToken": "string",
  "serviceArns": [ "string" ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[nextToken](#)

The `nextToken` value to include in a future `ListServicesByNamespace` request. When the results of a `ListServicesByNamespace` request exceed `maxResults`, this value can be used to retrieve the next page of results. When there are no more results to return, this value is `null`.

Type: String

serviceArns

The list of full ARN entries for each service that's associated with the specified namespace.

Type: Array of strings

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

NamespaceNotFoundException

The specified namespace wasn't found.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListTagsForResource

List the tags for an Amazon ECS resource.

Request Syntax

```
{  
  "resourceArn": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

resourceArn

The Amazon Resource Name (ARN) that identifies the resource to list the tags for. Currently, the supported resources are Amazon ECS tasks, services, task definitions, clusters, and container instances.

Type: String

Required: Yes

Response Syntax

```
{  
  "tags": [  
    {  
      "key": "string",  
      "value": "string"  
    }  
  ]  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

tags

The tags for the resource.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example lists the tags for the dev cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
X-Amz-Target: AmazonEC2ContainerServiceV20141113.ListTagsForResource
Content-Type: application/x-amz-json-1.1
X-Amz-Date: 20181026T195430Z
Authorization: AUTHPARAMS
Content-Length: 72

{
  "resourceArn": "arn:aws:ecs:us-west-2:012345678910:cluster/dev"
}
```

Sample Response

```
HTTP/1.1 200 OK
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f
Content-Type: application/x-amz-json-1.1
```

Content-Length: 39

Date: Fri, 26 Oct 2018 19:54:31 GMT

```
{
  "tags":[
    {
      "key":"team",
      "value":"dev"
    }
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListTaskDefinitionFamilies

Returns a list of task definition families that are registered to your account. This list includes task definition families that no longer have any ACTIVE task definition revisions.

You can filter out task definition families that don't contain any ACTIVE task definition revisions by setting the status parameter to ACTIVE. You can also filter the results with the familyPrefix parameter.

Request Syntax

```
{
  "familyPrefix": "string",
  "maxResults": number,
  "nextToken": "string",
  "status": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

familyPrefix

The familyPrefix is a string that's used to filter the results of ListTaskDefinitionFamilies. If you specify a familyPrefix, only task definition family names that begin with the familyPrefix string are returned.

Type: String

Required: No

maxResults

The maximum number of task definition family results that ListTaskDefinitionFamilies returned in paginated output. When this parameter is used, ListTaskDefinitions only returns maxResults results in a single page along with a nextToken response element. The remaining results of the initial request can be seen by sending another ListTaskDefinitionFamilies request with the returned nextToken value. This value can

be between 1 and 100. If this parameter isn't used, then `ListTaskDefinitionFamilies` returns up to 100 results and a `nextToken` value if applicable.

Type: Integer

Required: No

nextToken

The `nextToken` value returned from a `ListTaskDefinitionFamilies` request indicating that more results are available to fulfill the request and further calls will be needed. If `maxResults` was provided, it is possible the number of results to be fewer than `maxResults`.

Note

This token should be treated as an opaque identifier that is only used to retrieve the next items in a list and not for other programmatic purposes.

Type: String

Required: No

status

The task definition family status to filter the `ListTaskDefinitionFamilies` results with. By default, both `ACTIVE` and `INACTIVE` task definition families are listed. If this parameter is set to `ACTIVE`, only task definition families that have an `ACTIVE` task definition revision are returned. If this parameter is set to `INACTIVE`, only task definition families that do not have any `ACTIVE` task definition revisions are returned. If you paginate the resulting output, be sure to keep the `status` value constant in each subsequent request.

Type: String

Valid Values: `ACTIVE` | `INACTIVE` | `ALL`

Required: No

Response Syntax

```
{
```

```
"families": [ "string" ],  
"nextToken": "string"  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[families](#)

The list of task definition family names that match the `ListTaskDefinitionFamilies` request.

Type: Array of strings

[nextToken](#)

The `nextToken` value to include in a future `ListTaskDefinitionFamilies` request. When the results of a `ListTaskDefinitionFamilies` request exceed `maxResults`, this value can be used to retrieve the next page of results. This value is `null` when there are no more results to return.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request lists all of the task definition families in your account in the current Region.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 2
X-Amz-Target: AmazonEC2ContainerServiceV20141113.ListTaskDefinitionFamilies
X-Amz-Date: 20150429T191650Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
```

```
{}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 19:16:51 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 270
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f
```

```
{
  "families": [
    "console-sample-app",
    "ecs-demo",
    "ecs-private",
    "hello_world",
    "hpcc",
    "hpcc-t2-medium",
    "image-dedupe",
    "node-dedupe",
    "port-mappings",
    "redis-volumes-from",
    "sleep360",
    "terrible-timer",
    "test-volumes-from",
    "tt-empty",
    "tt-empty-2vol",
    "tt-empty-volumes",
    "web-timer"
  ]
}
```

Example

This example request lists all of the task definition families in your account in the current Region that begin with `hpcc`.

Sample Request

```
POST / HTTP/1.1
```

```
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 24
X-Amz-Target: AmazonEC2ContainerServiceV20141113.ListTaskDefinitionFamilies
X-Amz-Date: 20150429T191825Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "familyPrefix": "hpcc"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 19:18:25 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 38
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "families": [
    "hpcc",
    "hpcc-t2-medium"
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)

- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListTaskDefinitions

Returns a list of task definitions that are registered to your account. You can filter the results by family name with the `familyPrefix` parameter or by status with the `status` parameter.

Request Syntax

```
{  
  "familyPrefix": "string",  
  "maxResults": number,  
  "nextToken": "string",  
  "sort": "string",  
  "status": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

familyPrefix

The full family name to filter the `ListTaskDefinitions` results with. Specifying a `familyPrefix` limits the listed task definitions to task definition revisions that belong to that family.

Type: String

Required: No

maxResults

The maximum number of task definition results that `ListTaskDefinitions` returned in paginated output. When this parameter is used, `ListTaskDefinitions` only returns `maxResults` results in a single page along with a `nextToken` response element. The remaining results of the initial request can be seen by sending another `ListTaskDefinitions` request with the returned `nextToken` value. This value can be between 1 and 100. If this parameter isn't used, then `ListTaskDefinitions` returns up to 100 results and a `nextToken` value if applicable.

Type: Integer

Required: No

nextToken

The `nextToken` value returned from a `ListTaskDefinitions` request indicating that more results are available to fulfill the request and further calls will be needed. If `maxResults` was provided, it is possible the number of results to be fewer than `maxResults`.

Note

This token should be treated as an opaque identifier that is only used to retrieve the next items in a list and not for other programmatic purposes.

Type: String

Required: No

sort

The order to sort the results in. Valid values are `ASC` and `DESC`. By default, (`ASC`) task definitions are listed lexicographically by family name and in ascending numerical order by revision so that the newest task definitions in a family are listed last. Setting this parameter to `DESC` reverses the sort order on family name and revision. This is so that the newest task definitions in a family are listed first.

Type: String

Valid Values: `ASC` | `DESC`

Required: No

status

The task definition status to filter the `ListTaskDefinitions` results with. By default, only `ACTIVE` task definitions are listed. By setting this parameter to `INACTIVE`, you can view task definitions that are `INACTIVE` as long as an active task or service still references them. If you paginate the resulting output, be sure to keep the `status` value constant in each subsequent request.

Type: String

Valid Values: ACTIVE | INACTIVE | DELETE_IN_PROGRESS

Required: No

Response Syntax

```
{  
  "nextToken": "string",  
  "taskDefinitionArns": [ "string" ]  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

nextToken

The nextToken value to include in a future ListTaskDefinitions request. When the results of a ListTaskDefinitions request exceed maxResults, this value can be used to retrieve the next page of results. This value is null when there are no more results to return.

Type: String

taskDefinitionArns

The list of task definition Amazon Resource Name (ARN) entries for the ListTaskDefinitions request.

Type: Array of strings

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request lists all of the task definitions in the hello_world family.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
```



```
Content-Length: 31
X-Amz-Target: AmazonEC2ContainerServiceV20141113.ListTaskDefinitions
X-Amz-Date: 20150429T192041Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "familyPrefix": "hello_world"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 19:20:41 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 695
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "taskDefinitionArns": [
    "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:1",
    "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:2",
    "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:3",
    "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:4",
    "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:5",
    "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:6",
    "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:7",
    "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:8",
    "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:9",
    "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:10"
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)

- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

ListTasks

Returns a list of tasks. You can filter the results by cluster, task definition family, container instance, launch type, what IAM principal started the task, or by the desired status of the task.

Recently stopped tasks might appear in the returned results.

Request Syntax

```
{
  "cluster": "string",
  "containerInstance": "string",
  "desiredStatus": "string",
  "family": "string",
  "launchType": "string",
  "maxResults": number,
  "nextToken": "string",
  "serviceName": "string",
  "startedBy": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster to use when filtering the ListTasks results. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

containerInstance

The container instance ID or full ARN of the container instance to use when filtering the ListTasks results. Specifying a containerInstance limits the results to tasks that belong to that container instance.

Type: String

Required: No

desiredStatus

The task desired status to use when filtering the `ListTasks` results. Specifying a `desiredStatus` of `STOPPED` limits the results to tasks that Amazon ECS has set the desired status to `STOPPED`. This can be useful for debugging tasks that aren't starting properly or have died or finished. The default status filter is `RUNNING`, which shows tasks that Amazon ECS has set the desired status to `RUNNING`.

Note

Although you can filter results based on a desired status of `PENDING`, this doesn't return any results. Amazon ECS never sets the desired status of a task to that value (only a task's `lastStatus` may have a value of `PENDING`).

Type: String

Valid Values: `RUNNING` | `PENDING` | `STOPPED`

Required: No

family

The name of the task definition family to use when filtering the `ListTasks` results. Specifying a `family` limits the results to tasks that belong to that family.

Type: String

Required: No

launchType

The launch type to use when filtering the `ListTasks` results.

Type: String

Valid Values: `EC2` | `FARGATE` | `EXTERNAL` | `MANAGED_INSTANCES`

Required: No

maxResults

The maximum number of task results that `ListTasks` returned in paginated output. When this parameter is used, `ListTasks` only returns `maxResults` results in a single page along with a `nextToken` response element. The remaining results of the initial request can be seen by sending another `ListTasks` request with the returned `nextToken` value. This value can be between 1 and 100. If this parameter isn't used, then `ListTasks` returns up to 100 results and a `nextToken` value if applicable.

Type: Integer

Required: No

nextToken

The `nextToken` value returned from a `ListTasks` request indicating that more results are available to fulfill the request and further calls will be needed. If `maxResults` was provided, it's possible the number of results to be fewer than `maxResults`.

Note

This token should be treated as an opaque identifier that is only used to retrieve the next items in a list and not for other programmatic purposes.

Type: String

Required: No

serviceName

The name of the service to use when filtering the `ListTasks` results. Specifying a `serviceName` limits the results to tasks that belong to that service.

Type: String

Required: No

startedBy

The `startedBy` value to filter the task results with. Specifying a `startedBy` value limits the results to tasks that were started with that value.

When you specify `startedBy` as the filter, it must be the only filter that you use.

Type: String

Required: No

Response Syntax

```
{
  "nextToken": "string",
  "taskArns": [ "string" ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

nextToken

The nextToken value to include in a future ListTasks request. When the results of a ListTasks request exceed maxResults, this value can be used to retrieve the next page of results. This value is null when there are no more results to return.

Type: String

taskArns

The list of task ARN entries for the ListTasks request.

Type: Array of strings

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request lists all of the tasks in the default cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 2
X-Amz-Target: AmazonEC2ContainerServiceV20141113.ListTasks
X-Amz-Date: 20150429T192615Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 19:26:16 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 330
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "taskArns": [
    "arn:aws:ecs:us-east-1:012345678910:task/0b69d5c0-d655-4695-98cd-5d2d526d9d5a",
    "arn:aws:ecs:us-east-1:012345678910:task/51a01bdf-d00e-487e-ab14-7645330b6207",
    "arn:aws:ecs:us-east-1:012345678910:task/b0b28bb8-2be3-4810-b52b-88df129d893c",
    "arn:aws:ecs:us-east-1:012345678910:task/c09f0188-7f87-4b0f-bfc3-16296622b6fe"
  ]
}
```


See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutAccountSetting

Modifies an account setting. Account settings are set on a per-Region basis.

If you change the root user account setting, the default settings are reset for users and roles that do not have specified individual account settings. For more information, see [Account Settings](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{  
  "name": "string",  
  "principalArn": "string",  
  "value": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[name](#)

The Amazon ECS account setting name to modify.

The following are the valid values for the account setting name.

- `serviceLongArnFormat` - When modified, the Amazon Resource Name (ARN) and resource ID format of the resource type for a specified user, role, or the root user for an account is affected. The opt-in and opt-out account setting must be set for each Amazon ECS resource separately. The ARN and resource ID format of a resource is defined by the opt-in status of the user or role that created the resource. You must turn on this setting to use Amazon ECS features such as resource tagging.
- `taskLongArnFormat` - When modified, the Amazon Resource Name (ARN) and resource ID format of the resource type for a specified user, role, or the root user for an account is affected. The opt-in and opt-out account setting must be set for each Amazon ECS resource separately. The ARN and resource ID format of a resource is defined by the opt-in status of the user or role that created the resource. You must turn on this setting to use Amazon ECS features such as resource tagging.

- `containerInstanceLongArnFormat` - When modified, the Amazon Resource Name (ARN) and resource ID format of the resource type for a specified user, role, or the root user for an account is affected. The opt-in and opt-out account setting must be set for each Amazon ECS resource separately. The ARN and resource ID format of a resource is defined by the opt-in status of the user or role that created the resource. You must turn on this setting to use Amazon ECS features such as resource tagging.
- `awsvpcTrunking` - When modified, the elastic network interface (ENI) limit for any new container instances that support the feature is changed. If `awsvpcTrunking` is turned on, any new container instances that support the feature are launched have the increased ENI limits available to them. For more information, see [Elastic Network Interface Trunking](#) in the *Amazon Elastic Container Service Developer Guide*.
- `containerInsights` - Container Insights with enhanced observability provides all the Container Insights metrics, plus additional task and container metrics. This version supports enhanced observability for Amazon ECS clusters using the Amazon EC2 and Fargate launch types. After you configure Container Insights with enhanced observability on Amazon ECS, Container Insights auto-collects detailed infrastructure telemetry from the cluster level down to the container level in your environment and displays these critical performance data in curated dashboards removing the heavy lifting in observability set-up.

To use Container Insights with enhanced observability, set the `containerInsights` account setting to enhanced.

To use Container Insights, set the `containerInsights` account setting to enabled.

For more information, see [Monitor Amazon ECS containers using Container Insights with enhanced observability](#) in the *Amazon Elastic Container Service Developer Guide*.

- `dualStackIPv6` - When turned on, when using a VPC in dual stack mode, your tasks using the `awsvpc` network mode can have an IPv6 address assigned. For more information on using IPv6 with tasks launched on Amazon EC2 instances, see [Using a VPC in dual-stack mode](#). For more information on using IPv6 with tasks launched on Fargate, see [Using a VPC in dual-stack mode](#).
- `fargateTaskRetirementWaitPeriod` - When AWS determines that a security or infrastructure update is needed for an Amazon ECS task hosted on Fargate, the tasks need to be stopped and new tasks launched to replace them. Use `fargateTaskRetirementWaitPeriod` to configure the wait time to retire a Fargate task. For information about the Fargate tasks maintenance, see [AWS Fargate task maintenance](#) in the *Amazon ECS Developer Guide*.

- `tagResourceAuthorization` - Amazon ECS is introducing tagging authorization for resource creation. Users must have permissions for actions that create the resource, such as `ecsCreateCluster`. If tags are specified when you create a resource, AWS performs additional authorization to verify if users or roles have permissions to create tags. Therefore, you must grant explicit permissions to use the `ecs:TagResource` action. For more information, see [Grant permission to tag resources on creation](#) in the *Amazon ECS Developer Guide*.
- `defaultLogDriverMode` - Amazon ECS supports setting a default delivery mode of log messages from a container to the `logDriver` that you specify in the container's `logConfiguration`. The delivery mode affects application stability when the flow of logs from the container to the log driver is interrupted. The `defaultLogDriverMode` setting supports two values: `blocking` and `non-blocking`. If you don't specify a delivery mode in your container definition's `logConfiguration`, the mode you specify using this account setting will be used as the default. For more information about log delivery modes, see [LogConfiguration](#).

 Note

On June 25, 2025, Amazon ECS changed the default log driver mode from `blocking` to `non-blocking` to prioritize task availability over logging. To continue using the `blocking` mode after this change, do one of the following:

- Set the `mode` option in your container definition's `logConfiguration` as `blocking`.
 - Set the `defaultLogDriverMode` account setting to `blocking`.
- `guardDutyActivate` - The `guardDutyActivate` parameter is read-only in Amazon ECS and indicates whether Amazon ECS Runtime Monitoring is enabled or disabled by your security administrator in your Amazon ECS account. Amazon GuardDuty controls this account setting on your behalf. For more information, see [Protecting Amazon ECS workloads with Amazon ECS Runtime Monitoring](#).

Type: String

Valid Values: `serviceLongArnFormat` | `taskLongArnFormat`
| `containerInstanceLongArnFormat` | `awsvpcTrunking` |
`containerInsights` | `fargateFIPSMODE` | `tagResourceAuthorization`

| `fargateTaskRetirementWaitPeriod` | `guardDutyActivate` |
`defaultLogDriverMode`

Required: Yes

principalArn

The ARN of the principal, which can be a user, role, or the root user. If you specify the root user, it modifies the account setting for all users, roles, and the root user of the account unless a user or role explicitly overrides these settings. If this field is omitted, the setting is changed only for the authenticated user.

In order to use this parameter, you must be the root user, or the principal.

Note

You must use the root user when you set the Fargate wait time (`fargateTaskRetirementWaitPeriod`).

Federated users assume the account setting of the root user and can't have explicit account settings set for them.

Type: String

Required: No

value

The account setting value for the specified principal ARN. Accepted values are `enabled`, `disabled`, `enhanced`, `on`, and `off`.

When you specify `fargateTaskRetirementWaitPeriod` for the name, the following are the valid values:

- `0` - AWS sends the notification, and immediately retires the affected tasks.
- `7` - AWS sends the notification, and waits 7 calendar days to retire the tasks.
- `14` - AWS sends the notification, and waits 14 calendar days to retire the tasks.

Type: String

Required: Yes

Response Syntax

```
{
  "setting": {
    "name": "string",
    "principalArn": "string",
    "type": "string",
    "value": "string"
  }
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

setting

The current account setting for a resource.

Type: [Setting](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutAccountSettingDefault

Modifies an account setting for all users on an account for whom no individual account setting has been specified. Account settings are set on a per-Region basis.

Request Syntax

```
{
  "name": "string",
  "value": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

name

The resource name for which to modify the account setting.

The following are the valid values for the account setting name.

- `serviceLongArnFormat` - When modified, the Amazon Resource Name (ARN) and resource ID format of the resource type for a specified user, role, or the root user for an account is affected. The opt-in and opt-out account setting must be set for each Amazon ECS resource separately. The ARN and resource ID format of a resource is defined by the opt-in status of the user or role that created the resource. You must turn on this setting to use Amazon ECS features such as resource tagging.
- `taskLongArnFormat` - When modified, the Amazon Resource Name (ARN) and resource ID format of the resource type for a specified user, role, or the root user for an account is affected. The opt-in and opt-out account setting must be set for each Amazon ECS resource separately. The ARN and resource ID format of a resource is defined by the opt-in status of the user or role that created the resource. You must turn on this setting to use Amazon ECS features such as resource tagging.
- `containerInstanceLongArnFormat` - When modified, the Amazon Resource Name (ARN) and resource ID format of the resource type for a specified user, role, or the root user for an account is affected. The opt-in and opt-out account setting must be set for each Amazon

ECS resource separately. The ARN and resource ID format of a resource is defined by the opt-in status of the user or role that created the resource. You must turn on this setting to use Amazon ECS features such as resource tagging.

- `awsVpcTrunking` - When modified, the elastic network interface (ENI) limit for any new container instances that support the feature is changed. If `awsVpcTrunking` is turned on, any new container instances that support the feature are launched have the increased ENI limits available to them. For more information, see [Elastic Network Interface Trunking](#) in the *Amazon Elastic Container Service Developer Guide*.
- `containerInsights` - Container Insights with enhanced observability provides all the Container Insights metrics, plus additional task and container metrics. This version supports enhanced observability for Amazon ECS clusters using the Amazon EC2 and Fargate launch types. After you configure Container Insights with enhanced observability on Amazon ECS, Container Insights auto-collects detailed infrastructure telemetry from the cluster level down to the container level in your environment and displays these critical performance data in curated dashboards removing the heavy lifting in observability set-up.

To use Container Insights with enhanced observability, set the `containerInsights` account setting to `enhanced`.

To use Container Insights, set the `containerInsights` account setting to `enabled`.

For more information, see [Monitor Amazon ECS containers using Container Insights with enhanced observability](#) in the *Amazon Elastic Container Service Developer Guide*.

- `dualStackIPv6` - When turned on, when using a VPC in dual stack mode, your tasks using the `awsVpc` network mode can have an IPv6 address assigned. For more information on using IPv6 with tasks launched on Amazon EC2 instances, see [Using a VPC in dual-stack mode](#). For more information on using IPv6 with tasks launched on Fargate, see [Using a VPC in dual-stack mode](#).
- `fargateFIPSMODE` - If you specify `fargateFIPSMODE`, AWS Fargate FIPS 140 compliance is affected.
- `fargateTaskRetirementWaitPeriod` - When AWS determines that a security or infrastructure update is needed for an Amazon ECS task hosted on Fargate, the tasks need to be stopped and new tasks launched to replace them. Use `fargateTaskRetirementWaitPeriod` to configure the wait time to retire a Fargate task. For information about the Fargate tasks maintenance, see [AWS Fargate task maintenance](#) in the *Amazon ECS Developer Guide*.

- `tagResourceAuthorization` - Amazon ECS is introducing tagging authorization for resource creation. Users must have permissions for actions that create the resource, such as `ecsCreateCluster`. If tags are specified when you create a resource, AWS performs additional authorization to verify if users or roles have permissions to create tags. Therefore, you must grant explicit permissions to use the `ecs:TagResource` action. For more information, see [Grant permission to tag resources on creation](#) in the *Amazon ECS Developer Guide*.
- `defaultLogDriverMode` - Amazon ECS supports setting a default delivery mode of log messages from a container to the `logDriver` that you specify in the container's `logConfiguration`. The delivery mode affects application stability when the flow of logs from the container to the log driver is interrupted. The `defaultLogDriverMode` setting supports two values: `blocking` and `non-blocking`. If you don't specify a delivery mode in your container definition's `logConfiguration`, the mode you specify using this account setting will be used as the default. For more information about log delivery modes, see [LogConfiguration](#).

 Note

On June 25, 2025, Amazon ECS changed the default log driver mode from `blocking` to `non-blocking` to prioritize task availability over logging. To continue using the `blocking` mode after this change, do one of the following:

- Set the `mode` option in your container definition's `logConfiguration` as `blocking`.
 - Set the `defaultLogDriverMode` account setting to `blocking`.
- `guardDutyActivate` - The `guardDutyActivate` parameter is read-only in Amazon ECS and indicates whether Amazon ECS Runtime Monitoring is enabled or disabled by your security administrator in your Amazon ECS account. Amazon GuardDuty controls this account setting on your behalf. For more information, see [Protecting Amazon ECS workloads with Amazon ECS Runtime Monitoring](#).

Type: String

Valid Values: `serviceLongArnFormat` | `taskLongArnFormat`
| `containerInstanceLongArnFormat` | `awsvpcTrunking` |
`containerInsights` | `fargateFIPSMODE` | `tagResourceAuthorization`

| fargateTaskRetirementWaitPeriod | guardDutyActivate |
defaultLogDriverMode

Required: Yes

value

The account setting value for the specified principal ARN. Accepted values are enabled, disabled, on, enhanced, and off.

When you specify `fargateTaskRetirementWaitPeriod` for the name, the following are the valid values:

- 0 - AWS sends the notification, and immediately retires the affected tasks.
- 7 - AWS sends the notification, and waits 7 calendar days to retire the tasks.
- 14 - AWS sends the notification, and waits 14 calendar days to retire the tasks.

Type: String

Required: Yes

Response Syntax

```
{
  "setting": {
    "name": "string",
    "principalArn": "string",
    "type": "string",
    "value": "string"
  }
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

setting

The current setting for a resource.

Type: [Setting](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)

- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutAttributes

Create or update an attribute on an Amazon ECS resource. If the attribute doesn't exist, it's created. If the attribute exists, its value is replaced with the specified value. To delete an attribute, use [DeleteAttributes](#). For more information, see [Attributes](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{
  "attributes": [
    {
      "name": "string",
      "targetId": "string",
      "targetType": "string",
      "value": "string"
    }
  ],
  "cluster": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[attributes](#)

The attributes to apply to your resource. You can specify up to 10 custom attributes for each resource. You can specify up to 10 attributes in a single call.

Type: Array of [Attribute](#) objects

Required: Yes

[cluster](#)

The short name or full Amazon Resource Name (ARN) of the cluster that contains the resource to apply attributes. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

Response Syntax

```
{
  "attributes": [
    {
      "name": "string",
      "targetId": "string",
      "targetType": "string",
      "value": "string"
    }
  ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[attributes](#)

The attributes applied to your resource.

Type: Array of [Attribute](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AttributeLimitExceededException

You can apply up to 10 custom attributes for each resource. You can view the attributes of a resource with [ListAttributes](#). You can remove existing attributes on a resource with [DeleteAttributes](#).

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

TargetNotFoundException

The specified target wasn't found. You can view your available container instances with [ListContainerInstances](#). Amazon ECS container instances are cluster-specific and Region-specific.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example applies an attribute with the name `stack` and the value `production` to a container instance.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 192
```



```
X-Amz-Target: AmazonEC2ContainerServiceV20141113.PutAttributes
X-Amz-Date: 20161222T180005Z
User-Agent: aws-cli/1.11.30 Python/2.7.12 Darwin/16.3.0 botocore/1.4.87
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "cluster": "default",
  "attributes": [
    {
      "targetId": "arn:aws:ecs:us-west-2:123456789012:container-instance/1c3be8ed-
df30-47b4-8f1e-6e68ebd01f34",
      "name": "stack",
      "value": "production"
    }
  ]
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Thu, 22 Dec 2016 18:00:06 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 158
Connection: keep-alive
x-amzn-RequestId: 7835c1be-c870-11e6-a3b0-295902c79de2

{
  "attributes": [
    {
      "name": "stack",
      "targetId": "arn:aws:ecs:us-west-2:123456789012:container-instance/1c3be8ed-
df30-47b4-8f1e-6e68ebd01f34",
      "value": "production"
    }
  ]
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

PutClusterCapacityProviders

Modifies the available capacity providers and the default capacity provider strategy for a cluster.

You must specify both the available capacity providers and a default capacity provider strategy for the cluster. If the specified cluster has existing capacity providers associated with it, you must specify all existing capacity providers in addition to any new ones you want to add. Any existing capacity providers that are associated with a cluster that are omitted from a [PutClusterCapacityProviders](#) API call will be disassociated with the cluster. You can only disassociate an existing capacity provider from a cluster if it's not being used by any existing tasks.

When creating a service or running a task on a cluster, if no capacity provider or launch type is specified, then the cluster's default capacity provider strategy is used. We recommend that you define a default capacity provider strategy for your cluster. However, you must specify an empty array ([]) to bypass defining a default strategy.

Amazon ECS Managed Instances doesn't support this, because when you create a capacity provider with Amazon ECS Managed Instances, it becomes available only within the specified cluster.

Request Syntax

```
{
  "capacityProviders": [ "string" ],
  "cluster": "string",
  "defaultCapacityProviderStrategy": [
    {
      "base": number,
      "capacityProvider": "string",
      "weight": number
    }
  ]
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

capacityProviders

The name of one or more capacity providers to associate with the cluster.

If specifying a capacity provider that uses an Auto Scaling group, the capacity provider must already be created. New capacity providers can be created with the [CreateCapacityProvider](#) API operation.

To use a AWS Fargate capacity provider, specify either the FARGATE or FARGATE_SPOT capacity providers. The AWS Fargate capacity providers are available to all accounts and only need to be associated with a cluster to be used.

Type: Array of strings

Required: Yes

cluster

The short name or full Amazon Resource Name (ARN) of the cluster to modify the capacity provider settings for. If you don't specify a cluster, the default cluster is assumed.

Type: String

Required: Yes

defaultCapacityProviderStrategy

The capacity provider strategy to use by default for the cluster.

When creating a service or running a task on a cluster, if no capacity provider or launch type is specified then the default capacity provider strategy for the cluster is used.

A capacity provider strategy consists of one or more capacity providers along with the base and weight to assign to them. A capacity provider must be associated with the cluster to be used in a capacity provider strategy. The [PutClusterCapacityProviders](#) API is used to associate a capacity provider with a cluster. Only capacity providers with an ACTIVE or UPDATING status can be used.

If specifying a capacity provider that uses an Auto Scaling group, the capacity provider must already be created. New capacity providers can be created with the [CreateCapacityProvider](#) API operation.

To use a AWS Fargate capacity provider, specify either the FARGATE or FARGATE_SPOT capacity providers. The AWS Fargate capacity providers are available to all accounts and only need to be associated with a cluster to be used.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: Yes

Response Syntax

```
{
  "cluster": {
    "activeServicesCount": number,
    "attachments": [
      {
        "details": [
          {
            "name": "string",
            "value": "string"
          }
        ],
        "id": "string",
        "status": "string",
        "type": "string"
      }
    ],
    "attachmentsStatus": "string",
    "capacityProviders": [ "string" ],
    "clusterArn": "string",
    "clusterName": "string",
    "configuration": {
      "executeCommandConfiguration": {
        "kmsKeyId": "string",
        "logConfiguration": {
          "cloudWatchEncryptionEnabled": boolean,
          "cloudWatchLogGroupName": "string",
          "s3BucketName": "string",
          "s3EncryptionEnabled": boolean,
          "s3KeyPrefix": "string"
        },
        "logging": "string"
      }
    },
  },
}
```

```

    "managedStorageConfiguration": {
      "fargateEphemeralStorageKmsKeyId": "string",
      "kmsKeyId": "string"
    },
    "defaultCapacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "pendingTasksCount": number,
    "registeredContainerInstancesCount": number,
    "runningTasksCount": number,
    "serviceConnectDefaults": {
      "namespace": "string"
    },
    "settings": [
      {
        "name": "string",
        "value": "string"
      }
    ],
    "statistics": [
      {
        "name": "string",
        "value": "string"
      }
    ],
    "status": "string",
    "tags": [
      {
        "key": "string",
        "value": "string"
      }
    ]
  }
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

cluster

Details about the cluster.

Type: [Cluster](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ResourceInUseException

The specified resource is in-use and can't be removed.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UpdateInProgressException

There's already a current Amazon ECS container agent update in progress on the container instance that's specified. If the container agent becomes disconnected while it's in a transitional stage, such as PENDING or STAGING, the update process can get stuck in that state. However, when the agent reconnects, it resumes where it stopped previously.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

RegisterContainerInstance

Note

This action is only used by the Amazon ECS agent, and it is not intended for use outside of the agent.

Registers an EC2 instance into the specified cluster. This instance becomes available to place containers on.

Request Syntax

```
{
  "attributes": [
    {
      "name": "string",
      "targetId": "string",
      "targetType": "string",
      "value": "string"
    }
  ],
  "cluster": "string",
  "containerInstanceArn": "string",
  "instanceIdentityDocument": "string",
  "instanceIdentityDocumentSignature": "string",
  "platformDevices": [
    {
      "id": "string",
      "type": "string"
    }
  ],
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ],
  "totalResources": [
    {
      "doubleValue": number,
      "integerValue": number,
```

```
    "longValue": number,
    "name": "string",
    "stringSetValue": [ "string" ],
    "type": "string"
  }
],
"versionInfo": {
  "agentHash": "string",
  "agentVersion": "string",
  "dockerVersion": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[attributes](#)

The container instance attributes that this container instance supports.

Type: Array of [Attribute](#) objects

Required: No

[cluster](#)

The short name or full Amazon Resource Name (ARN) of the cluster to register your container instance with. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

[containerInstanceArn](#)

The ARN of the container instance (if it was previously registered).

Type: String

Required: No

instanceIdentityDocument

The instance identity document for the EC2 instance to register. This document can be found by running the following command from the instance: `curl http://169.254.169.254/latest/dynamic/instance-identity/document/`

Type: String

Required: No

instanceIdentityDocumentSignature

The instance identity document signature for the EC2 instance to register. This signature can be found by running the following command from the instance: `curl http://169.254.169.254/latest/dynamic/instance-identity/signature/`

Type: String

Required: No

platformDevices

The devices that are available on the container instance. The only supported device type is a GPU.

Type: Array of [PlatformDevice](#) objects

Required: No

tags

The metadata that you apply to the container instance to help you categorize and organize them. Each tag consists of a key and an optional value. You define both.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.

- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

[totalResources](#)

The resources available on the instance.

Type: Array of [Resource](#) objects

Required: No

[versionInfo](#)

The version information for the Amazon ECS container agent and Docker daemon that runs on the container instance.

Type: [VersionInfo](#) object

Required: No

Response Syntax

```
{
  "containerInstance": {
    "agentConnected": boolean,
    "agentUpdateStatus": "string",
    "attachments": [
      {
        "details": [
          {
            "name": "string",
            "value": "string"
          }
        ],
        "id": "string",
        "status": "string",
        "type": "string"
      }
    ]
  }
}
```

```

    }
  ],
  "attributes": [
    {
      "name": "string",
      "targetId": "string",
      "targetType": "string",
      "value": "string"
    }
  ],
  "capacityProviderName": "string",
  "containerInstanceArn": "string",
  "ec2InstanceId": "string",
  "healthStatus": {
    "details": [
      {
        "lastStatusChange": number,
        "lastUpdated": number,
        "status": "string",
        "type": "string"
      }
    ]
  },
  "overallStatus": "string"
},
"pendingTasksCount": number,
"registeredAt": number,
"registeredResources": [
  {
    "doubleValue": number,
    "integerValue": number,
    "longValue": number,
    "name": "string",
    "stringValue": [ "string" ],
    "type": "string"
  }
],
"remainingResources": [
  {
    "doubleValue": number,
    "integerValue": number,
    "longValue": number,
    "name": "string",
    "stringValue": [ "string" ],
    "type": "string"
  }
]

```

```
    }
  ],
  "runningTasksCount": number,
  "status": "string",
  "statusReason": "string",
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ],
  "version": number,
  "versionInfo": {
    "agentHash": "string",
    "agentVersion": "string",
    "dockerVersion": "string"
  }
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

containerInstance

The container instance that was registered.

Type: [ContainerInstance](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

RegisterTaskDefinition

Registers a new task definition from the supplied family and containerDefinitions. Optionally, you can add data volumes to your containers with the volumes parameter. For more information about task definition parameters and defaults, see [Amazon ECS Task Definitions](#) in the *Amazon Elastic Container Service Developer Guide*.

You can specify a role for your task with the taskRoleArn parameter. When you specify a role for a task, its containers can then use the latest versions of the AWS CLI or SDKs to make API requests to the AWS services that are specified in the policy that's associated with the role. For more information, see [IAM Roles for Tasks](#) in the *Amazon Elastic Container Service Developer Guide*.

You can specify a Docker networking mode for the containers in your task definition with the networkMode parameter. If you specify the awsvpc network mode, the task is allocated an elastic network interface, and you must specify a [NetworkConfiguration](#) when you create a service or run a task with the task definition. For more information, see [Task Networking](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{
  "containerDefinitions": [
    {
      "command": [ "string" ],
      "cpu": number,
      "credentialSpecs": [ "string" ],
      "dependsOn": [
        {
          "condition": "string",
          "containerName": "string"
        }
      ],
      "disableNetworking": boolean,
      "dnsSearchDomains": [ "string" ],
      "dnsServers": [ "string" ],
      "dockerLabels": {
        "string" : "string"
      },
      "dockerSecurityOptions": [ "string" ],
      "entryPoint": [ "string" ],
      "environment": [
```



```

    {
      "name": "string",
      "value": "string"
    }
  ],
  "environmentFiles": [
    {
      "type": "string",
      "value": "string"
    }
  ],
  "essential": boolean,
  "extraHosts": [
    {
      "hostname": "string",
      "ipAddress": "string"
    }
  ],
  "firelensConfiguration": {
    "options": {
      "string" : "string"
    },
    "type": "string"
  },
  "healthCheck": {
    "command": [ "string" ],
    "interval": number,
    "retries": number,
    "startPeriod": number,
    "timeout": number
  },
  "hostname": "string",
  "image": "string",
  "interactive": boolean,
  "links": [ "string" ],
  "linuxParameters": {
    "capabilities": {
      "add": [ "string" ],
      "drop": [ "string" ]
    },
    "devices": [
      {
        "containerPath": "string",
        "hostPath": "string",

```

```

        "permissions": [ "string" ]
    }
],
"initProcessEnabled": boolean,
"maxSwap": number,
"sharedMemorySize": number,
"swappiness": number,
"tmpfs": [
    {
        "containerPath": "string",
        "mountOptions": [ "string" ],
        "size": number
    }
]
},
"logConfiguration": {
    "logDriver": "string",
    "options": {
        "string" : "string"
    },
    "secretOptions": [
        {
            "name": "string",
            "valueFrom": "string"
        }
    ]
},
"memory": number,
"memoryReservation": number,
"mountPoints": [
    {
        "containerPath": "string",
        "readOnly": boolean,
        "sourceVolume": "string"
    }
],
"name": "string",
"portMappings": [
    {
        "appProtocol": "string",
        "containerPort": number,
        "containerPortRange": "string",
        "hostPort": number,
        "name": "string",

```

```
    "protocol": "string"
  },
],
"privileged": boolean,
"pseudoTerminal": boolean,
"readonlyRootFilesystem": boolean,
"repositoryCredentials": {
  "credentialsParameter": "string"
},
"resourceRequirements": [
  {
    "type": "string",
    "value": "string"
  }
],
"restartPolicy": {
  "enabled": boolean,
  "ignoredExitCodes": [ number ],
  "restartAttemptPeriod": number
},
"secrets": [
  {
    "name": "string",
    "valueFrom": "string"
  }
],
"startTimeout": number,
"stopTimeout": number,
"systemControls": [
  {
    "namespace": "string",
    "value": "string"
  }
],
"ulimits": [
  {
    "hardLimit": number,
    "name": "string",
    "softLimit": number
  }
],
"user": "string",
"versionConsistency": "string",
"volumesFrom": [
```

```

        {
            "readOnly": boolean,
            "sourceContainer": "string"
        }
    ],
    "workingDirectory": "string"
}
],
"cpu": "string",
"enableFaultInjection": boolean,
"ephemeralStorage": {
    "sizeInGiB": number
},
"executionRoleArn": "string",
"family": "string",
"inferenceAccelerators": [
    {
        "deviceName": "string",
        "deviceType": "string"
    }
],
"ipcMode": "string",
"memory": "string",
"networkMode": "string",
"pidMode": "string",
"placementConstraints": [
    {
        "expression": "string",
        "type": "string"
    }
],
"proxyConfiguration": {
    "containerName": "string",
    "properties": [
        {
            "name": "string",
            "value": "string"
        }
    ]
},
"type": "string"
},
"requiresCompatibilities": [ "string" ],
"runtimePlatform": {
    "cpuArchitecture": "string",

```

```

    "operatingSystemFamily": "string"
  },
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ],
  "taskRoleArn": "string",
  "volumes": [
    {
      "configuredAtLaunch": boolean,
      "dockerVolumeConfiguration": {
        "autoprovision": boolean,
        "driver": "string",
        "driverOpts": {
          "string" : "string"
        },
        "labels": {
          "string" : "string"
        },
        "scope": "string"
      },
      "efsVolumeConfiguration": {
        "authorizationConfig": {
          "accessPointId": "string",
          "iam": "string"
        },
        "fileSystemId": "string",
        "rootDirectory": "string",
        "transitEncryption": "string",
        "transitEncryptionPort": number
      },
      "fsxWindowsFileServerVolumeConfiguration": {
        "authorizationConfig": {
          "credentialsParameter": "string",
          "domain": "string"
        },
        "fileSystemId": "string",
        "rootDirectory": "string"
      },
      "host": {
        "sourcePath": "string"
      }
    },

```

```
    "name": "string"  
  }  
]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[containerDefinitions](#)

A list of container definitions in JSON format that describe the different containers that make up your task.

Type: Array of [ContainerDefinition](#) objects

Required: Yes

[cpu](#)

The number of CPU units used by the task. It can be expressed as an integer using CPU units (for example, 1024) or as a string using vCPUs (for example, 1 vCPU or 1 vcpu) in a task definition. String values are converted to an integer indicating the CPU units when the task definition is registered.

Note

Task-level CPU and memory parameters are ignored for Windows containers. We recommend specifying container-level resources for Windows containers.

If you're using the EC2 launch type or external launch type, this field is optional. Supported values are between 128 CPU units (0.125 vCPUs) and 196608 CPU units (192 vCPUs). If you do not specify a value, the parameter is ignored.

This field is required for Fargate. For information about the valid values, see [Task size](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

enableFaultInjection

Enables fault injection when you register your task definition and allows for fault injection requests to be accepted from the task's containers. The default value is `false`.

Type: Boolean

Required: No

ephemeralStorage

The amount of ephemeral storage to allocate for the task. This parameter is used to expand the total amount of ephemeral storage available, beyond the default amount, for tasks hosted on AWS Fargate. For more information, see [Using data volumes in tasks](#) in the *Amazon ECS Developer Guide*.

Note

For tasks using the Fargate launch type, the task requires the following platforms:

- Linux platform version 1.4.0 or later.
- Windows platform version 1.0.0 or later.

Type: [EphemeralStorage](#) object

Required: No

executionRoleArn

The Amazon Resource Name (ARN) of the task execution role that grants the Amazon ECS container agent permission to make AWS API calls on your behalf. For information about the required IAM roles for Amazon ECS, see [IAM roles for Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

family

You must specify a family for a task definition. You can use it track multiple versions of the same task definition. The family is used as a name for your task definition. Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed.

Type: String

Required: Yes

inferenceAccelerators

The Elastic Inference accelerators to use for the containers in the task.

Type: Array of [InferenceAccelerator](#) objects

Required: No

ipcMode

The IPC resource namespace to use for the containers in the task. The valid values are `host`, `task`, or `none`. If `host` is specified, then all containers within the tasks that specified the `host` IPC mode on the same container instance share the same IPC resources with the host Amazon EC2 instance. If `task` is specified, all containers within the specified task share the same IPC resources. If `none` is specified, then IPC resources within the containers of a task are private and not shared with other containers in a task or on the container instance. If no value is specified, then the IPC resource namespace sharing depends on the Docker daemon setting on the container instance.

If the `host` IPC mode is used, be aware that there is a heightened risk of undesired IPC namespace expose.

If you are setting namespaced kernel parameters using `systemControls` for the containers in the task, the following will apply to your IPC resource namespace. For more information, see [System Controls](#) in the *Amazon Elastic Container Service Developer Guide*.

- For tasks that use the `host` IPC mode, IPC namespace related `systemControls` are not supported.
- For tasks that use the `task` IPC mode, IPC namespace related `systemControls` will apply to all containers within a task.

Note

This parameter is not supported for Windows containers or tasks run on AWS Fargate.

Type: String

Valid Values: host | task | none

Required: No

memory

The amount of memory (in MiB) used by the task. It can be expressed as an integer using MiB (for example ,1024) or as a string using GB (for example, 1GB or 1 GB) in a task definition. String values are converted to an integer indicating the MiB when the task definition is registered.

Note

Task-level CPU and memory parameters are ignored for Windows containers. We recommend specifying container-level resources for Windows containers.

If using the EC2 launch type, this field is optional.

If using the Fargate launch type, this field is required and you must use one of the following values. This determines your range of supported values for the `cpu` parameter.

The CPU units cannot be less than 1 vCPU when you use Windows containers on Fargate.

- 512 (0.5 GB), 1024 (1 GB), 2048 (2 GB) - Available `cpu` values: 256 (.25 vCPU)
- 1024 (1 GB), 2048 (2 GB), 3072 (3 GB), 4096 (4 GB) - Available `cpu` values: 512 (.5 vCPU)
- 2048 (2 GB), 3072 (3 GB), 4096 (4 GB), 5120 (5 GB), 6144 (6 GB), 7168 (7 GB), 8192 (8 GB) - Available `cpu` values: 1024 (1 vCPU)
- Between 4096 (4 GB) and 16384 (16 GB) in increments of 1024 (1 GB) - Available `cpu` values: 2048 (2 vCPU)
- Between 8192 (8 GB) and 30720 (30 GB) in increments of 1024 (1 GB) - Available `cpu` values: 4096 (4 vCPU)
- Between 16 GB and 60 GB in 4 GB increments - Available `cpu` values: 8192 (8 vCPU)

This option requires Linux platform 1.4.0 or later.

- Between 32GB and 120 GB in 8 GB increments - Available `cpu` values: 16384 (16 vCPU)

This option requires Linux platform 1.4.0 or later.

Type: String

Required: No

networkMode

The Docker networking mode to use for the containers in the task. The valid values are `none`, `bridge`, `awsvpc`, and `host`. If no network mode is specified, the default is `bridge`.

For Amazon ECS tasks on Fargate, the `awsvpc` network mode is required. For Amazon ECS tasks on Amazon EC2 Linux instances, any network mode can be used. For Amazon ECS tasks on Amazon EC2 Windows instances, `<default>` or `awsvpc` can be used. If the network mode is set to `none`, you cannot specify port mappings in your container definitions, and the tasks containers do not have external connectivity. The `host` and `awsvpc` network modes offer the highest networking performance for containers because they use the EC2 network stack instead of the virtualized network stack provided by the `bridge` mode.

With the `host` and `awsvpc` network modes, exposed container ports are mapped directly to the corresponding host port (for the `host` network mode) or the attached elastic network interface port (for the `awsvpc` network mode), so you cannot take advantage of dynamic host port mappings.

Important

When using the `host` network mode, you should not run containers using the root user (UID 0). It is considered best practice to use a non-root user.

If the network mode is `awsvpc`, the task is allocated an elastic network interface, and you must specify a [NetworkConfiguration](#) value when you create a service or run a task with the task definition. For more information, see [Task Networking](#) in the *Amazon Elastic Container Service Developer Guide*.

If the network mode is `host`, you cannot run multiple instantiations of the same task on a single container instance when port mappings are used.

Type: String

Valid Values: `bridge` | `host` | `awsvpc` | `none`

Required: No

pidMode

The process namespace to use for the containers in the task. The valid values are `host` or `task`. On Fargate for Linux containers, the only valid value is `task`. For example, monitoring sidecars might need `pidMode` to access information about other containers running in the same task.

If `host` is specified, all containers within the tasks that specified the `host` PID mode on the same container instance share the same process namespace with the host Amazon EC2 instance.

If `task` is specified, all containers within the specified task share the same process namespace.

If no value is specified, the default is a private namespace for each container.

If the `host` PID mode is used, there's a heightened risk of undesired process namespace exposure.

Note

This parameter is not supported for Windows containers.

Note

This parameter is only supported for tasks that are hosted on AWS Fargate if the tasks are using platform version `1.4.0` or later (Linux). This isn't supported for Windows containers on Fargate.

Type: String

Valid Values: `host` | `task`

Required: No

placementConstraints

An array of placement constraint objects to use for the task. You can specify a maximum of 10 constraints for each task. This limit includes constraints in the task definition and those specified at runtime.

Type: Array of [TaskDefinitionPlacementConstraint](#) objects

Required: No

proxyConfiguration

The configuration details for the App Mesh proxy.

For tasks hosted on Amazon EC2 instances, the container instances require at least version 1.26.0 of the container agent and at least version 1.26.0-1 of the `ecs-init` package to use a proxy configuration. If your container instances are launched from the Amazon ECS-optimized AMI version 20190301 or later, then they contain the required versions of the container agent and `ecs-init`. For more information, see [Amazon ECS-optimized AMI versions](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [ProxyConfiguration](#) object

Required: No

requiresCompatibilities

The task launch type that Amazon ECS validates the task definition against. A client exception is returned if the task definition doesn't validate against the compatibilities specified. If no value is specified, the parameter is omitted from the response.

Type: Array of strings

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

runtimePlatform

The operating system that your tasks definitions run on.

Type: [RuntimePlatform](#) object

Required: No

tags

The metadata that you apply to the task definition to help you categorize and organize them. Each tag consists of a key and an optional value. You define both of them.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8

- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

[taskRoleArn](#)

The short name or full Amazon Resource Name (ARN) of the IAM role that containers in this task can assume. All containers in this task are granted the permissions that are specified in this role. For more information, see [IAM Roles for Tasks](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

[volumes](#)

A list of volume definitions in JSON format that containers in your task might use.

Type: Array of [Volume](#) objects

Required: No

Response Syntax

```
{
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ]
}
```

```

],
"taskDefinition": {
  "compatibilities": [ "string" ],
  "containerDefinitions": [
    {
      "command": [ "string" ],
      "cpu": number,
      "credentialSpecs": [ "string" ],
      "dependsOn": [
        {
          "condition": "string",
          "containerName": "string"
        }
      ],
      "disableNetworking": boolean,
      "dnsSearchDomains": [ "string" ],
      "dnsServers": [ "string" ],
      "dockerLabels": {
        "string" : "string"
      },
      "dockerSecurityOptions": [ "string" ],
      "entryPoint": [ "string" ],
      "environment": [
        {
          "name": "string",
          "value": "string"
        }
      ],
      "environmentFiles": [
        {
          "type": "string",
          "value": "string"
        }
      ],
      "essential": boolean,
      "extraHosts": [
        {
          "hostname": "string",
          "ipAddress": "string"
        }
      ],
      "firelensConfiguration": {
        "options": {
          "string" : "string"
        }
      }
    }
  ]
}

```

```

    },
    "type": "string"
  },
  "healthCheck": {
    "command": [ "string" ],
    "interval": number,
    "retries": number,
    "startPeriod": number,
    "timeout": number
  },
  "hostname": "string",
  "image": "string",
  "interactive": boolean,
  "links": [ "string" ],
  "linuxParameters": {
    "capabilities": {
      "add": [ "string" ],
      "drop": [ "string" ]
    },
    "devices": [
      {
        "containerPath": "string",
        "hostPath": "string",
        "permissions": [ "string" ]
      }
    ],
    "initProcessEnabled": boolean,
    "maxSwap": number,
    "sharedMemorySize": number,
    "swappiness": number,
    "tmpfs": [
      {
        "containerPath": "string",
        "mountOptions": [ "string" ],
        "size": number
      }
    ]
  },
  "logConfiguration": {
    "logDriver": "string",
    "options": {
      "string" : "string"
    },
    "secretOptions": [

```

```

        {
            "name": "string",
            "valueFrom": "string"
        }
    ],
    "memory": number,
    "memoryReservation": number,
    "mountPoints": [
        {
            "containerPath": "string",
            "readOnly": boolean,
            "sourceVolume": "string"
        }
    ],
    "name": "string",
    "portMappings": [
        {
            "appProtocol": "string",
            "containerPort": number,
            "containerPortRange": "string",
            "hostPort": number,
            "name": "string",
            "protocol": "string"
        }
    ],
    "privileged": boolean,
    "pseudoTerminal": boolean,
    "readonlyRootFilesystem": boolean,
    "repositoryCredentials": {
        "credentialsParameter": "string"
    },
    "resourceRequirements": [
        {
            "type": "string",
            "value": "string"
        }
    ],
    "restartPolicy": {
        "enabled": boolean,
        "ignoredExitCodes": [ number ],
        "restartAttemptPeriod": number
    },
    "secrets": [

```



```

        {
            "name": "string",
            "valueFrom": "string"
        }
    ],
    "startTimeout": number,
    "stopTimeout": number,
    "systemControls": [
        {
            "namespace": "string",
            "value": "string"
        }
    ],
    "ulimits": [
        {
            "hardLimit": number,
            "name": "string",
            "softLimit": number
        }
    ],
    "user": "string",
    "versionConsistency": "string",
    "volumesFrom": [
        {
            "readOnly": boolean,
            "sourceContainer": "string"
        }
    ],
    "workingDirectory": "string"
}
],
"cpu": "string",
"deregisteredAt": number,
"enableFaultInjection": boolean,
"ephemeralStorage": {
    "sizeInGiB": number
},
"executionRoleArn": "string",
"family": "string",
"inferenceAccelerators": [
    {
        "deviceName": "string",
        "deviceType": "string"
    }
]

```

```

],
  "ipcMode": "string",
  "memory": "string",
  "networkMode": "string",
  "pidMode": "string",
  "placementConstraints": [
    {
      "expression": "string",
      "type": "string"
    }
  ],
  "proxyConfiguration": {
    "containerName": "string",
    "properties": [
      {
        "name": "string",
        "value": "string"
      }
    ],
    "type": "string"
  },
  "registeredAt": number,
  "registeredBy": "string",
  "requiresAttributes": [
    {
      "name": "string",
      "targetId": "string",
      "targetType": "string",
      "value": "string"
    }
  ],
  "requiresCompatibilities": [ "string" ],
  "revision": number,
  "runtimePlatform": {
    "cpuArchitecture": "string",
    "operatingSystemFamily": "string"
  },
  "status": "string",
  "taskDefinitionArn": "string",
  "taskRoleArn": "string",
  "volumes": [
    {
      "configuredAtLaunch": boolean,
      "dockerVolumeConfiguration": {

```

```

        "autoprovision": boolean,
        "driver": "string",
        "driverOpts": {
            "string" : "string"
        },
        "labels": {
            "string" : "string"
        },
        "scope": "string"
    },
    "efsVolumeConfiguration": {
        "authorizationConfig": {
            "accessPointId": "string",
            "iam": "string"
        },
        "fileSystemId": "string",
        "rootDirectory": "string",
        "transitEncryption": "string",
        "transitEncryptionPort": number
    },
    "fsxWindowsFileServerVolumeConfiguration": {
        "authorizationConfig": {
            "credentialsParameter": "string",
            "domain": "string"
        },
        "fileSystemId": "string",
        "rootDirectory": "string"
    },
    "host": {
        "sourcePath": "string"
    },
    "name": "string"
}
]
}
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

tags

The list of tags associated with the task definition.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

taskDefinition

The full description of the registered task definition.

Type: [TaskDefinition](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request registers a task definition in the `hello_world` family with the host networking mode.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 486
X-Amz-Target: AmazonEC2ContainerServiceV20141113.RegisterTaskDefinition
X-Amz-Date: 20150429T193109Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "networkMode": "host",
  "containerDefinitions": [
    {
      "name": "wordpress",
      "links": [
        "mysql"
      ],
      "image": "public.ecr.aws/docker/library/wordpress:latest",
      "essential": true,
      "portMappings": [
        {
```

```
        "containerPort": 80,
        "hostPort": 80
      }
    ],
    "memory": 500,
    "cpu": 10
  },
  {
    "name": "mysql",
    "image": "public.ecr.aws/docker/library/mysql:latest",
    "cpu": 10,
    "environment": [
      {
        "name": "MYSQL_ROOT_PASSWORD",
        "value": "password"
      }
    ],
    "memory": 500,
    "healthCheck": {
      "retries": 3,
      "command": [
        "CMD-SHELL",
        "curl -f http://localhost:8080/ || exit 1"
      ],
      "timeout": 5,
      "interval": 30,
    },
    "essential": true
  }
],
"family": "hello_world"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Fri, 12 Aug 2016 22:17:20 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 714
Connection: keep-alive
x-amzn-RequestId: 896d7e0f-60da-11e6-8e21-55c97a4b6423
```

```
{
  "taskDefinition": {
    "containerDefinitions": [
      {
        "cpu": 10,
        "environment": [],
        "healthCheck": {
          "retries": 3,
          "command": [
            "CMD-SHELL",
            "curl -f http://localhost:8080/ || exit 1"
          ],
          "timeout": 5,
          "interval": 30,
          "startPeriod": null
        },
        "essential": true,
        "image": "public.ecr.aws/docker/library/wordpress:latest",
        "links": [
          "mysql"
        ],
        "memory": 500,
        "mountPoints": [],
        "name": "wordpress",
        "portMappings": [
          {
            "containerPort": 80,
            "hostPort": 80,
            "protocol": "tcp"
          }
        ],
        "volumesFrom": []
      },
      {
        "cpu": 10,
        "environment": [
          {
            "name": "MYSQL_ROOT_PASSWORD",
            "value": "password"
          }
        ],
        "essential": true,
        "image": "public.ecr.aws/docker/library/mysql:latest",
        "memory": 500,
```

```
        "mountPoints": [],
        "name": "mysql",
        "portMappings": [],
        "volumesFrom": []
    }
],
"family": "hello_world",
"networkMode": "host",
"runtimePlatform": "LINUX",
"requiresAttributes": [
    {
        "name": "com.amazonaws.ecs.capability.docker-remote-api.1.18"
    }
],
"revision": 4,
"status": "ACTIVE",
"taskDefinitionArn": "arn:aws:ecs:us-east-1:012345678910:task-definition/
hello_world:4",
"volumes": []
}
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

RunTask

Starts a new task using the specified task definition.

Note

On March 21, 2024, a change was made to resolve the task definition revision before authorization. When a task definition revision is not specified, authorization will occur using the latest revision of a task definition.

Note

Amazon Elastic Inference (EI) is no longer available to customers.

You can allow Amazon ECS to place tasks for you, or you can customize how Amazon ECS places tasks using placement constraints and placement strategies. For more information, see [Scheduling Tasks](#) in the *Amazon Elastic Container Service Developer Guide*.

Alternatively, you can use `StartTask` to use your own scheduler or place tasks manually on specific container instances.

You can attach Amazon EBS volumes to Amazon ECS tasks by configuring the volume when creating or updating a service. For more information, see [Amazon EBS volumes](#) in the *Amazon Elastic Container Service Developer Guide*.

The Amazon ECS API follows an eventual consistency model. This is because of the distributed nature of the system supporting the API. This means that the result of an API command you run that affects your Amazon ECS resources might not be immediately visible to all subsequent commands you run. Keep this in mind when you carry out an API command that immediately follows a previous API command.

To manage eventual consistency, you can do the following:

- Confirm the state of the resource before you run a command to modify it. Run the `DescribeTasks` command using an exponential backoff algorithm to ensure that you allow enough time for the previous command to propagate through the system. To do this, run the `DescribeTasks`

command repeatedly, starting with a couple of seconds of wait time and increasing gradually up to five minutes of wait time.

- Add wait time between subsequent commands, even if the DescribeTasks command returns an accurate response. Apply an exponential backoff algorithm starting with a couple of seconds of wait time, and increase gradually up to about five minutes of wait time.

If you get a `ConflictException` error, the `RunTask` request could not be processed due to conflicts. The provided `clientToken` is already in use with a different `RunTask` request. The `resourceIds` are the existing task ARNs which are already associated with the `clientToken`.

To fix this issue:

- Run `RunTask` with a unique `clientToken`.
- Run `RunTask` with the `clientToken` and the original set of parameters

If you get a `ClientException` error, the `RunTask` could not be processed because you use managed scaling and there is a capacity error because the quota of tasks in the `PROVISIONING` per cluster has been reached. For information about the service quotas, see [Amazon ECS service quotas](#).

Request Syntax

```
{
  "capacityProviderStrategy": [
    {
      "base": number,
      "capacityProvider": "string",
      "weight": number
    }
  ],
  "clientToken": "string",
  "cluster": "string",
  "count": number,
  "enableECSTags": boolean,
  "enableExecuteCommand": boolean,
  "group": "string",
  "launchType": "string",
  "networkConfiguration": {
    "awsVpcConfiguration": {
```

```

    "assignPublicIp": "string",
    "securityGroups": [ "string" ],
    "subnets": [ "string" ]
  }
},
"overrides": {
  "containerOverrides": [
    {
      "command": [ "string" ],
      "cpu": number,
      "environment": [
        {
          "name": "string",
          "value": "string"
        }
      ],
      "environmentFiles": [
        {
          "type": "string",
          "value": "string"
        }
      ],
      "memory": number,
      "memoryReservation": number,
      "name": "string",
      "resourceRequirements": [
        {
          "type": "string",
          "value": "string"
        }
      ]
    }
  ],
  "cpu": "string",
  "ephemeralStorage": {
    "sizeInGiB": number
  },
  "executionRoleArn": "string",
  "inferenceAcceleratorOverrides": [
    {
      "deviceName": "string",
      "deviceType": "string"
    }
  ],

```

```

    "memory": "string",
    "taskRoleArn": "string"
  },
  "placementConstraints": [
    {
      "expression": "string",
      "type": "string"
    }
  ],
  "placementStrategy": [
    {
      "field": "string",
      "type": "string"
    }
  ],
  "platformVersion": "string",
  "propagateTags": "string",
  "referenceId": "string",
  "startedBy": "string",
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ],
  "taskDefinition": "string",
  "volumeConfigurations": [
    {
      "managedEBSVolume": {
        "encrypted": boolean,
        "filesystemType": "string",
        "iops": number,
        "kmsKeyId": "string",
        "roleArn": "string",
        "sizeInGiB": number,
        "snapshotId": "string",
        "tagSpecifications": [
          {
            "propagateTags": "string",
            "resourceType": "string",
            "tags": [
              {
                "key": "string",
                "value": "string"
              }
            ]
          }
        ]
      }
    }
  ]
}

```

```
    }
  ]
}
],
"terminationPolicy": {
  "deleteOnTermination": boolean
},
"throughput": number,
"volumeInitializationRate": number,
"volumeType": "string"
},
"name": "string"
}
]
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[capacityProviderStrategy](#)

The capacity provider strategy to use for the task.

Note

If you want to use Amazon ECS Managed Instances, you must use the `capacityProviderStrategy` request parameter and omit the `launchType` request parameter.

If a `capacityProviderStrategy` is specified, the `launchType` parameter must be omitted. If no `capacityProviderStrategy` or `launchType` is specified, the `defaultCapacityProviderStrategy` for the cluster is used.

When you use cluster auto scaling, you must specify `capacityProviderStrategy` and not `launchType`.

A capacity provider strategy can contain a maximum of 20 capacity providers.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: No

[clientToken](#)

An identifier that you provide to ensure the idempotency of the request. It must be unique and is case sensitive. Up to 64 characters are allowed. The valid characters are characters in the range of 33-126, inclusive. For more information, see [Ensuring idempotency](#).

Type: String

Required: No

[cluster](#)

The short name or full Amazon Resource Name (ARN) of the cluster to run your task on. If you do not specify a cluster, the default cluster is assumed.

Each account receives a default cluster the first time you use the service, but you may also create other clusters.

Type: String

Required: No

[count](#)

The number of instantiations of the specified task to place on your cluster. You can specify up to 10 tasks for each call.

Type: Integer

Required: No

[enableECSTags](#)

Specifies whether to use Amazon ECS managed tags for the task. For more information, see [Tagging Your Amazon ECS Resources](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Boolean

Required: No

enableExecuteCommand

Determines whether to use the execute command functionality for the containers in this task. If `true`, this enables execute command functionality on all containers in the task.

If `true`, then the task definition must have a task role, or you must provide one as an override.

Type: Boolean

Required: No

group

The name of the task group to associate with the task. The default value is the family name of the task definition (for example, `family:my-family-name`).

Type: String

Required: No

launchType

The infrastructure to run your standalone task on. For more information, see [Amazon ECS launch types](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

If you want to use Amazon ECS Managed Instances, you must use the `capacityProviderStrategy` request parameter and omit the `launchType` request parameter.

The FARGATE launch type runs your tasks on AWS Fargate On-Demand infrastructure.

Note

Fargate Spot infrastructure is available for use but a capacity provider strategy must be used. For more information, see [AWS Fargate capacity providers](#) in the *Amazon ECS Developer Guide*.

The EC2 launch type runs your tasks on Amazon EC2 instances registered to your cluster.

The EXTERNAL launch type runs your tasks on your on-premises server or virtual machine (VM) capacity registered to your cluster.

A task can use either a launch type or a capacity provider strategy. If a `launchType` is specified, the `capacityProviderStrategy` parameter must be omitted.

When you use cluster auto scaling, you must specify `capacityProviderStrategy` and not `launchType`.

Type: String

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

networkConfiguration

The network configuration for the task. This parameter is required for task definitions that use the `awsvpc` network mode to receive their own elastic network interface, and it isn't supported for other network modes. For more information, see [Task networking](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [NetworkConfiguration](#) object

Required: No

overrides

A list of container overrides in JSON format that specify the name of a container in the specified task definition and the overrides it should receive. You can override the default command for a container (that's specified in the task definition or Docker image) with a command override. You can also override existing environment variables (that are specified in the task definition or Docker image) on a container or add new environment variables to it with an `environment` override.

A total of 8192 characters are allowed for overrides. This limit includes the JSON formatting characters of the override structure.

Type: [TaskOverride](#) object

Required: No

placementConstraints

An array of placement constraint objects to use for the task. You can specify up to 10 constraints for each task (including constraints in the task definition and those specified at runtime).

Type: Array of [PlacementConstraint](#) objects

Required: No

placementStrategy

The placement strategy objects to use for the task. You can specify a maximum of 5 strategy rules for each task.

Type: Array of [PlacementStrategy](#) objects

Required: No

platformVersion

The platform version the task uses. A platform version is only specified for tasks hosted on Fargate. If one isn't specified, the LATEST platform version is used. For more information, see [AWS Fargate platform versions](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

propagateTags

Specifies whether to propagate the tags from the task definition to the task. If no value is specified, the tags aren't propagated. Tags can only be propagated to the task during task creation. To add tags to a task after task creation, use the [TagResource](#) API action.

Note

An error will be received if you specify the SERVICE option when running a task.

Type: String

Valid Values: TASK_DEFINITION | SERVICE | NONE

Required: No

referenceId

This parameter is only used by Amazon ECS. It is not intended for use by customers.

Type: String

Required: No

startedBy

An optional tag specified when a task is started. For example, if you automatically trigger a task to run a batch process job, you could apply a unique identifier for that job to your task with the `startedBy` parameter. You can then identify which tasks belong to that job by filtering the results of a [ListTasks](#) call with the `startedBy` value. Up to 128 letters (uppercase and lowercase), numbers, hyphens (-), forward slash (/), and underscores (_) are allowed.

If a task is started by an Amazon ECS service, then the `startedBy` parameter contains the deployment ID of the service that starts it.

Type: String

Required: No

tags

The metadata that you apply to the task to help you categorize and organize them. Each tag consists of a key and an optional value, both of which you define.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

[taskDefinition](#)

The family and revision (family:revision) or full ARN of the task definition to run. If a revision isn't specified, the latest ACTIVE revision is used.

The full ARN value must match the value that you specified as the Resource of the principal's permissions policy.

When you specify a task definition, you must either specify a specific revision, or all revisions in the ARN.

To specify a specific revision, include the revision number in the ARN. For example, to specify revision 2, use `arn:aws:ecs:us-east-1:111122223333:task-definition/TaskFamilyName:2`.

To specify all revisions, use the wildcard (*) in the ARN. For example, to specify all revisions, use `arn:aws:ecs:us-east-1:111122223333:task-definition/TaskFamilyName:*`.

For more information, see [Policy Resources for Amazon ECS](#) in the Amazon Elastic Container Service Developer Guide.

Type: String

Required: Yes

[volumeConfigurations](#)

The details of the volume that was configuredAtLaunch. You can configure the size, volumeType, IOPS, throughput, snapshot and encryption in [TaskManagedEBSVolumeConfiguration](#). The name of the volume must match the name from the task definition.

Type: Array of [TaskVolumeConfiguration](#) objects

Required: No

Response Syntax

```
{
  "failures": [
    {
      "arn": "string",
      "detail": "string",
      "reason": "string"
    }
  ],
  "tasks": [
    {
      "attachments": [
        {
          "details": [
            {
              "name": "string",
              "value": "string"
            }
          ],
          "id": "string",
          "status": "string",
          "type": "string"
        }
      ],
      "attributes": [
        {
          "name": "string",
          "targetId": "string",
          "targetType": "string",
          "value": "string"
        }
      ],
      "availabilityZone": "string",
      "capacityProviderName": "string",
      "clusterArn": "string",
      "connectivity": "string",
      "connectivityAt": number,
      "containerInstanceArn": "string",
      "containers": [
        {
          "containerArn": "string",
          "cpu": "string",
```

```

    "exitCode": number,
    "gpuIds": [ "string" ],
    "healthStatus": "string",
    "image": "string",
    "imageDigest": "string",
    "lastStatus": "string",
    "managedAgents": [
      {
        "lastStartedAt": number,
        "lastStatus": "string",
        "name": "string",
        "reason": "string"
      }
    ],
    "memory": "string",
    "memoryReservation": "string",
    "name": "string",
    "networkBindings": [
      {
        "bindIP": "string",
        "containerPort": number,
        "containerPortRange": "string",
        "hostPort": number,
        "hostPortRange": "string",
        "protocol": "string"
      }
    ],
    "networkInterfaces": [
      {
        "attachmentId": "string",
        "ipv6Address": "string",
        "privateIpv4Address": "string"
      }
    ],
    "reason": "string",
    "runtimeId": "string",
    "taskArn": "string"
  }
],
"cpu": "string",
"createdAt": number,
"desiredStatus": "string",
"enableExecuteCommand": boolean,
"ephemeralStorage": {

```

```
    "sizeInGiB": number
  },
  "executionStoppedAt": number,
  "fargateEphemeralStorage": {
    "kmsKeyId": "string",
    "sizeInGiB": number
  },
  "group": "string",
  "healthStatus": "string",
  "inferenceAccelerators": [
    {
      "deviceName": "string",
      "deviceType": "string"
    }
  ],
  "lastStatus": "string",
  "launchType": "string",
  "memory": "string",
  "overrides": {
    "containerOverrides": [
      {
        "command": [ "string " ],
        "cpu": number,
        "environment": [
          {
            "name": "string",
            "value": "string"
          }
        ],
        "environmentFiles": [
          {
            "type": "string",
            "value": "string"
          }
        ],
        "memory": number,
        "memoryReservation": number,
        "name": "string",
        "resourceRequirements": [
          {
            "type": "string",
            "value": "string"
          }
        ]
      }
    ]
  }
}
```

```

    }
  ],
  "cpu": "string",
  "ephemeralStorage": {
    "sizeInGiB": number
  },
  "executionRoleArn": "string",
  "inferenceAcceleratorOverrides": [
    {
      "deviceName": "string",
      "deviceType": "string"
    }
  ],
  "memory": "string",
  "taskRoleArn": "string"
},
"platformFamily": "string",
"platformVersion": "string",
"pullStartedAt": number,
"pullStoppedAt": number,
"startedAt": number,
"startedBy": "string",
"stopCode": "string",
"stoppedAt": number,
"stoppedReason": "string",
"stoppingAt": number,
"tags": [
  {
    "key": "string",
    "value": "string"
  }
],
"taskArn": "string",
"taskDefinitionArn": "string",
"version": number
}
]
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

failures

Any failures associated with the call.

For information about how to address failures, see [Service event messages](#) and [API failure reasons](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Array of [Failure](#) objects

tasks

A full description of the tasks that were run. The tasks that were successfully placed on your cluster are described here.

Type: Array of [Task](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

BlockedException

Your AWS account was blocked. For more information, contact [AWS Support](#).

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

ConflictException

The request could not be processed because of conflict in the current state of the resource.

resourceIds

The existing task ARNs which are already associated with the `clientToken`.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

PlatformTaskDefinitionIncompatibilityException

The specified platform version doesn't satisfy the required capabilities of the task definition.

HTTP Status Code: 400

PlatformUnknownException

The specified platform version doesn't exist.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request runs the latest ACTIVE revision of the `hello_world` task definition family in the default cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 45
X-Amz-Target: AmazonEC2ContainerServiceV20141113.RunTask
X-Amz-Date: 20161121T215740Z
User-Agent: aws-cli/1.11.13 Python/2.7.12 Darwin/16.1.0 botocore/1.4.66
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "count": 1,
  "taskDefinition": "hello_world",
  "clientToken": "550e8400-e29b-41d4-a716-446655440000"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Mon, 21 Nov 2016 21:57:40 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 1025
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "failures": [],
  "tasks": [
    {
      "clusterArn": "arn:aws:ecs:us-east-1:012345678910:cluster/default",
      "containerInstanceArn": "arn:aws:ecs:us-east-1:012345678910:container-instance/default/4c543eed-f83f-47da-b1d8-3d23f1da4c64",
      "containers": [
        {
          "containerArn": "arn:aws:ecs:us-east-1:012345678910:container/default/e76594d4-27e1-4c74-98b5-46a6435eb769",
          "lastStatus": "PENDING",
          "name": "wordpress",
          "taskArn": "arn:aws:ecs:us-east-1:012345678910:task/default/fdf2c302-468c-4e55-b884-5331d816e7fb"
        },
        {
          "containerArn": "arn:aws:ecs:us-east-1:012345678910:container/b19106ea-4fa8-4f1d-9767-96922c82b070",
          "lastStatus": "PENDING",
          "name": "mysql",
          "taskArn": "arn:aws:ecs:us-east-1:012345678910:task/default/fdf2c302-468c-4e55-b884-5331d816e7fb"
        }
      ],
      "createdAt": 1479765460.842,
      "desiredStatus": "RUNNING",
      "lastStatus": "PENDING",
      "overrides": {
        "containerOverrides": [
          {
            "name": "wordpress"
          },
          {

```

```
        "name": "mysql"
      }
    ]
  },
  "taskArn": "arn:aws:ecs:us-east-1:012345678910:task/default/fdf2c302-468c-4e55-b884-5331d816e7fb",
  "taskDefinitionArn": "arn:aws:ecs:us-east-1:012345678910:task-definition/hello_world:6",
  "version": 1
}
]
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

StartTask

Starts a new task from the specified task definition on the specified container instance or instances.

Note

On March 21, 2024, a change was made to resolve the task definition revision before authorization. When a task definition revision is not specified, authorization will occur using the latest revision of a task definition.

Note

Amazon Elastic Inference (EI) is no longer available to customers.

Alternatively, you can use `RunTask` to place tasks for you. For more information, see [Scheduling Tasks](#) in the *Amazon Elastic Container Service Developer Guide*.

You can attach Amazon EBS volumes to Amazon ECS tasks by configuring the volume when creating or updating a service. For more information, see [Amazon EBS volumes](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{
  "cluster": "string",
  "containerInstances": [ "string" ],
  "enableECSTags": boolean,
  "enableExecuteCommand": boolean,
  "group": "string",
  "networkConfiguration": {
    "awsVpcConfiguration": {
      "assignPublicIp": "string",
      "securityGroups": [ "string" ],
      "subnets": [ "string" ]
    }
  },
  "overrides": {
    "containerOverrides": [
      {
```

```
    "command": [ "string" ],
    "cpu": number,
    "environment": [
      {
        "name": "string",
        "value": "string"
      }
    ],
    "environmentFiles": [
      {
        "type": "string",
        "value": "string"
      }
    ],
    "memory": number,
    "memoryReservation": number,
    "name": "string",
    "resourceRequirements": [
      {
        "type": "string",
        "value": "string"
      }
    ]
  }
],
"cpu": "string",
"ephemeralStorage": {
  "sizeInGiB": number
},
"executionRoleArn": "string",
"inferenceAcceleratorOverrides": [
  {
    "deviceName": "string",
    "deviceType": "string"
  }
],
"memory": "string",
"taskRoleArn": "string"
},
"propagateTags": "string",
"referenceId": "string",
"startedBy": "string",
"tags": [
  {
```

```

        "key": "string",
        "value": "string"
    }
],
"taskDefinition": "string",
"volumeConfigurations": [
    {
        "managedEBSVolume": {
            "encrypted": boolean,
            "filesystemType": "string",
            "iops": number,
            "kmsKeyId": "string",
            "roleArn": "string",
            "sizeInGiB": number,
            "snapshotId": "string",
            "tagSpecifications": [
                {
                    "propagateTags": "string",
                    "resourceType": "string",
                    "tags": [
                        {
                            "key": "string",
                            "value": "string"
                        }
                    ]
                }
            ]
        },
        "terminationPolicy": {
            "deleteOnTermination": boolean
        },
        "throughput": number,
        "volumeInitializationRate": number,
        "volumeType": "string"
    },
    "name": "string"
]
}

```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster where to start your task. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

containerInstances

The container instance IDs or full ARN entries for the container instances where you would like to place your task. You can specify up to 10 container instances.

Type: Array of strings

Required: Yes

enableECSTags

Specifies whether to use Amazon ECS managed tags for the task. For more information, see [Tagging Your Amazon ECS Resources](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Boolean

Required: No

enableExecuteCommand

Whether or not the execute command functionality is turned on for the task. If `true`, this turns on the execute command functionality on all containers in the task.

Type: Boolean

Required: No

group

The name of the task group to associate with the task. The default value is the family name of the task definition (for example, `family:my-family-name`).

Type: String

Required: No

networkConfiguration

The VPC subnet and security group configuration for tasks that receive their own elastic network interface by using the `awsvpc` networking mode.

Type: [NetworkConfiguration](#) object

Required: No

overrides

A list of container overrides in JSON format that specify the name of a container in the specified task definition and the overrides it receives. You can override the default command for a container (that's specified in the task definition or Docker image) with a `command` override. You can also override existing environment variables (that are specified in the task definition or Docker image) on a container or add new environment variables to it with an `environment` override.

Note

A total of 8192 characters are allowed for overrides. This limit includes the JSON formatting characters of the override structure.

Type: [TaskOverride](#) object

Required: No

propagateTags

Specifies whether to propagate the tags from the task definition or the service to the task. If no value is specified, the tags aren't propagated.

Type: String

Valid Values: `TASK_DEFINITION` | `SERVICE` | `NONE`

Required: No

referenceId

This parameter is only used by Amazon ECS. It is not intended for use by customers.

Type: String

Required: No

[startedBy](#)

An optional tag specified when a task is started. For example, if you automatically trigger a task to run a batch process job, you could apply a unique identifier for that job to your task with the `startedBy` parameter. You can then identify which tasks belong to that job by filtering the results of a [ListTasks](#) call with the `startedBy` value. Up to 36 letters (uppercase and lowercase), numbers, hyphens (-), forward slash (/), and underscores (_) are allowed.

If a task is started by an Amazon ECS service, the `startedBy` parameter contains the deployment ID of the service that starts it.

Type: String

Required: No

[tags](#)

The metadata that you apply to the task to help you categorize and organize them. Each tag consists of a key and an optional value, both of which you define.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

taskDefinition

The family and revision (family:revision) or full ARN of the task definition to start. If a revision isn't specified, the latest ACTIVE revision is used.

Type: String

Required: Yes

volumeConfigurations

The details of the volume that was configuredAtLaunch. You can configure the size, volumeType, IOPS, throughput, snapshot and encryption in [TaskManagedEBSVolumeConfiguration](#). The name of the volume must match the name from the task definition.

Type: Array of [TaskVolumeConfiguration](#) objects

Required: No

Response Syntax

```
{
  "failures": [
    {
      "arn": "string",
      "detail": "string",
      "reason": "string"
    }
  ],
  "tasks": [
    {
      "attachments": [
        {
          "details": [
            {
              "name": "string",
              "value": "string"
            }
          ],
          "id": "string",
          "status": "string",
          "type": "string"
        }
      ]
    }
  ]
}
```

```

    }
  ],
  "attributes": [
    {
      "name": "string",
      "targetId": "string",
      "targetType": "string",
      "value": "string"
    }
  ],
  "availabilityZone": "string",
  "capacityProviderName": "string",
  "clusterArn": "string",
  "connectivity": "string",
  "connectivityAt": number,
  "containerInstanceArn": "string",
  "containers": [
    {
      "containerArn": "string",
      "cpu": "string",
      "exitCode": number,
      "gpuIds": [ "string" ],
      "healthStatus": "string",
      "image": "string",
      "imageDigest": "string",
      "lastStatus": "string",
      "managedAgents": [
        {
          "lastStartedAt": number,
          "lastStatus": "string",
          "name": "string",
          "reason": "string"
        }
      ]
    }
  ],
  "memory": "string",
  "memoryReservation": "string",
  "name": "string",
  "networkBindings": [
    {
      "bindIP": "string",
      "containerPort": number,
      "containerPortRange": "string",
      "hostPort": number,
      "hostPortRange": "string",

```

```

        "protocol": "string"
    }
],
"networkInterfaces": [
    {
        "attachmentId": "string",
        "ipv6Address": "string",
        "privateIpv4Address": "string"
    }
],
"reason": "string",
"runtimeId": "string",
"taskArn": "string"
}
],
"cpu": "string",
"createdAt": number,
"desiredStatus": "string",
"enableExecuteCommand": boolean,
"ephemeralStorage": {
    "sizeInGiB": number
},
"executionStoppedAt": number,
"fargateEphemeralStorage": {
    "kmsKeyId": "string",
    "sizeInGiB": number
},
"group": "string",
"healthStatus": "string",
"inferenceAccelerators": [
    {
        "deviceName": "string",
        "deviceType": "string"
    }
],
"lastStatus": "string",
"launchType": "string",
"memory": "string",
"overrides": {
    "containerOverrides": [
        {
            "command": [ "string" ],
            "cpu": number,
            "environment": [

```

```

        {
            "name": "string",
            "value": "string"
        }
    ],
    "environmentFiles": [
        {
            "type": "string",
            "value": "string"
        }
    ],
    "memory": number,
    "memoryReservation": number,
    "name": "string",
    "resourceRequirements": [
        {
            "type": "string",
            "value": "string"
        }
    ]
    }
],
"cpu": "string",
"ephemeralStorage": {
    "sizeInGiB": number
},
"executionRoleArn": "string",
"inferenceAcceleratorOverrides": [
    {
        "deviceName": "string",
        "deviceType": "string"
    }
],
"memory": "string",
"taskRoleArn": "string"
},
"platformFamily": "string",
"platformVersion": "string",
"pullStartedAt": number,
"pullStoppedAt": number,
"startedAt": number,
"startedBy": "string",
"stopCode": "string",
"stoppedAt": number,

```

```
    "stoppedReason": "string",
    "stoppingAt": number,
    "tags": [
      {
        "key": "string",
        "value": "string"
      }
    ],
    "taskArn": "string",
    "taskDefinitionArn": "string",
    "version": number
  }
]
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

tasks

A full description of the tasks that were started. Each task that was successfully placed on your container instances is described.

Type: Array of [Task](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS,

these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request starts the latest ACTIVE revision of the `hello_world` task definition family in the default cluster on the container instance with the ID `4c543eed-f83f-47da-b1d8-3d23f1da4c64`.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 97
X-Amz-Target: AmazonEC2ContainerServiceV20141113.StartTask
X-Amz-Date: 20161121T220032Z
User-Agent: aws-cli/1.11.13 Python/2.7.12 Darwin/16.1.0 botocore/1.4.66
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "containerInstances": [
    "4c543eed-f83f-47da-b1d8-3d23f1da4c64"
  ],
  "taskDefinition": "hello_world"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Mon, 21 Nov 2016 22:00:32 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 1025
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "failures": [],
  "tasks": [
    {
      "clusterArn": "arn:aws:ecs:us-east-1:012345678910:cluster/default",
```

```

    "containerInstanceArn": "arn:aws:ecs:us-east-1:012345678910:container-instance/
default/4c543eed-f83f-47da-b1d8-3d23f1da4c64",
    "containers": [
      {
        "containerArn": "arn:aws:ecs:us-east-1:012345678910:container/
e76594d4-27e1-4c74-98b5-46a6435eb769",
        "lastStatus": "PENDING",
        "name": "wordpress",
        "taskArn": "arn:aws:ecs:us-east-1:012345678910:task/default/
fdf2c302-468c-4e55-b884-5331d816e7fb"
      },
      {
        "containerArn": "arn:aws:ecs:us-east-1:012345678910:container/default/
b19106ea-4fa8-4f1d-9767-96922c82b070",
        "lastStatus": "PENDING",
        "name": "mysql",
        "taskArn": "arn:aws:ecs:us-east-1:012345678910:task/default/
fdf2c302-468c-4e55-b884-5331d816e7fb"
      }
    ],
    "createdAt": 1479765460.842,
    "desiredStatus": "RUNNING",
    "lastStatus": "PENDING",
    "overrides": {
      "containerOverrides": [
        {
          "name": "wordpress"
        },
        {
          "name": "mysql"
        }
      ]
    },
    "taskArn": "arn:aws:ecs:us-east-1:012345678910:task/default/fdf2c302-468c-4e55-
b884-5331d816e7fb",
    "taskDefinitionArn": "arn:aws:ecs:us-east-1:012345678910:task-definition/
hello_world:6",
    "version": 1
  }
]
}

```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

StopServiceDeployment

Stops an ongoing service deployment.

The following stop types are available:

- **ROLLBACK** - This option rolls back the service deployment to the previous service revision.

You can use this option even if you didn't configure the service deployment for the rollback option.

For more information, see [Stopping Amazon ECS service deployments](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{  
  "serviceDeploymentArn": "string",  
  "stopType": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[serviceDeploymentArn](#)

The ARN of the service deployment that you want to stop.

Type: String

Required: Yes

[stopType](#)

How you want Amazon ECS to stop the service.

The valid values are ROLLBACK.

Type: String

Valid Values: ABORT | ROLLBACK

Required: No

Response Syntax

```
{  
  "serviceDeploymentArn": "string"  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[serviceDeploymentArn](#)

The ARN of the stopped service deployment.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ConflictException

The request could not be processed because of conflict in the current state of the resource.

resourceIds

The existing task ARNs which are already associated with the `clientToken`.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceDeploymentNotFoundException

The service deploy ARN that you specified in the `StopServiceDeployment` doesn't exist. You can use `ListServiceDeployments` to retrieve the service deployment ARNs.

HTTP Status Code: 400

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

Examples

Example

This example request stops the service deployment with the ARN of `arn:aws:ecs:us-east-1:123456789012:service-deployment/MyCluster/MyService/r9i43YFjvgF_xlg7m2eJ1` using the ROLLBACK stop type.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 55
X-Amz-Target: AmazonEC2ContainerServiceV20141113.StopServiceDeployment
X-Amz-Date: 20250407T133521Z
User-Agent: aws-cli/2.26 Python/3.12.6 Darwin/14.3.0
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "serviceDeploymentArn": "arn:aws:ecs:us-east-1:123456789012:service-deployment/
MyCluster/MyService/r9i43YFjvgF_xlg7m2eJ1",
  "stopType": "ROLLBACK"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Mon Apr 7, 2025 18:50:14 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 220
Connection: keep-alive
RequestId: 360c5551-123e-4e74-9914-7582d3a28807
{
  "serviceDeploymentArn": "arn:aws:ecs:us-east-1:123456789012:service-deployment/
MyCluster/MyService/r9i43YFjvgF_xlg7m2eJ1",
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

StopTask

Stops a running task. Any tags associated with the task will be deleted.

When you call `StopTask` on a task, the equivalent of `docker stop` is issued to the containers running in the task. This results in a `SIGTERM` value and a default 30-second timeout, after which the `SIGKILL` value is sent and the containers are forcibly stopped. If the container handles the `SIGTERM` value gracefully and exits within 30 seconds from receiving it, no `SIGKILL` value is sent.

For Windows containers, POSIX signals do not work and runtime stops the container by sending a `CTRL_SHUTDOWN_EVENT`. For more information, see [Unable to react to graceful shutdown of \(Windows\) container #25982](#) on GitHub.

Note

The default 30-second timeout can be configured on the Amazon ECS container agent with the `ECS_CONTAINER_STOP_TIMEOUT` variable. For more information, see [Amazon ECS Container Agent Configuration](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{
  "cluster": "string",
  "reason": "string",
  "task": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the task to stop. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

reason

An optional message specified when a task is stopped. For example, if you're using a custom scheduler, you can use this parameter to specify the reason for stopping the task here, and the message appears in subsequent [DescribeTasks](#) API operations on this task.

Type: String

Required: No

task

The full Amazon Resource Name (ARN) of the task.

Type: String

Required: Yes

Response Syntax

```
{
  "task": {
    "attachments": [
      {
        "details": [
          {
            "name": "string",
            "value": "string"
          }
        ],
        "id": "string",
        "status": "string",
        "type": "string"
      }
    ],
    "attributes": [
      {
        "name": "string",
        "targetId": "string",
        "targetType": "string",
        "value": "string"
      }
    ]
  }
}
```

```

],
"availabilityZone": "string",
"capacityProviderName": "string",
"clusterArn": "string",
"connectivity": "string",
"connectivityAt": number,
"containerInstanceArn": "string",
"containers": [
  {
    "containerArn": "string",
    "cpu": "string",
    "exitCode": number,
    "gpuIds": [ "string" ],
    "healthStatus": "string",
    "image": "string",
    "imageDigest": "string",
    "lastStatus": "string",
    "managedAgents": [
      {
        "lastStartedAt": number,
        "lastStatus": "string",
        "name": "string",
        "reason": "string"
      }
    ]
  },
  {
    "memory": "string",
    "memoryReservation": "string",
    "name": "string",
    "networkBindings": [
      {
        "bindIP": "string",
        "containerPort": number,
        "containerPortRange": "string",
        "hostPort": number,
        "hostPortRange": "string",
        "protocol": "string"
      }
    ],
    "networkInterfaces": [
      {
        "attachmentId": "string",
        "ipv6Address": "string",
        "privateIpv4Address": "string"
      }
    ]
  }
]

```

```

    ],
    "reason": "string",
    "runtimeId": "string",
    "taskArn": "string"
  }
],
"cpu": "string",
"createdAt": number,
"desiredStatus": "string",
"enableExecuteCommand": boolean,
"ephemeralStorage": {
  "sizeInGiB": number
},
"executionStoppedAt": number,
"fargateEphemeralStorage": {
  "kmsKeyId": "string",
  "sizeInGiB": number
},
"group": "string",
"healthStatus": "string",
"inferenceAccelerators": [
  {
    "deviceName": "string",
    "deviceType": "string"
  }
],
"lastStatus": "string",
"launchType": "string",
"memory": "string",
"overrides": {
  "containerOverrides": [
    {
      "command": [ "string" ],
      "cpu": number,
      "environment": [
        {
          "name": "string",
          "value": "string"
        }
      ]
    }
  ],
  "environmentFiles": [
    {
      "type": "string",
      "value": "string"
    }
  ]
}

```

```

        }
    ],
    "memory": number,
    "memoryReservation": number,
    "name": "string",
    "resourceRequirements": [
        {
            "type": "string",
            "value": "string"
        }
    ]
}
],
"cpu": "string",
"ephemeralStorage": {
    "sizeInGiB": number
},
"executionRoleArn": "string",
"inferenceAcceleratorOverrides": [
    {
        "deviceName": "string",
        "deviceType": "string"
    }
],
"memory": "string",
"taskRoleArn": "string"
},
"platformFamily": "string",
"platformVersion": "string",
"pullStartedAt": number,
"pullStoppedAt": number,
"startedAt": number,
"startedBy": "string",
"stopCode": "string",
"stoppedAt": number,
"stoppedReason": "string",
"stoppingAt": number,
"tags": [
    {
        "key": "string",
        "value": "string"
    }
],
"taskArn": "string",

```

```
    "taskDefinitionArn": "string",  
    "version": number  
  }  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

task

The task that was stopped.

Type: [Task](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request stops a task with the ID 1dc5c17a-422b-4dc4-b493-371970c6c4d6 in the default cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 88
X-Amz-Target: AmazonEC2ContainerServiceV20141113.StopTask
X-Amz-Date: 20250203T220318Z
User-Agent: aws-cli/2.2.26 Python/3.12.6 Darwin/16.1.0 botocore/1.4.66
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
```

```
{
  "task": "arn:aws:ecs:us-east-1:123456789012:task/
MyCluster/36ad4ef947bb470b89010d864574237e"
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Mon, 03 Feb 2025 22:03:18 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 1260
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f
```

```
{
  "task": {
    "attachments": [
      {
        "id": "9711cfc0-8c7d-470b-93e4-9d2041731db6",
        "type": "ElasticNetworkInterface",
        "status": "ATTACHED",
        "details": [
          {
            "name": "subnetId",
            "value": "subnet-0d0eab1bb38d5ca64"
          },
          {
            "name": "networkInterfaceId",
            "value": "eni-0f6b537b3ec7ba800"
          },
          {
            "name": "macAddress",
            "value": "0e:fd:16:ec:9a:01"
          },
          {
            "name": "privateDnsName",
            "value": "ip-10-0-1-114.ec2.internal"
          },
          {
            "name": "privateIPv4Address",
            "value": "10.0.1.114"
          }
        ]
      }
    ]
  }
}
```



```

    ]
  }
],
"attributes": [
  {
    "name": "ecs.cpu-architecture",
    "value": "x86_64"
  }
],
"availabilityZone": "us-east-1b",
"capacityProviderName": "FARGATE",
"clusterArn": "arn:aws:ecs:us-east-1:123456789012:cluster/MyCluster",
"connectivity": "CONNECTED",
"connectivityAt": "2025-02-03T15:26:37.435000-05:00",
"containers": [
  {
    "containerArn": "arn:aws:ecs:us-east-1:123456789012:container/
MyCluster/36ad4ef947bb470b89010d864574237e/0a7d1107-a0c8-45c4-ac05-f0aaab4c63db",
    "taskArn": "arn:aws:ecs:us-east-1:123456789012:task/
MyCluster/36ad4ef947bb470b89010d864574237e",
    "name": "web",
    "image": "nginx",
    "imageDigest":
"sha256:0a399eb16751829e1af26fea27b20c3ec28d7ab1fb72182879dcae1cca21206a",
    "runtimeId": "36ad4ef947bb470b89010d864574237e-0265927825",
    "lastStatus": "RUNNING",
    "networkBindings": [],
    "networkInterfaces": [
      {
        "attachmentId": "9711cfc0-8c7d-470b-93e4-9d2041731db6",
        "privateIpv4Address": "10.0.1.114"
      }
    ],
    "cpu": "99",
    "memory": "100"
  },
  {
    "containerArn": "arn:aws:ecs:us-east-1:123456789012:container/
MyCluster/36ad4ef947bb470b89010d864574237e/48dfdefe-b605-4747-b071-5f6bc8650b2d",
    "taskArn": "arn:aws:ecs:us-east-1:123456789012:task/
MyCluster/36ad4ef947bb470b89010d864574237e",
    "name": "aws-otel-collector",
    "image": "public.ecr.aws/aws-observability/aws-otel-collector:v0.32.0",

```

```

        "imageDigest":
"sha256:7a1b3560655071bcacd66902c20ebe9a69470d5691fe3bd36baace7c2f3c4640",
        "runtimeId": "36ad4ef947bb470b89010d864574237e-4027662657",
        "lastStatus": "RUNNING",
        "networkBindings": [],
        "networkInterfaces": [
            {
                "attachmentId": "9711cfc0-8c7d-470b-93e4-9d2041731db6",
                "privateIpv4Address": "10.0.1.114"
            }
        ],
        "cpu": "0"
    }
],
"cpu": "256",
"createdAt": "2025-02-03T15:26:33.277000-05:00",
"desiredStatus": "STOPPED",
"enableExecuteCommand": false,
"group": "family:webserver",
"lastStatus": "RUNNING",
"launchType": "FARGATE",
"memory": "512",
"overrides": {
    "containerOverrides": [
        {
            "name": "aws-otel-collector"
        },
        {
            "name": "web"
        }
    ],
    "inferenceAcceleratorOverrides": []
},
"platformVersion": "1.4.0",
"platformFamily": "Linux",
"pullStartedAt": "2025-02-03T15:26:44.970000-05:00",
"pullStoppedAt": "2025-02-03T15:26:51.926000-05:00",
"startedAt": "2025-02-03T15:26:53.096000-05:00",
"stopCode": "UserInitiated",
"stoppedReason": "Task stopped by user",
"stoppingAt": "2025-02-03T15:27:51.392000-05:00",
"tags": [],
"taskArn": "arn:aws:ecs:us-east-1:123456789012:task/
MyCluster/36ad4ef947bb470b89010d864574237e",

```

```
        "taskDefinitionArn": "arn:aws:ecs:us-east-1:123456789012:task-definition/
webserver:5",
        "version": 4,
        "ephemeralStorage": {
            "sizeInGiB": 20
        },
        "fargateEphemeralStorage": {
            "sizeInGiB": 20
        }
    }
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

SubmitAttachmentStateChanges

Note

This action is only used by the Amazon ECS agent, and it is not intended for use outside of the agent.

Sent to acknowledge that an attachment changed states.

Request Syntax

```
{
  "attachments": [
    {
      "attachmentArn": "string",
      "status": "string"
    }
  ],
  "cluster": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

attachments

Any attachments associated with the state change request.

Type: Array of [AttachmentStateChange](#) objects

Required: Yes

cluster

The short name or full ARN of the cluster that hosts the container instance the attachment belongs to.

Type: String

Required: No

Response Syntax

```
{  
  "acknowledgment": "string"  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

[acknowledgment](#)

Acknowledgement of the state change.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

SubmitContainerStateChange

Note

This action is only used by the Amazon ECS agent, and it is not intended for use outside of the agent.

Sent to acknowledge that a container changed states.

Request Syntax

```
{
  "cluster": "string",
  "containerName": "string",
  "exitCode": number,
  "networkBindings": [
    {
      "bindIP": "string",
      "containerPort": number,
      "containerPortRange": "string",
      "hostPort": number,
      "hostPortRange": "string",
      "protocol": "string"
    }
  ],
  "reason": "string",
  "runtimeId": "string",
  "status": "string",
  "task": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full ARN of the cluster that hosts the container.

Type: String

Required: No

containerName

The name of the container.

Type: String

Required: No

exitCode

The exit code that's returned for the state change request.

Type: Integer

Required: No

networkBindings

The network bindings of the container.

Type: Array of [NetworkBinding](#) objects

Required: No

reason

The reason for the state change request.

Type: String

Required: No

runtimeId

The ID of the Docker container.

Type: String

Required: No

status

The status of the state change request.

Type: String

Required: No

task

The task ID or full Amazon Resource Name (ARN) of the task that hosts the container.

Type: String

Required: No

Response Syntax

```
{  
  "acknowledgment": "string"  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

acknowledgment

Acknowledgement of the state change.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

SubmitTaskStateChange

Note

This action is only used by the Amazon ECS agent, and it is not intended for use outside of the agent.

Sent to acknowledge that a task changed states.

Request Syntax

```
{
  "attachments": [
    {
      "attachmentArn": "string",
      "status": "string"
    }
  ],
  "cluster": "string",
  "containers": [
    {
      "containerName": "string",
      "exitCode": number,
      "imageDigest": "string",
      "networkBindings": [
        {
          "bindIP": "string",
          "containerPort": number,
          "containerPortRange": "string",
          "hostPort": number,
          "hostPortRange": "string",
          "protocol": "string"
        }
      ],
      "reason": "string",
      "runtimeId": "string",
      "status": "string"
    }
  ],
  "executionStoppedAt": number,
  "managedAgents": [
```

```
{
  "containerName": "string",
  "managedAgentName": "string",
  "reason": "string",
  "status": "string"
},
"pullStartedAt": number,
"pullStoppedAt": number,
"reason": "string",
"status": "string",
"task": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[attachments](#)

Any attachments associated with the state change request.

Type: Array of [AttachmentStateChange](#) objects

Required: No

[cluster](#)

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the task.

Type: String

Required: No

[containers](#)

Any containers that's associated with the state change request.

Type: Array of [ContainerStateChange](#) objects

Required: No

executionStoppedAt

The Unix timestamp for the time when the task execution stopped.

Type: Timestamp

Required: No

managedAgents

The details for the managed agent that's associated with the task.

Type: Array of [ManagedAgentStateChange](#) objects

Required: No

pullStartedAt

The Unix timestamp for the time when the container image pull started.

Type: Timestamp

Required: No

pullStoppedAt

The Unix timestamp for the time when the container image pull completed.

Type: Timestamp

Required: No

reason

The reason for the state change request.

Type: String

Required: No

status

The status of the state change request.

Type: String

Required: No

task

The task ID or full ARN of the task in the state change request.

Type: String

Required: No

Response Syntax

```
{  
  "acknowledgment": "string"  
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

acknowledgment

Acknowledgement of the state change.

Type: String

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

TagResource

Associates the specified tags to a resource with the specified `resourceArn`. If existing tags on a resource aren't specified in the request parameters, they aren't changed. When a resource is deleted, the tags that are associated with that resource are deleted as well.

Request Syntax

```
{
  "resourceArn": "string",
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ]
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

resourceArn

The Amazon Resource Name (ARN) of the resource to add tags to. Currently, the supported resources are Amazon ECS capacity providers, tasks, services, task definitions, clusters, and container instances.

In order to tag a service that has the following ARN format, you need to migrate the service to the long ARN. For more information, see [Migrate an Amazon ECS short service ARN to a long ARN](#) in the *Amazon Elastic Container Service Developer Guide*.

arn:aws:ecs:region:aws_account_id:service/service-name

After the migration is complete, the service has the long ARN format, as shown below. Use this ARN to tag the service.

arn:aws:ecs:region:aws_account_id:service/cluster-name/service-name

If you try to tag a service with a short ARN, you receive an `InvalidParameterException` error.

Type: String

Required: Yes

tags

The tags to add to the resource. A tag is an array of key-value pairs.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws :`, `AWS :`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: Yes

Response Elements

If the action is successful, the service sends back an HTTP 200 response with an empty HTTP body.

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ResourceNotFoundException

The specified resource wasn't found.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example tags the test service in the dev cluster with key team and value dev.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
X-Amz-Target: AmazonEC2ContainerServiceV20141113.TagResource
Content-Type: application/x-amz-json-1.1
X-Amz-Date: 20241209T194744Z
Authorization: AUTHPARAMS
Content-Length: 115

{
  "resourceArn": "arn:aws:ecs:us-west-2:012345678910:service/dev/test",
  "tags": [
    {
      "key": "team",
      "value": "dev"
    }
  ]
}
```

Sample Response

```
HTTP/1.1 200 OK
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f
Content-Type: application/x-amz-json-1.1
Content-Length: 2
```

```
Date: Fri, 20 Dec 2024 20:01:34 GMT
```

```
{}
```

Example

This example tags the dev cluster with key team and value dev and the key second-key and value dev-key2.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
X-Amz-Target: AmazonEC2ContainerServiceV20141113.TagResource
Content-Type: application/x-amz-json-1.1
X-Amz-Date: 20181026T194744Z
Authorization: AUTHPARAMS
Content-Length: 115

{
  "resourceArn": "arn:aws:ecs:us-west-2:012345678910:cluster/dev",
  "tags": [
    {
      "key": "team",
      "value": "dev"
    },
    {
      "key": "second-key",
      "value": "dev-key2"
    }
  ]
}
```

Sample Response

```
HTTP/1.1 200 OK
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f
Content-Type: application/x-amz-json-1.1
Content-Length: 2
Date: Wed, 19 Oct 2022 20:01:34 GMT
```

```
{ }
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UntagResource

Deletes specified tags from a resource.

Request Syntax

```
{  
  "resourceArn": "string",  
  "tagKeys": [ "string" ]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

resourceArn

The Amazon Resource Name (ARN) of the resource to delete tags from. Currently, the supported resources are Amazon ECS capacity providers, tasks, services, task definitions, clusters, and container instances.

Type: String

Required: Yes

tagKeys

The keys of the tags to be removed.

Type: Array of strings

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `^([\p{L}\p{Z}\p{N}_.: /+=\ -@]* )$`

Required: Yes

Response Elements

If the action is successful, the service sends back an HTTP 200 response with an empty HTTP body.

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ResourceNotFoundException

The specified resource wasn't found.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example tags the dev cluster with key team and value dev.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
X-Amz-Target: AmazonEC2ContainerServiceV20141113.UntagResource
Content-Type: application/x-amz-json-1.1
X-Amz-Date: 20181026T200134Z
Authorization: AUTHPARAMS
Content-Length: 93

{
  "resourceArn": "arn:aws:ecs:us-west-2:012345678910:cluster/devcluster",
  "tagKeys": [
    "team"
  ]
}
```

Sample Response

```
HTTP/1.1 200 OK
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f
Content-Type: application/x-amz-json-1.1
Content-Length: 2
Date: Fri, 26 Oct 2018 20:01:34 GMT
```



```
{ }
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UpdateCapacityProvider

Modifies the parameters for a capacity provider.

These changes only apply to new Amazon ECS Managed Instances, or EC2 instances, not existing ones.

Request Syntax

```
{
  "autoScalingGroupProvider": {
    "managedDraining": "string",
    "managedScaling": {
      "instanceWarmupPeriod": number,
      "maximumScalingStepSize": number,
      "minimumScalingStepSize": number,
      "status": "string",
      "targetCapacity": number
    },
    "managedTerminationProtection": "string"
  },
  "cluster": "string",
  "managedInstancesProvider": {
    "infrastructureRoleArn": "string",
    "instanceLaunchTemplate": {
      "ec2InstanceProfileArn": "string",
      "instanceRequirements": {
        "acceleratorCount": {
          "max": number,
          "min": number
        },
        "acceleratorManufacturers": [ "string" ],
        "acceleratorNames": [ "string" ],
        "acceleratorTotalMemoryMiB": {
          "max": number,
          "min": number
        },
        "acceleratorTypes": [ "string" ],
        "allowedInstanceTypes": [ "string" ],
        "bareMetal": "string",
        "baselineEbsBandwidthMbps": {
          "max": number,
          "min": number
        }
      }
    }
  }
}
```

```

    },
    "burstablePerformance": "string",
    "cpuManufacturers": [ "string" ],
    "excludedInstanceTypes": [ "string" ],
    "instanceGenerations": [ "string" ],
    "localStorage": "string",
    "localStorageTypes": [ "string" ],
    "maxSpotPriceAsPercentageOfOptimalOnDemandPrice": number,
    "memoryGiBPerVCpu": {
        "max": number,
        "min": number
    },
    "memoryMiB": {
        "max": number,
        "min": number
    },
    "networkBandwidthGbps": {
        "max": number,
        "min": number
    },
    "networkInterfaceCount": {
        "max": number,
        "min": number
    },
    "onDemandMaxPricePercentageOverLowestPrice": number,
    "requireHibernateSupport": boolean,
    "spotMaxPricePercentageOverLowestPrice": number,
    "totalLocalStorageGB": {
        "max": number,
        "min": number
    },
    "vCpuCount": {
        "max": number,
        "min": number
    }
},
"monitoring": "string",
"networkConfiguration": {
    "securityGroups": [ "string" ],
    "subnets": [ "string" ]
},
"storageConfiguration": {
    "storageSizeGiB": number
}

```

```
    },  
    "propagateTags": "string"  
  },  
  "name": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[autoScalingGroupProvider](#)

An object that represent the parameters to update for the Auto Scaling group capacity provider.

Type: [AutoScalingGroupProviderUpdate](#) object

Required: No

[cluster](#)

The name of the cluster that contains the capacity provider to update. Managed instances capacity providers are cluster-scoped and can only be updated within their associated cluster.

Type: String

Required: No

[managedInstancesProvider](#)

The updated configuration for the Amazon ECS Managed Instances provider. You can modify the infrastructure role, instance launch template, and tag propagation settings. Changes take effect for new instances launched after the update.

Type: [UpdateManagedInstancesProviderConfiguration](#) object

Required: No

[name](#)

The name of the capacity provider to update.

Type: String

Required: Yes

Response Syntax

```
{
  "capacityProvider": {
    "autoScalingGroupProvider": {
      "autoScalingGroupArn": "string",
      "managedDraining": "string",
      "managedScaling": {
        "instanceWarmupPeriod": number,
        "maximumScalingStepSize": number,
        "minimumScalingStepSize": number,
        "status": "string",
        "targetCapacity": number
      },
      "managedTerminationProtection": "string"
    },
    "capacityProviderArn": "string",
    "cluster": "string",
    "managedInstancesProvider": {
      "infrastructureRoleArn": "string",
      "instanceLaunchTemplate": {
        "ec2InstanceProfileArn": "string",
        "instanceRequirements": {
          "acceleratorCount": {
            "max": number,
            "min": number
          },
          "acceleratorManufacturers": [ "string" ],
          "acceleratorNames": [ "string" ],
          "acceleratorTotalMemoryMiB": {
            "max": number,
            "min": number
          },
          "acceleratorTypes": [ "string" ],
          "allowedInstanceTypes": [ "string" ],
          "bareMetal": "string",
          "baselineEbsBandwidthMbps": {
            "max": number,
            "min": number
          },
          "burstablePerformance": "string",
          "cpuManufacturers": [ "string" ],
          "excludedInstanceTypes": [ "string" ],
```

```

    "instanceGenerations": [ "string" ],
    "localStorage": "string",
    "localStorageTypes": [ "string" ],
    "maxSpotPriceAsPercentageOfOptimalOnDemandPrice": number,
    "memoryGiBPerVCpu": {
        "max": number,
        "min": number
    },
    "memoryMiB": {
        "max": number,
        "min": number
    },
    "networkBandwidthGbps": {
        "max": number,
        "min": number
    },
    "networkInterfaceCount": {
        "max": number,
        "min": number
    },
    "onDemandMaxPricePercentageOverLowestPrice": number,
    "requireHibernateSupport": boolean,
    "spotMaxPricePercentageOverLowestPrice": number,
    "totalLocalStorageGB": {
        "max": number,
        "min": number
    },
    "vCpuCount": {
        "max": number,
        "min": number
    }
},
"monitoring": "string",
"networkConfiguration": {
    "securityGroups": [ "string" ],
    "subnets": [ "string" ]
},
"storageConfiguration": {
    "storageSizeGiB": number
}
},
"propagateTags": "string",
"name": "string",

```

```
    "status": "string",
    "tags": [
      {
        "key": "string",
        "value": "string"
      }
    ],
    "type": "string",
    "updateStatus": "string",
    "updateStatusReason": "string"
  }
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

capacityProvider

Details about the capacity provider.

Type: [CapacityProvider](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)

- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UpdateCluster

Updates the cluster.

Request Syntax

```
{
  "cluster": "string",
  "configuration": {
    "executeCommandConfiguration": {
      "kmsKeyId": "string",
      "logConfiguration": {
        "cloudWatchEncryptionEnabled": boolean,
        "cloudWatchLogGroupName": "string",
        "s3BucketName": "string",
        "s3EncryptionEnabled": boolean,
        "s3KeyPrefix": "string"
      },
      "logging": "string"
    },
    "managedStorageConfiguration": {
      "fargateEphemeralStorageKmsKeyId": "string",
      "kmsKeyId": "string"
    }
  },
  "serviceConnectDefaults": {
    "namespace": "string"
  },
  "settings": [
    {
      "name": "string",
      "value": "string"
    }
  ]
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The name of the cluster to modify the settings for.

Type: String

Required: Yes

configuration

The execute command configuration for the cluster.

Type: [ClusterConfiguration](#) object

Required: No

serviceConnectDefaults

Use this parameter to set a default Service Connect namespace. After you set a default Service Connect namespace, any new services with Service Connect turned on that are created in the cluster are added as client services in the namespace. This setting only applies to new services that set the enabled parameter to true in the ServiceConnectConfiguration. You can set the namespace of each service individually in the ServiceConnectConfiguration to override this default parameter.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [ClusterServiceConnectDefaultsRequest](#) object

Required: No

settings

The cluster settings for your cluster.

Type: Array of [ClusterSetting](#) objects

Required: No

Response Syntax

```
{
  "cluster": {
    "activeServicesCount": number,
    "attachments": [
      {
        "details": [
          {
            "name": "string",
            "value": "string"
          }
        ],
        "id": "string",
        "status": "string",
        "type": "string"
      }
    ],
    "attachmentsStatus": "string",
    "capacityProviders": [ "string" ],
    "clusterArn": "string",
    "clusterName": "string",
    "configuration": {
      "executeCommandConfiguration": {
        "kmsKeyId": "string",
        "logConfiguration": {
          "cloudWatchEncryptionEnabled": boolean,
          "cloudWatchLogGroupName": "string",
          "s3BucketName": "string",
          "s3EncryptionEnabled": boolean,
          "s3KeyPrefix": "string"
        },
        "logging": "string"
      },
      "managedStorageConfiguration": {
        "fargateEphemeralStorageKmsKeyId": "string",
        "kmsKeyId": "string"
      }
    },
    "defaultCapacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
```

```
    "weight": number
  },
],
"pendingTasksCount": number,
"registeredContainerInstancesCount": number,
"runningTasksCount": number,
"serviceConnectDefaults": {
  "namespace": "string"
},
"settings": [
  {
    "name": "string",
    "value": "string"
  }
],
"statistics": [
  {
    "name": "string",
    "value": "string"
  }
],
"status": "string",
"tags": [
  {
    "key": "string",
    "value": "string"
  }
]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

cluster

Details about the cluster.

Type: [Cluster](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

NamespaceNotFoundException

The specified namespace wasn't found.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UpdateClusterSettings

Modifies the settings to use for a cluster.

Request Syntax

```
{
  "cluster": "string",
  "settings": [
    {
      "name": "string",
      "value": "string"
    }
  ]
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The name of the cluster to modify the settings for.

Type: String

Required: Yes

settings

The setting to use by default for a cluster. This parameter is used to turn on CloudWatch Container Insights for a cluster. If this value is specified, it overrides the `containerInsights` value set with [PutAccountSetting](#) or [PutAccountSettingDefault](#).

Important

Currently, if you delete an existing cluster that does not have Container Insights turned on, and then create a new cluster with the same name with Container Insights tuned on, Container Insights will not actually be turned on. If you want to preserve the same

name for your existing cluster and turn on Container Insights, you must wait 7 days before you can re-create it.

Type: Array of [ClusterSetting](#) objects

Required: Yes

Response Syntax

```
{
  "cluster": {
    "activeServicesCount": number,
    "attachments": [
      {
        "details": [
          {
            "name": "string",
            "value": "string"
          }
        ],
        "id": "string",
        "status": "string",
        "type": "string"
      }
    ],
    "attachmentsStatus": "string",
    "capacityProviders": [ "string" ],
    "clusterArn": "string",
    "clusterName": "string",
    "configuration": {
      "executeCommandConfiguration": {
        "kmsKeyId": "string",
        "logConfiguration": {
          "cloudWatchEncryptionEnabled": boolean,
          "cloudWatchLogGroupName": "string",
          "s3BucketName": "string",
          "s3EncryptionEnabled": boolean,
          "s3KeyPrefix": "string"
        },
        "logging": "string"
      }
    },
  },
}
```

```

    "managedStorageConfiguration": {
      "fargateEphemeralStorageKmsKeyId": "string",
      "kmsKeyId": "string"
    },
    "defaultCapacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "pendingTasksCount": number,
    "registeredContainerInstancesCount": number,
    "runningTasksCount": number,
    "serviceConnectDefaults": {
      "namespace": "string"
    },
    "settings": [
      {
        "name": "string",
        "value": "string"
      }
    ],
    "statistics": [
      {
        "name": "string",
        "value": "string"
      }
    ],
    "status": "string",
    "tags": [
      {
        "key": "string",
        "value": "string"
      }
    ]
  }
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

cluster

Details about the cluster

Type: [Cluster](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UpdateContainerAgent

Updates the Amazon ECS container agent on a specified container instance. Updating the Amazon ECS container agent doesn't interrupt running tasks or services on the container instance. The process for updating the agent differs depending on whether your container instance was launched with the Amazon ECS-optimized AMI or another operating system.

Note

The UpdateContainerAgent API isn't supported for container instances using the Amazon ECS-optimized Amazon Linux 2 (arm64) AMI. To update the container agent, you can update the `ecs-init` package. This updates the agent. For more information, see [Updating the Amazon ECS container agent](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

Agent updates with the UpdateContainerAgent API operation do not apply to Windows container instances. We recommend that you launch new container instances to update the agent version in your Windows clusters.

The UpdateContainerAgent API requires an Amazon ECS-optimized AMI or Amazon Linux AMI with the `ecs-init` service installed and running. For help updating the Amazon ECS container agent on other operating systems, see [Manually updating the Amazon ECS container agent](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{  
  "cluster": "string",  
  "containerInstance": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that your container instance is running on. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

containerInstance

The container instance ID or full ARN entries for the container instance where you would like to update the Amazon ECS container agent.

Type: String

Required: Yes

Response Syntax

```
{
  "containerInstance": {
    "agentConnected": boolean,
    "agentUpdateStatus": "string",
    "attachments": [
      {
        "details": [
          {
            "name": "string",
            "value": "string"
          }
        ],
        "id": "string",
        "status": "string",
        "type": "string"
      }
    ],
    "attributes": [
      {
        "name": "string",
        "targetId": "string",
```

```

        "targetType": "string",
        "value": "string"
    }
],
"capacityProviderName": "string",
"containerInstanceArn": "string",
"ec2InstanceId": "string",
"healthStatus": {
    "details": [
        {
            "lastStatusChange": number,
            "lastUpdated": number,
            "status": "string",
            "type": "string"
        }
    ],
    "overallStatus": "string"
},
"pendingTasksCount": number,
"registeredAt": number,
"registeredResources": [
    {
        "doubleValue": number,
        "integerValue": number,
        "longValue": number,
        "name": "string",
        "stringValue": [ "string" ],
        "type": "string"
    }
],
"remainingResources": [
    {
        "doubleValue": number,
        "integerValue": number,
        "longValue": number,
        "name": "string",
        "stringValue": [ "string" ],
        "type": "string"
    }
],
"runningTasksCount": number,
"status": "string",
"statusReason": "string",
"tags": [

```

```
{
  {
    "key": "string",
    "value": "string"
  },
  "version": number,
  "versionInfo": {
    "agentHash": "string",
    "agentVersion": "string",
    "dockerVersion": "string"
  }
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

containerInstance

The container instance that the container agent was updated for.

Type: [ContainerInstance](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

MissingVersionException

Amazon ECS can't determine the current version of the Amazon ECS container agent on the container instance and doesn't have enough information to proceed with an update. This could be because the agent running on the container instance is a previous or custom version that doesn't use our version information.

HTTP Status Code: 400

NoUpdateAvailableException

There's no update available for this Amazon ECS container agent. This might be because the agent is already running the latest version or because it's so old that there's no update path to the current version.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UpdateInProgressException

There's already a current Amazon ECS container agent update in progress on the container instance that's specified. If the container agent becomes disconnected while it's in a transitional

stage, such as PENDING or STAGING, the update process can get stuck in that state. However, when the agent reconnects, it resumes where it stopped previously.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example updates the container agent version for the container instance with the ID 53ac7152-dcd1-4102-81f5-208962864132 in the update cluster.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 82
X-Amz-Target: AmazonEC2ContainerServiceV20141113.UpdateContainerAgent
X-Amz-Date: 20150528T152756Z
User-Agent: aws-cli/1.7.30 Python/2.7.9 Darwin/14.3.0
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "cluster": "update",
  "containerInstance": "53ac7152-dcd1-4102-81f5-208962864132"
}
```

Sample Response

```
HTTP/1.1 200 OK
```

```
Server: Server
Date: Thu, 28 May 2015 15:27:54 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 1033
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f
```

```
{
  "containerInstance": {
    "agentConnected": true,
    "agentUpdateStatus": "PENDING",
    ...
    "versionInfo": {
      "agentHash": "4023248",
      "agentVersion": "1.0.0",
      "dockerVersion": "DockerVersion: 1.5.0"
    }
  }
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UpdateContainerInstancesState

Modifies the status of an Amazon ECS container instance.

Once a container instance has reached an ACTIVE state, you can change the status of a container instance to DRAINING to manually remove an instance from a cluster, for example to perform system updates, update the Docker daemon, or scale down the cluster size.

Important

A container instance can't be changed to DRAINING until it has reached an ACTIVE status. If the instance is in any other status, an error will be received.

When you set a container instance to DRAINING, Amazon ECS prevents new tasks from being scheduled for placement on the container instance and replacement service tasks are started on other container instances in the cluster if the resources are available. Service tasks on the container instance that are in the PENDING state are stopped immediately.

Service tasks on the container instance that are in the RUNNING state are stopped and replaced according to the service's deployment configuration parameters, `minimumHealthyPercent` and `maximumPercent`. You can change the deployment configuration of your service using [UpdateService](#).

- If `minimumHealthyPercent` is below 100%, the scheduler can ignore `desiredCount` temporarily during task replacement. For example, `desiredCount` is four tasks, a minimum of 50% allows the scheduler to stop two existing tasks before starting two new tasks. If the minimum is 100%, the service scheduler can't remove existing tasks until the replacement tasks are considered healthy. Tasks for services that do not use a load balancer are considered healthy if they're in the RUNNING state. Tasks for services that use a load balancer are considered healthy if they're in the RUNNING state and are reported as healthy by the load balancer.
- The `maximumPercent` parameter represents an upper limit on the number of running tasks during task replacement. You can use this to define the replacement batch size. For example, if `desiredCount` is four tasks, a maximum of 200% starts four new tasks before stopping the four tasks to be drained, provided that the cluster resources required to do this are available. If the maximum is 100%, then replacement tasks can't start until the draining tasks have stopped.

Any PENDING or RUNNING tasks that do not belong to a service aren't affected. You must wait for them to finish or stop them manually.

A container instance has completed draining when it has no more RUNNING tasks. You can verify this using [ListTasks](#).

When a container instance has been drained, you can set a container instance to ACTIVE status and once it has reached that status the Amazon ECS scheduler can begin scheduling tasks on the instance again.

Request Syntax

```
{  
  "cluster": "string",  
  "containerInstances": [ "string" ],  
  "status": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[cluster](#)

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the container instance to update. If you do not specify a cluster, the default cluster is assumed.

Type: String

Required: No

[containerInstances](#)

A list of up to 10 container instance IDs or full ARN entries.

Type: Array of strings

Required: Yes

status

The container instance state to update the container instance with. The only valid values for this action are ACTIVE and DRAINING. A container instance can only be updated to DRAINING status once it has reached an ACTIVE state. If a container instance is in REGISTERING, DEREGISTERING, or REGISTRATION_FAILED state you can describe the container instance but can't update the container instance state.

Type: String

Valid Values: ACTIVE | DRAINING | REGISTERING | DEREGISTERING | REGISTRATION_FAILED

Required: Yes

Response Syntax

```
{
  "containerInstances": [
    {
      "agentConnected": boolean,
      "agentUpdateStatus": "string",
      "attachments": [
        {
          "details": [
            {
              "name": "string",
              "value": "string"
            }
          ],
          "id": "string",
          "status": "string",
          "type": "string"
        }
      ],
      "attributes": [
        {
          "name": "string",
          "targetId": "string",
          "targetType": "string",
          "value": "string"
        }
      ]
    }
  ]
}
```

```

],
"capacityProviderName": "string",
"containerInstanceArn": "string",
"ec2InstanceId": "string",
"healthStatus": {
  "details": [
    {
      "lastStatusChange": number,
      "lastUpdated": number,
      "status": "string",
      "type": "string"
    }
  ],
  "overallStatus": "string"
},
"pendingTasksCount": number,
"registeredAt": number,
"registeredResources": [
  {
    "doubleValue": number,
    "integerValue": number,
    "longValue": number,
    "name": "string",
    "stringSetValue": [ "string" ],
    "type": "string"
  }
],
"remainingResources": [
  {
    "doubleValue": number,
    "integerValue": number,
    "longValue": number,
    "name": "string",
    "stringSetValue": [ "string" ],
    "type": "string"
  }
],
"runningTasksCount": number,
"status": "string",
"statusReason": "string",
"tags": [
  {
    "key": "string",
    "value": "string"
  }
]

```

```
    }
  ],
  "version": number,
  "versionInfo": {
    "agentHash": "string",
    "agentVersion": "string",
    "dockerVersion": "string"
  }
},
"failures": [
  {
    "arn": "string",
    "detail": "string",
    "reason": "string"
  }
]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

containerInstances

The list of container instances.

Type: Array of [ContainerInstance](#) objects

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS,

these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example sets a container instance in the default cluster with the ID 1c3be8ed-df30-47b4-8f1e-6e68ebd01f34 to the DRAINING status so that it can't receive tasks for placement.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 114
X-Amz-Target: AmazonEC2ContainerServiceV20141113.UpdateContainerInstancesState
X-Amz-Date: 20161220T221142Z
User-Agent: aws-cli/1.11.31 Python/2.7.12 Darwin/16.3.0 botocore/1.4.88
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "status": "DRAINING",
  "cluster": "default",
  "containerInstances": [
    "1c3be8ed-df30-47b4-8f1e-6e68ebd01f34"
  ]
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Tue, 20 Dec 2016 22:11:42 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 2344
Connection: keep-alive
x-amzn-RequestId: 49d68928-c701-11e6-8f99-6103d648cdad

{
  "containerInstances": [
    {
      "agentConnected": true,
```

```
"attributes": [  
  {  
    "name": "ecs.availability-zone",  
    "value": "us-west-2b"  
  },  
  {  
    "name": "com.amazonaws.ecs.capability.logging-driver.syslog"  
  },  
  {  
    "name": "ecs.instance-type",  
    "value": "c4.xlarge"  
  },  
  {  
    "name": "ecs.ami-id",  
    "value": "ami-a2ca61c2"  
  },  
  {  
    "name": "com.amazonaws.ecs.capability.task-iam-role-network-host"  
  },  
  {  
    "name": "com.amazonaws.ecs.capability.logging-driver.awslogs"  
  },  
  {  
    "name": "com.amazonaws.ecs.capability.logging-driver.json-file"  
  },  
  {  
    "name": "com.amazonaws.ecs.capability.docker-remote-api.1.17"  
  },  
  {  
    "name": "com.amazonaws.ecs.capability.privileged-container"  
  },  
  {  
    "name": "com.amazonaws.ecs.capability.docker-remote-api.1.18"  
  },  
  {  
    "name": "com.amazonaws.ecs.capability.docker-remote-api.1.19"  
  },  
  {  
    "name": "com.amazonaws.ecs.capability.ecr-auth"  
  },  
  {  
    "name": "ecs.os-type",  
    "value": "linux"  
  },  
],
```

```
{
  "name": "com.amazonaws.ecs.capability.docker-remote-api.1.20"
},
{
  "name": "com.amazonaws.ecs.capability.docker-remote-api.1.21"
},
{
  "name": "com.amazonaws.ecs.capability.docker-remote-api.1.22"
},
{
  "name": "com.amazonaws.ecs.capability.task-iam-role"
},
{
  "name": "com.amazonaws.ecs.capability.docker-remote-api.1.23"
}
],
"containerInstanceArn": "arn:aws:ecs:us-west-2:012345678910:container-instance/
default/1c3be8ed-df30-47b4-8f1e-6e68ebd01f34",
"ec2InstanceId": "i-05d99c76955727ec6",
"pendingTasksCount": 0,
"registeredResources": [
  {
    "doubleValue": 0,
    "integerValue": 4096,
    "longValue": 0,
    "name": "CPU",
    "type": "INTEGER"
  },
  {
    "doubleValue": 0,
    "integerValue": 7482,
    "longValue": 0,
    "name": "MEMORY",
    "type": "INTEGER"
  },
  {
    "doubleValue": 0,
    "integerValue": 0,
    "longValue": 0,
    "name": "PORTS",
    "stringSetValue": [
      "22",
      "2376",
      "2375",
```

```
        "51678",
        "51679"
    ],
    "type": "STRINGSET"
},
{
    "doubleValue": 0,
    "integerValue": 0,
    "longValue": 0,
    "name": "PORTS_UDP",
    "stringSetValue": [],
    "type": "STRINGSET"
}
],
"remainingResources": [
    {
        "doubleValue": 0,
        "integerValue": 4096,
        "longValue": 0,
        "name": "CPU",
        "type": "INTEGER"
    },
    {
        "doubleValue": 0,
        "integerValue": 7482,
        "longValue": 0,
        "name": "MEMORY",
        "type": "INTEGER"
    },
    {
        "doubleValue": 0,
        "integerValue": 0,
        "longValue": 0,
        "name": "PORTS",
        "stringSetValue": [
            "22",
            "2376",
            "2375",
            "51678",
            "51679"
        ],
        "type": "STRINGSET"
    },
    {
```

```
        "doubleValue": 0,
        "integerValue": 0,
        "longValue": 0,
        "name": "PORTS_UDP",
        "stringSetValue": [],
        "type": "STRINGSET"
    }
],
"runningTasksCount": 0,
"status": "DRAINING",
"version": 30,
"versionInfo": {
    "agentHash": "efe53c6",
    "agentVersion": "1.13.1",
    "dockerVersion": "DockerVersion: 1.11.2"
}
},
"failures": []
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UpdateService

Modifies the parameters of a service.

Note

On March 21, 2024, a change was made to resolve the task definition revision before authorization. When a task definition revision is not specified, authorization will occur using the latest revision of a task definition.

For services using the rolling update (ECS) you can update the desired count, deployment configuration, network configuration, load balancers, service registries, enable ECS managed tags option, propagate tags option, task placement constraints and strategies, and task definition. When you update any of these parameters, Amazon ECS starts new tasks with the new configuration.

You can attach Amazon EBS volumes to Amazon ECS tasks by configuring the volume when starting or running a task, or when creating or updating a service. For more information, see [Amazon EBS volumes](#) in the *Amazon Elastic Container Service Developer Guide*. You can update your volume configurations and trigger a new deployment. `volumeConfigurations` is only supported for REPLICA service and not DAEMON service. If you leave `volumeConfigurations` null, it doesn't trigger a new deployment. For more information on volumes, see [Amazon EBS volumes](#) in the *Amazon Elastic Container Service Developer Guide*.

For services using the blue/green (CODE_DEPLOY) deployment controller, only the desired count, deployment configuration, health check grace period, task placement constraints and strategies, enable ECS managed tags option, and propagate tags can be updated using this API. If the network configuration, platform version, task definition, or load balancer need to be updated, create a new AWS CodeDeploy deployment. For more information, see [CreateDeployment](#) in the *AWS CodeDeploy API Reference*.

For services using an external deployment controller, you can update only the desired count, task placement constraints and strategies, health check grace period, enable ECS managed tags option, and propagate tags option, using this API. If the launch type, load balancer, network configuration, platform version, or task definition need to be updated, create a new task set For more information, see [CreateTaskSet](#).

You can add to or subtract from the number of instantiations of a task definition in a service by specifying the cluster that the service is running in and a new `desiredCount` parameter.

You can attach Amazon EBS volumes to Amazon ECS tasks by configuring the volume when starting or running a task, or when creating or updating a service. For more information, see [Amazon EBS volumes](#) in the *Amazon Elastic Container Service Developer Guide*.

If you have updated the container image of your application, you can create a new task definition with that image and deploy it to your service. The service scheduler uses the minimum healthy percent and maximum percent parameters (in the service's deployment configuration) to determine the deployment strategy.

 Note

If your updated Docker image uses the same tag as what is in the existing task definition for your service (for example, `my_image:latest`), you don't need to create a new revision of your task definition. You can update the service using the `forceNewDeployment` option. The new tasks launched by the deployment pull the current image/tag combination from your repository when they start.

You can also update the deployment configuration of a service. When a deployment is triggered by updating the task definition of a service, the service scheduler uses the deployment configuration parameters, `minimumHealthyPercent` and `maximumPercent`, to determine the deployment strategy.

- If `minimumHealthyPercent` is below 100%, the scheduler can ignore `desiredCount` temporarily during a deployment. For example, if `desiredCount` is four tasks, a minimum of 50% allows the scheduler to stop two existing tasks before starting two new tasks. Tasks for services that don't use a load balancer are considered healthy if they're in the `RUNNING` state. Tasks for services that use a load balancer are considered healthy if they're in the `RUNNING` state and are reported as healthy by the load balancer.
- The `maximumPercent` parameter represents an upper limit on the number of running tasks during a deployment. You can use it to define the deployment batch size. For example, if `desiredCount` is four tasks, a maximum of 200% starts four new tasks before stopping the four older tasks (provided that the cluster resources required to do this are available).

When [UpdateService](#) stops a task during a deployment, the equivalent of `docker stop` is issued to the containers running in the task. This results in a `SIGTERM` and a 30-second timeout. After this, `SIGKILL` is sent and the containers are forcibly stopped. If the container handles the `SIGTERM` gracefully and exits within 30 seconds from receiving it, no `SIGKILL` is sent.

When the service scheduler launches new tasks, it determines task placement in your cluster with the following logic.

- Determine which of the container instances in your cluster can support your service's task definition. For example, they have the required CPU, memory, ports, and container instance attributes.
- By default, the service scheduler attempts to balance tasks across Availability Zones in this manner even though you can choose a different placement strategy.
 - Sort the valid container instances by the fewest number of running tasks for this service in the same Availability Zone as the instance. For example, if zone A has one running service task and zones B and C each have zero, valid container instances in either zone B or C are considered optimal for placement.
 - Place the new service task on a valid container instance in an optimal Availability Zone (based on the previous steps), favoring container instances with the fewest number of running tasks for this service.

When the service scheduler stops running tasks, it attempts to maintain balance across the Availability Zones in your cluster using the following logic:

- Sort the container instances by the largest number of running tasks for this service in the same Availability Zone as the instance. For example, if zone A has one running service task and zones B and C each have two, container instances in either zone B or C are considered optimal for termination.
- Stop the task on a container instance in an optimal Availability Zone (based on the previous steps), favoring container instances with the largest number of running tasks for this service.

Request Syntax

```
{  
  "availabilityZoneRebalancing": "string",  
  "capacityProviderStrategy": [  
    {
```

```

        "base": number,
        "capacityProvider": "string",
        "weight": number
    }
],
"cluster": "string",
"deploymentConfiguration": {
    "alarms": {
        "alarmNames": [ "string" ],
        "enable": boolean,
        "rollback": boolean
    },
    "bakeTimeInMinutes": number,
    "canaryConfiguration": {
        "canaryBakeTimeInMinutes": number,
        "canaryPercent": number
    },
    "deploymentCircuitBreaker": {
        "enable": boolean,
        "rollback": boolean
    },
    "lifecycleHooks": [
        {
            "hookDetails": JSON value,
            "hookTargetArn": "string",
            "lifecycleStages": [ "string" ],
            "roleArn": "string"
        }
    ],
    "linearConfiguration": {
        "stepBakeTimeInMinutes": number,
        "stepPercent": number
    },
    "maximumPercent": number,
    "minimumHealthyPercent": number,
    "strategy": "string"
},
"deploymentController": {
    "type": "string"
},
"desiredCount": number,
"enableECSTags": boolean,
"enableExecuteCommand": boolean,
"forceNewDeployment": boolean,

```

```
"healthCheckGracePeriodSeconds": number,
"loadBalancers": [
  {
    "advancedConfiguration": {
      "alternateTargetGroupArn": "string",
      "productionListenerRule": "string",
      "roleArn": "string",
      "testListenerRule": "string"
    },
    "containerName": "string",
    "containerPort": number,
    "loadBalancerName": "string",
    "targetGroupArn": "string"
  }
],
"networkConfiguration": {
  "awsvpcConfiguration": {
    "assignPublicIp": "string",
    "securityGroups": [ "string " ],
    "subnets": [ "string " ]
  }
},
"placementConstraints": [
  {
    "expression": "string",
    "type": "string"
  }
],
"placementStrategy": [
  {
    "field": "string",
    "type": "string"
  }
],
"platformVersion": "string",
"propagateTags": "string",
"service": "string",
"serviceConnectConfiguration": {
  "accessLogConfiguration": {
    "format": "string",
    "includeQueryParameters": "string"
  },
  "enabled": boolean,
  "logConfiguration": {
```

```

    "logDriver": "string",
    "options": {
      "string": "string"
    },
    "secretOptions": [
      {
        "name": "string",
        "valueFrom": "string"
      }
    ],
    "namespace": "string",
    "services": [
      {
        "clientAliases": [
          {
            "dnsName": "string",
            "port": number,
            "testTrafficRules": {
              "header": {
                "name": "string",
                "value": {
                  "exact": "string"
                }
              }
            }
          }
        ],
        "discoveryName": "string",
        "ingressPortOverride": number,
        "portName": "string",
        "timeout": {
          "idleTimeoutSeconds": number,
          "perRequestTimeoutSeconds": number
        },
        "tls": {
          "issuerCertificateAuthority": {
            "awsPcaAuthorityArn": "string"
          },
          "kmsKey": "string",
          "roleArn": "string"
        }
      }
    ]
  ]

```

```
},
"serviceRegistries": [
  {
    "containerName": "string",
    "containerPort": number,
    "port": number,
    "registryArn": "string"
  }
],
"taskDefinition": "string",
"volumeConfigurations": [
  {
    "managedEBSVolume": {
      "encrypted": boolean,
      "filesystemType": "string",
      "iops": number,
      "kmsKeyId": "string",
      "roleArn": "string",
      "sizeInGiB": number,
      "snapshotId": "string",
      "tagSpecifications": [
        {
          "propagateTags": "string",
          "resourceType": "string",
          "tags": [
            {
              "key": "string",
              "value": "string"
            }
          ]
        }
      ],
      "throughput": number,
      "volumeInitializationRate": number,
      "volumeType": "string"
    },
    "name": "string"
  }
],
"vpcLatticeConfigurations": [
  {
    "portName": "string",
    "roleArn": "string",
    "targetGroupArn": "string"
```

```
}  
]  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[availabilityZoneRebalancing](#)

Indicates whether to use Availability Zone rebalancing for the service.

For more information, see [Balancing an Amazon ECS service across Availability Zones](#) in the Amazon Elastic Container Service Developer Guide .

The default behavior of AvailabilityZoneRebalancing differs between create and update requests:

- For create service requests, when no value is specified for AvailabilityZoneRebalancing, Amazon ECS defaults the value to ENABLED.
- For update service requests, when no value is specified for AvailabilityZoneRebalancing, Amazon ECS defaults to the existing service's AvailabilityZoneRebalancing value. If the service never had an AvailabilityZoneRebalancing value set, Amazon ECS treats this as DISABLED.

This parameter doesn't trigger a new service deployment.

Type: String

Valid Values: ENABLED | DISABLED

Required: No

[capacityProviderStrategy](#)

The details of a capacity provider strategy. You can set a capacity provider when you create a cluster, run a task, or update a service.

Note

If you want to use Amazon ECS Managed Instances, you must use the `capacityProviderStrategy` request parameter.

When you use Fargate, the capacity providers are FARGATE or FARGATE_SPOT.

When you use Amazon EC2, the capacity providers are Auto Scaling groups.

You can change capacity providers for rolling deployments and blue/green deployments.

The following list provides the valid transitions:

- Update the Fargate launch type to an Auto Scaling group capacity provider.
- Update the Amazon EC2 launch type to a Fargate capacity provider.
- Update the Fargate capacity provider to an Auto Scaling group capacity provider.
- Update the Amazon EC2 capacity provider to a Fargate capacity provider.
- Update the Auto Scaling group or Fargate capacity provider back to the launch type.

Pass an empty list in the `capacityProviderStrategy` parameter.

For information about AWS CDK considerations, see [AWS CDK considerations](#).

This parameter doesn't trigger a new service deployment.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: No

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that your service runs on. If you do not specify a cluster, the default cluster is assumed.

You can't change the cluster name.

Type: String

Required: No

deploymentConfiguration

Optional deployment parameters that control how many tasks run during the deployment and the ordering of stopping and starting tasks.

This parameter doesn't trigger a new service deployment.

Type: [DeploymentConfiguration](#) object

Required: No

deploymentController

The deployment controller to use for the service.

Type: [DeploymentController](#) object

Required: No

desiredCount

The number of instantiations of the task to place and keep running in your service.

This parameter doesn't trigger a new service deployment.

Type: Integer

Required: No

enableECSTags

Determines whether to turn on Amazon ECS managed tags for the tasks in the service. For more information, see [Tagging Your Amazon ECS Resources](#) in the *Amazon Elastic Container Service Developer Guide*.

Only tasks launched after the update will reflect the update. To update the tags on all tasks, set `forceNewDeployment` to `true`, so that Amazon ECS starts new tasks with the updated tags.

This parameter doesn't trigger a new service deployment.

Type: Boolean

Required: No

enableExecuteCommand

If `true`, this enables execute command functionality on all task containers.

If you do not want to override the value that was set when the service was created, you can set this to `null` when performing this action.

This parameter doesn't trigger a new service deployment.

Type: Boolean

Required: No

forceNewDeployment

Determines whether to force a new deployment of the service. By default, deployments aren't forced. You can use this option to start a new deployment with no service definition changes. For example, you can update a service's tasks to use a newer Docker image with the same image/tag combination (`my_image:latest`) or to roll Fargate tasks onto a newer platform version.

Type: Boolean

Required: No

healthCheckGracePeriodSeconds

The period of time, in seconds, that the Amazon ECS service scheduler ignores unhealthy Elastic Load Balancing, VPC Lattice, and container health checks after a task has first started. If you don't specify a health check grace period value, the default value of 0 is used. If you don't use any of the health checks, then `healthCheckGracePeriodSeconds` is unused.

If your service's tasks take a while to start and respond to health checks, you can specify a health check grace period of up to 2,147,483,647 seconds (about 69 years). During that time, the Amazon ECS service scheduler ignores health check status. This grace period can prevent the service scheduler from marking tasks as unhealthy and stopping them before they have time to come up.

If your service has more running tasks than desired, unhealthy tasks in the grace period might be stopped to reach the desired count.

This parameter doesn't trigger a new service deployment.

Type: Integer

Required: No

loadBalancers

Note

You must have a service-linked role when you update this property

A list of Elastic Load Balancing load balancer objects. It contains the load balancer name, the container name, and the container port to access from the load balancer. The container name is as it appears in a container definition.

When you add, update, or remove a load balancer configuration, Amazon ECS starts new tasks with the updated Elastic Load Balancing configuration, and then stops the old tasks when the new tasks are running.

For services that use rolling updates, you can add, update, or remove Elastic Load Balancing target groups. You can update from a single target group to multiple target groups and from multiple target groups to a single target group.

For services that use blue/green deployments, you can update Elastic Load Balancing target groups by using [CreateDeployment](#) through AWS CodeDeploy. Note that multiple target groups are not supported for blue/green deployments. For more information see [Register multiple target groups with a service](#) in the *Amazon Elastic Container Service Developer Guide*.

For services that use the external deployment controller, you can add, update, or remove load balancers by using [CreateTaskSet](#). Note that multiple target groups are not supported for external deployments. For more information see [Register multiple target groups with a service](#) in the *Amazon Elastic Container Service Developer Guide*.

You can remove existing `loadBalancers` by passing an empty list.

This parameter triggers a new service deployment.

Type: Array of [LoadBalancer](#) objects

Required: No

networkConfiguration

An object representing the network configuration for the service.

This parameter triggers a new service deployment.

Type: [NetworkConfiguration](#) object

Required: No

[placementConstraints](#)

An array of task placement constraint objects to update the service to use. If no value is specified, the existing placement constraints for the service will remain unchanged. If this value is specified, it will override any existing placement constraints defined for the service. To remove all existing placement constraints, specify an empty array.

You can specify a maximum of 10 constraints for each task. This limit includes constraints in the task definition and those specified at runtime.

This parameter doesn't trigger a new service deployment.

Type: Array of [PlacementConstraint](#) objects

Required: No

[placementStrategy](#)

The task placement strategy objects to update the service to use. If no value is specified, the existing placement strategy for the service will remain unchanged. If this value is specified, it will override the existing placement strategy defined for the service. To remove an existing placement strategy, specify an empty object.

You can specify a maximum of five strategy rules for each service.

This parameter doesn't trigger a new service deployment.

Type: Array of [PlacementStrategy](#) objects

Required: No

[platformVersion](#)

The platform version that your tasks in the service run on. A platform version is only specified for tasks using the Fargate launch type. If a platform version is not specified, the LATEST platform version is used. For more information, see [AWS Fargate Platform Versions](#) in the *Amazon Elastic Container Service Developer Guide*.

This parameter triggers a new service deployment.

Type: String

Required: No

propagateTags

Determines whether to propagate the tags from the task definition or the service to the task. If no value is specified, the tags aren't propagated.

Only tasks launched after the update will reflect the update. To update the tags on all tasks, set `forceNewDeployment` to `true`, so that Amazon ECS starts new tasks with the updated tags.

This parameter doesn't trigger a new service deployment.

Type: String

Valid Values: `TASK_DEFINITION` | `SERVICE` | `NONE`

Required: No

service

The name of the service to update.

Type: String

Required: Yes

serviceConnectConfiguration

The configuration for this service to discover and connect to services, and be discovered by, and connected from, other services within a namespace.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

This parameter triggers a new service deployment.

Type: [ServiceConnectConfiguration](#) object

Required: No

serviceRegistries

Note

You must have a service-linked role when you update this property.
For more information about the role see the `CreateService` request parameter [role](#).

The details for the service discovery registries to assign to this service. For more information, see [Service Discovery](#).

When you add, update, or remove the service registries configuration, Amazon ECS starts new tasks with the updated service registries configuration, and then stops the old tasks when the new tasks are running.

You can remove existing `serviceRegistries` by passing an empty list.

This parameter triggers a new service deployment.

Type: Array of [ServiceRegistry](#) objects

Required: No

taskDefinition

The family and revision (`family:revision`) or full ARN of the task definition to run in your service. If a `revision` is not specified, the latest `ACTIVE` revision is used. If you modify the task definition with `UpdateService`, Amazon ECS spawns a task with the new version of the task definition and then stops an old task after the new version is running.

This parameter triggers a new service deployment.

Type: String

Required: No

volumeConfigurations

The details of the volume that was `configuredAtLaunch`. You can configure the size, `volumeType`, IOPS, throughput, snapshot and encryption in [ServiceManagedEBSVolumeConfiguration](#). The name of the volume must match the name from the task definition. If set to null, no new deployment is triggered. Otherwise, if this configuration differs from the existing one, it triggers a new deployment.

This parameter triggers a new service deployment.

Type: Array of [ServiceVolumeConfiguration](#) objects

Required: No

[vpcLatticeConfigurations](#)

An object representing the VPC Lattice configuration for the service being updated.

This parameter triggers a new service deployment.

Type: Array of [VpcLatticeConfiguration](#) objects

Required: No

Response Syntax

```
{
  "service": {
    "availabilityZoneRebalancing": "string",
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "clusterArn": "string",
    "createdAt": number,
    "createdBy": "string",
    "deploymentConfiguration": {
      "alarms": {
        "alarmNames": [ "string" ],
        "enable": boolean,
        "rollback": boolean
      },
      "bakeTimeInMinutes": number,
      "canaryConfiguration": {
        "canaryBakeTimeInMinutes": number,
        "canaryPercent": number
      },
      "deploymentCircuitBreaker": {
        "enable": boolean,
```

```

    "rollback": boolean
  },
  "lifecycleHooks": [
    {
      "hookDetails": JSON value,
      "hookTargetArn": "string",
      "lifecycleStages": [ "string" ],
      "roleArn": "string"
    }
  ],
  "linearConfiguration": {
    "stepBakeTimeInMinutes": number,
    "stepPercent": number
  },
  "maximumPercent": number,
  "minimumHealthyPercent": number,
  "strategy": "string"
},
"deploymentController": {
  "type": "string"
},
"deployments": [
  {
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "createdAt": number,
    "desiredCount": number,
    "failedTasks": number,
    "fargateEphemeralStorage": {
      "kmsKeyId": "string"
    },
    "id": "string",
    "launchType": "string",
    "networkConfiguration": {
      "awsvpcConfiguration": {
        "assignPublicIp": "string",
        "securityGroups": [ "string" ],
        "subnets": [ "string" ]
      }
    }
  }
]

```

```

    },
    "pendingCount": number,
    "platformFamily": "string",
    "platformVersion": "string",
    "rolloutState": "string",
    "rolloutStateReason": "string",
    "runningCount": number,
    "serviceConnectConfiguration": {
      "accessLogConfiguration": {
        "format": "string",
        "includeQueryParameters": "string"
      },
      "enabled": boolean,
      "logConfiguration": {
        "logDriver": "string",
        "options": {
          "string" : "string"
        },
        "secretOptions": [
          {
            "name": "string",
            "valueFrom": "string"
          }
        ]
      },
      "namespace": "string",
      "services": [
        {
          "clientAliases": [
            {
              "dnsName": "string",
              "port": number,
              "testTrafficRules": {
                "header": {
                  "name": "string",
                  "value": {
                    "exact": "string"
                  }
                }
              }
            }
          ]
        }
      ],
      "discoveryName": "string",
      "ingressPortOverride": number,

```



```

    "portName": "string",
    "timeout": {
      "idleTimeoutSeconds": number,
      "perRequestTimeoutSeconds": number
    },
    "tls": {
      "issuerCertificateAuthority": {
        "awsPcaAuthorityArn": "string"
      },
      "kmsKey": "string",
      "roleArn": "string"
    }
  }
],
"serviceConnectResources": [
  {
    "discoveryArn": "string",
    "discoveryName": "string"
  }
],
"status": "string",
"taskDefinition": "string",
"updatedAt": number,
"volumeConfigurations": [
  {
    "managedEBSVolume": {
      "encrypted": boolean,
      "filesystemType": "string",
      "iops": number,
      "kmsKeyId": "string",
      "roleArn": "string",
      "sizeInGiB": number,
      "snapshotId": "string",
      "tagSpecifications": [
        {
          "propagateTags": "string",
          "resourceType": "string",
          "tags": [
            {
              "key": "string",
              "value": "string"
            }
          ]
        }
      ]
    }
  }
]

```

```

        }
      ],
      "throughput": number,
      "volumeInitializationRate": number,
      "volumeType": "string"
    },
    "name": "string"
  }
],
"vpcLatticeConfigurations": [
  {
    "portName": "string",
    "roleArn": "string",
    "targetGroupArn": "string"
  }
]
},
"desiredCount": number,
"enableECSManagedTags": boolean,
"enableExecuteCommand": boolean,
"events": [
  {
    "createdAt": number,
    "id": "string",
    "message": "string"
  }
],
"healthCheckGracePeriodSeconds": number,
"launchType": "string",
"loadBalancers": [
  {
    "advancedConfiguration": {
      "alternateTargetGroupArn": "string",
      "productionListenerRule": "string",
      "roleArn": "string",
      "testListenerRule": "string"
    },
    "containerName": "string",
    "containerPort": number,
    "loadBalancerName": "string",
    "targetGroupArn": "string"
  }
],

```

```

    "networkConfiguration": {
      "awsvpcConfiguration": {
        "assignPublicIp": "string",
        "securityGroups": [ "string" ],
        "subnets": [ "string" ]
      }
    },
    "pendingCount": number,
    "placementConstraints": [
      {
        "expression": "string",
        "type": "string"
      }
    ],
    "placementStrategy": [
      {
        "field": "string",
        "type": "string"
      }
    ],
    "platformFamily": "string",
    "platformVersion": "string",
    "propagateTags": "string",
    "roleArn": "string",
    "runningCount": number,
    "schedulingStrategy": "string",
    "serviceArn": "string",
    "serviceName": "string",
    "serviceRegistries": [
      {
        "containerName": "string",
        "containerPort": number,
        "port": number,
        "registryArn": "string"
      }
    ],
    "status": "string",
    "tags": [
      {
        "key": "string",
        "value": "string"
      }
    ],
    "taskDefinition": "string",

```

```

"taskSets": [
  {
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "clusterArn": "string",
    "computedDesiredCount": number,
    "createdAt": number,
    "externalId": "string",
    "fargateEphemeralStorage": {
      "kmsKeyId": "string"
    },
    "id": "string",
    "launchType": "string",
    "loadBalancers": [
      {
        "advancedConfiguration": {
          "alternateTargetGroupArn": "string",
          "productionListenerRule": "string",
          "roleArn": "string",
          "testListenerRule": "string"
        },
        "containerName": "string",
        "containerPort": number,
        "loadBalancerName": "string",
        "targetGroupArn": "string"
      }
    ],
    "networkConfiguration": {
      "awsvpcConfiguration": {
        "assignPublicIp": "string",
        "securityGroups": [ "string " ],
        "subnets": [ "string " ]
      }
    },
    "pendingCount": number,
    "platformFamily": "string",
    "platformVersion": "string",
    "runningCount": number,
    "scale": {

```

```

        "unit": "string",
        "value": number
    },
    "serviceArn": "string",
    "serviceRegistries": [
        {
            "containerName": "string",
            "containerPort": number,
            "port": number,
            "registryArn": "string"
        }
    ],
    "stabilityStatus": "string",
    "stabilityStatusAt": number,
    "startedBy": "string",
    "status": "string",
    "tags": [
        {
            "key": "string",
            "value": "string"
        }
    ],
    "taskDefinition": "string",
    "taskSetArn": "string",
    "updatedAt": number
}
]
}
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

service

The full description of your service following the update call.

Type: [Service](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

NamespaceNotFoundException

The specified namespace wasn't found.

HTTP Status Code: 400

PlatformTaskDefinitionIncompatibilityException

The specified platform version doesn't satisfy the required capabilities of the task definition.

HTTP Status Code: 400

PlatformUnknownException

The specified platform version doesn't exist.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotActiveException

The specified service isn't active. You can't update a service that's inactive. If you have previously deleted a service, you can re-create it with [CreateService](#).

HTTP Status Code: 400

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS,

these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example

This example request updates the `hello_world` service to a desired count of 3.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 45
X-Amz-Target: AmazonEC2ContainerServiceV20141113.UpdateService
X-Amz-Date: 20150429T194543Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "service": "hello_world",
  "desiredCount": 3
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 29 Apr 2015 19:45:43 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 13376
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "service": {
    "clusterArn": "arn:aws:ecs:us-east-1:012345678910:cluster/default",
    "deploymentConfiguration": {
      "maximumPercent": 200,
      "minimumHealthyPercent": 100
    },
    "deployments": [
      {
        "createdAt": 1430333711.033,
```



```
    "desiredCount": 3,
    "id": "ecs-svc/9223370606521064774",
    "pendingCount": 0,
    "runningCount": 0,
    "status": "PRIMARY",
    "taskDefinition": "arn:aws:ecs:us-east-1:012345678910:task-definition/
hello_world:10",
    "updatedAt": 1430336267.173
  }
],
"desiredCount": 3,
"events": [],
"loadBalancers": [],
"pendingCount": 0,
"runningCount": 0,
"serviceArn": "arn:aws:ecs:us-east-1:012345678910:service/default/hello_world",
"serviceName": "hello_world",
"status": "ACTIVE",
"taskDefinition": "arn:aws:ecs:us-east-1:012345678910:task-definition/
hello_world:10"
}
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UpdateServicePrimaryTaskSet

Modifies which task set in a service is the primary task set. Any parameters that are updated on the primary task set in a service will transition to the service. This is used when a service uses the EXTERNAL deployment controller type. For more information, see [Amazon ECS Deployment Types](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{  
  "cluster": "string",  
  "primaryTaskSet": "string",  
  "service": "string"  
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[cluster](#)

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the service that the task set exists in.

Type: String

Required: Yes

[primaryTaskSet](#)

The short name or full Amazon Resource Name (ARN) of the task set to set as the primary task set in the deployment.

Type: String

Required: Yes

[service](#)

The short name or full Amazon Resource Name (ARN) of the service that the task set exists in.

Type: String

Required: Yes

Response Syntax

```
{
  "taskSet": {
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "clusterArn": "string",
    "computedDesiredCount": number,
    "createdAt": number,
    "externalId": "string",
    "fargateEphemeralStorage": {
      "kmsKeyId": "string"
    },
    "id": "string",
    "launchType": "string",
    "loadBalancers": [
      {
        "advancedConfiguration": {
          "alternateTargetGroupArn": "string",
          "productionListenerRule": "string",
          "roleArn": "string",
          "testListenerRule": "string"
        },
        "containerName": "string",
        "containerPort": number,
        "loadBalancerName": "string",
        "targetGroupArn": "string"
      }
    ],
    "networkConfiguration": {
      "awsVpcConfiguration": {
        "assignPublicIp": "string",
        "securityGroups": [ "string " ],
        "subnets": [ "string " ]
      }
    }
  }
}
```

```

    }
  },
  "pendingCount": number,
  "platformFamily": "string",
  "platformVersion": "string",
  "runningCount": number,
  "scale": {
    "unit": "string",
    "value": number
  },
  "serviceArn": "string",
  "serviceRegistries": [
    {
      "containerName": "string",
      "containerPort": number,
      "port": number,
      "registryArn": "string"
    }
  ],
  "stabilityStatus": "string",
  "stabilityStatusAt": number,
  "startedBy": "string",
  "status": "string",
  "tags": [
    {
      "key": "string",
      "value": "string"
    }
  ],
  "taskDefinition": "string",
  "taskSetArn": "string",
  "updatedAt": number
}
}

```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

taskSet

The details about the task set.

Type: [TaskSet](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotActiveException

The specified service isn't active. You can't update a service that's inactive. If you have previously deleted a service, you can re-create it with [CreateService](#).

HTTP Status Code: 400

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

TaskSetNotFoundException

The specified task set wasn't found. You can view your available task sets with [DescribeTaskSets](#). Task sets are specific to each cluster, service and Region.

HTTP Status Code: 400

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)

- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UpdateTaskProtection

Updates the protection status of a task. You can set `protectionEnabled` to `true` to protect your task from termination during scale-in events from [Service Autoscaling](#) or [deployments](#).

Task-protection, by default, expires after 2 hours at which point Amazon ECS clears the `protectionEnabled` property making the task eligible for termination by a subsequent scale-in event.

You can specify a custom expiration period for task protection from 1 minute to up to 2,880 minutes (48 hours). To specify the custom expiration period, set the `expiresInMinutes` property. The `expiresInMinutes` property is always reset when you invoke this operation for a task that already has `protectionEnabled` set to `true`. You can keep extending the protection expiration period of a task by invoking this operation repeatedly.

To learn more about Amazon ECS task protection, see [Task scale-in protection](#) in the Amazon Elastic Container Service Developer Guide .

Note

This operation is only supported for tasks belonging to an Amazon ECS service. Invoking this operation for a standalone task will result in an `TASK_NOT_VALID` failure. For more information, see [API failure reasons](#).

Important

If you prefer to set task protection from within the container, we recommend using the [Task scale-in protection endpoint](#).

Request Syntax

```
{  
  "cluster": "string",  
  "expiresInMinutes": number,  
  "protectionEnabled": boolean,  
  "tasks": [ "string" ]  
}
```

```
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

[cluster](#)

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the service that the task sets exist in.

Type: String

Required: Yes

[expiresInMinutes](#)

If you set `protectionEnabled` to `true`, you can specify the duration for task protection in minutes. You can specify a value from 1 minute to up to 2,880 minutes (48 hours). During this time, your task will not be terminated by scale-in events from Service Auto Scaling or deployments. After this time period lapses, `protectionEnabled` will be reset to `false`.

If you don't specify the time, then the task is automatically protected for 120 minutes (2 hours).

Type: Integer

Required: No

[protectionEnabled](#)

Specify `true` to mark a task for protection and `false` to unset protection, making it eligible for termination.

Type: Boolean

Required: Yes

[tasks](#)

A list of up to 10 task IDs or full ARN entries.

Type: Array of strings

Required: Yes

Response Syntax

```
{
  "failures": [
    {
      "arn": "string",
      "detail": "string",
      "reason": "string"
    }
  ],
  "protectedTasks": [
    {
      "expirationDate": number,
      "protectionEnabled": boolean,
      "taskArn": "string"
    }
  ]
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

failures

Any failures associated with the call.

Type: Array of [Failure](#) objects

protectedTasks

A list of tasks with the following information.

- **taskArn**: The task ARN.
- **protectionEnabled**: The protection status of the task. If scale-in protection is turned on for a task, the value is true. Otherwise, it is false.
- **expirationDate**: The epoch time when protection for the task will expire.

Type: Array of [ProtectedTask](#) objects

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ResourceNotFoundException

The specified resource wasn't found.

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

Examples

In the following example or examples, the Authorization header contents (AUTHPARAMS) must be replaced with an AWS Signature Version 4 signature. For more information, see [Signature Version 4 Signing Process](#) in the *AWS General Reference*.

You only need to learn how to sign HTTP requests if you intend to create them manually. When you use the [AWS Command Line Interface](#) or one of the [AWS SDKs](#) to make requests to AWS, these tools automatically sign the requests for you, with the access key that you specify when you configure the tools. When you use these tools, you don't have to sign requests yourself.

Example 1

This example request enables scale-in protection for a task for 60 minutes.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 132
X-Amz-Target: AmazonEC2ContainerServiceV20141113.UpdateTaskProtection
X-Amz-Date: 20221102T184632Z Content-Type:
application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
```

```
"cluster": "test-task-protection",
"tasks": [
    "b8b1cf532d0e46ba8d44a40d1de16772"
],
"protectionEnabled": true,
"expiresInMinutes": 60
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Wed, 02 Nov 2022 17:56:32 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 180
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "protectedTasks": [
    {
      "taskArn": "arn:aws:ecs:us-west-2:012345678910:task/default/
b8b1cf532d0e46ba8d44a40d1de16772",
      "protectionEnabled": true,
      "expirationDate": 1667415392.553
    }
  ],
  "failures": []
}
```

Example 2

This example request enables scale-in protection for a task without specifying the `expiresInMinutes` parameter. By default, the task will be protected from scale-in events for 120 minutes.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
Accept-Encoding: identity
Content-Length: 132
```

```
X-Amz-Target: AmazonEC2ContainerServiceV20141113.UpdateTaskProtection
X-Amz-Date: 20221103T212558Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "cluster": "test-task-protection",
  "tasks": [
    "b8b1cf532d0e46ba8d44a40d1de16772"
  ],
  "protectionEnabled": true
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Thu, 03 Nov 2022 21:25:57 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 180
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "protectedTasks": [
    {
      "taskArn": "arn:aws:ecs:us-west-2:012345678910:task/default/
b8b1cf532d0e46ba8d44a40d1de16772",
      "protectionEnabled": true,
      "expirationDate": 1667517958.355
    }
  ],
  "failures": []
}
```

Example 3

This example request turns off scale-in protection for a task.

Sample Request

```
POST / HTTP/1.1
Host: ecs.us-west-2.amazonaws.com
```

```
Accept-Encoding: identity
Content-Length:109
X-Amz-Target: AmazonEC2ContainerServiceV20141113.UpdateTaskProtection
X-Amz-Date: 20221103T213134Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
```

```
{
  "cluster": "test-task-protection",
  "tasks": [
    "b8b1cf532d0e46ba8d44a40d1de16772"
  ],
  "protectionEnabled":false
}
```

Sample Response

```
HTTP/1.1 200 OK
Server: Server
Date: Thu, 03 Nov 2022 21:31:34 GMT
Content-Type: application/x-amz-json-1.1
Content-Length: 180
Connection: keep-alive
x-amzn-RequestId: 123a4b56-7c89-01d2-3ef4-example5678f

{
  "protectedTasks": [
    {
      "taskArn": "arn:aws:ecs:us-west-2:012345678910:task/default/
b8b1cf532d0e46ba8d44a40d1de16772",
      "protectionEnabled": false
    }
  ],
  "failures": []
}
```

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)

- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

UpdateTaskSet

Modifies a task set. This is used when a service uses the EXTERNAL deployment controller type. For more information, see [Amazon ECS Deployment Types](#) in the *Amazon Elastic Container Service Developer Guide*.

Request Syntax

```
{
  "cluster": "string",
  "scale": {
    "unit": "string",
    "value": number
  },
  "service": "string",
  "taskSet": "string"
}
```

Request Parameters

For information about the parameters that are common to all actions, see [Common Parameters](#).

The request accepts the following data in JSON format.

cluster

The short name or full Amazon Resource Name (ARN) of the cluster that hosts the service that the task set is found in.

Type: String

Required: Yes

scale

A floating-point percentage of the desired number of tasks to place and keep running in the task set.

Type: [Scale](#) object

Required: Yes

service

The short name or full Amazon Resource Name (ARN) of the service that the task set is found in.

Type: String

Required: Yes

taskSet

The short name or full Amazon Resource Name (ARN) of the task set to update.

Type: String

Required: Yes

Response Syntax

```
{
  "taskSet": {
    "capacityProviderStrategy": [
      {
        "base": number,
        "capacityProvider": "string",
        "weight": number
      }
    ],
    "clusterArn": "string",
    "computedDesiredCount": number,
    "createdAt": number,
    "externalId": "string",
    "fargateEphemeralStorage": {
      "kmsKeyId": "string"
    },
    "id": "string",
    "launchType": "string",
    "loadBalancers": [
      {
        "advancedConfiguration": {
          "alternateTargetGroupArn": "string",
          "productionListenerRule": "string",
          "roleArn": "string",
          "testListenerRule": "string"
        }
      }
    ]
  }
}
```

```

        "containerName": "string",
        "containerPort": number,
        "loadBalancerName": "string",
        "targetGroupArn": "string"
    }
],
"networkConfiguration": {
    "awsVpcConfiguration": {
        "assignPublicIp": "string",
        "securityGroups": [ "string" ],
        "subnets": [ "string" ]
    }
},
"pendingCount": number,
"platformFamily": "string",
"platformVersion": "string",
"runningCount": number,
"scale": {
    "unit": "string",
    "value": number
},
"serviceArn": "string",
"serviceRegistries": [
    {
        "containerName": "string",
        "containerPort": number,
        "port": number,
        "registryArn": "string"
    }
],
"stabilityStatus": "string",
"stabilityStatusAt": number,
"startedBy": "string",
"status": "string",
"tags": [
    {
        "key": "string",
        "value": "string"
    }
],
"taskDefinition": "string",
"taskSetArn": "string",
"updatedAt": number
}

```

```
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The following data is returned in JSON format by the service.

taskSet

Details about the task set.

Type: [TaskSet](#) object

Errors

For information about the errors that are common to all actions, see [Common Errors](#).

AccessDeniedException

You don't have authorization to perform the requested action.

HTTP Status Code: 400

ClientException

These errors are usually caused by a client action. This client action might be using an action or resource on behalf of a user that doesn't have permissions to use the action or resource. Or, it might be specifying an identifier that isn't valid.

message

Message that describes the cause of the exception.

HTTP Status Code: 400

ClusterNotFoundException

The specified cluster wasn't found. You can view your available clusters with [ListClusters](#). Amazon ECS clusters are Region specific.

HTTP Status Code: 400

InvalidParameterException

The specified parameter isn't valid. Review the available parameters for the API request.

For more information about service event errors, see [Amazon ECS service event messages](#).

HTTP Status Code: 400

ServerException

These errors are usually caused by a server issue.

message

Message that describes the cause of the exception.

HTTP Status Code: 500

ServiceNotActiveException

The specified service isn't active. You can't update a service that's inactive. If you have previously deleted a service, you can re-create it with [CreateService](#).

HTTP Status Code: 400

ServiceNotFoundException

The specified service wasn't found. You can view your available services with [ListServices](#). Amazon ECS services are cluster specific and Region specific.

HTTP Status Code: 400

TaskSetNotFoundException

The specified task set wasn't found. You can view your available task sets with [DescribeTaskSets](#). Task sets are specific to each cluster, service and Region.

HTTP Status Code: 400

UnsupportedFeatureException

The specified task isn't supported in this Region.

HTTP Status Code: 400

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS Command Line Interface V2](#)
- [AWS SDK for .NET](#)
- [AWS SDK for C++](#)
- [AWS SDK for Go v2](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for JavaScript V3](#)
- [AWS SDK for Kotlin](#)
- [AWS SDK for PHP V3](#)
- [AWS SDK for Python](#)
- [AWS SDK for Ruby V3](#)

Data Types

The Amazon EC2 Container Service API contains several data types that various actions use. This section describes each data type in detail.

Note

The order of each element in a data type structure is not guaranteed. Applications should not assume a particular order.

The following data types are supported:

- [AcceleratorCountRequest](#)
- [AcceleratorTotalMemoryMiBRequest](#)
- [AdvancedConfiguration](#)
- [Attachment](#)
- [AttachmentStateChange](#)
- [Attribute](#)
- [AutoScalingGroupProvider](#)
- [AutoScalingGroupProviderUpdate](#)
- [AwsVpcConfiguration](#)
- [BaselineEbsBandwidthMbpsRequest](#)
- [CanaryConfiguration](#)
- [CapacityProvider](#)
- [CapacityProviderStrategyItem](#)
- [Cluster](#)
- [ClusterConfiguration](#)
- [ClusterServiceConnectDefaults](#)
- [ClusterServiceConnectDefaultsRequest](#)
- [ClusterSetting](#)
- [Container](#)
- [ContainerDefinition](#)

- [ContainerDependency](#)
- [ContainerImage](#)
- [ContainerInstance](#)
- [ContainerInstanceHealthStatus](#)
- [ContainerOverride](#)
- [ContainerRestartPolicy](#)
- [ContainerStateChange](#)
- [CreatedAt](#)
- [CreateManagedInstancesProviderConfiguration](#)
- [Deployment](#)
- [DeploymentAlarms](#)
- [DeploymentCircuitBreaker](#)
- [DeploymentConfiguration](#)
- [DeploymentController](#)
- [DeploymentEphemeralStorage](#)
- [DeploymentLifecycleHook](#)
- [Device](#)
- [DockerVolumeConfiguration](#)
- [EBSTagSpecification](#)
- [EFSAuthorizationConfig](#)
- [EFSVolumeConfiguration](#)
- [EnvironmentFile](#)
- [EphemeralStorage](#)
- [ExecuteCommandConfiguration](#)
- [ExecuteCommandLogConfiguration](#)
- [Failure](#)
- [FirelensConfiguration](#)
- [FSxWindowsFileServerAuthorizationConfig](#)
- [FSxWindowsFileServerVolumeConfiguration](#)
- [HealthCheck](#)

- [HostEntry](#)
- [HostVolumeProperties](#)
- [InferenceAccelerator](#)
- [InferenceAcceleratorOverride](#)
- [InstanceHealthCheckResult](#)
- [InstanceLaunchTemplate](#)
- [InstanceLaunchTemplateUpdate](#)
- [InstanceRequirementsRequest](#)
- [KernelCapabilities](#)
- [KeyValuePair](#)
- [LinearConfiguration](#)
- [LinuxParameters](#)
- [LoadBalancer](#)
- [LogConfiguration](#)
- [ManagedAgent](#)
- [ManagedAgentStateChange](#)
- [ManagedInstancesNetworkConfiguration](#)
- [ManagedInstancesProvider](#)
- [ManagedInstancesStorageConfiguration](#)
- [ManagedScaling](#)
- [ManagedStorageConfiguration](#)
- [MemoryGiBPerVCpuRequest](#)
- [MemoryMiBRequest](#)
- [MountPoint](#)
- [NetworkBandwidthGbpsRequest](#)
- [NetworkBinding](#)
- [NetworkConfiguration](#)
- [NetworkInterface](#)
- [NetworkInterfaceCountRequest](#)
- [PlacementConstraint](#)

- [PlacementStrategy](#)
- [PlatformDevice](#)
- [PortMapping](#)
- [ProtectedTask](#)
- [ProxyConfiguration](#)
- [RepositoryCredentials](#)
- [ResolvedConfiguration](#)
- [Resource](#)
- [ResourceRequirement](#)
- [Rollback](#)
- [RuntimePlatform](#)
- [Scale](#)
- [Secret](#)
- [Service](#)
- [ServiceConnectAccessLogConfiguration](#)
- [ServiceConnectClientAlias](#)
- [ServiceConnectConfiguration](#)
- [ServiceConnectService](#)
- [ServiceConnectServiceResource](#)
- [ServiceConnectTestTrafficHeaderMatchRules](#)
- [ServiceConnectTestTrafficHeaderRules](#)
- [ServiceConnectTestTrafficRules](#)
- [ServiceConnectTlsCertificateAuthority](#)
- [ServiceConnectTlsConfiguration](#)
- [ServiceDeployment](#)
- [ServiceDeploymentAlarms](#)
- [ServiceDeploymentBrief](#)
- [ServiceDeploymentCircuitBreaker](#)
- [ServiceEvent](#)
- [ServiceManagedEBSVolumeConfiguration](#)

- [ServiceRegistry](#)
- [ServiceRevision](#)
- [ServiceRevisionLoadBalancer](#)
- [ServiceRevisionSummary](#)
- [ServiceVolumeConfiguration](#)
- [Session](#)
- [Setting](#)
- [SystemControl](#)
- [Tag](#)
- [Task](#)
- [TaskDefinition](#)
- [TaskDefinitionPlacementConstraint](#)
- [TaskEphemeralStorage](#)
- [TaskManagedEBSVolumeConfiguration](#)
- [TaskManagedEBSVolumeTerminationPolicy](#)
- [TaskOverride](#)
- [TaskSet](#)
- [TaskVolumeConfiguration](#)
- [TimeoutConfiguration](#)
- [Tmpfs](#)
- [TotalLocalStorageGBRequest](#)
- [Ulimit](#)
- [UpdateManagedInstancesProviderConfiguration](#)
- [VCpuCountRangeRequest](#)
- [VersionInfo](#)
- [Volume](#)
- [VolumeFrom](#)
- [VpcLatticeConfiguration](#)

AcceleratorCountRequest

The minimum and maximum number of accelerators (such as GPUs) for instance type selection. This is used for workloads that require specific numbers of accelerators.

Contents

max

The maximum number of accelerators. Instance types with more accelerators are excluded from selection.

Type: Integer

Required: No

min

The minimum number of accelerators. Instance types with fewer accelerators are excluded from selection.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AcceleratorTotalMemoryMiBRequest

The minimum and maximum total accelerator memory in mebibytes (MiB) for instance type selection. This is important for GPU workloads that require specific amounts of video memory.

Contents

max

The maximum total accelerator memory in MiB. Instance types with more accelerator memory are excluded from selection.

Type: Integer

Required: No

min

The minimum total accelerator memory in MiB. Instance types with less accelerator memory are excluded from selection.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AdvancedConfiguration

The advanced settings for a load balancer used in blue/green deployments. Specify the alternate target group, listener rules, and IAM role required for traffic shifting during blue/green deployments. For more information, see [Required resources for Amazon ECS blue/green deployments](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

alternateTargetGroupArn

The Amazon Resource Name (ARN) of the alternate target group for Amazon ECS blue/green deployments.

Type: String

Required: No

productionListenerRule

The Amazon Resource Name (ARN) that identifies the production listener rule (in the case of an Application Load Balancer) or listener (in the case for an Network Load Balancer) for routing production traffic.

Type: String

Required: No

roleArn

The Amazon Resource Name (ARN) of the IAM role that grants Amazon ECS permission to call the Elastic Load Balancing APIs for you.

Type: String

Required: No

testListenerRule

The Amazon Resource Name (ARN) that identifies) that identifies the test listener rule (in the case of an Application Load Balancer) or listener (in the case for an Network Load Balancer) for routing test traffic.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Attachment

An object representing a container instance or task attachment.

Contents

details

Details of the attachment.

For elastic network interfaces, this includes the network interface ID, the MAC address, the subnet ID, and the private IPv4 address.

For Service Connect services, this includes `portName`, `clientAliases`, `discoveryName`, and `ingressPortOverride`.

For Elastic Block Storage, this includes `roleArn`, `deleteOnTermination`, `volumeName`, `volumeId`, and `statusReason` (only when the attachment fails to create or attach).

Type: Array of [KeyValuePair](#) objects

Required: No

id

The unique identifier for the attachment.

Type: String

Required: No

status

The status of the attachment. Valid values are PRECREATED, CREATED, ATTACHING, ATTACHED, DETACHING, DETACHED, DELETED, and FAILED.

Type: String

Required: No

type

The type of the attachment, such as `ElasticNetworkInterface`, `Service Connect`, and `AmazonElasticBlockStorage`.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AttachmentStateChange

An object representing a change in state for a task attachment.

Contents

attachmentArn

The Amazon Resource Name (ARN) of the attachment.

Type: String

Required: Yes

status

The status of the attachment.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Attribute

An attribute is a name-value pair that's associated with an Amazon ECS object. Use attributes to extend the Amazon ECS data model by adding custom metadata to your resources. For more information, see [Attributes](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

name

The name of the attribute. The name must contain between 1 and 128 characters. The name may contain letters (uppercase and lowercase), numbers, hyphens (-), underscores (_), forward slashes (/), back slashes (\), or periods (.).

Type: String

Required: Yes

targetId

The ID of the target. You can specify the short form ID for a resource or the full Amazon Resource Name (ARN).

Type: String

Required: No

targetType

The type of the target to attach the attribute with. This parameter is required if you use the short form ID for a resource instead of the full ARN.

Type: String

Valid Values: container-instance

Required: No

value

The value of the attribute. The value must contain between 1 and 128 characters. It can contain letters (uppercase and lowercase), numbers, hyphens (-), underscores (_), periods (.), at signs (@), forward slashes (/), back slashes (\), colons (:), or spaces. The value can't start or end with a space.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AutoScalingGroupProvider

The details of the Auto Scaling group for the capacity provider.

Contents

autoScalingGroupArn

The Amazon Resource Name (ARN) that identifies the Auto Scaling group, or the Auto Scaling group name.

Type: String

Required: Yes

managedDraining

The managed draining option for the Auto Scaling group capacity provider. When you enable this, Amazon ECS manages and gracefully drains the EC2 container instances that are in the Auto Scaling group capacity provider.

Type: String

Valid Values: ENABLED | DISABLED

Required: No

managedScaling

The managed scaling settings for the Auto Scaling group capacity provider.

Type: [ManagedScaling](#) object

Required: No

managedTerminationProtection

The managed termination protection setting to use for the Auto Scaling group capacity provider. This determines whether the Auto Scaling group has managed termination protection. The default is off.

⚠ Important

When using managed termination protection, managed scaling must also be used otherwise managed termination protection doesn't work.

When managed termination protection is on, Amazon ECS prevents the Amazon EC2 instances in an Auto Scaling group that contain tasks from being terminated during a scale-in action. The Auto Scaling group and each instance in the Auto Scaling group must have instance protection from scale-in actions on as well. For more information, see [Instance Protection](#) in the *AWS Auto Scaling User Guide*.

When managed termination protection is off, your Amazon EC2 instances aren't protected from termination when the Auto Scaling group scales in.

Type: String

Valid Values: ENABLED | DISABLED

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AutoScalingGroupProviderUpdate

The details of the Auto Scaling group capacity provider to update.

Contents

managedDraining

The managed draining option for the Auto Scaling group capacity provider. When you enable this, Amazon ECS manages and gracefully drains the EC2 container instances that are in the Auto Scaling group capacity provider.

Type: String

Valid Values: ENABLED | DISABLED

Required: No

managedScaling

The managed scaling settings for the Auto Scaling group capacity provider.

Type: [ManagedScaling](#) object

Required: No

managedTerminationProtection

The managed termination protection setting to use for the Auto Scaling group capacity provider. This determines whether the Auto Scaling group has managed termination protection.

Important

When using managed termination protection, managed scaling must also be used otherwise managed termination protection doesn't work.

When managed termination protection is on, Amazon ECS prevents the Amazon EC2 instances in an Auto Scaling group that contain tasks from being terminated during a scale-in action. The Auto Scaling group and each instance in the Auto Scaling group must have instance protection from scale-in actions on. For more information, see [Instance Protection](#) in the *AWS Auto Scaling User Guide*.

When managed termination protection is off, your Amazon EC2 instances aren't protected from termination when the Auto Scaling group scales in.

Type: String

Valid Values: ENABLED | DISABLED

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

AwsVpcConfiguration

An object representing the networking details for a task or service. For example

```
awsVpcConfiguration={subnets=["subnet-12344321"],securityGroups=["sg-12344321"]}
```

Contents

subnets

The IDs of the subnets associated with the task or service. There's a limit of 16 subnets that can be specified.

Note

All specified subnets must be from the same VPC.

Type: Array of strings

Required: Yes

assignPublicIp

Whether the task's elastic network interface receives a public IP address.

Consider the following when you set this value:

- When you use `create-service` or `update-service`, the default is `DISABLED`.
- When the service deploymentController is ECS, the value must be `DISABLED`.

Type: String

Valid Values: `ENABLED` | `DISABLED`

Required: No

securityGroups

The IDs of the security groups associated with the task or service. If you don't specify a security group, the default security group for the VPC is used. There's a limit of 5 security groups that can be specified.

Note

All specified security groups must be from the same VPC.

Type: Array of strings

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

BaselineEbsBandwidthMbpsRequest

The minimum and maximum baseline Amazon EBS bandwidth in megabits per second (Mbps) for instance type selection. This is important for workloads with high storage I/O requirements.

Contents

max

The maximum baseline Amazon EBS bandwidth in Mbps. Instance types with higher Amazon EBS bandwidth are excluded from selection.

Type: Integer

Required: No

min

The minimum baseline Amazon EBS bandwidth in Mbps. Instance types with lower Amazon EBS bandwidth are excluded from selection.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

CanaryConfiguration

Configuration for a canary deployment strategy that shifts a fixed percentage of traffic to the new service revision, waits for a specified bake time, then shifts the remaining traffic.

This is only valid when you run `CreateService` or `UpdateService` with `deploymentController` set to `ECS` and a `deploymentConfiguration` with a `strategy` set to `CANARY`.

Contents

canaryBakeTimeInMinutes

The amount of time in minutes to wait during the canary phase before shifting the remaining production traffic to the new service revision. Valid values are 0 to 1440 minutes (24 hours). The default value is 10.

Type: Integer

Required: No

canaryPercent

The percentage of production traffic to shift to the new service revision during the canary phase. Valid values are multiples of 0.1 from 0.1 to 100.0. The default value is 5.0.

Type: Double

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

CapacityProvider

The details for a capacity provider.

Contents

autoScalingGroupProvider

The Auto Scaling group settings for the capacity provider.

Type: [AutoScalingGroupProvider](#) object

Required: No

capacityProviderArn

The Amazon Resource Name (ARN) that identifies the capacity provider.

Type: String

Required: No

cluster

The cluster that this capacity provider is associated with. Managed instances capacity providers are cluster-scoped, meaning they can only be used within their associated cluster.

This is required for Managed instances.

Type: String

Required: No

managedInstancesProvider

The configuration for the Amazon ECS Managed Instances provider. This includes the infrastructure role, the launch template configuration, and tag propagation settings.

Type: [ManagedInstancesProvider](#) object

Required: No

name

The name of the capacity provider.

Type: String

Required: No

status

The current status of the capacity provider. Only capacity providers in an ACTIVE state can be used in a cluster. When a capacity provider is successfully deleted, it has an INACTIVE status.

Type: String

Valid Values: PROVISIONING | ACTIVE | DEPROVISIONING | INACTIVE

Required: No

tags

The metadata that you apply to the capacity provider to help you categorize and organize it. Each tag consists of a key and an optional value. You define both.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

type

The type of capacity provider. For Amazon ECS Managed Instances, this value is `MANAGED_INSTANCES`, indicating that Amazon ECS manages the underlying Amazon EC2 instances on your behalf.

Type: String

Valid Values: EC2_AUTOSCALING | MANAGED_INSTANCES | FARGATE | FARGATE_SPOT

Required: No

updateStatus

The update status of the capacity provider. The following are the possible states that is returned.

DELETE_IN_PROGRESS

The capacity provider is in the process of being deleted.

DELETE_COMPLETE

The capacity provider was successfully deleted and has an INACTIVE status.

DELETE_FAILED

The capacity provider can't be deleted. The update status reason provides further details about why the delete failed.

Type: String

Valid Values: CREATE_IN_PROGRESS | CREATE_COMPLETE | CREATE_FAILED
| DELETE_IN_PROGRESS | DELETE_COMPLETE | DELETE_FAILED |
UPDATE_IN_PROGRESS | UPDATE_COMPLETE | UPDATE_FAILED

Required: No

updateStatusReason

The update status reason. This provides further details about the update status for the capacity provider.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

CapacityProviderStrategyItem

The details of a capacity provider strategy. A capacity provider strategy can be set when using the [RunTask](#) or [CreateCluster](#) APIs or as the default capacity provider strategy for a cluster with the [CreateCluster](#) API.

Only capacity providers that are already associated with a cluster and have an ACTIVE or UPDATING status can be used in a capacity provider strategy. The [PutClusterCapacityProviders](#) API is used to associate a capacity provider with a cluster.

If specifying a capacity provider that uses an Auto Scaling group, the capacity provider must already be created. New Auto Scaling group capacity providers can be created with the [CreateClusterCapacityProvider](#) API operation.

To use a AWS Fargate capacity provider, specify either the FARGATE or FARGATE_SPOT capacity providers. The AWS Fargate capacity providers are available to all accounts and only need to be associated with a cluster to be used in a capacity provider strategy.

With FARGATE_SPOT, you can run interruption tolerant tasks at a rate that's discounted compared to the FARGATE price. FARGATE_SPOT runs tasks on spare compute capacity. When AWS needs the capacity back, your tasks are interrupted with a two-minute warning. FARGATE_SPOT supports Linux tasks with the X86_64 architecture on platform version 1.3.0 or later. FARGATE_SPOT supports Linux tasks with the ARM64 architecture on platform version 1.4.0 or later.

A capacity provider strategy can contain a maximum of 20 capacity providers.

Contents

capacityProvider

The short name of the capacity provider.

Type: String

Required: Yes

base

The *base* value designates how many tasks, at a minimum, to run on the specified capacity provider for each service. Only one capacity provider in a capacity provider strategy can have a *base* defined. If no value is specified, the default value of 0 is used.

Base value characteristics:

- Only one capacity provider in a strategy can have a base defined
- The default value is 0 if not specified
- The valid range is 0 to 100,000
- Base requirements are satisfied first before weight distribution

Type: Integer

Valid Range: Minimum value of 0. Maximum value of 100000.

Required: No

weight

The *weight* value designates the relative percentage of the total number of tasks launched that should use the specified capacity provider. The weight value is taken into consideration after the base value, if defined, is satisfied.

If no weight value is specified, the default value of 0 is used. When multiple capacity providers are specified within a capacity provider strategy, at least one of the capacity providers must have a weight value greater than zero and any capacity providers with a weight of 0 can't be used to place tasks. If you specify multiple capacity providers in a strategy that all have a weight of 0, any RunTask or CreateService actions using the capacity provider strategy will fail.

Weight value characteristics:

- Weight is considered after the base value is satisfied
- The default value is 0 if not specified
- The valid range is 0 to 1,000
- At least one capacity provider must have a weight greater than zero
- Capacity providers with weight of 0 cannot place tasks

Task distribution logic:

1. Base satisfaction: The minimum number of tasks specified by the base value are placed on that capacity provider
2. Weight distribution: After base requirements are met, additional tasks are distributed according to weight ratios

Examples:

Equal Distribution: Two capacity providers both with weight 1 will split tasks evenly after base requirements are met.

Weighted Distribution: If capacityProviderA has weight 1 and capacityProviderB has weight 4, then for every 1 task on A, 4 tasks will run on B.

Type: Integer

Valid Range: Minimum value of 0. Maximum value of 1000.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Cluster

A regional grouping of one or more container instances where you can run task requests. Each account receives a default cluster the first time you use the Amazon ECS service, but you may also create other clusters. Clusters may contain more than one instance type simultaneously.

Contents

activeServicesCount

The number of services that are running on the cluster in an ACTIVE state. You can view these services with [ListServices](#).

Type: Integer

Required: No

attachments

The resources attached to a cluster. When using a capacity provider with a cluster, the capacity provider and associated resources are returned as cluster attachments.

Type: Array of [Attachment](#) objects

Required: No

attachmentsStatus

The status of the capacity providers associated with the cluster. The following are the states that are returned.

UPDATE_IN_PROGRESS

The available capacity providers for the cluster are updating.

UPDATE_COMPLETE

The capacity providers have successfully updated.

UPDATE_FAILED

The capacity provider updates failed.

Type: String

Required: No

capacityProviders

The capacity providers associated with the cluster.

Type: Array of strings

Required: No

clusterArn

The Amazon Resource Name (ARN) that identifies the cluster. For more information about the ARN format, see [Amazon Resource Name \(ARN\)](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: No

clusterName

A user-generated string that you use to identify your cluster.

Type: String

Required: No

configuration

The execute command and managed storage configuration for the cluster.

Type: [ClusterConfiguration](#) object

Required: No

defaultCapacityProviderStrategy

The default capacity provider strategy for the cluster. When services or tasks are run in the cluster with no launch type or capacity provider strategy specified, the default capacity provider strategy is used.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: No

pendingTasksCount

The number of tasks in the cluster that are in the PENDING state.

Type: Integer

Required: No

registeredContainerInstancesCount

The number of container instances registered into the cluster. This includes container instances in both ACTIVE and DRAINING status.

Type: Integer

Required: No

runningTasksCount

The number of tasks in the cluster that are in the RUNNING state.

Type: Integer

Required: No

serviceConnectDefaults

Use this parameter to set a default Service Connect namespace. After you set a default Service Connect namespace, any new services with Service Connect turned on that are created in the cluster are added as client services in the namespace. This setting only applies to new services that set the `enabled` parameter to `true` in the `ServiceConnectConfiguration`. You can set the namespace of each service individually in the `ServiceConnectConfiguration` to override this default parameter.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [ClusterServiceConnectDefaults](#) object

Required: No

settings

The settings for the cluster. This parameter indicates whether CloudWatch Container Insights is on or off for a cluster.

Type: Array of [ClusterSetting](#) objects

Required: No

statistics

Additional information about your clusters that are separated by launch type. They include the following:

- `runningEC2TasksCount`
- `RunningFargateTasksCount`
- `pendingEC2TasksCount`
- `pendingFargateTasksCount`
- `activeEC2ServiceCount`
- `activeFargateServiceCount`
- `drainingEC2ServiceCount`
- `drainingFargateServiceCount`

Type: Array of [KeyValuePair](#) objects

Required: No

status

The status of the cluster. The following are the possible states that are returned.

ACTIVE

The cluster is ready to accept tasks and if applicable you can register container instances with the cluster.

PROVISIONING

The cluster has capacity providers that are associated with it and the resources needed for the capacity provider are being created.

DEPROVISIONING

The cluster has capacity providers that are associated with it and the resources needed for the capacity provider are being deleted.

FAILED

The cluster has capacity providers that are associated with it and the resources needed for the capacity provider have failed to create.

INACTIVE

The cluster has been deleted. Clusters with an INACTIVE status may remain discoverable in your account for a period of time. However, this behavior is subject to change in the future. We don't recommend that you rely on INACTIVE clusters persisting.

Type: String

Required: No

tags

The metadata that you apply to the cluster to help you categorize and organize them. Each tag consists of a key and an optional value. You define both.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ClusterConfiguration

The execute command and managed storage configuration for the cluster.

Contents

executeCommandConfiguration

The details of the execute command configuration.

Type: [ExecuteCommandConfiguration](#) object

Required: No

managedStorageConfiguration

The details of the managed storage configuration.

Type: [ManagedStorageConfiguration](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ClusterServiceConnectDefaults

Use this parameter to set a default Service Connect namespace. After you set a default Service Connect namespace, any new services with Service Connect turned on that are created in the cluster are added as client services in the namespace. This setting only applies to new services that set the `enabled` parameter to `true` in the `ServiceConnectConfiguration`. You can set the namespace of each service individually in the `ServiceConnectConfiguration` to override this default parameter.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

namespace

The namespace name or full Amazon Resource Name (ARN) of the AWS Cloud Map namespace. When you create a service and don't specify a Service Connect configuration, this namespace is used.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ClusterServiceConnectDefaultsRequest

Use this parameter to set a default Service Connect namespace. After you set a default Service Connect namespace, any new services with Service Connect turned on that are created in the cluster are added as client services in the namespace. This setting only applies to new services that set the `enabled` parameter to `true` in the `ServiceConnectConfiguration`. You can set the namespace of each service individually in the `ServiceConnectConfiguration` to override this default parameter.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

namespace

The namespace name or full Amazon Resource Name (ARN) of the AWS Cloud Map namespace that's used when you create a service and don't specify a Service Connect configuration. The namespace name can include up to 1024 characters. The name is case-sensitive. The name can't include greater than (>), less than (<), double quotation marks ("), or slash (/).

If you enter an existing namespace name or ARN, then that namespace will be used. Any namespace type is supported. The namespace must be in this account and this AWS Region.

If you enter a new name, a AWS Cloud Map namespace will be created. Amazon ECS creates a AWS Cloud Map namespace with the "API calls" method of instance discovery only. This instance discovery method is the "HTTP" namespace type in the AWS Command Line Interface. Other types of instance discovery aren't used by Service Connect.

If you update the cluster with an empty string "" for the namespace name, the cluster configuration for Service Connect is removed. Note that the namespace will remain in AWS Cloud Map and must be deleted separately.

For more information about AWS Cloud Map, see [Working with Services](#) in the *AWS Cloud Map Developer Guide*.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ClusterSetting

The settings to use when creating a cluster. This parameter is used to turn on CloudWatch Container Insights with enhanced observability or CloudWatch Container Insights for a cluster.

Container Insights with enhanced observability provides all the Container Insights metrics, plus additional task and container metrics. This version supports enhanced observability for Amazon ECS clusters using the Amazon EC2 and Fargate launch types. After you configure Container Insights with enhanced observability on Amazon ECS, Container Insights auto-collects detailed infrastructure telemetry from the cluster level down to the container level in your environment and displays these critical performance data in curated dashboards removing the heavy lifting in observability set-up.

For more information, see [Monitor Amazon ECS containers using Container Insights with enhanced observability](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

name

The name of the cluster setting. The value is `containerInsights`.

Type: String

Valid Values: `containerInsights`

Required: No

value

The value to set for the cluster setting. The supported values are `enhanced`, `enabled`, and `disabled`.

To use Container Insights with enhanced observability, set the `containerInsights` account setting to `enhanced`.

To use Container Insights, set the `containerInsights` account setting to `enabled`.

If a cluster value is specified, it will override the `containerInsights` value set with [PutAccountSetting](#) or [PutAccountSettingDefault](#).

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Container

A Docker container that's part of a task.

Contents

containerArn

The Amazon Resource Name (ARN) of the container.

Type: String

Required: No

cpu

The number of CPU units set for the container. The value is 0 if no value was specified in the container definition when the task definition was registered.

Type: String

Required: No

exitCode

The exit code returned from the container.

Type: Integer

Required: No

gpuIds

The IDs of each GPU assigned to the container.

Type: Array of strings

Required: No

healthStatus

The health status of the container. If health checks aren't configured for this container in its task definition, then it reports the health status as UNKNOWN.

Type: String

Valid Values: HEALTHY | UNHEALTHY | UNKNOWN

Required: No

image

The image used for the container.

Type: String

Required: No

imageDigest

The container image manifest digest.

Type: String

Required: No

lastStatus

The last known status of the container.

Type: String

Required: No

managedAgents

The details of any Amazon ECS managed agents associated with the container.

Type: Array of [ManagedAgent](#) objects

Required: No

memory

The hard limit (in MiB) of memory set for the container.

Type: String

Required: No

memoryReservation

The soft limit (in MiB) of memory set for the container.

Type: String

Required: No

name

The name of the container.

Type: String

Required: No

networkBindings

The network bindings associated with the container.

Type: Array of [NetworkBinding](#) objects

Required: No

networkInterfaces

The network interfaces associated with the container.

Type: Array of [NetworkInterface](#) objects

Required: No

reason

A short (1024 max characters) human-readable string to provide additional details about a running or stopped container.

Type: String

Required: No

runtimeId

The ID of the Docker container.

Type: String

Required: No

taskArn

The ARN of the task.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ContainerDefinition

Container definitions are used in task definitions to describe the different containers that are launched as part of a task.

Contents

command

The command that's passed to the container. This parameter maps to `Cmd` in the docker container create command and the `COMMAND` parameter to docker run. If there are multiple arguments, each argument is a separated string in the array.

Type: Array of strings

Required: No

cpu

The number of cpu units reserved for the container. This parameter maps to `CpuShares` in the docker container create command and the `--cpu-shares` option to docker run.

This field is optional for tasks using the Fargate launch type, and the only requirement is that the total amount of CPU reserved for all containers within a task be lower than the task-level `cpu` value.

Note

You can determine the number of CPU units that are available per EC2 instance type by multiplying the vCPUs listed for that instance type on the [Amazon EC2 Instances](#) detail page by 1,024.

Linux containers share unallocated CPU units with other containers on the container instance with the same ratio as their allocated amount. For example, if you run a single-container task on a single-core instance type with 512 CPU units specified for that container, and that's the only task running on the container instance, that container could use the full 1,024 CPU unit share at any given time. However, if you launched another copy of the same task on that container instance, each task is guaranteed a minimum of 512 CPU units when needed. Moreover, each container could float to higher CPU usage if the other container was not using it. If both tasks were 100% active all of the time, they would be limited to 512 CPU units.

On Linux container instances, the Docker daemon on the container instance uses the CPU value to calculate the relative CPU share ratios for running containers. The minimum valid CPU share value that the Linux kernel allows is 2, and the maximum valid CPU share value that the Linux kernel allows is 262144. However, the CPU parameter isn't required, and you can use CPU values below 2 or above 262144 in your container definitions. For CPU values below 2 (including null) or above 262144, the behavior varies based on your Amazon ECS container agent version:

- **Agent versions less than or equal to 1.1.0:** Null and zero CPU values are passed to Docker as 0, which Docker then converts to 1,024 CPU shares. CPU values of 1 are passed to Docker as 1, which the Linux kernel converts to two CPU shares.
- **Agent versions greater than or equal to 1.2.0:** Null, zero, and CPU values of 1 are passed to Docker as 2.
- **Agent versions greater than or equal to 1.84.0:** CPU values greater than 256 vCPU are passed to Docker as 256, which is equivalent to 262144 CPU shares.

On Windows container instances, the CPU limit is enforced as an absolute limit, or a quota. Windows containers only have access to the specified amount of CPU that's described in the task definition. A null or zero CPU value is passed to Docker as 0, which Windows interprets as 1% of one CPU.

Type: Integer

Required: No

credentialSpecs

A list of ARNs in SSM or Amazon S3 to a credential spec (CredSpec) file that configures the container for Active Directory authentication. We recommend that you use this parameter instead of the `dockerSecurityOptions`. The maximum number of ARNs is 1.

There are two formats for each ARN.

`credentialSpecdomainless:MyARN`

You use `credentialSpecdomainless:MyARN` to provide a CredSpec with an additional section for a secret in AWS Secrets Manager. You provide the login credentials to the domain in the secret.

Each task that runs on any container instance can join different domains.

You can use this format without joining the container instance to a domain.

credentialspec:MyARN

You use `credentialspec:MyARN` to provide a CredSpec for a single domain.

You must join the container instance to the domain before you start any tasks that use this task definition.

In both formats, replace MyARN with the ARN in SSM or Amazon S3.

If you provide a `credentialspecdomainless:MyARN`, the `credspec` must provide a ARN in AWS Secrets Manager for a secret containing the username, password, and the domain to connect to. For better security, the instance isn't joined to the domain for domainless authentication. Other applications on the instance can't use the domainless credentials. You can use this parameter to run tasks on the same instance, even if the tasks need to join different domains. For more information, see [Using gMSAs for Windows Containers](#) and [Using gMSAs for Linux Containers](#).

Type: Array of strings

Required: No

dependsOn

The dependencies defined for container startup and shutdown. A container can contain multiple dependencies on other containers in a task definition. When a dependency is defined for container startup, for container shutdown it is reversed.

For tasks using the EC2 launch type, the container instances require at least version 1.26.0 of the container agent to turn on container dependencies. However, we recommend using the latest container agent version. For information about checking your agent version and updating to the latest version, see [Updating the Amazon ECS Container Agent](#) in the *Amazon Elastic Container Service Developer Guide*. If you're using an Amazon ECS-optimized Linux AMI, your instance needs at least version 1.26.0-1 of the `ecs-init` package. If your container instances are launched from version 20190301 or later, then they contain the required versions of the container agent and `ecs-init`. For more information, see [Amazon ECS-optimized Linux AMI](#) in the *Amazon Elastic Container Service Developer Guide*.

For tasks using the Fargate launch type, the task or service requires the following platforms:

- Linux platform version 1.3.0 or later.
- Windows platform version 1.0.0 or later.

Type: Array of [ContainerDependency](#) objects

Required: No

disableNetworking

When this parameter is true, networking is off within the container. This parameter maps to `NetworkDisabled` in the docker container create command.

Note

This parameter is not supported for Windows containers.

Type: Boolean

Required: No

dnsSearchDomains

A list of DNS search domains that are presented to the container. This parameter maps to `DnsSearch` in the docker container create command and the `--dns-search` option to docker run.

Note

This parameter is not supported for Windows containers.

Type: Array of strings

Required: No

dnsServers

A list of DNS servers that are presented to the container. This parameter maps to `Dns` in the docker container create command and the `--dns` option to docker run.

Note

This parameter is not supported for Windows containers.

Type: Array of strings

Required: No

dockerLabels

A key/value map of labels to add to the container. This parameter maps to `Labels` in the docker container create command and the `--label` option to docker run. This parameter requires version 1.18 of the Docker Remote API or greater on your container instance. To check the Docker Remote API version on your container instance, log in to your container instance and run the following command: `sudo docker version --format '{{.Server.APIVersion}}'`

Type: String to string map

Required: No

dockerSecurityOptions

A list of strings to provide custom configuration for multiple security systems. This field isn't valid for containers in tasks using the Fargate launch type.

For Linux tasks on EC2, this parameter can be used to reference custom labels for SELinux and AppArmor multi-level security systems.

For any tasks on EC2, this parameter can be used to reference a credential spec file that configures a container for Active Directory authentication. For more information, see [Using gMSAs for Windows Containers](#) and [Using gMSAs for Linux Containers](#) in the *Amazon Elastic Container Service Developer Guide*.

This parameter maps to `SecurityOpt` in the docker container create command and the `--security-opt` option to docker run.

Note

The Amazon ECS container agent running on a container instance must register with the `ECS_SELINUX_CAPABLE=true` or `ECS_APPARMOR_CAPABLE=true` environment variables before containers placed on that instance can use these security options. For more information, see [Amazon ECS Container Agent Configuration](#) in the *Amazon Elastic Container Service Developer Guide*.

Valid values: "no-new-privileges" | "apparmor:PROFILE" | "label:value" | "credentialSpec:CredentialSpecFilePath"

Type: Array of strings

Required: No

entryPoint

Important

Early versions of the Amazon ECS container agent don't properly handle `entryPoint` parameters. If you have problems using `entryPoint`, update your container agent or enter your commands and arguments as command array items instead.

The entry point that's passed to the container. This parameter maps to `Entrypoint` in the docker container create command and the `--entrypoint` option to docker run.

Type: Array of strings

Required: No

environment

The environment variables to pass to a container. This parameter maps to `Env` in the docker container create command and the `--env` option to docker run.

Important

We don't recommend that you use plaintext environment variables for sensitive information, such as credential data.

Type: Array of [KeyValuePair](#) objects

Required: No

environmentFiles

A list of files containing the environment variables to pass to a container. This parameter maps to the `--env-file` option to docker run.

You can specify up to ten environment files. The file must have a `.env` file extension. Each line in an environment file contains an environment variable in `VARIABLE=VALUE` format. Lines beginning with `#` are treated as comments and are ignored.

If there are environment variables specified using the `environment` parameter in a container definition, they take precedence over the variables contained within an environment file. If multiple environment files are specified that contain the same variable, they're processed from the top down. We recommend that you use unique variable names. For more information, see [Specifying Environment Variables](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Array of [EnvironmentFile](#) objects

Required: No

essential

If the `essential` parameter of a container is marked as `true`, and that container fails or stops for any reason, all other containers that are part of the task are stopped. If the `essential` parameter of a container is marked as `false`, its failure doesn't affect the rest of the containers in a task. If this parameter is omitted, a container is assumed to be essential.

All tasks must have at least one essential container. If you have an application that's composed of multiple containers, group containers that are used for a common purpose into components, and separate the different components into multiple task definitions. For more information, see [Application Architecture](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Boolean

Required: No

extraHosts

A list of hostnames and IP address mappings to append to the `/etc/hosts` file on the container. This parameter maps to `ExtraHosts` in the `docker container create` command and the `--add-host` option to `docker run`.

Note

This parameter isn't supported for Windows containers or tasks that use the `awsvpc` network mode.

Type: Array of [HostEntry](#) objects

Required: No

firelensConfiguration

The FireLens configuration for the container. This is used to specify and configure a log router for container logs. For more information, see [Custom Log Routing](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [FirelensConfiguration](#) object

Required: No

healthCheck

The container health check command and associated configuration parameters for the container. This parameter maps to HealthCheck in the docker container create command and the HEALTHCHECK parameter of docker run.

Type: [HealthCheck](#) object

Required: No

hostname

The hostname to use for your container. This parameter maps to Hostname in the docker container create command and the `--hostname` option to docker run.

Note

The `hostname` parameter is not supported if you're using the `awsvpc` network mode.

Type: String

Required: No

image

The image used to start a container. This string is passed directly to the Docker daemon. By default, images in the Docker Hub registry are available. Other repositories are specified with either `repository-url/image:tag` or `repository-url/image@digest`. For images using tags (`repository-url/image:tag`), up to 255 characters total are allowed, including

letters (uppercase and lowercase), numbers, hyphens, underscores, colons, periods, forward slashes, and number signs (#). For images using digests (repository-url/image@digest), the 255 character limit applies only to the repository URL and image name (everything before the @ sign). The only supported hash function is sha256, and the hash value after sha256: must be exactly 64 characters (only letters A-F, a-f, and numbers 0-9 are allowed). This parameter maps to Image in the docker container create command and the IMAGE parameter of docker run.

- When a new task starts, the Amazon ECS container agent pulls the latest version of the specified image and tag for the container to use. However, subsequent updates to a repository image aren't propagated to already running tasks.
- Images in Amazon ECR repositories can be specified by either using the full registry/repository:tag or registry/repository@digest. For example, 012345678910.dkr.ecr.<region-name>.amazonaws.com/<repository-name>:latest or 012345678910.dkr.ecr.<region-name>.amazonaws.com/<repository-name>@sha256:94afd1f2e64d908bc90dbca0035a5b567EXAMPLE.
- Images in official repositories on Docker Hub use a single name (for example, ubuntu or mongo).
- Images in other repositories on Docker Hub are qualified with an organization name (for example, amazon/amazon-ecs-agent).
- Images in other online repositories are qualified further by a domain name (for example, quay.io/assemblyline/ubuntu).

Type: String

Required: No

interactive

When this parameter is true, you can deploy containerized applications that require stdin or a tty to be allocated. This parameter maps to OpenStdin in the docker container create command and the --interactive option to docker run.

Type: Boolean

Required: No

links

The links parameter allows containers to communicate with each other without the need for port mappings. This parameter is only supported if the network mode of a task definition is

`bridge`. The `name:internalName` construct is analogous to `name:alias` in Docker links. Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed.. This parameter maps to `Links` in the docker container create command and the `--link` option to `docker run`.

 Note

This parameter is not supported for Windows containers.

 Important

Containers that are collocated on a single container instance may be able to communicate with each other without requiring links or host port mappings. Network isolation is achieved on the container instance using security groups and VPC settings.

Type: Array of strings

Required: No

linuxParameters

Linux-specific modifications that are applied to the default Docker container configuration, such as Linux kernel capabilities. For more information see [KernelCapabilities](#).

 Note

This parameter is not supported for Windows containers.

Type: [LinuxParameters](#) object

Required: No

logConfiguration

The log configuration specification for the container.

This parameter maps to `LogConfig` in the docker container create command and the `--log-driver` option to `docker run`. By default, containers use the same logging driver that

the Docker daemon uses. However the container can use a different logging driver than the Docker daemon by specifying a log driver with this parameter in the container definition. To use a different logging driver for a container, the log system must be configured properly on the container instance (or on a different log server for remote logging options).

 Note

Amazon ECS currently supports a subset of the logging drivers available to the Docker daemon (shown in the [LogConfiguration](#) data type). Additional log drivers may be available in future releases of the Amazon ECS container agent.

This parameter requires version 1.18 of the Docker Remote API or greater on your container instance. To check the Docker Remote API version on your container instance, log in to your container instance and run the following command: `sudo docker version --format '{{.Server.APIVersion}}'`

 Note

The Amazon ECS container agent running on a container instance must register the logging drivers available on that instance with the `ECS_AVAILABLE_LOGGING_DRIVERS` environment variable before containers placed on that instance can use these log configuration options. For more information, see [Amazon ECS Container Agent Configuration](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [LogConfiguration](#) object

Required: No

memory

The amount (in MiB) of memory to present to the container. If your container attempts to exceed the memory specified here, the container is killed. The total amount of memory reserved for all containers within a task must be lower than the task memory value, if one is specified. This parameter maps to `Memory` in the `docker container create` command and the `--memory` option to `docker run`.

If using the Fargate launch type, this parameter is optional.

If using the EC2 launch type, you must specify either a task-level memory value or a container-level memory value. If you specify both a container-level memory and `memoryReservation` value, memory must be greater than `memoryReservation`. If you specify `memoryReservation`, then that value is subtracted from the available memory resources for the container instance where the container is placed. Otherwise, the value of memory is used.

The Docker 20.10.0 or later daemon reserves a minimum of 6 MiB of memory for a container. So, don't specify less than 6 MiB of memory for your containers.

The Docker 19.03.13-ce or earlier daemon reserves a minimum of 4 MiB of memory for a container. So, don't specify less than 4 MiB of memory for your containers.

Type: Integer

Required: No

memoryReservation

The soft limit (in MiB) of memory to reserve for the container. When system memory is under heavy contention, Docker attempts to keep the container memory to this soft limit. However, your container can consume more memory when it needs to, up to either the hard limit specified with the `memory` parameter (if applicable), or all of the available memory on the container instance, whichever comes first. This parameter maps to `MemoryReservation` in the `docker container create` command and the `--memory-reservation` option to `docker run`.

If a task-level memory value is not specified, you must specify a non-zero integer for one or both of `memory` or `memoryReservation` in a container definition. If you specify both, memory must be greater than `memoryReservation`. If you specify `memoryReservation`, then that value is subtracted from the available memory resources for the container instance where the container is placed. Otherwise, the value of memory is used.

For example, if your container normally uses 128 MiB of memory, but occasionally bursts to 256 MiB of memory for short periods of time, you can set a `memoryReservation` of 128 MiB, and a `memory` hard limit of 300 MiB. This configuration would allow the container to only reserve 128 MiB of memory from the remaining resources on the container instance, but also allow the container to consume more memory resources when needed.

The Docker 20.10.0 or later daemon reserves a minimum of 6 MiB of memory for a container. So, don't specify less than 6 MiB of memory for your containers.

The Docker 19.03.13-ce or earlier daemon reserves a minimum of 4 MiB of memory for a container. So, don't specify less than 4 MiB of memory for your containers.

Type: Integer

Required: No

mountPoints

The mount points for data volumes in your container.

This parameter maps to `Volumes` in the docker container create command and the `--volume` option to docker run.

Windows containers can mount whole directories on the same drive as `$env:ProgramData`. Windows containers can't mount directories on a different drive, and mount point can't be across drives.

Type: Array of [MountPoint](#) objects

Required: No

name

The name of a container. If you're linking multiple containers together in a task definition, the name of one container can be entered in the `links` of another container to connect the containers. Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed. This parameter maps to `name` in the docker container create command and the `--name` option to docker run.

Type: String

Required: No

portMappings

The list of port mappings for the container. Port mappings allow containers to access ports on the host container instance to send or receive traffic.

For task definitions that use the `awsvpc` network mode, only specify the `containerPort`. The `hostPort` can be left blank or it must be the same value as the `containerPort`.

Port mappings on Windows use the `NetNAT` gateway address rather than `localhost`. There's no loopback for port mappings on Windows, so you can't access a container's mapped port from the host itself.

This parameter maps to `PortBindings` in the docker container create command and the `--publish` option to docker run. If the network mode of a task definition is set to `none`, then you can't specify port mappings. If the network mode of a task definition is set to `host`, then host ports must either be undefined or they must match the container port in the port mapping.

 Note

After a task reaches the `RUNNING` status, manual and automatic host and container port assignments are visible in the **Network Bindings** section of a container description for a selected task in the Amazon ECS console. The assignments are also visible in the `networkBindings` section [DescribeTasks](#) responses.

Type: Array of [PortMapping](#) objects

Required: No

privileged

When this parameter is `true`, the container is given elevated privileges on the host container instance (similar to the `root` user). This parameter maps to `Privileged` in the docker container create command and the `--privileged` option to docker run

 Note

This parameter is not supported for Windows containers or tasks run on AWS Fargate.

Type: Boolean

Required: No

pseudoTerminal

When this parameter is `true`, a TTY is allocated. This parameter maps to `Tty` in the docker container create command and the `--tty` option to docker run.

Type: Boolean

Required: No

readonlyRootFilesystem

When this parameter is true, the container is given read-only access to its root file system. This parameter maps to `ReadonlyRootfs` in the docker container create command and the `--read-only` option to docker run.

Note

This parameter is not supported for Windows containers.

Type: Boolean

Required: No

repositoryCredentials

The private repository authentication credentials to use.

Type: [RepositoryCredentials](#) object

Required: No

resourceRequirements

The type and amount of a resource to assign to a container. The only supported resource is a GPU.

Type: Array of [ResourceRequirement](#) objects

Required: No

restartPolicy

The restart policy for a container. When you set up a restart policy, Amazon ECS can restart the container without needing to replace the task. For more information, see [Restart individual containers in Amazon ECS tasks with container restart policies](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [ContainerRestartPolicy](#) object

Required: No

secrets

The secrets to pass to the container. For more information, see [Specifying Sensitive Data](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Array of [Secret](#) objects

Required: No

startTimeout

Time duration (in seconds) to wait before giving up on resolving dependencies for a container. For example, you specify two containers in a task definition with containerA having a dependency on containerB reaching a COMPLETE, SUCCESS, or HEALTHY status. If a startTimeout value is specified for containerB and it doesn't reach the desired status within that time then containerA gives up and not start. This results in the task transitioning to a STOPPED state.

Note

When the ECS_CONTAINER_START_TIMEOUT container agent configuration variable is used, it's enforced independently from this start timeout value.

For tasks using the Fargate launch type, the task or service requires the following platforms:

- Linux platform version 1.3.0 or later.
- Windows platform version 1.0.0 or later.

For tasks using the EC2 launch type, your container instances require at least version 1.26.0 of the container agent to use a container start timeout value. However, we recommend using the latest container agent version. For information about checking your agent version and updating to the latest version, see [Updating the Amazon ECS Container Agent](#) in the *Amazon Elastic Container Service Developer Guide*. If you're using an Amazon ECS-optimized Linux AMI, your instance needs at least version 1.26.0-1 of the ecs-init package. If your container instances are launched from version 20190301 or later, then they contain the required versions of the container agent and ecs-init. For more information, see [Amazon ECS-optimized Linux AMI](#) in the *Amazon Elastic Container Service Developer Guide*.

The valid values for Fargate are 2-120 seconds.

Type: Integer

Required: No

stopTimeout

Time duration (in seconds) to wait before the container is forcefully killed if it doesn't exit normally on its own.

For tasks using the Fargate launch type, the task or service requires the following platforms:

- Linux platform version 1.3.0 or later.
- Windows platform version 1.0.0 or later.

For tasks that use the Fargate launch type, the max stop timeout value is 120 seconds and if the parameter is not specified, the default value of 30 seconds is used.

For tasks that use the EC2 launch type, if the `stopTimeout` parameter isn't specified, the value set for the Amazon ECS container agent configuration variable `ECS_CONTAINER_STOP_TIMEOUT` is used. If neither the `stopTimeout` parameter or the `ECS_CONTAINER_STOP_TIMEOUT` agent configuration variable are set, then the default values of 30 seconds for Linux containers and 30 seconds on Windows containers are used. Your container instances require at least version 1.26.0 of the container agent to use a container stop timeout value. However, we recommend using the latest container agent version. For information about checking your agent version and updating to the latest version, see [Updating the Amazon ECS Container Agent](#) in the *Amazon Elastic Container Service Developer Guide*. If you're using an Amazon ECS-optimized Linux AMI, your instance needs at least version 1.26.0-1 of the `ecs-init` package. If your container instances are launched from version 20190301 or later, then they contain the required versions of the container agent and `ecs-init`. For more information, see [Amazon ECS-optimized Linux AMI](#) in the *Amazon Elastic Container Service Developer Guide*.

The valid values for Fargate are 2-120 seconds.

Type: Integer

Required: No

systemControls

A list of namespaced kernel parameters to set in the container. This parameter maps to `Sysctl`s in the docker container create command and the `--sysctl` option to docker run. For example, you can configure `net.ipv4.tcp_keepalive_time` setting to maintain longer lived connections.

Type: Array of [SystemControl](#) objects

Required: No

ulimits

A list of `ulimits` to set in the container. If a `ulimit` value is specified in a task definition, it overrides the default values set by Docker. This parameter maps to `Ulimits` in the docker container create command and the `--ulimit` option to docker run. Valid naming values are displayed in the [Ulimit](#) data type.

Amazon ECS tasks hosted on AWS Fargate use the default resource limit values set by the operating system with the exception of the `nofile` resource limit parameter which AWS Fargate overrides. The `nofile` resource limit sets a restriction on the number of open files that a container can use. The default `nofile` soft limit is 65535 and the default hard limit is 65535.

This parameter requires version 1.18 of the Docker Remote API or greater on your container instance. To check the Docker Remote API version on your container instance, log in to your container instance and run the following command: `sudo docker version --format '{{.Server.APIVersion}}'`

Note

This parameter is not supported for Windows containers.

Type: Array of [Ulimit](#) objects

Required: No

user

The user to use inside the container. This parameter maps to `User` in the docker container create command and the `--user` option to docker run.

Important

When running tasks using the host network mode, don't run containers using the root user (UID 0). We recommend using a non-root user for better security.

You can specify the user using the following formats. If specifying a UID or GID, you must specify it as a positive integer.

- `user`
- `user:group`
- `uid`
- `uid:gid`
- `user:gid`
- `uid:group`

 Note

This parameter is not supported for Windows containers.

Type: String

Required: No

versionConsistency

Specifies whether Amazon ECS will resolve the container image tag provided in the container definition to an image digest. By default, the value is enabled. If you set the value for a container as `disabled`, Amazon ECS will not resolve the provided container image tag to a digest and will use the original image URI specified in the container definition for deployment. For more information about container image resolution, see [Container image resolution](#) in the *Amazon ECS Developer Guide*.

Type: String

Valid Values: `enabled` | `disabled`

Required: No

volumesFrom

Data volumes to mount from another container. This parameter maps to `VolumesFrom` in the `docker container create` command and the `--volumes-from` option to `docker run`.

Type: Array of [VolumeFrom](#) objects

Required: No

workingDirectory

The working directory to run commands inside the container in. This parameter maps to `WorkingDir` in the `docker container create` command and the `--workdir` option to `docker run`.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ContainerDependency

The dependencies defined for container startup and shutdown. A container can contain multiple dependencies. When a dependency is defined for container startup, for container shutdown it is reversed.

Your Amazon ECS container instances require at least version 1.26.0 of the container agent to use container dependencies. However, we recommend using the latest container agent version. For information about checking your agent version and updating to the latest version, see [Updating the Amazon ECS Container Agent](#) in the *Amazon Elastic Container Service Developer Guide*. If you're using an Amazon ECS-optimized Linux AMI, your instance needs at least version 1.26.0-1 of the `ecs-init` package. If your container instances are launched from version 20190301 or later, then they contain the required versions of the container agent and `ecs-init`. For more information, see [Amazon ECS-optimized Linux AMI](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

For tasks that use the Fargate launch type, the task or service requires the following platforms:

- Linux platform version 1.3.0 or later.
- Windows platform version 1.0.0 or later.

For more information about how to create a container dependency, see [Container dependency](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

condition

The dependency condition of the container. The following are the available conditions and their behavior:

- **START** - This condition emulates the behavior of links and volumes today. It validates that a dependent container is started before permitting other containers to start.
- **COMPLETE** - This condition validates that a dependent container runs to completion (exits) before permitting other containers to start. This can be useful for nonessential containers that run a script and then exit. This condition can't be set on an essential container.

- **SUCCESS** - This condition is the same as **COMPLETE**, but it also requires that the container exits with a zero status. This condition can't be set on an essential container.
- **HEALTHY** - This condition validates that the dependent container passes its Docker health check before permitting other containers to start. This requires that the dependent container has health checks configured. This condition is confirmed only at task startup.

Type: String

Valid Values: **START** | **COMPLETE** | **SUCCESS** | **HEALTHY**

Required: Yes

containerName

The name of a container.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ContainerImage

The details about the container image a service revision uses.

To ensure that all tasks in a service use the same container image, Amazon ECS resolves container image names and any image tags specified in the task definition to container image digests.

After the container image digest has been established, Amazon ECS uses the digest to start any other desired tasks, and for any future service and service revision updates. This leads to all tasks in a service always running identical container images, resulting in version consistency for your software. For more information, see [Container image resolution](#) in the Amazon ECS Developer Guide.

Contents

containerName

The name of the container.

Type: String

Required: No

image

The container image.

Type: String

Required: No

imageDigest

The container image digest.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ContainerInstance

An Amazon EC2 or External instance that's running the Amazon ECS agent and has been registered with a cluster.

Contents

agentConnected

This parameter returns `true` if the agent is connected to Amazon ECS. An instance with an agent that may be unhealthy or stopped return `false`. Only instances connected to an agent can accept task placement requests.

Type: Boolean

Required: No

agentUpdateStatus

The status of the most recent agent update. If an update wasn't ever requested, this value is `NULL`.

Type: String

Valid Values: `PENDING` | `STAGING` | `STAGED` | `UPDATING` | `UPDATED` | `FAILED`

Required: No

attachments

The resources attached to a container instance, such as an elastic network interface.

Type: Array of [Attachment](#) objects

Required: No

attributes

The attributes set for the container instance, either by the Amazon ECS container agent at instance registration or manually with the [PutAttributes](#) operation.

Type: Array of [Attribute](#) objects

Required: No

capacityProviderName

The capacity provider that's associated with the container instance.

Type: String

Required: No

containerInstanceArn

The Amazon Resource Name (ARN) of the container instance. For more information about the ARN format, see [Amazon Resource Name \(ARN\)](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: No

ec2InstanceId

The ID of the container instance. For Amazon EC2 instances, this value is the Amazon EC2 instance ID. For external instances, this value is the AWS Systems Manager managed instance ID.

Type: String

Required: No

healthStatus

An object representing the health status of the container instance.

Type: [ContainerInstanceHealthStatus](#) object

Required: No

pendingTasksCount

The number of tasks on the container instance that are in the PENDING status.

Type: Integer

Required: No

registeredAt

The Unix timestamp for the time when the container instance was registered.

Type: Timestamp

Required: No

registeredResources

For CPU and memory resource types, this parameter describes the amount of each resource that was available on the container instance when the container agent registered it with Amazon ECS. This value represents the total amount of CPU and memory that can be allocated on this container instance to tasks. For port resource types, this parameter describes the ports that were reserved by the Amazon ECS container agent when it registered the container instance with Amazon ECS.

Type: Array of [Resource](#) objects

Required: No

remainingResources

For CPU and memory resource types, this parameter describes the remaining CPU and memory that wasn't already allocated to tasks and is therefore available for new tasks. For port resource types, this parameter describes the ports that were reserved by the Amazon ECS container agent (at instance registration time) and any task containers that have reserved port mappings on the host (with the host or bridge network mode). Any port that's not specified here is available for new tasks.

Type: Array of [Resource](#) objects

Required: No

runningTasksCount

The number of tasks on the container instance that have a desired status (`desiredStatus`) of `RUNNING`.

Type: Integer

Required: No

status

The status of the container instance. The valid values are `REGISTERING`, `REGISTRATION_FAILED`, `ACTIVE`, `INACTIVE`, `DEREGISTERING`, or `DRAINING`.

If your account has opted in to the `awsvpcTrunking` account setting, then any newly registered container instance will transition to a `REGISTERING` status while the trunk elastic

network interface is provisioned for the instance. If the registration fails, the instance will transition to a `REGISTRATION_FAILED` status. You can describe the container instance and see the reason for failure in the `statusReason` parameter. Once the container instance is terminated, the instance transitions to a `DEREGISTERING` status while the trunk elastic network interface is deprovisioned. The instance then transitions to an `INACTIVE` status.

The `ACTIVE` status indicates that the container instance can accept tasks. The `DRAINING` indicates that new tasks aren't placed on the container instance and any service tasks running on the container instance are removed if possible. For more information, see [Container instance draining](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

statusReason

The reason that the container instance reached its current status.

Type: String

Required: No

tags

The metadata that you apply to the container instance to help you categorize and organize them. Each tag consists of a key and an optional value. You define both.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: `+ - = . _ : / @`.
- Tag keys and values are case-sensitive.
- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

version

The version counter for the container instance. Every time a container instance experiences a change that triggers a CloudWatch event, the version counter is incremented. If you're replicating your Amazon ECS container instance state with CloudWatch Events, you can compare the version of a container instance reported by the Amazon ECS APIs with the version reported in CloudWatch Events for the container instance (inside the `detail` object) to verify that the version in your event stream is current.

Type: Long

Required: No

versionInfo

The version information for the Amazon ECS container agent and Docker daemon running on the container instance.

Type: [VersionInfo](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ContainerInstanceHealthStatus

An object representing the health status of the container instance.

Contents

details

An array of objects representing the details of the container instance health status.

Type: Array of [InstanceHealthCheckResult](#) objects

Required: No

overallStatus

The overall health status of the container instance. This is an aggregate status of all container instance health checks.

Type: String

Valid Values: OK | IMPAIRED | INSUFFICIENT_DATA | INITIALIZING

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ContainerOverride

The overrides that are sent to a container. An empty container override can be passed in. An example of an empty container override is `{"containerOverrides": [ ] }`. If a non-empty container override is specified, the name parameter must be included.

You can use Secrets Manager or AWS Systems Manager Parameter Store to store the sensitive data. For more information, see [Retrieve secrets through environment variables](#) in the Amazon ECS Developer Guide.

Contents

command

The command to send to the container that overrides the default command from the Docker image or the task definition. You must also specify a container name.

Type: Array of strings

Required: No

cpu

The number of cpu units reserved for the container, instead of the default value from the task definition. You must also specify a container name.

Type: Integer

Required: No

environment

The environment variables to send to the container. You can add new environment variables, which are added to the container at launch, or you can override the existing environment variables from the Docker image or the task definition. You must also specify a container name.

Type: Array of [KeyValuePair](#) objects

Required: No

environmentFiles

A list of files containing the environment variables to pass to a container, instead of the value from the container definition.

Type: Array of [EnvironmentFile](#) objects

Required: No

memory

The hard limit (in MiB) of memory to present to the container, instead of the default value from the task definition. If your container attempts to exceed the memory specified here, the container is killed. You must also specify a container name.

Type: Integer

Required: No

memoryReservation

The soft limit (in MiB) of memory to reserve for the container, instead of the default value from the task definition. You must also specify a container name.

Type: Integer

Required: No

name

The name of the container that receives the override. This parameter is required if any override is specified.

Type: String

Required: No

resourceRequirements

The type and amount of a resource to assign to a container, instead of the default value from the task definition. The only supported resource is a GPU.

Type: Array of [ResourceRequirement](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ContainerRestartPolicy

You can enable a restart policy for each container defined in your task definition, to overcome transient failures faster and maintain task availability. When you enable a restart policy for a container, Amazon ECS can restart the container if it exits, without needing to replace the task. For more information, see [Restart individual containers in Amazon ECS tasks with container restart policies](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

enabled

Specifies whether a restart policy is enabled for the container.

Type: Boolean

Required: Yes

ignoredExitCodes

A list of exit codes that Amazon ECS will ignore and not attempt a restart on. You can specify a maximum of 50 container exit codes. By default, Amazon ECS does not ignore any exit codes.

Type: Array of integers

Required: No

restartAttemptPeriod

A period of time (in seconds) that the container must run for before a restart can be attempted. A container can be restarted only once every `restartAttemptPeriod` seconds. If a container isn't able to run for this time period and exits early, it will not be restarted. You can set a minimum `restartAttemptPeriod` of 60 seconds and a maximum `restartAttemptPeriod` of 1800 seconds. By default, a container must run for 300 seconds before it can be restarted.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ContainerStateChange

An object that represents a change in state for a container.

Contents

containerName

The name of the container.

Type: String

Required: No

exitCode

The exit code for the container, if the state change is a result of the container exiting.

Type: Integer

Required: No

imageDigest

The container image SHA 256 digest.

Type: String

Required: No

networkBindings

Any network bindings that are associated with the container.

Type: Array of [NetworkBinding](#) objects

Required: No

reason

The reason for the state change.

Type: String

Required: No

runtimeId

The ID of the Docker container.

Type: String

Required: No

status

The status of the container.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

CreatedAt

The optional filter to narrow the `ListServiceDeployment` results.

If you do not specify a value, service deployments that were created before the current time are included in the result.

Contents

after

Include service deployments in the result that were created after this time. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: Timestamp

Required: No

before

Include service deployments in the result that were created before this time. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: Timestamp

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

CreateManagedInstancesProviderConfiguration

The configuration for creating a Amazon ECS Managed Instances provider. This specifies how Amazon ECS should manage Amazon EC2 instances, including the infrastructure role, instance launch template, and whether to propagate tags from the capacity provider to the instances.

Contents

infrastructureRoleArn

The Amazon Resource Name (ARN) of the infrastructure role that Amazon ECS uses to manage instances on your behalf. This role must have permissions to launch, terminate, and manage Amazon EC2 instances, as well as access to other AWS services required for Amazon ECS Managed Instances functionality.

For more information, see [Amazon ECS infrastructure IAM role](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: Yes

instanceLaunchTemplate

The launch template configuration that specifies how Amazon ECS should launch Amazon EC2 instances. This includes the instance profile, network configuration, storage settings, and instance requirements for attribute-based instance type selection.

For more information, see [Store instance launch parameters in Amazon EC2 launch templates](#) in the *Amazon EC2 User Guide*.

Type: [InstanceLaunchTemplate](#) object

Required: Yes

propagateTags

Specifies whether to propagate tags from the capacity provider to the Amazon ECS Managed Instances. When enabled, tags applied to the capacity provider are automatically applied to all instances launched by this provider.

Type: String

Valid Values: CAPACITY_PROVIDER | NONE

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Deployment

The details of an Amazon ECS service deployment. This is used only when a service uses the ECS deployment controller type.

Contents

capacityProviderStrategy

The capacity provider strategy that the deployment is using.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: No

createdAt

The Unix timestamp for the time when the service deployment was created.

Type: Timestamp

Required: No

desiredCount

The most recent desired count of tasks that was specified for the service to deploy or maintain.

Type: Integer

Required: No

failedTasks

The number of consecutively failed tasks in the deployment. A task is considered a failure if the service scheduler can't launch the task, the task doesn't transition to a RUNNING state, or if it fails any of its defined health checks and is stopped.

Note

Once a service deployment has one or more successfully running tasks, the failed task count resets to zero and stops being evaluated.

Type: Integer

Required: No

fargateEphemeralStorage

The Fargate ephemeral storage settings for the deployment.

Type: [DeploymentEphemeralStorage](#) object

Required: No

id

The ID of the deployment.

Type: String

Required: No

launchType

The launch type the tasks in the service are using. For more information, see [Amazon ECS Launch Types](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

networkConfiguration

The VPC subnet and security group configuration for tasks that receive their own elastic network interface by using the awsvpc networking mode.

Type: [NetworkConfiguration](#) object

Required: No

pendingCount

The number of tasks in the deployment that are in the PENDING status.

Type: Integer

Required: No

platformFamily

The operating system that your tasks in the service, or tasks are running on. A platform family is specified only for tasks using the Fargate launch type.

All tasks that run as part of this service must use the same `platformFamily` value as the service, for example, `LINUX`.

Type: String

Required: No

platformVersion

The platform version that your tasks in the service run on. A platform version is only specified for tasks using the Fargate launch type. If one isn't specified, the LATEST platform version is used. For more information, see [AWS Fargate Platform Versions](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

rolloutState

Note

The `rolloutState` of a service is only returned for services that use the rolling update (ECS) deployment type that aren't behind a Classic Load Balancer.

The rollout state of the deployment. When a service deployment is started, it begins in an `IN_PROGRESS` state. When the service reaches a steady state, the deployment transitions to a `COMPLETED` state. If the service fails to reach a steady state and circuit breaker is turned on, the deployment transitions to a `FAILED` state. A deployment in `FAILED` state doesn't launch any new tasks. For more information, see [DeploymentCircuitBreaker](#).

Type: String

Valid Values: `COMPLETED` | `FAILED` | `IN_PROGRESS`

Required: No

rolloutStateReason

A description of the rollout state of a deployment.

Type: String

Required: No

runningCount

The number of tasks in the deployment that are in the RUNNING status.

Type: Integer

Required: No

serviceConnectConfiguration

The details of the Service Connect configuration that's used by this deployment. Compare the configuration between multiple deployments when troubleshooting issues with new deployments.

The configuration for this service to discover and connect to services, and be discovered by, and connected from, other services within a namespace.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [ServiceConnectConfiguration](#) object

Required: No

serviceConnectResources

The list of Service Connect resources that are associated with this deployment. Each list entry maps a discovery name to a AWS Cloud Map service name.

Type: Array of [ServiceConnectServiceResource](#) objects

Required: No

status

The status of the deployment. The following describes each state.

PRIMARY

The most recent deployment of a service.

ACTIVE

A service deployment that still has running tasks, but are in the process of being replaced with a new PRIMARY deployment.

INACTIVE

A deployment that has been completely replaced.

Type: String

Required: No

taskDefinition

The most recent task definition that was specified for the tasks in the service to use.

Type: String

Required: No

updatedAt

The Unix timestamp for the time when the service deployment was last updated.

Type: Timestamp

Required: No

volumeConfigurations

The details of the volume that was configuredAtLaunch. You can configure different settings like the size, throughput, volumeType, and encryption in [ServiceManagedEBSVolumeConfiguration](#). The name of the volume must match the name from the task definition.

Type: Array of [ServiceVolumeConfiguration](#) objects

Required: No

vpcLatticeConfigurations

The VPC Lattice configuration for the service deployment.

Type: Array of [VpcLatticeConfiguration](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

DeploymentAlarms

One of the methods which provide a way for you to quickly identify when a deployment has failed, and then to optionally roll back the failure to the last working deployment.

When the alarms are generated, Amazon ECS sets the service deployment to failed. Set the rollback parameter to have Amazon ECS to roll back your service to the last completed deployment after a failure.

You can only use the DeploymentAlarms method to detect failures when the DeploymentController is set to ECS.

For more information, see [Rolling update](#) in the Amazon Elastic Container Service Developer Guide .

Contents

alarmNames

One or more CloudWatch alarm names. Use a "," to separate the alarms.

Type: Array of strings

Required: Yes

enable

Determines whether to use the CloudWatch alarm option in the service deployment process.

Type: Boolean

Required: Yes

rollback

Determines whether to configure Amazon ECS to roll back the service if a service deployment fails. If rollback is used, when a service deployment fails, the service is rolled back to the last deployment that completed successfully.

Type: Boolean

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

DeploymentCircuitBreaker

Note

The deployment circuit breaker can only be used for services using the rolling update (ECS) deployment type.

The **deployment circuit breaker** determines whether a service deployment will fail if the service can't reach a steady state. If it is turned on, a service deployment will transition to a failed state and stop launching new tasks. You can also configure Amazon ECS to roll back your service to the last completed deployment after a failure. For more information, see [Rolling update](#) in the *Amazon Elastic Container Service Developer Guide*.

For more information about API failure reasons, see [API failure reasons](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

enable

Determines whether to use the deployment circuit breaker logic for the service.

Type: Boolean

Required: Yes

rollback

Determines whether to configure Amazon ECS to roll back the service if a service deployment fails. If rollback is on, when a service deployment fails, the service is rolled back to the last deployment that completed successfully.

Type: Boolean

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

DeploymentConfiguration

Optional deployment parameters that control how many tasks run during a deployment and the ordering of stopping and starting tasks.

Contents

alarms

Information about the CloudWatch alarms.

Type: [DeploymentAlarms](#) object

Required: No

bakeTimeInMinutes

The time period when both blue and green service revisions are running simultaneously after the production traffic has shifted.

You must provide this parameter when you use the BLUE_GREEN deployment strategy.

Type: Integer

Required: No

canaryConfiguration

Configuration for canary deployment strategy. Only valid when the deployment strategy is CANARY. This configuration enables shifting a fixed percentage of traffic for testing, followed by shifting the remaining traffic after a bake period.

Type: [CanaryConfiguration](#) object

Required: No

deploymentCircuitBreaker

Note

The deployment circuit breaker can only be used for services using the rolling update (ECS) deployment type.

The **deployment circuit breaker** determines whether a service deployment will fail if the service can't reach a steady state. If you use the deployment circuit breaker, a service deployment will transition to a failed state and stop launching new tasks. If you use the rollback option, when a service deployment fails, the service is rolled back to the last deployment that completed successfully. For more information, see [Rolling update](#) in the *Amazon Elastic Container Service Developer Guide*

Type: [DeploymentCircuitBreaker](#) object

Required: No

lifecycleHooks

An array of deployment lifecycle hook objects to run custom logic at specific stages of the deployment lifecycle.

Type: Array of [DeploymentLifecycleHook](#) objects

Required: No

linearConfiguration

Configuration for linear deployment strategy. Only valid when the deployment strategy is LINEAR. This configuration enables progressive traffic shifting in equal percentage increments with configurable bake times between each step.

Type: [LinearConfiguration](#) object

Required: No

maximumPercent

If a service is using the rolling update (ECS) deployment type, the `maximumPercent` parameter represents an upper limit on the number of your service's tasks that are allowed in the RUNNING or PENDING state during a deployment, as a percentage of the `desiredCount` (rounded down to the nearest integer). This parameter enables you to define the deployment batch size. For example, if your service is using the REPLICA service scheduler and has a `desiredCount` of four tasks and a `maximumPercent` value of 200%, the scheduler may start four new tasks before stopping the four older tasks (provided that the cluster resources required to do this are available). The default `maximumPercent` value for a service using the REPLICA service scheduler is 200%.

The Amazon ECS scheduler uses this parameter to replace unhealthy tasks by starting replacement tasks first and then stopping the unhealthy tasks, as long as cluster resources for starting replacement tasks are available. For more information about how the scheduler replaces unhealthy tasks, see [Amazon ECS services](#).

If a service is using either the blue/green (CODE_DEPLOY) or EXTERNAL deployment types, and tasks in the service use the EC2 launch type, the **maximum percent** value is set to the default value. The **maximum percent** value is used to define the upper limit on the number of the tasks in the service that remain in the RUNNING state while the container instances are in the DRAINING state.

 Note

You can't specify a custom `maximumPercent` value for a service that uses either the blue/green (CODE_DEPLOY) or EXTERNAL deployment types and has tasks that use the EC2 launch type.

If the service uses either the blue/green (CODE_DEPLOY) or EXTERNAL deployment types, and the tasks in the service use the Fargate launch type, the `maximum percent` value is not used. The value is still returned when describing your service.

Type: Integer

Required: No

minimumHealthyPercent

If a service is using the rolling update (ECS) deployment type, the `minimumHealthyPercent` represents a lower limit on the number of your service's tasks that must remain in the RUNNING state during a deployment, as a percentage of the `desiredCount` (rounded up to the nearest integer). This parameter enables you to deploy without using additional cluster capacity. For example, if your service has a `desiredCount` of four tasks and a `minimumHealthyPercent` of 50%, the service scheduler may stop two existing tasks to free up cluster capacity before starting two new tasks.

If any tasks are unhealthy and if `maximumPercent` doesn't allow the Amazon ECS scheduler to start replacement tasks, the scheduler stops the unhealthy tasks one-by-one — using the `minimumHealthyPercent` as a constraint — to clear up capacity to launch replacement tasks.

For more information about how the scheduler replaces unhealthy tasks, see [Amazon ECS services](#).

For services that *do not* use a load balancer, the following should be noted:

- A service is considered healthy if all essential containers within the tasks in the service pass their health checks.
- If a task has no essential containers with a health check defined, the service scheduler will wait for 40 seconds after a task reaches a RUNNING state before the task is counted towards the minimum healthy percent total.
- If a task has one or more essential containers with a health check defined, the service scheduler will wait for the task to reach a healthy status before counting it towards the minimum healthy percent total. A task is considered healthy when all essential containers within the task have passed their health checks. The amount of time the service scheduler can wait for is determined by the container health check settings.

For services that *do* use a load balancer, the following should be noted:

- If a task has no essential containers with a health check defined, the service scheduler will wait for the load balancer target group health check to return a healthy status before counting the task towards the minimum healthy percent total.
- If a task has an essential container with a health check defined, the service scheduler will wait for both the task to reach a healthy status and the load balancer target group health check to return a healthy status before counting the task towards the minimum healthy percent total.

The default value for a replica service for `minimumHealthyPercent` is 100%. The default `minimumHealthyPercent` value for a service using the DAEMON service schedule is 0% for the AWS CLI, the AWS SDKs, and the APIs and 50% for the AWS Management Console.

The minimum number of healthy tasks during a deployment is the `desiredCount` multiplied by the `minimumHealthyPercent/100`, rounded up to the nearest integer value.

If a service is using either the blue/green (CODE_DEPLOY) or EXTERNAL deployment types and is running tasks that use the EC2 launch type, the **minimum healthy percent** value is set to the default value. The **minimum healthy percent** value is used to define the lower limit on the number of the tasks in the service that remain in the RUNNING state while the container instances are in the DRAINING state.

Note

You can't specify a custom `minimumHealthyPercent` value for a service that uses either the blue/green (`CODE_DEPLOY`) or `EXTERNAL` deployment types and has tasks that use the EC2 launch type.

If a service is using either the blue/green (`CODE_DEPLOY`) or `EXTERNAL` deployment types and is running tasks that use the Fargate launch type, the minimum healthy percent value is not used, although it is returned when describing your service.

Type: Integer

Required: No

strategy

The deployment strategy for the service. Choose from these valid values:

- `ROLLING` - When you create a service which uses the rolling update (`ROLLING`) deployment strategy, the Amazon ECS service scheduler replaces the currently running tasks with new tasks. The number of tasks that Amazon ECS adds or removes from the service during a rolling update is controlled by the service deployment configuration.
- `BLUE_GREEN` - A blue/green deployment strategy (`BLUE_GREEN`) is a release methodology that reduces downtime and risk by running two identical production environments called blue and green. With Amazon ECS blue/green deployments, you can validate new service revisions before directing production traffic to them. This approach provides a safer way to deploy changes with the ability to quickly roll back if needed.

Type: String

Valid Values: `ROLLING` | `BLUE_GREEN` | `LINEAR` | `CANARY`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

DeploymentController

The deployment controller to use for the service.

Contents

type

The deployment controller type to use.

The deployment controller is the mechanism that determines how tasks are deployed for your service. The valid options are:

- ECS

When you create a service which uses the ECS deployment controller, you can choose between the following deployment strategies:

- **ROLLING:** When you create a service which uses the *rolling update* (ROLLING) deployment strategy, the Amazon ECS service scheduler replaces the currently running tasks with new tasks. The number of tasks that Amazon ECS adds or removes from the service during a rolling update is controlled by the service deployment configuration.

Rolling update deployments are best suited for the following scenarios:

- **Gradual service updates:** You need to update your service incrementally without taking the entire service offline at once.
- **Limited resource requirements:** You want to avoid the additional resource costs of running two complete environments simultaneously (as required by blue/green deployments).
- **Acceptable deployment time:** Your application can tolerate a longer deployment process, as rolling updates replace tasks one by one.
- **No need for instant roll back:** Your service can tolerate a rollback process that takes minutes rather than seconds.
- **Simple deployment process:** You prefer a straightforward deployment approach without the complexity of managing multiple environments, target groups, and listeners.
- **No load balancer requirement:** Your service doesn't use or require a load balancer, Application Load Balancer, Network Load Balancer, or Service Connect (which are required for blue/green deployments).

- **Stateful applications:** Your application maintains state that makes it difficult to run two parallel environments.
- **Cost sensitivity:** You want to minimize deployment costs by not running duplicate environments during deployment.

Rolling updates are the default deployment strategy for services and provide a balance between deployment safety and resource efficiency for many common application scenarios.

- **BLUE_GREEN:** A *blue/green* deployment strategy (BLUE_GREEN) is a release methodology that reduces downtime and risk by running two identical production environments called blue and green. With Amazon ECS blue/green deployments, you can validate new service revisions before directing production traffic to them. This approach provides a safer way to deploy changes with the ability to quickly roll back if needed.

Amazon ECS blue/green deployments are best suited for the following scenarios:

- **Service validation:** When you need to validate new service revisions before directing production traffic to them
 - **Zero downtime:** When your service requires zero-downtime deployments
 - **Instant roll back:** When you need the ability to quickly roll back if issues are detected
 - **Load balancer requirement:** When your service uses Application Load Balancer, Network Load Balancer, or Service Connect
- **External**

Use a third-party deployment controller.

- **Blue/green deployment (powered by AWS CodeDeploy)**

AWS CodeDeploy installs an updated version of the application as a new replacement task set and reroutes production traffic from the original application task set to the replacement task set. The original task set is terminated after a successful deployment. Use this deployment controller to verify a new deployment of a service before sending production traffic to it.

Type: String

Valid Values: ECS | CODE_DEPLOY | EXTERNAL

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

DeploymentEphemeralStorage

The amount of ephemeral storage to allocate for the deployment.

Contents

kmsKeyId

Specify an AWS Key Management Service key ID to encrypt the ephemeral storage for deployment.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

DeploymentLifecycleHook

A deployment lifecycle hook runs custom logic at specific stages of the deployment process. Currently, you can use Lambda functions as hook targets.

For more information, see [Lifecycle hooks for Amazon ECS service deployments](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

hookDetails

Use this field to specify custom parameters that Amazon ECS will pass to your hook target invocations (such as a Lambda function).

Type: JSON value

Required: No

hookTargetArn

The Amazon Resource Name (ARN) of the hook target. Currently, only Lambda function ARNs are supported.

You must provide this parameter when configuring a deployment lifecycle hook.

Type: String

Required: No

lifecycleStages

The lifecycle stages at which to run the hook. Choose from these valid values:

- RECONCILE_SERVICE

The reconciliation stage that only happens when you start a new service deployment with more than 1 service revision in an ACTIVE state.

You can use a lifecycle hook for this stage.

- PRE_SCALE_UP

The green service revision has not started. The blue service revision is handling 100% of the production traffic. There is no test traffic.

You can use a lifecycle hook for this stage.

- `POST_SCALE_UP`

The green service revision has started. The blue service revision is handling 100% of the production traffic. There is no test traffic.

You can use a lifecycle hook for this stage.

- `TEST_TRAFFIC_SHIFT`

The blue and green service revisions are running. The blue service revision handles 100% of the production traffic. The green service revision is migrating from 0% to 100% of test traffic.

You can use a lifecycle hook for this stage.

- `POST_TEST_TRAFFIC_SHIFT`

The test traffic shift is complete. The green service revision handles 100% of the test traffic.

You can use a lifecycle hook for this stage.

- `PRODUCTION_TRAFFIC_SHIFT`

Production traffic is shifting to the green service revision. The green service revision is migrating from 0% to 100% of production traffic.

You can use a lifecycle hook for this stage.

- `POST_PRODUCTION_TRAFFIC_SHIFT`

The production traffic shift is complete.

You can use a lifecycle hook for this stage.

You must provide this parameter when configuring a deployment lifecycle hook.

Type: Array of strings

Valid Values: `RECONCILE_SERVICE` | `PRE_SCALE_UP` | `POST_SCALE_UP` | `TEST_TRAFFIC_SHIFT` | `POST_TEST_TRAFFIC_SHIFT` | `PRODUCTION_TRAFFIC_SHIFT` | `POST_PRODUCTION_TRAFFIC_SHIFT`

Required: No

roleArn

The Amazon Resource Name (ARN) of the IAM role that grants Amazon ECS permission to call Lambda functions on your behalf.

For more information, see [Permissions required for Lambda functions in Amazon ECS blue/green deployments](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Device

An object representing a container instance host device.

Contents

hostPath

The path for the device on the host container instance.

Type: String

Required: Yes

containerPath

The path inside the container at which to expose the host device.

Type: String

Required: No

permissions

The explicit permissions to provide to the container for the device. By default, the container has permissions for `read`, `write`, and `mknod` for the device.

Type: Array of strings

Valid Values: `read` | `write` | `mknod`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

DockerVolumeConfiguration

This parameter is specified when you're using Docker volumes. Docker volumes are only supported when you're using the EC2 launch type. Windows containers only support the use of the `local` driver. To use bind mounts, specify a host instead.

Contents

autoprovision

If this value is `true`, the Docker volume is created if it doesn't already exist.

Note

This field is only used if the scope is shared.

Type: Boolean

Required: No

driver

The Docker volume driver to use. The driver value must match the driver name provided by Docker because it is used for task placement. If the driver was installed using the Docker plugin CLI, use `docker plugin ls` to retrieve the driver name from your container instance. If the driver was installed using another method, use Docker plugin discovery to retrieve the driver name. This parameter maps to `Driver` in the `docker container create` command and the `xxdriver` option to `docker volume create`.

Type: String

Required: No

driverOpts

A map of Docker driver-specific options passed through. This parameter maps to `DriverOpts` in the `docker create-volume` command and the `xxopt` option to `docker volume create`.

Type: String to string map

Required: No

labels

Custom metadata to add to your Docker volume. This parameter maps to `Labels` in the docker container create command and the `xxlabel` option to docker volume create.

Type: String to string map

Required: No

scope

The scope for the Docker volume that determines its lifecycle. Docker volumes that are scoped to a task are automatically provisioned when the task starts and destroyed when the task stops. Docker volumes that are scoped as `shared` persist after the task stops.

Type: String

Valid Values: `task` | `shared`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

EBSTagSpecification

The tag specifications of an Amazon EBS volume.

Contents

resourceType

The type of volume resource.

Type: String

Valid Values: volume

Required: Yes

propagateTags

Determines whether to propagate the tags from the task definition to the Amazon EBS volume. Tags can only propagate to a SERVICE specified in ServiceVolumeConfiguration. If no value is specified, the tags aren't propagated.

Type: String

Valid Values: TASK_DEFINITION | SERVICE | NONE

Required: No

tags

The tags applied to this Amazon EBS volume. AmazonECSCreated and AmazonECSManaged are reserved tags that can't be used.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

EFSAuthorizationConfig

The authorization configuration details for the Amazon EFS file system.

Contents

accessPointId

The Amazon EFS access point ID to use. If an access point is specified, the root directory value specified in the `EFSVolumeConfiguration` must either be omitted or set to `/` which will enforce the path set on the EFS access point. If an access point is used, transit encryption must be on in the `EFSVolumeConfiguration`. For more information, see [Working with Amazon EFS access points](#) in the *Amazon Elastic File System User Guide*.

Type: String

Required: No

iam

Determines whether to use the Amazon ECS task role defined in a task definition when mounting the Amazon EFS file system. If it is turned on, transit encryption must be turned on in the `EFSVolumeConfiguration`. If this parameter is omitted, the default value of `DISABLED` is used. For more information, see [Using Amazon EFS access points](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Valid Values: `ENABLED` | `DISABLED`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

EFSVolumeConfiguration

This parameter is specified when you're using an Amazon Elastic File System file system for task storage. For more information, see [Amazon EFS volumes](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

fileSystemId

The Amazon EFS file system ID to use.

Type: String

Required: Yes

authorizationConfig

The authorization configuration details for the Amazon EFS file system.

Type: [EFSAuthorizationConfig](#) object

Required: No

rootDirectory

The directory within the Amazon EFS file system to mount as the root directory inside the host. If this parameter is omitted, the root of the Amazon EFS volume will be used. Specifying / will have the same effect as omitting this parameter.

Important

If an EFS access point is specified in the `authorizationConfig`, the `rootDirectory` parameter must either be omitted or set to / which will enforce the path set on the EFS access point.

Type: String

Required: No

transitEncryption

Determines whether to use encryption for Amazon EFS data in transit between the Amazon ECS host and the Amazon EFS server. Transit encryption must be turned on if Amazon EFS IAM authorization is used. If this parameter is omitted, the default value of DISABLED is used. For more information, see [Encrypting data in transit](#) in the *Amazon Elastic File System User Guide*.

Type: String

Valid Values: ENABLED | DISABLED

Required: No

transitEncryptionPort

The port to use when sending encrypted data between the Amazon ECS host and the Amazon EFS server. If you do not specify a transit encryption port, it will use the port selection strategy that the Amazon EFS mount helper uses. For more information, see [EFS mount helper](#) in the *Amazon Elastic File System User Guide*.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

EnvironmentFile

A list of files containing the environment variables to pass to a container. You can specify up to ten environment files. The file must have a `.env` file extension. Each line in an environment file should contain an environment variable in `VARIABLE=VALUE` format. Lines beginning with `#` are treated as comments and are ignored.

If there are environment variables specified using the `environment` parameter in a container definition, they take precedence over the variables contained within an environment file. If multiple environment files are specified that contain the same variable, they're processed from the top down. We recommend that you use unique variable names. For more information, see [Use a file to pass environment variables to a container](#) in the *Amazon Elastic Container Service Developer Guide*.

Environment variable files are objects in Amazon S3 and all Amazon S3 security considerations apply.

You must use the following platforms for the Fargate launch type:

- Linux platform version `1.4.0` or later.
- Windows platform version `1.0.0` or later.

Consider the following when using the Fargate launch type:

- The file is handled like a native Docker `env-file`.
- There is no support for shell escape handling.
- The container entry point interprets the `VARIABLE` values.

Contents

type

The file type to use. Environment files are objects in Amazon S3. The only supported value is `s3`.

Type: String

Valid Values: `s3`

Required: Yes

value

The Amazon Resource Name (ARN) of the Amazon S3 object containing the environment variable file.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

EphemeralStorage

The amount of ephemeral storage to allocate for the task. This parameter is used to expand the total amount of ephemeral storage available, beyond the default amount, for tasks hosted on AWS Fargate. For more information, see [Using data volumes in tasks](#) in the *Amazon ECS Developer Guide*.

Note

For tasks using the Fargate launch type, the task requires the following platforms:

- Linux platform version 1.4.0 or later.
- Windows platform version 1.0.0 or later.

Contents

sizeInGiB

The total amount, in GiB, of ephemeral storage to set for the task. The minimum supported value is 21 GiB and the maximum supported value is 200 GiB.

Type: Integer

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ExecuteCommandConfiguration

The details of the execute command configuration.

Contents

kmsKeyId

Specify an AWS Key Management Service key ID to encrypt the data between the local client and the container.

Type: String

Required: No

logConfiguration

The log configuration for the results of the execute command actions. The logs can be sent to CloudWatch Logs or an Amazon S3 bucket. When `logging=OVERRIDE` is specified, a `logConfiguration` must be provided.

Type: [ExecuteCommandLogConfiguration](#) object

Required: No

logging

The log setting to use for redirecting logs for your execute command results. The following log settings are available.

- **NONE:** The execute command session is not logged.
- **DEFAULT:** The `awslogs` configuration in the task definition is used. If no logging parameter is specified, it defaults to this value. If no `awslogs` log driver is configured in the task definition, the output won't be logged.
- **OVERRIDE:** Specify the logging details as a part of `logConfiguration`. If the `OVERRIDE` logging option is specified, the `logConfiguration` is required.

Type: String

Valid Values: NONE | DEFAULT | OVERRIDE

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ExecuteCommandLogConfiguration

The log configuration for the results of the execute command actions. The logs can be sent to CloudWatch Logs or an Amazon S3 bucket.

Contents

cloudWatchEncryptionEnabled

Determines whether to use encryption on the CloudWatch logs. If not specified, encryption will be off.

Type: Boolean

Required: No

cloudWatchLogGroupName

The name of the CloudWatch log group to send logs to.

Note

The CloudWatch log group must already be created.

Type: String

Required: No

s3BucketName

The name of the S3 bucket to send logs to.

Note

The S3 bucket must already be created.

Type: String

Required: No

s3EncryptionEnabled

Determines whether to use encryption on the S3 logs. If not specified, encryption is not used.

Type: Boolean

Required: No

s3KeyPrefix

An optional folder in the S3 bucket to place logs in.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Failure

A failed resource. For a list of common causes, see [API failure reasons](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

arn

The Amazon Resource Name (ARN) of the failed resource.

Type: String

Required: No

detail

The details of the failure.

Type: String

Required: No

reason

The reason for the failure.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

FirelensConfiguration

The FireLens configuration for the container. This is used to specify and configure a log router for container logs. For more information, see [Custom log routing](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

type

The log router to use. The valid values are `fluentd` or `fluentbit`.

Type: String

Valid Values: `fluentd` | `fluentbit`

Required: Yes

options

The options to use when configuring the log router. This field is optional and can be used to specify a custom configuration file or to add additional metadata, such as the task, task definition, cluster, and container instance details to the log event. If specified, the syntax to use is `"options":{"enable-ecs-log-metadata":"true|false", "config-file-type":"s3|file", "config-file-value":"arn:aws:s3:::mybucket/fluent.conf|filepath"}`. For more information, see [Creating a task definition that uses a FireLens configuration](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

Tasks hosted on AWS Fargate only support the file configuration file type.

Type: String to string map

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

FSxWindowsFileServerAuthorizationConfig

The authorization configuration details for Amazon FSx for Windows File Server file system. See [FSxWindowsFileServerVolumeConfiguration](#) in the *Amazon ECS API Reference*.

For more information and the input format, see [Amazon FSx for Windows File Server Volumes](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

credentialsParameter

The authorization credential option to use. The authorization credential options can be provided using either the Amazon Resource Name (ARN) of an AWS Secrets Manager secret or SSM Parameter Store parameter. The ARN refers to the stored credentials.

Type: String

Required: Yes

domain

A fully qualified domain name hosted by an [AWS Directory Service](#) Managed Microsoft AD (Active Directory) or self-hosted AD on Amazon EC2.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

FSxWindowsFileServerVolumeConfiguration

This parameter is specified when you're using [Amazon FSx for Windows File Server](#) file system for task storage.

For more information and the input format, see [Amazon FSx for Windows File Server volumes](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

authorizationConfig

The authorization configuration details for the Amazon FSx for Windows File Server file system.

Type: [FSxWindowsFileServerAuthorizationConfig](#) object

Required: Yes

fileSystemId

The Amazon FSx for Windows File Server file system ID to use.

Type: String

Required: Yes

rootDirectory

The directory within the Amazon FSx for Windows File Server file system to mount as the root directory inside the host.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)

- [AWS SDK for Ruby V3](#)

HealthCheck

An object representing a container health check. Health check parameters that are specified in a container definition override any Docker health checks that exist in the container image (such as those specified in a parent image or from the image's Dockerfile). This configuration maps to the HEALTHCHECK parameter of docker run.

Note

The Amazon ECS container agent only monitors and reports on the health checks specified in the task definition. Amazon ECS does not monitor Docker health checks that are embedded in a container image and not specified in the container definition. Health check parameters that are specified in a container definition override any Docker health checks that exist in the container image.

You can view the health status of both individual containers and a task with the DescribeTasks API operation or when viewing the task details in the console.

The health check is designed to make sure that your containers survive agent restarts, upgrades, or temporary unavailability.

Amazon ECS performs health checks on containers with the default that launched the container instance or the task.

The following describes the possible `healthStatus` values for a container:

- **HEALTHY**-The container health check has passed successfully.
- **UNHEALTHY**-The container health check has failed.
- **UNKNOWN**-The container health check is being evaluated, there's no container health check defined, or Amazon ECS doesn't have the health status of the container.

The following describes the possible `healthStatus` values based on the container health checker status of essential containers in the task with the following priority order (high to low):

- **UNHEALTHY**-One or more essential containers have failed their health check.
- **UNKNOWN**-Any essential container running within the task is in an UNKNOWN state and no other essential containers have an UNHEALTHY state.

- HEALTHY-All essential containers within the task have passed their health checks.

Consider the following task health example with 2 containers.

- If Container1 is UNHEALTHY and Container2 is UNKNOWN, the task health is UNHEALTHY.
- If Container1 is UNHEALTHY and Container2 is HEALTHY, the task health is UNHEALTHY.
- If Container1 is HEALTHY and Container2 is UNKNOWN, the task health is UNKNOWN.
- If Container1 is HEALTHY and Container2 is HEALTHY, the task health is HEALTHY.

Consider the following task health example with 3 containers.

- If Container1 is UNHEALTHY and Container2 is UNKNOWN, and Container3 is UNKNOWN, the task health is UNHEALTHY.
- If Container1 is UNHEALTHY and Container2 is UNKNOWN, and Container3 is HEALTHY, the task health is UNHEALTHY.
- If Container1 is UNHEALTHY and Container2 is HEALTHY, and Container3 is HEALTHY, the task health is UNHEALTHY.
- If Container1 is HEALTHY and Container2 is UNKNOWN, and Container3 is HEALTHY, the task health is UNKNOWN.
- If Container1 is HEALTHY and Container2 is UNKNOWN, and Container3 is UNKNOWN, the task health is UNKNOWN.
- If Container1 is HEALTHY and Container2 is HEALTHY, and Container3 is HEALTHY, the task health is HEALTHY.

If a task is run manually, and not as part of a service, the task will continue its lifecycle regardless of its health status. For tasks that are part of a service, if the task reports as unhealthy then the task will be stopped and the service scheduler will replace it.

When a container health check fails for a task that is part of a service, the following process occurs:

1. The task is marked as UNHEALTHY.
2. The unhealthy task will be stopped, and during the stopping process, it will go through the following states:

- **DEACTIVATING** - In this state, Amazon ECS performs additional steps before stopping the task. For example, for tasks that are part of services configured to use Elastic Load Balancing target groups, target groups will be deregistered in this state.
 - **STOPPING** - The task is in the process of being stopped.
 - **DEPROVISIONING** - Resources associated with the task are being cleaned up.
 - **STOPPED** - The task has been completely stopped.
3. After the old task stops, a new task will be launched to ensure service operation, and the new task will go through the following lifecycle:
- **PROVISIONING** - Resources required for the task are being provisioned.
 - **PENDING** - The task is waiting to be placed on a container instance.
 - **ACTIVATING** - In this state, Amazon ECS pulls container images, creates containers, configures task networking, registers load balancer target groups, and configures service discovery status.
 - **RUNNING** - The task is running and performing its work.

For more detailed information about task lifecycle states, see [Task lifecycle](#) in the *Amazon Elastic Container Service Developer Guide*.

The following are notes about container health check support:

- If the Amazon ECS container agent becomes disconnected from the Amazon ECS service, this won't cause a container to transition to an UNHEALTHY status. This is by design, to ensure that containers remain running during agent restarts or temporary unavailability. The health check status is the "last heard from" response from the Amazon ECS agent, so if the container was considered HEALTHY prior to the disconnect, that status will remain until the agent reconnects and another health check occurs. There are no assumptions made about the status of the container health checks.
- Container health checks require version 1.17.0 or greater of the Amazon ECS container agent. For more information, see [Updating the Amazon ECS container agent](#).
- Container health checks are supported for Fargate tasks if you're using platform version 1.1.0 or greater. For more information, see [AWS Fargate platform versions](#).
- Container health checks aren't supported for tasks that are part of a service that's configured to use a Classic Load Balancer.

For an example of how to specify a task definition with multiple containers where container dependency is specified, see [Container dependency](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

command

A string array representing the command that the container runs to determine if it is healthy. The string array must start with CMD to run the command arguments directly, or CMD-SHELL to run the command with the container's default shell.

When you use the AWS Management Console JSON panel, the AWS Command Line Interface, or the APIs, enclose the list of commands in double quotes and brackets.

```
[ "CMD-SHELL", "curl -f http://localhost/ || exit 1" ]
```

You don't include the double quotes and brackets when you use the AWS Management Console.

```
CMD-SHELL, curl -f http://localhost/ || exit 1
```

An exit code of 0 indicates success, and non-zero exit code indicates failure. For more information, see `HealthCheck` in the `docker container create` command.

Type: Array of strings

Required: Yes

interval

The time period in seconds between each health check execution. You may specify between 5 and 300 seconds. The default value is 30 seconds. This value applies only when you specify a command.

Type: Integer

Required: No

retries

The number of times to retry a failed health check before the container is considered unhealthy. You may specify between 1 and 10 retries. The default value is 3. This value applies only when you specify a command.

Type: Integer

Required: No

startPeriod

The optional grace period to provide containers time to bootstrap before failed health checks count towards the maximum number of retries. You can specify between 0 and 300 seconds. By default, the `startPeriod` is off. This value applies only when you specify a command.

Note

If a health check succeeds within the `startPeriod`, then the container is considered healthy and any subsequent failures count toward the maximum number of retries.

Type: Integer

Required: No

timeout

The time period in seconds to wait for a health check to succeed before it is considered a failure. You may specify between 2 and 60 seconds. The default value is 5. This value applies only when you specify a command.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

HostEntry

Hostnames and IP address entries that are added to the `/etc/hosts` file of a container via the `extraHosts` parameter of its [ContainerDefinition](#).

Contents

hostname

The hostname to use in the `/etc/hosts` entry.

Type: String

Required: Yes

ipAddress

The IP address to use in the `/etc/hosts` entry.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

HostVolumeProperties

Details on a container instance bind mount host volume.

Contents

sourcePath

When the host parameter is used, specify a `sourcePath` to declare the path on the host container instance that's presented to the container. If this parameter is empty, then the Docker daemon has assigned a host path for you. If the host parameter contains a `sourcePath` file location, then the data volume persists at the specified location on the host container instance until you delete it manually. If the `sourcePath` value doesn't exist on the host container instance, the Docker daemon creates it. If the location does exist, the contents of the source path folder are exported.

If you're using the Fargate launch type, the `sourcePath` parameter is not supported.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

InferenceAccelerator

Details on an Elastic Inference accelerator. For more information, see [Working with Amazon Elastic Inference on Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

deviceName

The Elastic Inference accelerator device name. The deviceName must also be referenced in a container definition as a [ResourceRequirement](#).

Type: String

Required: Yes

deviceType

The Elastic Inference accelerator type to use.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

InferenceAcceleratorOverride

Details on an Elastic Inference accelerator task override. This parameter is used to override the Elastic Inference accelerator specified in the task definition. For more information, see [Working with Amazon Elastic Inference on Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

deviceName

The Elastic Inference accelerator device name to override for the task. This parameter must match a `deviceName` specified in the task definition.

Type: String

Required: No

deviceType

The Elastic Inference accelerator type to use.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

InstanceHealthCheckResult

An object representing the result of a container instance health status check.

Contents

lastStatusChange

The Unix timestamp for when the container instance health status last changed.

Type: Timestamp

Required: No

lastUpdated

The Unix timestamp for when the container instance health status was last updated.

Type: Timestamp

Required: No

status

The container instance health status.

Type: String

Valid Values: OK | IMPAIRED | INSUFFICIENT_DATA | INITIALIZING

Required: No

type

The type of container instance health status that was verified.

Type: String

Valid Values: CONTAINER_RUNTIME

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

InstanceLaunchTemplate

The launch template configuration for Amazon ECS Managed Instances. This defines how Amazon ECS launches Amazon EC2 instances, including the instance profile for your tasks, network and storage configuration, capacity options, and instance requirements for flexible instance type selection.

Contents

ec2InstanceProfileArn

The Amazon Resource Name (ARN) of the instance profile that Amazon ECS applies to Amazon ECS Managed Instances. This instance profile must include the necessary permissions for your tasks to access AWS services and resources.

For more information, see [Amazon ECS instance profile for Managed Instances](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: Yes

networkConfiguration

The network configuration for Amazon ECS Managed Instances. This specifies the subnets and security groups that instances use for network connectivity.

Type: [ManagedInstancesNetworkConfiguration](#) object

Required: Yes

instanceRequirements

The instance requirements. You can specify:

- The instance types
- Instance requirements such as vCPU count, memory, network performance, and accelerator specifications

Amazon ECS automatically selects the instances that match the specified criteria.

Type: [InstanceRequirementsRequest](#) object

Required: No

monitoring

CloudWatch provides two categories of monitoring: basic monitoring and detailed monitoring. By default, your managed instance is configured for basic monitoring. You can optionally enable detailed monitoring to help you more quickly identify and act on operational issues. You can enable or turn off detailed monitoring at launch or when the managed instance is running or stopped. For more information, see [Detailed monitoring for Amazon ECS Managed Instances](#) in the Amazon ECS Developer Guide.

Type: String

Valid Values: BASIC | DETAILED

Required: No

storageConfiguration

The storage configuration for Amazon ECS Managed Instances. This defines the root volume size and type for the instances.

Type: [ManagedInstancesStorageConfiguration](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

InstanceLaunchTemplateUpdate

The updated launch template configuration for Amazon ECS Managed Instances. You can modify the instance profile, network configuration, storage settings, and instance requirements. Changes apply to new instances launched after the update.

For more information, see [Store instance launch parameters in Amazon EC2 launch templates](#) in the *Amazon EC2 User Guide*.

Contents

ec2InstanceProfileArn

The updated Amazon Resource Name (ARN) of the instance profile. The new instance profile must have the necessary permissions for your tasks.

For more information, see [Amazon ECS instance profile for Managed Instances](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: No

instanceRequirements

The updated instance requirements for attribute-based instance type selection. Changes to instance requirements affect which instance types Amazon ECS selects for new instances.

Type: [InstanceRequirementsRequest](#) object

Required: No

monitoring

CloudWatch provides two categories of monitoring: basic monitoring and detailed monitoring. By default, your managed instance is configured for basic monitoring. You can optionally enable detailed monitoring to help you more quickly identify and act on operational issues. You can enable or turn off detailed monitoring at launch or when the managed instance is running or stopped. For more information, see [Detailed monitoring for Amazon ECS Managed Instances](#) in the Amazon ECS Developer Guide.

Type: String

Valid Values: BASIC | DETAILED

Required: No

networkConfiguration

The updated network configuration for Amazon ECS Managed Instances. Changes to subnets and security groups affect new instances launched after the update.

Type: [ManagedInstancesNetworkConfiguration](#) object

Required: No

storageConfiguration

The updated storage configuration for Amazon ECS Managed Instances. Changes to storage settings apply to new instances launched after the update.

Type: [ManagedInstancesStorageConfiguration](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

InstanceRequirementsRequest

The instance requirements for attribute-based instance type selection. Instead of specifying exact instance types, you define requirements such as vCPU count, memory size, network performance, and accelerator specifications. Amazon ECS automatically selects Amazon EC2 instance types that match these requirements, providing flexibility and helping to mitigate capacity constraints.

Contents

memoryMiB

The minimum and maximum amount of memory in mebibytes (MiB) for the instance types. Amazon ECS selects instance types that have memory within this range.

Type: [MemoryMiBRequest](#) object

Required: Yes

vCpuCount

The minimum and maximum number of vCPUs for the instance types. Amazon ECS selects instance types that have vCPU counts within this range.

Type: [VCpuCountRangeRequest](#) object

Required: Yes

acceleratorCount

The minimum and maximum number of accelerators for the instance types. This is used when you need instances with specific numbers of GPUs or other accelerators.

Type: [AcceleratorCountRequest](#) object

Required: No

acceleratorManufacturers

The accelerator manufacturers to include. You can specify `nvidia`, `amd`, `amazon-web-services`, or `xilinx` depending on your accelerator requirements.

Type: Array of strings

Valid Values: `amazon-web-services` | `amd` | `nvidia` | `xilinx` | `habana`

Required: No

acceleratorNames

The specific accelerator names to include. For example, you can specify `a100`, `v100`, `k80`, or other specific accelerator models.

Type: Array of strings

Valid Values: `a100` | `inferentia` | `k520` | `k80` | `m60` | `radeon-pro-v520` | `t4` | `vu9p` | `v100` | `a10g` | `h100` | `t4g`

Required: No

acceleratorTotalMemoryMiB

The minimum and maximum total accelerator memory in mebibytes (MiB). This is important for GPU workloads that require specific amounts of video memory.

Type: [AcceleratorTotalMemoryMiBRequest](#) object

Required: No

acceleratorTypes

The accelerator types to include. You can specify `gpu` for graphics processing units, `fpga` for field programmable gate arrays, or `inference` for machine learning inference accelerators.

Type: Array of strings

Valid Values: `gpu` | `fpga` | `inference`

Required: No

allowedInstanceTypes

The instance types to include in the selection. When specified, Amazon ECS only considers these instance types, subject to the other requirements specified.

Type: Array of strings

Array Members: Maximum number of 400 items.

Length Constraints: Minimum length of 1. Maximum length of 30.

Pattern: `[a-zA-Z0-9\.\*\-]+`

Required: No

bareMetal

Indicates whether to include bare metal instance types. Set to `included` to allow bare metal instances, `excluded` to exclude them, or `required` to use only bare metal instances.

Type: String

Valid Values: `included` | `required` | `excluded`

Required: No

baselineEbsBandwidthMbps

The minimum and maximum baseline Amazon EBS bandwidth in megabits per second (Mbps). This is important for workloads with high storage I/O requirements.

Type: [BaselineEbsBandwidthMbpsRequest](#) object

Required: No

burstablePerformance

Indicates whether to include burstable performance instance types (T2, T3, T3a, T4g). Set to `included` to allow burstable instances, `excluded` to exclude them, or `required` to use only burstable instances.

Type: String

Valid Values: `included` | `required` | `excluded`

Required: No

cpuManufacturers

The CPU manufacturers to include or exclude. You can specify `intel`, `amd`, or `amazon-web-services` to control which CPU types are used for your workloads.

Type: Array of strings

Valid Values: `intel` | `amd` | `amazon-web-services`

Required: No

excludedInstanceTypes

The instance types to exclude from selection. Use this to prevent Amazon ECS from selecting specific instance types that may not be suitable for your workloads.

Type: Array of strings

Array Members: Maximum number of 400 items.

Length Constraints: Minimum length of 1. Maximum length of 30.

Pattern: `[a-zA-Z0-9\.\*\-]+`

Required: No

instanceGenerations

The instance generations to include. You can specify `current` to use the latest generation instances, or `previous` to include previous generation instances for cost optimization.

Type: Array of strings

Valid Values: `current` | `previous`

Required: No

localStorage

Indicates whether to include instance types with local storage. Set to `included` to allow local storage, `excluded` to exclude it, or `required` to use only instances with local storage.

Type: String

Valid Values: `included` | `required` | `excluded`

Required: No

localStorageTypes

The local storage types to include. You can specify `hdd` for hard disk drives, `ssd` for solid state drives, or `both`.

Type: Array of strings

Valid Values: `hdd` | `ssd`

Required: No

maxSpotPriceAsPercentageOfOptimalOnDemandPrice

The maximum price for Spot instances as a percentage of the optimal On-Demand price. This provides more precise cost control for Spot instance selection.

Type: Integer

Required: No

memoryGiBPerVCpu

The minimum and maximum amount of memory per vCPU in gibibytes (GiB). This helps ensure that instance types have the appropriate memory-to-CPU ratio for your workloads.

Type: [MemoryGiBPerVCpuRequest](#) object

Required: No

networkBandwidthGbps

The minimum and maximum network bandwidth in gigabits per second (Gbps). This is crucial for network-intensive workloads that require high throughput.

Type: [NetworkBandwidthGbpsRequest](#) object

Required: No

networkInterfaceCount

The minimum and maximum number of network interfaces for the instance types. This is useful for workloads that require multiple network interfaces.

Type: [NetworkInterfaceCountRequest](#) object

Required: No

onDemandMaxPricePercentageOverLowestPrice

The price protection threshold for On-Demand Instances, as a percentage higher than an identified On-Demand price. The identified On-Demand price is the price of the lowest priced current generation C, M, or R instance type with your specified attributes. If no current generation C, M, or R instance type matches your attributes, then the identified price is from either the lowest priced current generation instance types or, failing that, the lowest priced

previous generation instance types that match your attributes. When Amazon ECS selects instance types with your attributes, we will exclude instance types whose price exceeds your specified threshold.

Type: Integer

Required: No

requireHibernateSupport

Indicates whether the instance types must support hibernation. When set to `true`, only instance types that support hibernation are selected.

Type: Boolean

Required: No

spotMaxPricePercentageOverLowestPrice

The maximum price for Spot instances as a percentage over the lowest priced On-Demand instance. This helps control Spot instance costs while maintaining access to capacity.

Type: Integer

Required: No

totalLocalStorageGB

The minimum and maximum total local storage in gigabytes (GB) for instance types with local storage.

Type: [TotalLocalStorageGBRequest](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

KernelCapabilities

The Linux capabilities to add or remove from the default Docker configuration for a container defined in the task definition. For more detailed information about these Linux capabilities, see the [capabilities\(7\)](#) Linux manual page.

The following describes how Docker processes the Linux capabilities specified in the add and drop request parameters. For information about the latest behavior, see [Docker Compose: order of cap_drop and cap_add](#) in the Docker Community Forum.

- When the container is a privileged container, the container capabilities are all of the default Docker capabilities. The capabilities specified in the add request parameter, and the drop request parameter are ignored.
- When the add request parameter is set to ALL, the container capabilities are all of the default Docker capabilities, excluding those specified in the drop request parameter.
- When the drop request parameter is set to ALL, the container capabilities are the capabilities specified in the add request parameter.
- When the add request parameter and the drop request parameter are both empty, the container capabilities are all of the default Docker capabilities.
- The default is to first drop the capabilities specified in the drop request parameter, and then add the capabilities specified in the add request parameter.

Contents

add

The Linux capabilities for the container that have been added to the default configuration provided by Docker. This parameter maps to CapAdd in the docker container create command and the --cap-add option to docker run.

Note

Tasks launched on AWS Fargate only support adding the SYS_PTRACE kernel capability.

Valid values: "ALL" | "AUDIT_CONTROL" | "AUDIT_WRITE" | "BLOCK_SUSPEND"
| "CHOWN" | "DAC_OVERRIDE" | "DAC_READ_SEARCH" | "FOWNER" | "FSETID"

| "IPC_LOCK" | "IPC_OWNER" | "KILL" | "LEASE" | "LINUX_IMMUTABLE"
| "MAC_ADMIN" | "MAC_OVERRIDE" | "MKNOD" | "NET_ADMIN" |
"NET_BIND_SERVICE" | "NET_BROADCAST" | "NET_RAW" | "SETFCAP" | "SETGID"
| "SETPCAP" | "SETUID" | "SYS_ADMIN" | "SYS_BOOT" | "SYS_CHROOT" |
"SYS_MODULE" | "SYS_NICE" | "SYS_PACCT" | "SYS_PTRACE" | "SYS_RAWIO" |
"SYS_RESOURCE" | "SYS_TIME" | "SYS_TTY_CONFIG" | "SYSLOG" | "WAKE_ALARM"

Type: Array of strings

Required: No

drop

The Linux capabilities for the container that have been removed from the default configuration provided by Docker. This parameter maps to CapDrop in the docker container create command and the --cap-drop option to docker run.

Valid values: "ALL" | "AUDIT_CONTROL" | "AUDIT_WRITE" | "BLOCK_SUSPEND"
| "CHOWN" | "DAC_OVERRIDE" | "DAC_READ_SEARCH" | "FOWNER" | "FSETID"
| "IPC_LOCK" | "IPC_OWNER" | "KILL" | "LEASE" | "LINUX_IMMUTABLE"
| "MAC_ADMIN" | "MAC_OVERRIDE" | "MKNOD" | "NET_ADMIN" |
"NET_BIND_SERVICE" | "NET_BROADCAST" | "NET_RAW" | "SETFCAP" | "SETGID"
| "SETPCAP" | "SETUID" | "SYS_ADMIN" | "SYS_BOOT" | "SYS_CHROOT" |
"SYS_MODULE" | "SYS_NICE" | "SYS_PACCT" | "SYS_PTRACE" | "SYS_RAWIO" |
"SYS_RESOURCE" | "SYS_TIME" | "SYS_TTY_CONFIG" | "SYSLOG" | "WAKE_ALARM"

Type: Array of strings

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

KeyValuePair

A key-value pair object.

Contents

name

The name of the key-value pair. For environment variables, this is the name of the environment variable.

Type: String

Required: No

value

The value of the key-value pair. For environment variables, this is the value of the environment variable.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

LinearConfiguration

Configuration for linear deployment strategy that shifts production traffic in equal percentage increments with configurable wait times between each step until 100% of traffic is shifted to the new service revision. This is only valid when you run `CreateService` or `UpdateService` with `deploymentController` set to `ECS` and a `deploymentConfiguration` with a strategy set to `LINEAR`.

Contents

stepBakeTimeInMinutes

The amount of time in minutes to wait between each traffic shifting step during a linear deployment. Valid values are 0 to 1440 minutes (24 hours). The default value is 6. This bake time is not applied after reaching 100 percent traffic.

Type: Integer

Required: No

stepPercent

The percentage of production traffic to shift in each step during a linear deployment. Valid values are multiples of 0.1 from 3.0 to 100.0. The default value is 10.0.

Type: Double

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

LinuxParameters

The Linux-specific options that are applied to the container, such as Linux [KernelCapabilities](#).

Contents

capabilities

The Linux capabilities for the container that are added to or dropped from the default configuration provided by Docker.

Note

For tasks that use the Fargate launch type, `capabilities` is supported for all platform versions but the `add` parameter is only supported if using platform version 1.4.0 or later.

Type: [KernelCapabilities](#) object

Required: No

devices

Any host devices to expose to the container. This parameter maps to `Devices` in the `docker container create` command and the `--device` option to `docker run`.

Note

If you're using tasks that use the Fargate launch type, the `devices` parameter isn't supported.

Type: Array of [Device](#) objects

Required: No

initProcessEnabled

Run an `init` process inside the container that forwards signals and reaps processes. This parameter maps to the `--init` option to `docker run`. This parameter requires version 1.25

of the Docker Remote API or greater on your container instance. To check the Docker Remote API version on your container instance, log in to your container instance and run the following command: `sudo docker version --format '{{.Server.APIVersion}}'`

Type: Boolean

Required: No

maxSwap

The total amount of swap memory (in MiB) a container can use. This parameter will be translated to the `--memory-swap` option to `docker run` where the value would be the sum of the container memory plus the `maxSwap` value.

If a `maxSwap` value of 0 is specified, the container will not use swap. Accepted values are 0 or any positive integer. If the `maxSwap` parameter is omitted, the container will use the swap configuration for the container instance it is running on. A `maxSwap` value must be set for the `swappiness` parameter to be used.

Note

If you're using tasks that use the Fargate launch type, the `maxSwap` parameter isn't supported.

If you're using tasks on Amazon Linux 2023 the `swappiness` parameter isn't supported.

Type: Integer

Required: No

sharedMemorySize

The value for the size (in MiB) of the `/dev/shm` volume. This parameter maps to the `--shm-size` option to `docker run`.

Note

If you are using tasks that use the Fargate launch type, the `sharedMemorySize` parameter is not supported.

Type: Integer

Required: No

swappiness

This allows you to tune a container's memory swappiness behavior. A swappiness value of 0 will cause swapping to not happen unless absolutely necessary. A swappiness value of 100 will cause pages to be swapped very aggressively. Accepted values are whole numbers between 0 and 100. If the swappiness parameter is not specified, a default value of 60 is used. If a value is not specified for maxSwap then this parameter is ignored. This parameter maps to the `--memory-swappiness` option to `docker run`.

Note

If you're using tasks that use the Fargate launch type, the swappiness parameter isn't supported.

If you're using tasks on Amazon Linux 2023 the swappiness parameter isn't supported.

Type: Integer

Required: No

tmpfs

The container path, mount options, and size (in MiB) of the tmpfs mount. This parameter maps to the `--tmpfs` option to `docker run`.

Note

If you're using tasks that use the Fargate launch type, the tmpfs parameter isn't supported.

Type: Array of [Tmpfs](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

LoadBalancer

The load balancer configuration to use with a service or task set.

When you add, update, or remove a load balancer configuration, Amazon ECS starts a new deployment with the updated Elastic Load Balancing configuration. This causes tasks to register to and deregister from load balancers.

We recommend that you verify this on a test environment before you update the Elastic Load Balancing configuration.

A service-linked role is required for services that use multiple target groups. For more information, see [Using service-linked roles](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

advancedConfiguration

The advanced settings for the load balancer used in blue/green deployments. Specify the alternate target group, listener rules, and IAM role required for traffic shifting during blue/green deployments.

Type: [AdvancedConfiguration](#) object

Required: No

containerName

The name of the container (as it appears in a container definition) to associate with the load balancer.

You need to specify the container name when configuring the target group for an Amazon ECS load balancer.

Type: String

Required: No

containerPort

The port on the container to associate with the load balancer. This port must correspond to a `containerPort` in the task definition the tasks in the service are using. For tasks that use the

EC2 launch type, the container instance they're launched on must allow ingress traffic on the `hostPort` of the port mapping.

Type: Integer

Required: No

loadBalancerName

The name of the load balancer to associate with the Amazon ECS service or task set.

If you are using an Application Load Balancer or a Network Load Balancer the load balancer name parameter should be omitted.

Type: String

Required: No

targetGroupArn

The full Amazon Resource Name (ARN) of the Elastic Load Balancing target group or groups associated with a service or task set.

A target group ARN is only specified when using an Application Load Balancer or Network Load Balancer.

For services using the ECS deployment controller, you can specify one or multiple target groups. For more information, see [Registering multiple target groups with a service](#) in the *Amazon Elastic Container Service Developer Guide*.

For services using the CODE_DEPLOY deployment controller, you're required to define two target groups for the load balancer. For more information, see [Blue/green deployment with CodeDeploy](#) in the *Amazon Elastic Container Service Developer Guide*.

Important

If your service's task definition uses the `awsvpc` network mode, you must choose `ip` as the target type, not `instance`. Do this when creating your target groups because tasks that use the `awsvpc` network mode are associated with an elastic network interface, not an Amazon EC2 instance. This network mode is required for the Fargate launch type.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

LogConfiguration

The log configuration for the container. This parameter maps to LogConfig in the docker container create command and the `--log-driver` option to docker run.

By default, containers use the same logging driver that the Docker daemon uses. However, the container might use a different logging driver than the Docker daemon by specifying a log driver configuration in the container definition.

Understand the following when specifying a log configuration for your containers.

- Amazon ECS currently supports a subset of the logging drivers available to the Docker daemon. Additional log drivers may be available in future releases of the Amazon ECS container agent.

For tasks on AWS Fargate, the supported log drivers are `awslogs`, `splunk`, and `awsfirelens`.

For tasks hosted on Amazon EC2 instances, the supported log drivers are `awslogs`, `fluentd`, `gelf`, `json-file`, `journald`, `syslog`, `splunk`, and `awsfirelens`.

- This parameter requires version 1.18 of the Docker Remote API or greater on your container instance.
- For tasks that are hosted on Amazon EC2 instances, the Amazon ECS container agent must register the available logging drivers with the `ECS_AVAILABLE_LOGGING_DRIVERS` environment variable before containers placed on that instance can use these log configuration options. For more information, see [Amazon ECS container agent configuration](#) in the *Amazon Elastic Container Service Developer Guide*.
- For tasks that are on AWS Fargate, because you don't have access to the underlying infrastructure your tasks are hosted on, any additional software needed must be installed outside of the task. For example, the Fluentd output aggregators or a remote host running Logstash to send Gelf logs to.

Contents

logDriver

The log driver to use for the container.

For tasks on AWS Fargate, the supported log drivers are `awslogs`, `splunk`, and `awsfirelens`.

For tasks hosted on Amazon EC2 instances, the supported log drivers are `awslogs`, `fluentd`, `gelf`, `json-file`, `journald`, `syslog`, `splunk`, and `awsfirelens`.

For more information about using the `awslogs` log driver, see [Send Amazon ECS logs to CloudWatch](#) in the *Amazon Elastic Container Service Developer Guide*.

For more information about using the `awsfirelens` log driver, see [Send Amazon ECS logs to an AWS service or AWS Partner](#).

 Note

If you have a custom driver that isn't listed, you can fork the Amazon ECS container agent project that's [available on GitHub](#) and customize it to work with that driver. We encourage you to submit pull requests for changes that you would like to have included. However, we don't currently provide support for running modified copies of this software.

Type: String

Valid Values: `json-file` | `syslog` | `journald` | `gelf` | `fluentd` | `awslogs` | `splunk` | `awsfirelens`

Required: Yes

options

The configuration options to send to the log driver.

The options you can specify depend on the log driver. Some of the options you can specify when you use the `awslogs` log driver to route logs to Amazon CloudWatch include the following:

`awslogs-create-group`

Required: No

Specify whether you want the log group to be created automatically. If this option isn't specified, it defaults to `false`.

 Note

Your IAM policy must include the `logs:CreateLogGroup` permission before you attempt to use `awslogs-create-group`.

`awslogs-region`

Required: Yes

Specify the AWS Region that the `awslogs` log driver is to send your Docker logs to. You can choose to send all of your logs from clusters in different Regions to a single region in CloudWatch Logs. This is so that they're all visible in one location. Otherwise, you can separate them by Region for more granularity. Make sure that the specified log group exists in the Region that you specify with this option.

`awslogs-group`

Required: Yes

Make sure to specify a log group that the `awslogs` log driver sends its log streams to.

`awslogs-stream-prefix`

Required: Yes, when using Fargate. Optional when using EC2.

Use the `awslogs-stream-prefix` option to associate a log stream with the specified prefix, the container name, and the ID of the Amazon ECS task that the container belongs to. If you specify a prefix with this option, then the log stream takes the format `prefix-name/container-name/ecs-task-id`.

If you don't specify a prefix with this option, then the log stream is named after the container ID that's assigned by the Docker daemon on the container instance. Because it's difficult to trace logs back to the container that sent them with just the Docker container ID (which is only available on the container instance), we recommend that you specify a prefix with this option.

For Amazon ECS services, you can use the service name as the prefix. Doing so, you can trace log streams to the service that the container belongs to, the name of the container that sent them, and the ID of the task that the container belongs to.

You must specify a stream-prefix for your logs to have your logs appear in the Log pane when using the Amazon ECS console.

awslogs-datetime-format

Required: No

This option defines a multiline start pattern in Python `strftime` format. A log message consists of a line that matches the pattern and any following lines that don't match the pattern. The matched line is the delimiter between log messages.

One example of a use case for using this format is for parsing output such as a stack dump, which might otherwise be logged in multiple entries. The correct pattern allows it to be captured in a single entry.

For more information, see [awslogs-datetime-format](#).

You cannot configure both the `awslogs-datetime-format` and `awslogs-multiline-pattern` options.

Note

Multiline logging performs regular expression parsing and matching of all log messages. This might have a negative impact on logging performance.

awslogs-multiline-pattern

Required: No

This option defines a multiline start pattern that uses a regular expression. A log message consists of a line that matches the pattern and any following lines that don't match the pattern. The matched line is the delimiter between log messages.

For more information, see [awslogs-multiline-pattern](#).

This option is ignored if `awslogs-datetime-format` is also configured.

You cannot configure both the `awslogs-datetime-format` and `awslogs-multiline-pattern` options.

Note

Multiline logging performs regular expression parsing and matching of all log messages. This might have a negative impact on logging performance.

The following options apply to all supported log drivers.

mode

Required: No

Valid values: non-blocking | blocking

This option defines the delivery mode of log messages from the container to the log driver specified using `logDriver`. The delivery mode you choose affects application availability when the flow of logs from container is interrupted.

If you use the blocking mode and the flow of logs is interrupted, calls from container code to write to the `stdout` and `stderr` streams will block. The logging thread of the application will block as a result. This may cause the application to become unresponsive and lead to container healthcheck failure.

If you use the non-blocking mode, the container's logs are instead stored in an in-memory intermediate buffer configured with the `max-buffer-size` option. This prevents the application from becoming unresponsive when logs cannot be sent. We recommend using this mode if you want to ensure service availability and are okay with some log loss. For more information, see [Preventing log loss with non-blocking mode in the awslogs container log driver](#).

You can set a default mode for all containers in a specific AWS Region by using the `defaultLogDriverMode` account setting. If you don't specify the mode option or configure the account setting, Amazon ECS will default to the non-blocking mode. For more information about the account setting, see [Default log driver mode](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

On June 25, 2025, Amazon ECS changed the default log driver mode from blocking to non-blocking to prioritize task availability over logging. To continue using the blocking mode after this change, do one of the following:

- Set the mode option in your container definition's logConfiguration as blocking.
- Set the defaultLogDriverMode account setting to blocking.

max-buffer-size

Required: No

Default value: 10m

When non-blocking mode is used, the max-buffer-size log option controls the size of the buffer that's used for intermediate message storage. Make sure to specify an adequate buffer size based on your application. When the buffer fills up, further logs cannot be stored. Logs that cannot be stored are lost.

To route logs using the splunk log router, you need to specify a splunk-token and a splunk-url.

When you use the awsfirelens log router to route logs to an AWS Service or AWS Partner Network destination for log storage and analytics, you can set the log-driver-buffer-limit option to limit the number of events that are buffered in memory, before being sent to the log router container. It can help to resolve potential log loss issue because high throughput might result in memory running out for the buffer inside of Docker.

Other options you can specify when using awsfirelens to route logs depend on the destination. When you export logs to Amazon Data Firehose, you can specify the AWS Region with region and a name for the log stream with delivery_stream.

When you export logs to Amazon Kinesis Data Streams, you can specify an AWS Region with region and a data stream name with stream.

When you export logs to Amazon OpenSearch Service, you can specify options like Name, Host (OpenSearch Service endpoint without protocol), Port, Index, Type, Aws_auth, Aws_region, Suppress_Type_Name, and tls. For more information, see [Under the hood: FireLens for Amazon ECS Tasks](#).

When you export logs to Amazon S3, you can specify the bucket using the bucket option. You can also specify region, total_file_size, upload_timeout, and use_put_object as options.

This parameter requires version 1.19 of the Docker Remote API or greater on your container instance. To check the Docker Remote API version on your container instance, log in to your container instance and run the following command: `sudo docker version --format '{{.Server.APIVersion}}'`

Type: String to string map

Required: No

secretOptions

The secrets to pass to the log configuration. For more information, see [Specifying sensitive data](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Array of [Secret](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ManagedAgent

Details about the managed agent status for the container.

Contents

lastStartedAt

The Unix timestamp for the time when the managed agent was last started.

Type: Timestamp

Required: No

lastStatus

The last known status of the managed agent.

Type: String

Required: No

name

The name of the managed agent. When the execute command feature is turned on, the managed agent name is `ExecuteCommandAgent`.

Type: String

Valid Values: `ExecuteCommandAgent`

Required: No

reason

The reason for why the managed agent is in the state it is in.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ManagedAgentStateChange

An object representing a change in state for a managed agent.

Contents

containerName

The name of the container that's associated with the managed agent.

Type: String

Required: Yes

managedAgentName

The name of the managed agent.

Type: String

Valid Values: ExecuteCommandAgent

Required: Yes

status

The status of the managed agent.

Type: String

Required: Yes

reason

The reason for the status of the managed agent.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ManagedInstancesNetworkConfiguration

The network configuration for Amazon ECS Managed Instances. This specifies the VPC subnets and security groups that instances use for network connectivity. Amazon ECS Managed Instances support multiple network modes including `awsvpc` (instances receive ENIs for task isolation), `host` (instances share network namespace with tasks), and `none` (no external network connectivity), ensuring backward compatibility for migrating workloads from Fargate or Amazon EC2.

Contents

securityGroups

The list of security group IDs to apply to Amazon ECS Managed Instances. These security groups control the network traffic allowed to and from the instances.

Type: Array of strings

Required: No

subnets

The list of subnet IDs where Amazon ECS can launch Amazon ECS Managed Instances. Instances are distributed across the specified subnets for high availability. All subnets must be in the same VPC.

Type: Array of strings

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ManagedInstancesProvider

The configuration for a Amazon ECS Managed Instances provider. Amazon ECS uses this configuration to automatically launch, manage, and terminate Amazon EC2 instances on your behalf. Managed instances provide access to the full range of Amazon EC2 instance types and features while offloading infrastructure management to AWS.

Contents

infrastructureRoleArn

The Amazon Resource Name (ARN) of the infrastructure role that Amazon ECS assumes to manage instances. This role must include permissions for Amazon EC2 instance lifecycle management, networking, and any additional AWS services required for your workloads.

For more information, see [Amazon ECS infrastructure IAM role](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: No

instanceLaunchTemplate

The launch template that defines how Amazon ECS launches Amazon ECS Managed Instances. This includes the instance profile for your tasks, network and storage configuration, and instance requirements that determine which Amazon EC2 instance types can be used.

For more information, see [Store instance launch parameters in Amazon EC2 launch templates](#) in the *Amazon EC2 User Guide*.

Type: [InstanceLaunchTemplate](#) object

Required: No

propagateTags

Determines whether tags from the capacity provider are automatically applied to Amazon ECS Managed Instances. This helps with cost allocation and resource management by ensuring consistent tagging across your infrastructure.

Type: String

Valid Values: CAPACITY_PROVIDER | NONE

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ManagedInstancesStorageConfiguration

The storage configuration for Amazon ECS Managed Instances. This defines the root volume configuration for the instances.

Contents

storageSizeGiB

The size of the tasks volume.

Type: Integer

Valid Range: Minimum value of 1.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ManagedScaling

The managed scaling settings for the Auto Scaling group capacity provider.

When managed scaling is turned on, Amazon ECS manages the scale-in and scale-out actions of the Auto Scaling group. Amazon ECS manages a target tracking scaling policy using an Amazon ECS managed CloudWatch metric with the specified `targetCapacity` value as the target value for the metric. For more information, see [Using managed scaling](#) in the *Amazon Elastic Container Service Developer Guide*.

If managed scaling is off, the user must manage the scaling of the Auto Scaling group.

Contents

instanceWarmupPeriod

The period of time, in seconds, after a newly launched Amazon EC2 instance can contribute to CloudWatch metrics for Auto Scaling group. If this parameter is omitted, the default value of 300 seconds is used.

Type: Integer

Valid Range: Minimum value of 0. Maximum value of 10000.

Required: No

maximumScalingStepSize

The maximum number of Amazon EC2 instances that Amazon ECS will scale out at one time. If this parameter is omitted, the default value of 10000 is used.

Type: Integer

Valid Range: Minimum value of 1. Maximum value of 10000.

Required: No

minimumScalingStepSize

The minimum number of Amazon EC2 instances that Amazon ECS will scale out at one time. The scale in process is not affected by this parameter. If this parameter is omitted, the default value of 1 is used.

When additional capacity is required, Amazon ECS will scale up the minimum scaling step size even if the actual demand is less than the minimum scaling step size.

Type: Integer

Valid Range: Minimum value of 1. Maximum value of 10000.

Required: No

status

Determines whether to use managed scaling for the capacity provider.

Type: String

Valid Values: ENABLED | DISABLED

Required: No

targetCapacity

The target capacity utilization as a percentage for the capacity provider. The specified value must be greater than 0 and less than or equal to 100. For example, if you want the capacity provider to maintain 10% spare capacity, then that means the utilization is 90%, so use a `targetCapacity` of 90. The default value of 100 percent results in the Amazon EC2 instances in your Auto Scaling group being completely used.

Type: Integer

Valid Range: Minimum value of 1. Maximum value of 100.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ManagedStorageConfiguration

The managed storage configuration for the cluster.

Contents

fargateEphemeralStorageKmsKeyId

Specify the AWS Key Management Service key ID for Fargate ephemeral storage.

When you specify a `fargateEphemeralStorageKmsKeyId`, AWS Fargate uses the key to encrypt data at rest in ephemeral storage. For more information about Fargate ephemeral storage encryption, see [Customer managed keys for AWS Fargate ephemeral storage for Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*.

The key must be a single Region key.

Type: String

Required: No

kmsKeyId

Specify a AWS Key Management Service key ID to encrypt Amazon ECS managed storage.

When you specify a `kmsKeyId`, Amazon ECS uses the key to encrypt data volumes managed by Amazon ECS that are attached to tasks in the cluster. The following data volumes are managed by Amazon ECS: Amazon EBS. For more information about encryption of Amazon EBS volumes attached to Amazon ECS tasks, see [Encrypt data stored in Amazon EBS volumes for Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*.

The key must be a single Region key.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MemoryGiBPerVCpuRequest

The minimum and maximum amount of memory per vCPU in gibibytes (GiB). This helps ensure that instance types have the appropriate memory-to-CPU ratio for your workloads.

Contents

max

The maximum amount of memory per vCPU in GiB. Instance types with a higher memory-to-vCPU ratio are excluded from selection.

Type: Double

Required: No

min

The minimum amount of memory per vCPU in GiB. Instance types with a lower memory-to-vCPU ratio are excluded from selection.

Type: Double

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MemoryMiBRequest

The minimum and maximum amount of memory in mebibytes (MiB) for instance type selection. This ensures that selected instance types have adequate memory for your workloads.

Contents

min

The minimum amount of memory in MiB. Instance types with less memory than this value are excluded from selection.

Type: Integer

Required: Yes

max

The maximum amount of memory in MiB. Instance types with more memory than this value are excluded from selection.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

MountPoint

The details for a volume mount point that's used in a container definition.

Contents

containerPath

The path on the container to mount the host volume at.

Type: String

Required: No

readOnly

If this value is `true`, the container has read-only access to the volume. If this value is `false`, then the container can write to the volume. The default value is `false`.

Type: Boolean

Required: No

sourceVolume

The name of the volume to mount. Must be a volume name referenced in the `name` parameter of task definition `volume`.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

NetworkBandwidthGbpsRequest

The minimum and maximum network bandwidth in gigabits per second (Gbps) for instance type selection. This is important for network-intensive workloads.

Contents

max

The maximum network bandwidth in Gbps. Instance types with higher network bandwidth are excluded from selection.

Type: Double

Required: No

min

The minimum network bandwidth in Gbps. Instance types with lower network bandwidth are excluded from selection.

Type: Double

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

NetworkBinding

Details on the network bindings between a container and its host container instance. After a task reaches the RUNNING status, manual and automatic host and container port assignments are visible in the networkBindings section of [DescribeTasks](#) API responses.

Contents

bindIP

The IP address that the container is bound to on the container instance.

Type: String

Required: No

containerPort

The port number on the container that's used with the network binding.

Type: Integer

Required: No

containerPortRange

The port number range on the container that's bound to the dynamically mapped host port range.

The following rules apply when you specify a containerPortRange:

- You must use either the bridge network mode or the awsvpc network mode.
- This parameter is available for both the EC2 and AWS Fargate launch types.
- This parameter is available for both the Linux and Windows operating systems.
- The container instance must have at least version 1.67.0 of the container agent and at least version 1.67.0-1 of the ecs-init package
- You can specify a maximum of 100 port ranges per container.
- You do not specify a hostPortRange. The value of the hostPortRange is set as follows:
 - For containers in a task with the awsvpc network mode, the hostPortRange is set to the same value as the containerPortRange. This is a static mapping strategy.

- For containers in a task with the bridge network mode, the Amazon ECS agent finds open host ports from the default ephemeral range and passes it to docker to bind them to the container ports.
- The containerPortRange valid values are between 1 and 65535.
- A port can only be included in one port mapping per container.
- You cannot specify overlapping port ranges.
- The first port in the range must be less than last port in the range.
- Docker recommends that you turn off the docker-proxy in the Docker daemon config file when you have a large number of ports.

For more information, see [Issue #11185](#) on the Github website.

For information about how to turn off the docker-proxy in the Docker daemon config file, see [Docker daemon](#) in the *Amazon ECS Developer Guide*.

You can call [DescribeTasks](#) to view the hostPortRange which are the host ports that are bound to the container ports.

Type: String

Required: No

hostPort

The port number on the host that's used with the network binding.

Type: Integer

Required: No

hostPortRange

The port number range on the host that's used with the network binding. This is assigned is assigned by Docker and delivered by the Amazon ECS agent.

Type: String

Required: No

protocol

The protocol used for the network binding.

Type: String

Valid Values: tcp | udp

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

NetworkConfiguration

The network configuration for a task or service.

Contents

awsvpcConfiguration

The VPC subnets and security groups that are associated with a task.

Note

All specified subnets and security groups must be from the same VPC.

Type: [AwsVpcConfiguration](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

NetworkInterface

An object representing the elastic network interface for tasks that use the awsvpc network mode.

Contents

attachmentId

The attachment ID for the network interface.

Type: String

Required: No

ipv6Address

The private IPv6 address for the network interface.

Type: String

Required: No

privateIpv4Address

The private IPv4 address for the network interface.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

NetworkInterfaceCountRequest

The minimum and maximum number of network interfaces for instance type selection. This is useful for workloads that require multiple network interfaces.

Contents

max

The maximum number of network interfaces. Instance types that support more network interfaces are excluded from selection.

Type: Integer

Required: No

min

The minimum number of network interfaces. Instance types that support fewer network interfaces are excluded from selection.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

PlacementConstraint

An object representing a constraint on task placement. For more information, see [Task placement constraints](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

If you're using the Fargate launch type, task placement constraints aren't supported.

Contents

expression

A cluster query language expression to apply to the constraint. The expression can have a maximum length of 2000 characters. You can't specify an expression if the constraint type is `distinctInstance`. For more information, see [Cluster query language](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

type

The type of constraint. Use `distinctInstance` to ensure that each task in a particular group is running on a different container instance. Use `memberOf` to restrict the selection to a group of valid candidates.

Type: String

Valid Values: `distinctInstance` | `memberOf`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

PlacementStrategy

The task placement strategy for a task or service. For more information, see [Task placement strategies](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

field

The field to apply the placement strategy against. For the spread placement strategy, valid values are `instanceId` (or `host`, which has the same effect), or any platform or custom attribute that's applied to a container instance, such as `attribute:ecs.availability-zone`. For the binpack placement strategy, valid values are `cpu` and `memory`. For the random placement strategy, this field is not used.

Type: String

Required: No

type

The type of placement strategy. The `random` placement strategy randomly places tasks on available candidates. The `spread` placement strategy spreads placement across available candidates evenly based on the `field` parameter. The `binpack` strategy places tasks on available candidates that have the least available amount of the resource that's specified with the `field` parameter. For example, if you binpack on `memory`, a task is placed on the instance with the least amount of remaining memory but still enough to run the task.

Type: String

Valid Values: `random` | `spread` | `binpack`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

PlatformDevice

The devices that are available on the container instance. The only supported device type is a GPU.

Contents

id

The ID for the GPUs on the container instance. The available GPU IDs can also be obtained on the container instance in the `/var/lib/ecs/gpu/nvidia_gpu_info.json` file.

Type: String

Required: Yes

type

The type of device that's available on the container instance. The only supported value is GPU.

Type: String

Valid Values: GPU

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

PortMapping

Port mappings allow containers to access ports on the host container instance to send or receive traffic. Port mappings are specified as part of the container definition.

If you use containers in a task with the `awsvpc` or `host` network mode, specify the exposed ports using `containerPort`. The `hostPort` can be left blank or it must be the same value as the `containerPort`.

Most fields of this parameter (`containerPort`, `hostPort`, `protocol`) maps to `PortBindings` in the `docker` container `create` command and the `--publish` option to `docker run`. If the network mode of a task definition is set to `host`, host ports must either be undefined or match the container port in the port mapping.

Note

You can't expose the same container port for multiple protocols. If you attempt this, an error is returned.

After a task reaches the `RUNNING` status, manual and automatic host and container port assignments are visible in the `networkBindings` section of [DescribeTasks](#) API responses.

Contents

appProtocol

The application protocol that's used for the port mapping. This parameter only applies to Service Connect. We recommend that you set this parameter to be consistent with the protocol that your application uses. If you set this parameter, Amazon ECS adds protocol-specific connection handling to the Service Connect proxy. If you set this parameter, Amazon ECS adds protocol-specific telemetry in the Amazon ECS console and CloudWatch.

If you don't set a value for this parameter, then TCP is used. However, Amazon ECS doesn't add protocol-specific telemetry for TCP.

`appProtocol` is immutable in a Service Connect service. Updating this field requires a service deletion and redeployment.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Valid Values: `http` | `http2` | `grpc`

Required: No

containerPort

The port number on the container that's bound to the user-specified or automatically assigned host port.

If you use containers in a task with the `awsvpc` or `host` network mode, specify the exposed ports using `containerPort`.

If you use containers in a task with the `bridge` network mode and you specify a container port and not a host port, your container automatically receives a host port in the ephemeral port range. For more information, see `hostPort`. Port mappings that are automatically assigned in this way do not count toward the 100 reserved ports limit of a container instance.

Type: Integer

Required: No

containerPortRange

The port number range on the container that's bound to the dynamically mapped host port range.

The following rules apply when you specify a `containerPortRange`:

- You must use either the `bridge` network mode or the `awsvpc` network mode.
- This parameter is available for both the EC2 and AWS Fargate launch types.
- This parameter is available for both the Linux and Windows operating systems.
- The container instance must have at least version 1.67.0 of the container agent and at least version 1.67.0-1 of the `ecs-init` package
- You can specify a maximum of 100 port ranges per container.

- You do not specify a `hostPortRange`. The value of the `hostPortRange` is set as follows:
 - For containers in a task with the `awsvpc` network mode, the `hostPortRange` is set to the same value as the `containerPortRange`. This is a static mapping strategy.
 - For containers in a task with the `bridge` network mode, the Amazon ECS agent finds open host ports from the default ephemeral range and passes it to docker to bind them to the container ports.
- The `containerPortRange` valid values are between 1 and 65535.
- A port can only be included in one port mapping per container.
- You cannot specify overlapping port ranges.
- The first port in the range must be less than last port in the range.
- Docker recommends that you turn off the `docker-proxy` in the Docker daemon config file when you have a large number of ports.

For more information, see [Issue #11185](#) on the Github website.

For information about how to turn off the `docker-proxy` in the Docker daemon config file, see [Docker daemon](#) in the *Amazon ECS Developer Guide*.

You can call [DescribeTasks](#) to view the `hostPortRange` which are the host ports that are bound to the container ports.

Type: String

Required: No

hostPort

The port number on the container instance to reserve for your container.

If you specify a `containerPortRange`, leave this field empty and the value of the `hostPort` is set as follows:

- For containers in a task with the `awsvpc` network mode, the `hostPort` is set to the same value as the `containerPort`. This is a static mapping strategy.
- For containers in a task with the `bridge` network mode, the Amazon ECS agent finds open ports on the host and automatically binds them to the container ports. This is a dynamic mapping strategy.

If you use containers in a task with the `awsvpc` or `host` network mode, the `hostPort` can either be left blank or set to the same value as the `containerPort`.

If you use containers in a task with the bridge network mode, you can specify a non-reserved host port for your container port mapping, or you can omit the `hostPort` (or set it to 0) while specifying a `containerPort` and your container automatically receives a port in the ephemeral port range for your container instance operating system and Docker version.

The default ephemeral port range for Docker version 1.6.0 and later is listed on the instance under `/proc/sys/net/ipv4/ip_local_port_range`. If this kernel parameter is unavailable, the default ephemeral port range from 49153 through 65535 (Linux) or 49152 through 65535 (Windows) is used. Do not attempt to specify a host port in the ephemeral port range as these are reserved for automatic assignment. In general, ports below 32768 are outside of the ephemeral port range.

The default reserved ports are 22 for SSH, the Docker ports 2375 and 2376, and the Amazon ECS container agent ports 51678-51680. Any host port that was previously specified in a running task is also reserved while the task is running. That is, after a task stops, the host port is released. The current reserved ports are displayed in the `remainingResources` of [DescribeContainerInstances](#) output. A container instance can have up to 100 reserved ports at a time. This number includes the default reserved ports. Automatically assigned ports aren't included in the 100 reserved ports quota.

Type: Integer

Required: No

name

The name that's used for the port mapping. This parameter is the name that you use in the `serviceConnectConfiguration` and the `vpcLatticeConfigurations` of a service. The name can include up to 64 characters. The characters can include lowercase letters, numbers, underscores (`_`), and hyphens (`-`). The name can't start with a hyphen.

Type: String

Required: No

protocol

The protocol used for the port mapping. Valid values are `tcp` and `udp`. The default is `tcp`. `protocol` is immutable in a Service Connect service. Updating this field requires a service deletion and redeployment.

Type: String

Valid Values: tcp | udp

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ProtectedTask

An object representing the protection status details for a task. You can set the protection status with the [UpdateTaskProtection](#) API and get the status of tasks with the [GetTaskProtection](#) API.

Contents

expirationDate

The epoch time when protection for the task will expire.

Type: Timestamp

Required: No

protectionEnabled

The protection status of the task. If scale-in protection is on for a task, the value is `true`. Otherwise, it is `false`.

Type: Boolean

Required: No

taskArn

The task ARN.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ProxyConfiguration

The configuration details for the App Mesh proxy.

For tasks that use the EC2 launch type, the container instances require at least version 1.26.0 of the container agent and at least version 1.26.0-1 of the `ecs-init` package to use a proxy configuration. If your container instances are launched from the Amazon ECS optimized AMI version 20190301 or later, then they contain the required versions of the container agent and `ecs-init`. For more information, see [Amazon ECS-optimized Linux AMI](#)

Contents

containerName

The name of the container that will serve as the App Mesh proxy.

Type: String

Required: Yes

properties

The set of network configuration parameters to provide the Container Network Interface (CNI) plugin, specified as key-value pairs.

- **IgnoredUID** - (Required) The user ID (UID) of the proxy container as defined by the `user` parameter in a container definition. This is used to ensure the proxy ignores its own traffic. If **IgnoredGID** is specified, this field can be empty.
- **IgnoredGID** - (Required) The group ID (GID) of the proxy container as defined by the `user` parameter in a container definition. This is used to ensure the proxy ignores its own traffic. If **IgnoredUID** is specified, this field can be empty.
- **AppPorts** - (Required) The list of ports that the application uses. Network traffic to these ports is forwarded to the **ProxyIngressPort** and **ProxyEgressPort**.
- **ProxyIngressPort** - (Required) Specifies the port that incoming traffic to the **AppPorts** is directed to.
- **ProxyEgressPort** - (Required) Specifies the port that outgoing traffic from the **AppPorts** is directed to.
- **EgressIgnoredPorts** - (Required) The egress traffic going to the specified ports is ignored and not redirected to the **ProxyEgressPort**. It can be an empty list.

- **EgressIgnoredIPs** - (Required) The egress traffic going to the specified IP addresses is ignored and not redirected to the `ProxyEgressPort`. It can be an empty list.

Type: Array of [KeyValuePair](#) objects

Required: No

type

The proxy type. The only supported value is `APPMESH`.

Type: String

Valid Values: `APPMESH`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

RepositoryCredentials

The repository credentials for private registry authentication.

Contents

credentialsParameter

The Amazon Resource Name (ARN) of the secret containing the private repository credentials.

Note

When you use the Amazon ECS API, AWS CLI, or AWS SDK, if the secret exists in the same Region as the task that you're launching then you can use either the full ARN or the name of the secret. When you use the AWS Management Console, you must specify the full ARN of the secret.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ResolvedConfiguration

The resolved configuration for a service revision, which contains the actual resources your service revision uses, such as which target groups serve traffic.

Contents

loadBalancers

The resolved load balancer configuration for the service revision. This includes information about which target groups serve traffic and which listener rules direct traffic to them.

Type: Array of [ServiceRevisionLoadBalancer](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Resource

Describes the resources available for a container instance.

Contents

doubleValue

When the `doubleValue` type is set, the value of the resource must be a double precision floating-point type.

Type: Double

Required: No

integerValue

When the `integerValue` type is set, the value of the resource must be an integer.

Type: Integer

Required: No

longValue

When the `longValue` type is set, the value of the resource must be an extended precision floating-point type.

Type: Long

Required: No

name

The name of the resource, such as CPU, MEMORY, PORTS, PORTS_UDP, or a user-defined resource.

Type: String

Required: No

stringSetValue

When the `stringSetValue` type is set, the value of the resource must be a string type.

Type: Array of strings

Required: No

type

The type of the resource. Valid values: INTEGER, DOUBLE, LONG, or STRINGSET.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ResourceRequirement

The type and amount of a resource to assign to a container. The supported resource types are GPUs and Elastic Inference accelerators. For more information, see [Working with GPUs on Amazon ECS](#) or [Working with Amazon Elastic Inference on Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*

Contents

type

The type of resource to assign to a container.

Type: String

Valid Values: GPU | InferenceAccelerator

Required: Yes

value

The value for the specified resource type.

When the type is GPU, the value is the number of physical GPUs the Amazon ECS container agent reserves for the container. The number of GPUs that's reserved for all containers in a task can't exceed the number of available GPUs on the container instance that the task is launched on.

When the type is InferenceAccelerator, the value matches the `deviceName` for an [InferenceAccelerator](#) specified in a task definition.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Rollback

Information about the service deployment rollback.

Contents

reason

The reason the rollback happened. For example, the circuit breaker initiated the rollback operation.

Type: String

Required: No

serviceRevisionArn

The ARN of the service revision deployed as part of the rollback.

Type: String

Required: No

startedAt

Time time that the rollback started. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: Timestamp

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

RuntimePlatform

Information about the platform for the Amazon ECS service or task.

For more information about RuntimePlatform, see [RuntimePlatform](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

cpuArchitecture

The CPU architecture.

You can run your Linux tasks on an ARM-based platform by setting the value to ARM64. This option is available for tasks that run on Linux Amazon EC2 instance, Amazon ECS Managed Instances, or Linux containers on Fargate.

Type: String

Valid Values: X86_64 | ARM64

Required: No

operatingSystemFamily

The operating system.

Type: String

Valid Values: WINDOWS_SERVER_2019_FULL | WINDOWS_SERVER_2019_CORE
| WINDOWS_SERVER_2016_FULL | WINDOWS_SERVER_2004_CORE
| WINDOWS_SERVER_2022_CORE | WINDOWS_SERVER_2022_FULL
| WINDOWS_SERVER_2025_CORE | WINDOWS_SERVER_2025_FULL |
WINDOWS_SERVER_20H2_CORE | LINUX

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Scale

A floating-point percentage of the desired number of tasks to place and keep running in the task set.

Contents

unit

The unit of measure for the scale value.

Type: String

Valid Values: PERCENT

Required: No

value

The value, specified as a percent total of a service's `desiredCount`, to scale the task set. Accepted values are numbers between 0 and 100.

Type: Double

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Secret

An object representing the secret to expose to your container. Secrets can be exposed to a container in the following ways:

- To inject sensitive data into your containers as environment variables, use the `secrets` container definition parameter.
- To reference sensitive information in the log configuration of a container, use the `secretOptions` container definition parameter.

For more information, see [Specifying sensitive data](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

name

The name of the secret.

Type: String

Required: Yes

valueFrom

The secret to expose to the container. The supported values are either the full ARN of the AWS Secrets Manager secret or the full ARN of the parameter in the SSM Parameter Store.

For information about the required AWS Identity and Access Management permissions, see [Required IAM permissions for Amazon ECS secrets](#) (for Secrets Manager) or [Required IAM permissions for Amazon ECS secrets](#) (for Systems Manager Parameter store) in the *Amazon Elastic Container Service Developer Guide*.

Note

If the SSM Parameter Store parameter exists in the same Region as the task you're launching, then you can use either the full ARN or name of the parameter. If the parameter exists in a different Region, then the full ARN must be specified.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Service

Details on a service within a cluster.

Contents

availabilityZoneRebalancing

Indicates whether to use Availability Zone rebalancing for the service.

For more information, see [Balancing an Amazon ECS service across Availability Zones](#) in the Amazon Elastic Container Service Developer Guide .

The default behavior of AvailabilityZoneRebalancing differs between create and update requests:

- For create service requests, when no value is specified for AvailabilityZoneRebalancing, Amazon ECS defaults the value to ENABLED.
- For update service requests, when no value is specified for AvailabilityZoneRebalancing, Amazon ECS defaults to the existing service's AvailabilityZoneRebalancing value. If the service never had an AvailabilityZoneRebalancing value set, Amazon ECS treats this as DISABLED.

Type: String

Valid Values: ENABLED | DISABLED

Required: No

capacityProviderStrategy

The capacity provider strategy the service uses. When using the DescribeServices API, this field is omitted if the service was created using a launch type.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: No

clusterArn

The Amazon Resource Name (ARN) of the cluster that hosts the service.

Type: String

Required: No

createdAt

The Unix timestamp for the time when the service was created.

Type: Timestamp

Required: No

createdBy

The principal that created the service.

Type: String

Required: No

deploymentConfiguration

Optional deployment parameters that control how many tasks run during the deployment and the ordering of stopping and starting tasks.

Type: [DeploymentConfiguration](#) object

Required: No

deploymentController

The deployment controller type the service is using.

Type: [DeploymentController](#) object

Required: No

deployments

The current state of deployments for the service.

Type: Array of [Deployment](#) objects

Required: No

desiredCount

The desired number of instantiations of the task definition to keep running on the service. This value is specified when the service is created with [CreateService](#) , and it can be modified with [UpdateService](#).

Type: Integer

Required: No

enableECSTags

Determines whether to use Amazon ECS managed tags for the tasks in the service. For more information, see [Tagging Your Amazon ECS Resources](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Boolean

Required: No

enableExecuteCommand

Determines whether the execute command functionality is turned on for the service. If `true`, the execute command functionality is turned on for all containers in tasks as part of the service.

Type: Boolean

Required: No

events

The event stream for your service. A maximum of 100 of the latest events are displayed.

Type: Array of [ServiceEvent](#) objects

Required: No

healthCheckGracePeriodSeconds

The period of time, in seconds, that the Amazon ECS service scheduler ignores unhealthy Elastic Load Balancing, VPC Lattice, and container health checks after a task has first started.

If your service has more running tasks than desired, unhealthy tasks in the grace period might be stopped to reach the desired count.

Type: Integer

Required: No

launchType

The launch type the service is using. When using the `DescribeServices` API, this field is omitted if the service was created using a capacity provider strategy.

Type: String

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

loadBalancers

A list of Elastic Load Balancing load balancer objects. It contains the load balancer name, the container name, and the container port to access from the load balancer. The container name is as it appears in a container definition.

Type: Array of [LoadBalancer](#) objects

Required: No

networkConfiguration

The VPC subnet and security group configuration for tasks that receive their own elastic network interface by using the awsvpc networking mode.

Type: [NetworkConfiguration](#) object

Required: No

pendingCount

The number of tasks in the cluster that are in the PENDING state.

Type: Integer

Required: No

placementConstraints

The placement constraints for the tasks in the service.

Type: Array of [PlacementConstraint](#) objects

Required: No

placementStrategy

The placement strategy that determines how tasks for the service are placed.

Type: Array of [PlacementStrategy](#) objects

Required: No

platformFamily

The operating system that your tasks in the service run on. A platform family is specified only for tasks using the Fargate launch type.

All tasks that run as part of this service must use the same `platformFamily` value as the service (for example, `LINUX`).

Type: String

Required: No

platformVersion

The platform version to run your service on. A platform version is only specified for tasks that are hosted on AWS Fargate. If one isn't specified, the `LATEST` platform version is used. For more information, see [AWS Fargate Platform Versions](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

propagateTags

Determines whether to propagate the tags from the task definition or the service to the task. If no value is specified, the tags aren't propagated.

Type: String

Valid Values: `TASK_DEFINITION` | `SERVICE` | `NONE`

Required: No

roleArn

The ARN of the IAM role that's associated with the service. It allows the Amazon ECS container agent to register container instances with an Elastic Load Balancing load balancer.

Type: String

Required: No

runningCount

The number of tasks in the cluster that are in the `RUNNING` state.

Type: Integer

Required: No

schedulingStrategy

The scheduling strategy to use for the service. For more information, see [Services](#).

There are two service scheduler strategies available.

- **REPLICA**-The replica scheduling strategy places and maintains the desired number of tasks across your cluster. By default, the service scheduler spreads tasks across Availability Zones. You can use task placement strategies and constraints to customize task placement decisions.
- **DAEMON**-The daemon scheduling strategy deploys exactly one task on each active container instance. This task meets all of the task placement constraints that you specify in your cluster. The service scheduler also evaluates the task placement constraints for running tasks. It stops tasks that don't meet the placement constraints.



Note

Fargate tasks don't support the DAEMON scheduling strategy.

Type: String

Valid Values: REPLICA | DAEMON

Required: No

serviceArn

The ARN that identifies the service. For more information about the ARN format, see [Amazon Resource Name \(ARN\)](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: No

serviceName

The name of your service. Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed. Service names must be unique within a cluster. However, you can have similarly named services in multiple clusters within a Region or across multiple Regions.

Type: String

Required: No

serviceRegistries

The details for the service discovery registries to assign to this service. For more information, see [Service Discovery](#).

Type: Array of [ServiceRegistry](#) objects

Required: No

status

The status of the service. The valid values are ACTIVE, DRAINING, or INACTIVE.

Type: String

Required: No

tags

The metadata that you apply to the service to help you categorize and organize them. Each tag consists of a key and an optional value. You define both the key and value.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws :`, `AWS :`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

taskDefinition

The task definition to use for tasks in the service. This value is specified when the service is created with [CreateService](#), and it can be modified with [UpdateService](#).

Type: String

Required: No

taskSets

Information about a set of Amazon ECS tasks in either an AWS CodeDeploy or an EXTERNAL deployment. An Amazon ECS task set includes details such as the desired number of tasks, how many tasks are running, and whether the task set serves production traffic.

Type: Array of [TaskSet](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceConnectAccessLogConfiguration

Configuration for Service Connect access logging. Access logs provide detailed information about requests made to your service, including request patterns, response codes, and timing data for debugging and monitoring purposes.

Note

To enable access logs, you must also specify a `logConfiguration` in the `serviceConnectConfiguration`.

Contents

format

The format for Service Connect access log output. Choose `TEXT` for human-readable logs or `JSON` for structured data that integrates well with log analysis tools.

Type: String

Valid Values: `TEXT` | `JSON`

Required: Yes

includeQueryParameters

Specifies whether to include query parameters in Service Connect access logs.

When enabled, query parameters from HTTP requests are included in the access logs. Consider security and privacy implications when enabling this feature, as query parameters may contain sensitive information such as request IDs and tokens. By default, this parameter is `DISABLED`.

Type: String

Valid Values: `DISABLED` | `ENABLED`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceConnectClientAlias

Each alias ("endpoint") is a fully-qualified name and port number that other tasks ("clients") can use to connect to this service.

Each name and port mapping must be unique within the namespace.

Tasks that run in a namespace can use short names to connect to services in the namespace.

Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

port

The listening port number for the Service Connect proxy. This port is available inside of all of the tasks within the same namespace.

To avoid changing your applications in client Amazon ECS services, set this to the same port that the client application uses by default. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Integer

Valid Range: Minimum value of 0. Maximum value of 65535.

Required: Yes

dnsName

The `dnsName` is the name that you use in the applications of client tasks to connect to this service. The name must be a valid DNS name but doesn't need to be fully-qualified. The name can include up to 127 characters. The name can include lowercase letters, numbers, underscores (`_`), hyphens (`-`), and periods (`.`). The name can't start with a hyphen.

If this parameter isn't specified, the default value of `discoveryName.namespace` is used. If the `discoveryName` isn't specified, the port mapping name from the task definition is used in `portName.namespace`.

To avoid changing your applications in client Amazon ECS services, set this to the same name that the client application uses by default. For example, a few common names are database, db, or the lowercase name of a database, such as `mysql` or `redis`. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

testTrafficRules

The configuration for test traffic routing rules used during blue/green deployments with Amazon ECS Service Connect. This allows you to route a portion of traffic to the new service revision of your service for testing before shifting all production traffic.

Type: [ServiceConnectTestTrafficRules](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceConnectConfiguration

The Service Connect configuration of your Amazon ECS service. The configuration for this service to discover and connect to services, and be discovered by, and connected from, other services within a namespace.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

enabled

Specifies whether to use Service Connect with this service.

Type: Boolean

Required: Yes

accessLogConfiguration

The configuration for Service Connect access logging. Access logs capture detailed information about requests made to your service, including request patterns, response codes, and timing data. They can be useful for debugging connectivity issues, monitoring service performance, and auditing service-to-service communication for security and compliance purposes.

Note

To enable access logs, you must also specify a `logConfiguration` in the `serviceConnectConfiguration`.

Type: [ServiceConnectAccessLogConfiguration](#) object

Required: No

logConfiguration

The log configuration for the container. This parameter maps to LogConfig in the docker container create command and the `--log-driver` option to docker run.

By default, containers use the same logging driver that the Docker daemon uses. However, the container might use a different logging driver than the Docker daemon by specifying a log driver configuration in the container definition.

Understand the following when specifying a log configuration for your containers.

- Amazon ECS currently supports a subset of the logging drivers available to the Docker daemon. Additional log drivers may be available in future releases of the Amazon ECS container agent.

For tasks on AWS Fargate, the supported log drivers are `awslogs`, `splunk`, and `awsfirelens`.

For tasks hosted on Amazon EC2 instances, the supported log drivers are `awslogs`, `fluentd`, `gelf`, `json-file`, `journald`, `syslog`, `splunk`, and `awsfirelens`.

- This parameter requires version 1.18 of the Docker Remote API or greater on your container instance.
- For tasks that are hosted on Amazon EC2 instances, the Amazon ECS container agent must register the available logging drivers with the `ECS_AVAILABLE_LOGGING_DRIVERS` environment variable before containers placed on that instance can use these log configuration options. For more information, see [Amazon ECS container agent configuration](#) in the *Amazon Elastic Container Service Developer Guide*.
- For tasks that are on AWS Fargate, because you don't have access to the underlying infrastructure your tasks are hosted on, any additional software needed must be installed outside of the task. For example, the Fluentd output aggregators or a remote host running Logstash to send Gelf logs to.

Type: [LogConfiguration](#) object

Required: No

namespace

The namespace name or full Amazon Resource Name (ARN) of the AWS Cloud Map namespace for use with Service Connect. The namespace must be in the same AWS Region as the Amazon

ECS service and cluster. The type of namespace doesn't affect Service Connect. For more information about AWS Cloud Map, see [Working with Services](#) in the *AWS Cloud Map Developer Guide*.

Type: String

Required: No

services

The list of Service Connect service objects. These are names and aliases (also known as endpoints) that are used by other Amazon ECS services to connect to this service.

This field is not required for a "client" Amazon ECS service that's a member of a namespace only to connect to other services within the namespace. An example of this would be a frontend application that accepts incoming requests from either a load balancer that's attached to the service or by other means.

An object selects a port from the task definition, assigns a name for the AWS Cloud Map service, and a list of aliases (endpoints) and ports for client applications to refer to this service.

Type: Array of [ServiceConnectService](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceConnectService

The Service Connect service object configuration. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

portName

The `portName` must match the name of one of the `portMappings` from all the containers in the task definition of this Amazon ECS service.

Type: String

Required: Yes

clientAliases

The list of client aliases for this Service Connect service. You use these to assign names that can be used by client applications. The maximum number of client aliases that you can have in this list is 1.

Each alias ("endpoint") is a fully-qualified name and port number that other Amazon ECS tasks ("clients") can use to connect to this service.

Each name and port mapping must be unique within the namespace.

For each `ServiceConnectService`, you must provide at least one `clientAlias` with one port.

Type: Array of [ServiceConnectClientAlias](#) objects

Required: No

discoveryName

The `discoveryName` is the name of the new AWS Cloud Map service that Amazon ECS creates for this Amazon ECS service. This must be unique within the AWS Cloud Map namespace. The name can contain up to 64 characters. The name can include lowercase letters, numbers, underscores (`_`), and hyphens (`-`). The name can't start with a hyphen.

If the `discoveryName` isn't specified, the port mapping name from the task definition is used in `portName.namespace`.

Type: String

Required: No

ingressPortOverride

The port number for the Service Connect proxy to listen on.

Use the value of this field to bypass the proxy for traffic on the port number specified in the named `portMapping` in the task definition of this application, and then use it in your VPC security groups to allow traffic into the proxy for this Amazon ECS service.

In `awsvpc` mode and Fargate, the default value is the container port number. The container port number is in the `portMapping` in the task definition. In bridge mode, the default value is the ephemeral port of the Service Connect proxy.

Type: Integer

Valid Range: Minimum value of 0. Maximum value of 65535.

Required: No

timeout

A reference to an object that represents the configured timeouts for Service Connect.

Type: [TimeoutConfiguration](#) object

Required: No

tls

A reference to an object that represents a Transport Layer Security (TLS) configuration.

Type: [ServiceConnectTlsConfiguration](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceConnectServiceResource

The Service Connect resource. Each configuration maps a discovery name to a AWS Cloud Map service name. The data is stored in AWS Cloud Map as part of the Service Connect configuration for each discovery name of this Amazon ECS service.

A task can resolve the `dnsName` for each of the `clientAliases` of a service. However a task can't resolve the discovery names. If you want to connect to a service, refer to the `ServiceConnectConfiguration` of that service for the list of `clientAliases` that you can use.

Contents

`discoveryArn`

The Amazon Resource Name (ARN) for the service in AWS Cloud Map that matches the discovery name for this Service Connect resource. You can use this ARN in other integrations with AWS Cloud Map. However, Service Connect can't ensure connectivity outside of Amazon ECS.

Type: String

Required: No

`discoveryName`

The discovery name of this Service Connect resource.

The `discoveryName` is the name of the new AWS Cloud Map service that Amazon ECS creates for this Amazon ECS service. This must be unique within the AWS Cloud Map namespace. The name can contain up to 64 characters. The name can include lowercase letters, numbers, underscores (`_`), and hyphens (`-`). The name can't start with a hyphen.

If the `discoveryName` isn't specified, the port mapping name from the task definition is used in `portName.namespace`.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceConnectTestTrafficHeaderMatchRules

The header matching rules for test traffic routing in Amazon ECS blue/green deployments. These rules determine how incoming requests are matched based on HTTP headers to route test traffic to the new service revision.

Contents

exact

The exact value that the HTTP header must match for the test traffic routing rule to apply. This provides precise control over which requests are routed to the new service revision during blue/green deployments.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceConnectTestTrafficHeaderRules

The HTTP header rules used to identify and route test traffic during Amazon ECS blue/green deployments. These rules specify which HTTP headers to examine and what values to match for routing decisions.

For more information, see [Service Connect for Amazon ECS blue/green deployments](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

name

The name of the HTTP header to examine for test traffic routing. Common examples include custom headers like X-Test-Version or X-Canary-Request that can be used to identify test traffic.

Type: String

Required: Yes

value

The header value matching configuration that determines how the HTTP header value is evaluated for test traffic routing decisions.

Type: [ServiceConnectTestTrafficHeaderMatchRules](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceConnectTestTrafficRules

The test traffic routing configuration for Amazon ECS blue/green deployments. This configuration allows you to define rules for routing specific traffic to the new service revision during the deployment process, allowing for safe testing before full production traffic shift.

For more information, see [Service Connect for Amazon ECS blue/green deployments](#) in the *Amazon Elastic Container Service Developer Guide*.

Contents

header

The HTTP header-based routing rules that determine which requests should be routed to the new service version during blue/green deployment testing. These rules provide fine-grained control over test traffic routing based on request headers.

Type: [ServiceConnectTestTrafficHeaderRules](#) object

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceConnectTlsCertificateAuthority

The certificate root authority that secures your service.

Contents

awsPcaAuthorityArn

The ARN of the AWS Private Certificate Authority certificate.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceConnectTlsConfiguration

The key that encrypts and decrypts your resources for Service Connect TLS.

Contents

issuerCertificateAuthority

The signer certificate authority.

Type: [ServiceConnectTlsCertificateAuthority](#) object

Required: Yes

kmsKey

The AWS Key Management Service key.

Type: String

Required: No

roleArn

The Amazon Resource Name (ARN) of the IAM role that's associated with the Service Connect TLS.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceDeployment

Information about the service deployment.

Service deployments provide a comprehensive view of your deployments. For information about service deployments, see [View service history using Amazon ECS service deployments](#) in the Amazon Elastic Container Service Developer Guide .

Contents

alarms

The CloudWatch alarms that determine when a service deployment fails.

Type: [ServiceDeploymentAlarms](#) object

Required: No

clusterArn

The ARN of the cluster that hosts the service.

Type: String

Required: No

createdAt

The time the service deployment was created. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: Timestamp

Required: No

deploymentCircuitBreaker

The circuit breaker configuration that determines a service deployment failed.

Type: [ServiceDeploymentCircuitBreaker](#) object

Required: No

deploymentConfiguration

Optional deployment parameters that control how many tasks run during a deployment and the ordering of stopping and starting tasks.

Type: [DeploymentConfiguration](#) object

Required: No

finishedAt

The time the service deployment finished. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: Timestamp

Required: No

lifecycleStage

The current lifecycle stage of the deployment. Possible values include:

- RECONCILE_SERVICE

The reconciliation stage that only happens when you start a new service deployment with more than 1 service revision in an ACTIVE state.

- PRE_SCALE_UP

The green service revision has not started. The blue service revision is handling 100% of the production traffic. There is no test traffic.

- SCALE_UP

The stage when the green service revision scales up to 100% and launches new tasks. The green service revision is not serving any traffic at this point.

- POST_SCALE_UP

The green service revision has started. The blue service revision is handling 100% of the production traffic. There is no test traffic.

- TEST_TRAFFIC_SHIFT

The blue and green service revisions are running. The blue service revision handles 100% of the production traffic. The green service revision is migrating from 0% to 100% of test traffic.

- POST_TEST_TRAFFIC_SHIFT

The test traffic shift is complete. The green service revision handles 100% of the test traffic.

- PRODUCTION_TRAFFIC_SHIFT

Production traffic is shifting to the green service revision. The green service revision is migrating from 0% to 100% of production traffic.

- `POST_PRODUCTION_TRAFFIC_SHIFT`

The production traffic shift is complete.

- `BAKE_TIME`

The stage when both blue and green service revisions are running simultaneously after the production traffic has shifted.

- `CLEAN_UP`

The stage when the blue service revision has completely scaled down to 0 running tasks. The green service revision is now the production service revision after this stage.

Type: String

Valid Values: `RECONCILE_SERVICE` | `PRE_SCALE_UP` | `SCALE_UP` | `POST_SCALE_UP` | `TEST_TRAFFIC_SHIFT` | `POST_TEST_TRAFFIC_SHIFT` | `PRODUCTION_TRAFFIC_SHIFT` | `POST_PRODUCTION_TRAFFIC_SHIFT` | `BAKE_TIME` | `CLEAN_UP`

Required: No

rollback

The rollback options the service deployment uses when the deployment fails.

Type: [Rollback](#) object

Required: No

serviceArn

The ARN of the service for this service deployment.

Type: String

Required: No

serviceDeploymentArn

The ARN of the service deployment.

Type: String

Required: No

sourceServiceRevisions

The currently deployed workload configuration.

Type: Array of [ServiceRevisionSummary](#) objects

Required: No

startedAt

The time the service deployment started. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: Timestamp

Required: No

status

The service deployment state.

Type: String

Valid Values: PENDING | SUCCESSFUL | STOPPED | STOP_REQUESTED | IN_PROGRESS
| ROLLBACK_REQUESTED | ROLLBACK_IN_PROGRESS | ROLLBACK_SUCCESSFUL |
ROLLBACK_FAILED

Required: No

statusReason

Information about why the service deployment is in the current status. For example, the circuit breaker detected a failure.

Type: String

Required: No

stoppedAt

The time the service deployment stopped. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

The service deployment stops when any of the following actions happen:

- A user manually stops the deployment

- The rollback option is not in use for the failure detection mechanism (the circuit breaker or alarm-based) and the service fails.

Type: Timestamp

Required: No

targetServiceRevision

The workload configuration being deployed.

Type: [ServiceRevisionSummary](#) object

Required: No

updatedAt

The time that the service deployment was last updated. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: Timestamp

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceDeploymentAlarms

The CloudWatch alarms used to determine a service deployment failed.

Amazon ECS considers the service deployment as failed when any of the alarms move to the ALARM state. For more information, see [How CloudWatch alarms detect Amazon ECS deployment failures](#) in the Amazon ECS Developer Guide.

Contents

alarmNames

The name of the CloudWatch alarms that determine when a service deployment failed. A "," separates the alarms.

Type: Array of strings

Required: No

status

The status of the alarms check. Amazon ECS is not using alarms for service deployment failures when the status is DISABLED.

Type: String

Valid Values: TRIGGERED | MONITORING | MONITORING_COMPLETE | DISABLED

Required: No

triggeredAlarmNames

One or more CloudWatch alarm names that have been triggered during the service deployment. A "," separates the alarm names.

Type: Array of strings

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceDeploymentBrief

The service deployment properties that are returned when you call `ListServiceDeployments`.

This provides a high-level overview of the service deployment.

Contents

clusterArn

The ARN of the cluster that hosts the service.

Type: String

Required: No

createdAt

The time that the service deployment was created. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: Timestamp

Required: No

finishedAt

The time that the service deployment completed. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: Timestamp

Required: No

serviceArn

The ARN of the service for this service deployment.

Type: String

Required: No

serviceDeploymentArn

The ARN of the service deployment.

Type: String

Required: No

startedAt

The time that the service deployment started. The format is yyyy-MM-dd HH:mm:ss.SSSSSS.

Type: Timestamp

Required: No

status

The status of the service deployment

Type: String

Valid Values: PENDING | SUCCESSFUL | STOPPED | STOP_REQUESTED | IN_PROGRESS
| ROLLBACK_REQUESTED | ROLLBACK_IN_PROGRESS | ROLLBACK_SUCCESSFUL |
ROLLBACK_FAILED

Required: No

statusReason

Information about why the service deployment is in the current status. For example, the circuit breaker detected a deployment failure.

Type: String

Required: No

targetServiceRevisionArn

The ARN of the service revision being deployed.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceDeploymentCircuitBreaker

Information about the circuit breaker used to determine when a service deployment has failed.

The deployment circuit breaker is the rolling update mechanism that determines if the tasks reach a steady state. The deployment circuit breaker has an option that will automatically roll back a failed deployment to the last completed service revision. For more information, see [How the Amazon ECS deployment circuit breaker detects failures](#) in the *Amazon ECS Developer Guide*.

Contents

failureCount

The number of times the circuit breaker detected a service deployment failure.

Type: Integer

Required: No

status

The circuit breaker status. Amazon ECS is not using the circuit breaker for service deployment failures when the status is DISABLED.

Type: String

Valid Values: TRIGGERED | MONITORING | MONITORING_COMPLETE | DISABLED

Required: No

threshold

The threshold which determines that the service deployment failed.

The deployment circuit breaker calculates the threshold value, and then uses the value to determine when to move the deployment to a FAILED state. The deployment circuit breaker has a minimum threshold of 3 and a maximum threshold of 200. and uses the values in the following formula to determine the deployment failure.

$0.5 * \text{desired task count}$

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceEvent

The details for an event that's associated with a service.

Contents

createdAt

The Unix timestamp for the time when the event was triggered.

Type: Timestamp

Required: No

id

The ID string for the event.

Type: String

Required: No

message

The event message.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceManagedEBSVolumeConfiguration

The configuration for the Amazon EBS volume that Amazon ECS creates and manages on your behalf. These settings are used to create each Amazon EBS volume, with one volume created for each task in the service. For information about the supported launch types and operating systems, see [Supported operating systems and launch types](#) in the *Amazon Elastic Container Service Developer Guide*.

Many of these parameters map 1:1 with the Amazon EBS CreateVolume API request parameters.

Contents

roleArn

The ARN of the IAM role to associate with this volume. This is the Amazon ECS infrastructure IAM role that is used to manage your AWS infrastructure. We recommend using the Amazon ECS-managed AmazonECSInfrastructureRolePolicyForVolumes IAM policy with this role. For more information, see [Amazon ECS infrastructure IAM role](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: Yes

encrypted

Indicates whether the volume should be encrypted. If you turn on Region-level Amazon EBS encryption by default but set this value as `false`, the setting is overridden and the volume is encrypted with the KMS key specified for Amazon EBS encryption by default. This parameter maps 1:1 with the Encrypted parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

Type: Boolean

Required: No

filesystemType

The filesystem type for the volume. For volumes created from a snapshot, you must specify the same filesystem type that the volume was using when the snapshot was created. If there is a filesystem type mismatch, the tasks will fail to start.

The available Linux filesystem types are `ext3`, `ext4`, and `xfs`. If no value is specified, the `xfs` filesystem type is used by default.

The available Windows filesystem types are NTFS.

Type: String

Valid Values: `ext3` | `ext4` | `xfs` | `ntfs`

Required: No

iops

The number of I/O operations per second (IOPS). For `gp3`, `io1`, and `io2` volumes, this represents the number of IOPS that are provisioned for the volume. For `gp2` volumes, this represents the baseline performance of the volume and the rate at which the volume accumulates I/O credits for bursting.

The following are the supported values for each volume type.

- `gp3`: 3,000 - 16,000 IOPS
- `io1`: 100 - 64,000 IOPS
- `io2`: 100 - 256,000 IOPS

This parameter is required for `io1` and `io2` volume types. The default for `gp3` volumes is 3,000 IOPS. This parameter is not supported for `st1`, `sc1`, or standard volume types.

This parameter maps 1:1 with the `Iops` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

Type: Integer

Required: No

kmsKeyId

The Amazon Resource Name (ARN) identifier of the AWS Key Management Service key to use for Amazon EBS encryption. When a key is specified using this parameter, it overrides Amazon EBS default encryption or any KMS key that you specified for cluster-level managed storage encryption. This parameter maps 1:1 with the `KmsKeyId` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*. For more information about encrypting Amazon EBS volumes

attached to tasks, see [Encrypt data stored in Amazon EBS volumes attached to Amazon ECS tasks](#).

⚠ Important

AWS authenticates the AWS Key Management Service key asynchronously. Therefore, if you specify an ID, alias, or ARN that is invalid, the action can appear to complete, but eventually fails.

Type: String

Required: No

sizeInGiB

The size of the volume in GiB. You must specify either a volume size or a snapshot ID. If you specify a snapshot ID, the snapshot size is used for the volume size by default. You can optionally specify a volume size greater than or equal to the snapshot size. This parameter maps 1:1 with the Size parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

The following are the supported volume size values for each volume type.

- gp2 and gp3: 1-16,384
- io1 and io2: 4-16,384
- st1 and sc1: 125-16,384
- standard: 1-1,024

Type: Integer

Required: No

snapshotId

The snapshot that Amazon ECS uses to create volumes for attachment to tasks maintained by the service. You must specify either snapshotId or sizeInGiB in your volume configuration. This parameter maps 1:1 with the SnapshotId parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

Type: String

Required: No

tagSpecifications

The tags to apply to the volume. Amazon ECS applies service-managed tags by default. This parameter maps 1:1 with the `TagSpecifications.N` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

Type: Array of [EBSTagSpecification](#) objects

Required: No

throughput

The throughput to provision for a volume, in MiB/s, with a maximum of 1,000 MiB/s. This parameter maps 1:1 with the `Throughput` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

Important

This parameter is only supported for the gp3 volume type.

Type: Integer

Required: No

volumeInitializationRate

The rate, in MiB/s, at which data is fetched from a snapshot of an existing EBS volume to create new volumes for attachment to the tasks maintained by the service. This property can be specified only if you specify a `snapshotId`. For more information, see [Initialize Amazon EBS volumes](#) in the *Amazon EBS User Guide*.

Type: Integer

Required: No

volumeType

The volume type. This parameter maps 1:1 with the `VolumeType` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*. For more information, see [Amazon EBS volume types](#) in the *Amazon EC2 User Guide*.

The following are the supported volume types.

- General Purpose SSD: gp2|gp3
- Provisioned IOPS SSD: io1|io2
- Throughput Optimized HDD: st1
- Cold HDD: sc1
- Magnetic: standard

**Note**

The magnetic volume type is not supported on Fargate.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceRegistry

The details for the service registry.

Each service may be associated with one service registry. Multiple service registries for each service are not supported.

When you add, update, or remove the service registries configuration, Amazon ECS starts a new deployment. New tasks are registered and deregistered to the updated service registry configuration.

Contents

containerName

The container name value to be used for your service discovery service. It's already specified in the task definition. If the task definition that your service task specifies uses the bridge or host network mode, you must specify a `containerName` and `containerPort` combination from the task definition. If the task definition that your service task specifies uses the `awsvpc` network mode and a type SRV DNS record is used, you must specify either a `containerName` and `containerPort` combination or a `port` value. However, you can't specify both.

Type: String

Required: No

containerPort

The port value to be used for your service discovery service. It's already specified in the task definition. If the task definition your service task specifies uses the bridge or host network mode, you must specify a `containerName` and `containerPort` combination from the task definition. If the task definition your service task specifies uses the `awsvpc` network mode and a type SRV DNS record is used, you must specify either a `containerName` and `containerPort` combination or a `port` value. However, you can't specify both.

Type: Integer

Required: No

port

The port value used if your service discovery service specified an SRV record. This field might be used if both the `awsvpc` network mode and SRV records are used.

Type: Integer

Required: No

registryArn

The Amazon Resource Name (ARN) of the service registry. The currently supported service registry is AWS Cloud Map. For more information, see [CreateService](#).

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceRevision

Information about the service revision.

A service revision contains a record of the workload configuration Amazon ECS is attempting to deploy. Whenever you create or deploy a service, Amazon ECS automatically creates and captures the configuration that you're trying to deploy in the service revision. For information about service revisions, see [Amazon ECS service revisions](#) in the Amazon Elastic Container Service Developer Guide .

Contents

capacityProviderStrategy

The capacity provider strategy the service revision uses.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: No

clusterArn

The ARN of the cluster that hosts the service.

Type: String

Required: No

containerImages

The container images the service revision uses.

Type: Array of [ContainerImage](#) objects

Required: No

createdAt

The time that the service revision was created. The format is yyyy-mm-dd HH:mm:ss.SSSSS.

Type: Timestamp

Required: No

fargateEphemeralStorage

The amount of ephemeral storage to allocate for the deployment.

Type: [DeploymentEphemeralStorage](#) object

Required: No

guardDutyEnabled

Indicates whether Runtime Monitoring is turned on.

Type: Boolean

Required: No

launchType

The launch type the service revision uses.

Type: String

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

loadBalancers

The load balancers the service revision uses.

Type: Array of [LoadBalancer](#) objects

Required: No

networkConfiguration

The network configuration for a task or service.

Type: [NetworkConfiguration](#) object

Required: No

platformFamily

The platform family the service revision uses.

Type: String

Required: No

platformVersion

For the Fargate launch type, the platform version the service revision uses.

Type: String

Required: No

resolvedConfiguration

The resolved configuration for the service revision which contains the actual resources your service revision uses, such as which target groups serve traffic.

Type: [ResolvedConfiguration](#) object

Required: No

serviceArn

The ARN of the service for the service revision.

Type: String

Required: No

serviceConnectConfiguration

The Service Connect configuration of your Amazon ECS service. The configuration for this service to discover and connect to services, and be discovered by, and connected from, other services within a namespace.

Tasks that run in a namespace can use short names to connect to services in the namespace. Tasks can connect to services across all of the clusters in the namespace. Tasks connect through a managed proxy container that collects logs and metrics for increased visibility. Only the tasks that Amazon ECS services create are supported with Service Connect. For more information, see [Service Connect](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [ServiceConnectConfiguration](#) object

Required: No

serviceRegistries

The service registries (for Service Discovery) the service revision uses.

Type: Array of [ServiceRegistry](#) objects

Required: No

serviceRevisionArn

The ARN of the service revision.

Type: String

Required: No

taskDefinition

The task definition the service revision uses.

Type: String

Required: No

volumeConfigurations

The volumes that are configured at deployment that the service revision uses.

Type: Array of [ServiceVolumeConfiguration](#) objects

Required: No

vpcLatticeConfigurations

The VPC Lattice configuration for the service revision.

Type: Array of [VpcLatticeConfiguration](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceRevisionLoadBalancer

The resolved load balancer configuration for a service revision. This includes information about which target groups serve traffic and which listener rules direct traffic to them.

Contents

productionListenerRule

The Amazon Resource Name (ARN) of the production listener rule or listener that directs traffic to the target group associated with the service revision.

Type: String

Required: No

targetGroupArn

The Amazon Resource Name (ARN) of the target group associated with the service revision.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceRevisionSummary

The information about the number of requested, pending, and running tasks for a service revision.

Contents

arn

The ARN of the service revision.

Type: String

Required: No

pendingTaskCount

The number of pending tasks for the service revision.

Type: Integer

Required: No

requestedProductionTrafficWeight

The percentage of production traffic that is directed to this service revision. This value represents a snapshot of the traffic distribution and may not reflect real-time changes during active deployments. Valid values are 0.0 to 100.0.

Type: Double

Required: No

requestedTaskCount

The number of requested tasks for the service revision.

Type: Integer

Required: No

requestedTestTrafficWeight

The percentage of test traffic that is directed to this service revision. This value represents a snapshot of the traffic distribution and may not reflect real-time changes during active deployments. Valid values are 0.0 to 100.0.

Type: Double

Required: No

runningTaskCount

The number of running tasks for the service revision.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

ServiceVolumeConfiguration

The configuration for a volume specified in the task definition as a volume that is configured at launch time. Currently, the only supported volume type is an Amazon EBS volume.

Contents

name

The name of the volume. This value must match the volume name from the `Volume` object in the task definition.

Type: String

Required: Yes

managedEBSVolume

The configuration for the Amazon EBS volume that Amazon ECS creates and manages on your behalf. These settings are used to create each Amazon EBS volume, with one volume created for each task in the service. The Amazon EBS volumes are visible in your account in the Amazon EC2 console once they are created.

Type: [ServiceManagedEBSVolumeConfiguration](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Session

The details for the execute command session.

Contents

sessionId

The ID of the execute command session.

Type: String

Required: No

streamUrl

A URL to the managed agent on the container that the SSM Session Manager client uses to send commands and receive output from the container.

Type: String

Required: No

tokenValue

An encrypted token value containing session and caller information. It's used to authenticate the connection to the container.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Setting

The current account setting for a resource.

Contents

name

The Amazon ECS resource name.

Type: String

Valid Values: `serviceLongArnFormat` | `taskLongArnFormat` | `containerInstanceLongArnFormat` | `awsvpcTrunking` | `containerInsights` | `fargateFIPSMODE` | `tagResourceAuthorization` | `fargateTaskRetirementWaitPeriod` | `guardDutyActivate` | `defaultLogDriverMode`

Required: No

principalArn

The ARN of the principal. It can be a user, role, or the root user. If this field is omitted, the authenticated user is assumed.

Type: String

Required: No

type

Indicates whether AWS manages the account setting, or if the user manages it.

`aws_managed` account settings are read-only, as AWS manages such on the customer's behalf. Currently, the `guardDutyActivate` account setting is the only one AWS manages.

Type: String

Valid Values: `user` | `aws_managed`

Required: No

value

Determines whether the account setting is on or off for the specified resource.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

SystemControl

A list of namespaced kernel parameters to set in the container. This parameter maps to `Sysctl`s in the docker container create command and the `--sysctl` option to docker run. For example, you can configure `net.ipv4.tcp_keepalive_time` setting to maintain longer lived connections.

We don't recommend that you specify network-related `systemControls` parameters for multiple containers in a single task that also uses either the `awsvpc` or `host` network mode. Doing this has the following disadvantages:

- For tasks that use the `awsvpc` network mode including Fargate, if you set `systemControls` for any container, it applies to all containers in the task. If you set different `systemControls` for multiple containers in a single task, the container that's started last determines which `systemControls` take effect.
- For tasks that use the `host` network mode, the network namespace `systemControls` aren't supported.

If you're setting an IPC resource namespace to use for the containers in the task, the following conditions apply to your system controls. For more information, see [IPC mode](#).

- For tasks that use the `host` IPC mode, IPC namespace `systemControls` aren't supported.
- For tasks that use the `task` IPC mode, IPC namespace `systemControls` values apply to all containers within a task.

Note

This parameter is not supported for Windows containers.

Note

This parameter is only supported for tasks that are hosted on AWS Fargate if the tasks are using platform version `1.4.0` or later (Linux). This isn't supported for Windows containers on Fargate.

Contents

namespace

The namespaced kernel parameter to set a value for.

Type: String

Required: No

value

The namespaced kernel parameter to set a value for.

Valid IPC namespace values: "kernel.msgmax" | "kernel.msgmnb" | "kernel.msgmni" | "kernel.sem" | "kernel.shmall" | "kernel.shmmax" | "kernel.shmmni" | "kernel.shm_rmid_forced", and Sysctls that start with "fs.mqueue.*"

Valid network namespace values: Sysctls that start with "net.*". Only namespaced Sysctls that exist within the container starting with "net.*" are accepted.

All of these values are supported by Fargate.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Tag

The metadata that you apply to a resource to help you categorize and organize them. Each tag consists of a key and an optional value. You define them.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Contents

key

One part of a key-value pair that make up a tag. A key is a general label that acts like a category for more specific tag values.

Type: String

Length Constraints: Minimum length of 1. Maximum length of 128.

Pattern: `^([\p{L}\p{Z}\p{N}_.: /+=\ -@]*)$`

Required: No

value

The optional part of a key-value pair that make up a tag. A value acts as a descriptor within a tag category (key).

Type: String

Length Constraints: Minimum length of 0. Maximum length of 256.

Pattern: `^([\p{L}\p{Z}\p{N}_.: /+=\ -@]*)$`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Task

Details on a task in a cluster.

Contents

attachments

The Elastic Network Adapter that's associated with the task if the task uses the awsvpc network mode.

Type: Array of [Attachment](#) objects

Required: No

attributes

The attributes of the task

Type: Array of [Attribute](#) objects

Required: No

availabilityZone

The Availability Zone for the task.

Type: String

Required: No

capacityProviderName

The capacity provider that's associated with the task.

Type: String

Required: No

clusterArn

The ARN of the cluster that hosts the task.

Type: String

Required: No

connectivity

The connectivity status of a task.

Type: String

Valid Values: CONNECTED | DISCONNECTED

Required: No

connectivityAt

The Unix timestamp for the time when the task last went into CONNECTED status.

Type: Timestamp

Required: No

containerInstanceArn

The ARN of the container instances that host the task.

Type: String

Required: No

containers

The containers that's associated with the task.

Type: Array of [Container](#) objects

Required: No

cpu

The number of CPU units used by the task as expressed in a task definition. It can be expressed as an integer using CPU units (for example, 1024). It can also be expressed as a string using vCPUs (for example, 1 vCPU or 1 vcpu). String values are converted to an integer that indicates the CPU units when the task definition is registered.

If you're using the EC2 launch type or the external launch type, this field is optional. Supported values are between 128 CPU units (0.125 vCPUs) and 196608 CPU units (192 vCPUs). If you do not specify a value, the parameter is ignored.

This field is required for Fargate. For information about the valid values, see [Task size](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

createdAt

The Unix timestamp for the time when the task was created. More specifically, it's for the time when the task entered the PENDING state.

Type: Timestamp

Required: No

desiredStatus

The desired status of the task. For more information, see [Task Lifecycle](#).

Type: String

Required: No

enableExecuteCommand

Determines whether execute command functionality is turned on for this task. If `true`, execute command functionality is turned on all the containers in the task.

Type: Boolean

Required: No

ephemeralStorage

The ephemeral storage settings for the task.

Type: [EphemeralStorage](#) object

Required: No

executionStoppedAt

The Unix timestamp for the time when the task execution stopped.

Type: Timestamp

Required: No

fargateEphemeralStorage

The Fargate ephemeral storage settings for the task.

Type: [TaskEphemeralStorage](#) object

Required: No

group

The name of the task group that's associated with the task.

Type: String

Required: No

healthStatus

The health status for the task. It's determined by the health of the essential containers in the task. If all essential containers in the task are reporting as `HEALTHY`, the task status also reports as `HEALTHY`. If any essential containers in the task are reporting as `UNHEALTHY` or `UNKNOWN`, the task status also reports as `UNHEALTHY` or `UNKNOWN`.

Note

The Amazon ECS container agent doesn't monitor or report on Docker health checks that are embedded in a container image and not specified in the container definition. For example, this includes those specified in a parent image or from the image's Dockerfile. Health check parameters that are specified in a container definition override any Docker health checks that are found in the container image.

Type: String

Valid Values: `HEALTHY` | `UNHEALTHY` | `UNKNOWN`

Required: No

inferenceAccelerators

The Elastic Inference accelerator that's associated with the task.

Type: Array of [InferenceAccelerator](#) objects

Required: No

lastStatus

The last known status for the task. For more information, see [Task Lifecycle](#).

Type: String

Required: No

launchType

The infrastructure where your task runs on. For more information, see [Amazon ECS launch types](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

memory

The amount of memory (in MiB) that the task uses as expressed in a task definition. It can be expressed as an integer using MiB (for example, 1024). If it's expressed as a string using GB (for example, 1GB or 1 GB), it's converted to an integer indicating the MiB when the task definition is registered.

If you use the EC2 launch type, this field is optional.

If you use the Fargate launch type, this field is required. You must use one of the following values. The value that you choose determines the range of supported values for the `cpu` parameter.

- 512 (0.5 GB), 1024 (1 GB), 2048 (2 GB) - Available `cpu` values: 256 (.25 vCPU)
- 1024 (1 GB), 2048 (2 GB), 3072 (3 GB), 4096 (4 GB) - Available `cpu` values: 512 (.5 vCPU)
- 2048 (2 GB), 3072 (3 GB), 4096 (4 GB), 5120 (5 GB), 6144 (6 GB), 7168 (7 GB), 8192 (8 GB) - Available `cpu` values: 1024 (1 vCPU)
- Between 4096 (4 GB) and 16384 (16 GB) in increments of 1024 (1 GB) - Available `cpu` values: 2048 (2 vCPU)

- Between 8192 (8 GB) and 30720 (30 GB) in increments of 1024 (1 GB) - Available cpu values: 4096 (4 vCPU)
- Between 16 GB and 60 GB in 4 GB increments - Available cpu values: 8192 (8 vCPU)

This option requires Linux platform 1.4.0 or later.

- Between 32GB and 120 GB in 8 GB increments - Available cpu values: 16384 (16 vCPU)

This option requires Linux platform 1.4.0 or later.

Type: String

Required: No

overrides

One or more container overrides.

Type: [TaskOverride](#) object

Required: No

platformFamily

The operating system that your tasks are running on. A platform family is specified only for tasks that use the Fargate launch type.

All tasks that run as part of this service must use the same `platformFamily` value as the service (for example, `LINUX`).

Type: String

Required: No

platformVersion

The platform version where your task runs on. A platform version is only specified for tasks that use the Fargate launch type. If you didn't specify one, the LATEST platform version is used. For more information, see [AWS Fargate Platform Versions](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

pullStartedAt

The Unix timestamp for the time when the container image pull began.

Type: Timestamp

Required: No

pullStoppedAt

The Unix timestamp for the time when the container image pull completed.

Type: Timestamp

Required: No

startedAt

The Unix timestamp for the time when the task started. More specifically, it's for the time when the task transitioned from the PENDING state to the RUNNING state.

Type: Timestamp

Required: No

startedBy

The tag specified when a task is started. If an Amazon ECS service started the task, the `startedBy` parameter contains the deployment ID of that service.

Type: String

Required: No

stopCode

The stop code indicating why a task was stopped. The `stoppedReason` might contain additional details.

For more information about stop code, see [Stopped tasks error codes](#) in the *Amazon ECS Developer Guide*.

Type: String

Valid Values: TaskFailedToStart | EssentialContainerExited | UserInitiated | ServiceSchedulerInitiated | SpotInterruption | TerminationNotice

Required: No

stoppedAt

The Unix timestamp for the time when the task was stopped. More specifically, it's for the time when the task transitioned from the RUNNING state to the STOPPED state.

Type: Timestamp

Required: No

stoppedReason

The reason that the task was stopped.

Type: String

Required: No

stoppingAt

The Unix timestamp for the time when the task stops. More specifically, it's for the time when the task transitions from the RUNNING state to STOPPING.

Type: Timestamp

Required: No

tags

The metadata that you apply to the task to help you categorize and organize the task. Each tag consists of a key and an optional value. You define both the key and value.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.

- Do not use `aws:`, `AWS:`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

taskArn

The Amazon Resource Name (ARN) of the task.

Type: String

Required: No

taskDefinitionArn

The ARN of the task definition that creates the task.

Type: String

Required: No

version

The version counter for the task. Every time a task experiences a change that starts a CloudWatch event, the version counter is incremented. If you replicate your Amazon ECS task state with CloudWatch Events, you can compare the version of a task reported by the Amazon ECS API actions with the version reported in CloudWatch Events for the task (inside the `detail` object) to verify that the version in your event stream is current.

Type: Long

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)

- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

TaskDefinition

The details of a task definition which describes the container and volume definitions of an Amazon Elastic Container Service task. You can specify which Docker images to use, the required resources, and other configurations related to launching the task definition through an Amazon ECS service or task.

Contents

compatibilities

Amazon ECS validates the task definition parameters with those supported by the launch type. For more information, see [Amazon ECS launch types](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Array of strings

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

containerDefinitions

A list of container definitions in JSON format that describe the different containers that make up your task. For more information about container definition parameters and defaults, see [Amazon ECS Task Definitions](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Array of [ContainerDefinition](#) objects

Required: No

cpu

The number of cpu units used by the task. If you use the EC2 launch type, this field is optional. Any value can be used. If you use the Fargate launch type, this field is required. You must use one of the following values. The value that you choose determines your range of valid values for the memory parameter.

If you're using the EC2 launch type or the external launch type, this field is optional. Supported values are between 128 CPU units (0.125 vCPUs) and 196608 CPU units (192 vCPUs).

This field is required for Fargate. For information about the valid values, see [Task size](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

deregisteredAt

The Unix timestamp for the time when the task definition was deregistered.

Type: Timestamp

Required: No

enableFaultInjection

Enables fault injection and allows for fault injection requests to be accepted from the task's containers. The default value is `false`.

Type: Boolean

Required: No

ephemeralStorage

The ephemeral storage settings to use for tasks run with the task definition.

Type: [EphemeralStorage](#) object

Required: No

executionRoleArn

The Amazon Resource Name (ARN) of the task execution role that grants the Amazon ECS container agent permission to make AWS API calls on your behalf. For information about the required IAM roles for Amazon ECS, see [IAM roles for Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

family

The name of a family that this task definition is registered to. Up to 255 characters are allowed. Letters (both uppercase and lowercase letters), numbers, hyphens (-), and underscores (_) are allowed.

A family groups multiple versions of a task definition. Amazon ECS gives the first task definition that you registered to a family a revision number of 1. Amazon ECS gives sequential revision numbers to each task definition that you add.

Type: String

Required: No

inferenceAccelerators

The Elastic Inference accelerator that's associated with the task.

Type: Array of [InferenceAccelerator](#) objects

Required: No

ipcMode

The IPC resource namespace to use for the containers in the task. The valid values are `host`, `task`, or `none`. If `host` is specified, then all containers within the tasks that specified the `host` IPC mode on the same container instance share the same IPC resources with the host Amazon EC2 instance. If `task` is specified, all containers within the specified task share the same IPC resources. If `none` is specified, then IPC resources within the containers of a task are private and not shared with other containers in a task or on the container instance. If no value is specified, then the IPC resource namespace sharing depends on the Docker daemon setting on the container instance.

If the `host` IPC mode is used, be aware that there is a heightened risk of undesired IPC namespace expose.

If you are setting namespaced kernel parameters using `systemControls` for the containers in the task, the following will apply to your IPC resource namespace. For more information, see [System Controls](#) in the *Amazon Elastic Container Service Developer Guide*.

- For tasks that use the `host` IPC mode, IPC namespace related `systemControls` are not supported.
- For tasks that use the `task` IPC mode, IPC namespace related `systemControls` will apply to all containers within a task.

Note

This parameter is not supported for Windows containers or tasks run on AWS Fargate.

Type: String

Valid Values: host | task | none

Required: No

memory

The amount (in MiB) of memory used by the task.

If your tasks runs on Amazon EC2 instances, you must specify either a task-level memory value or a container-level memory value. This field is optional and any value can be used. If a task-level memory value is specified, the container-level memory value is optional. For more information regarding container-level memory and memory reservation, see [ContainerDefinition](#).

If your tasks runs on AWS Fargate, this field is required. You must use one of the following values. The value you choose determines your range of valid values for the `cpu` parameter.

- 512 (0.5 GB), 1024 (1 GB), 2048 (2 GB) - Available `cpu` values: 256 (.25 vCPU)
- 1024 (1 GB), 2048 (2 GB), 3072 (3 GB), 4096 (4 GB) - Available `cpu` values: 512 (.5 vCPU)
- 2048 (2 GB), 3072 (3 GB), 4096 (4 GB), 5120 (5 GB), 6144 (6 GB), 7168 (7 GB), 8192 (8 GB) - Available `cpu` values: 1024 (1 vCPU)
- Between 4096 (4 GB) and 16384 (16 GB) in increments of 1024 (1 GB) - Available `cpu` values: 2048 (2 vCPU)
- Between 8192 (8 GB) and 30720 (30 GB) in increments of 1024 (1 GB) - Available `cpu` values: 4096 (4 vCPU)
- Between 16 GB and 60 GB in 4 GB increments - Available `cpu` values: 8192 (8 vCPU)

This option requires Linux platform 1.4.0 or later.

- Between 32GB and 120 GB in 8 GB increments - Available `cpu` values: 16384 (16 vCPU)

This option requires Linux platform 1.4.0 or later.

Type: String

Required: No

networkMode

The Docker networking mode to use for the containers in the task. The valid values are `none`, `bridge`, `awsvpc`, and `host`. If no network mode is specified, the default is `bridge`.

For Amazon ECS tasks on Fargate, the `awsvpc` network mode is required. For Amazon ECS tasks on Amazon EC2 Linux instances, any network mode can be used. For Amazon ECS tasks on Amazon EC2 Windows instances, `<default>` or `awsvpc` can be used. If the network mode is set to `none`, you cannot specify port mappings in your container definitions, and the tasks containers do not have external connectivity. The `host` and `awsvpc` network modes offer the highest networking performance for containers because they use the EC2 network stack instead of the virtualized network stack provided by the `bridge` mode.

With the `host` and `awsvpc` network modes, exposed container ports are mapped directly to the corresponding host port (for the `host` network mode) or the attached elastic network interface port (for the `awsvpc` network mode), so you cannot take advantage of dynamic host port mappings.

Important

When using the `host` network mode, you should not run containers using the root user (UID 0). It is considered best practice to use a non-root user.

If the network mode is `awsvpc`, the task is allocated an elastic network interface, and you must specify a [NetworkConfiguration](#) value when you create a service or run a task with the task definition. For more information, see [Task Networking](#) in the *Amazon Elastic Container Service Developer Guide*.

If the network mode is `host`, you cannot run multiple instantiations of the same task on a single container instance when port mappings are used.

Type: String

Valid Values: `bridge` | `host` | `awsvpc` | `none`

Required: No

pidMode

The process namespace to use for the containers in the task. The valid values are `host` or `task`. On Fargate for Linux containers, the only valid value is `task`. For example, monitoring sidecars might need `pidMode` to access information about other containers running in the same task.

If `host` is specified, all containers within the tasks that specified the `host` PID mode on the same container instance share the same process namespace with the host Amazon EC2 instance.

If `task` is specified, all containers within the specified task share the same process namespace.

If no value is specified, the default is a private namespace for each container.

If the host PID mode is used, there's a heightened risk of undesired process namespace exposure.

 Note

This parameter is not supported for Windows containers.

 Note

This parameter is only supported for tasks that are hosted on AWS Fargate if the tasks are using platform version 1.4.0 or later (Linux). This isn't supported for Windows containers on Fargate.

Type: String

Valid Values: `host` | `task`

Required: No

placementConstraints

An array of placement constraint objects to use for tasks.

 Note

This parameter isn't supported for tasks run on AWS Fargate.

Type: Array of [TaskDefinitionPlacementConstraint](#) objects

Required: No

proxyConfiguration

The configuration details for the App Mesh proxy.

Your Amazon ECS container instances require at least version 1.26.0 of the container agent and at least version 1.26.0-1 of the `ecs-init` package to use a proxy configuration. If your container instances are launched from the Amazon ECS optimized AMI version 20190301 or later, they contain the required versions of the container agent and `ecs-init`. For more information, see [Amazon ECS-optimized Linux AMI](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: [ProxyConfiguration](#) object

Required: No

registeredAt

The Unix timestamp for the time when the task definition was registered.

Type: Timestamp

Required: No

registeredBy

The principal that registered the task definition.

Type: String

Required: No

requiresAttributes

The container instance attributes required by your task. When an Amazon EC2 instance is registered to your cluster, the Amazon ECS container agent assigns some standard attributes to the instance. You can apply custom attributes. These are specified as key-value pairs using the Amazon ECS console or the [PutAttributes](#) API. These attributes are used when determining task placement for tasks hosted on Amazon EC2 instances. For more information, see [Attributes](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

This parameter isn't supported for tasks run on AWS Fargate.

Type: Array of [Attribute](#) objects

Required: No

requiresCompatibilities

The task launch types the task definition was validated against. The valid values are `MANAGED_INSTANCES`, `EC2`, `FARGATE`, and `EXTERNAL`. For more information, see [Amazon ECS launch types](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: Array of strings

Valid Values: `EC2` | `FARGATE` | `EXTERNAL` | `MANAGED_INSTANCES`

Required: No

revision

The revision of the task in a particular family. The revision is a version number of a task definition in a family. When you register a task definition for the first time, the revision is 1. Each time that you register a new revision of a task definition in the same family, the revision value always increases by one. This is even if you deregistered previous revisions in this family.

Type: Integer

Required: No

runtimePlatform

The operating system that your task definitions are running on. A platform family is specified only for tasks using the Fargate launch type.

When you specify a task in a service, this value must match the `runtimePlatform` value of the service.

Type: [RuntimePlatform](#) object

Required: No

status

The status of the task definition.

Type: String

Valid Values: `ACTIVE` | `INACTIVE` | `DELETE_IN_PROGRESS`

Required: No

taskDefinitionArn

The full Amazon Resource Name (ARN) of the task definition.

Type: String

Required: No

taskRoleArn

The short name or full Amazon Resource Name (ARN) of the AWS Identity and Access Management role that grants containers in the task permission to call AWS APIs on your behalf. For information about the required IAM roles for Amazon ECS, see [IAM roles for Amazon ECS](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

volumes

The list of data volume definitions for the task. For more information, see [Using data volumes in tasks](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

The `host` and `sourcePath` parameters aren't supported for tasks run on AWS Fargate.

Type: Array of [Volume](#) objects

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)

- [AWS SDK for Ruby V3](#)

TaskDefinitionPlacementConstraint

The constraint on task placement in the task definition. For more information, see [Task placement constraints](#) in the *Amazon Elastic Container Service Developer Guide*.

Note

Task placement constraints aren't supported for tasks run on AWS Fargate.

Contents

expression

A cluster query language expression to apply to the constraint. For more information, see [Cluster query language](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

type

The type of constraint. The `memberOf` constraint restricts selection to be from a group of valid candidates.

Type: String

Valid Values: `memberOf`

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

TaskEphemeralStorage

The amount of ephemeral storage to allocate for the task.

Contents

kmsKeyId

Specify an AWS Key Management Service key ID to encrypt the ephemeral storage for the task.

Type: String

Required: No

sizeInGiB

The total amount, in GiB, of the ephemeral storage to set for the task. The minimum supported value is 20 GiB and the maximum supported value is 200 GiB.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

TaskManagedEBSVolumeConfiguration

The configuration for the Amazon EBS volume that Amazon ECS creates and manages on your behalf. These settings are used to create each Amazon EBS volume, with one volume created for each task.

Contents

roleArn

The ARN of the IAM role to associate with this volume. This is the Amazon ECS infrastructure IAM role that is used to manage your AWS infrastructure. We recommend using the Amazon ECS-managed `AmazonECSInfrastructureRolePolicyForVolumes` IAM policy with this role. For more information, see [Amazon ECS infrastructure IAM role](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: Yes

encrypted

Indicates whether the volume should be encrypted. If you turn on Region-level Amazon EBS encryption by default but set this value as `false`, the setting is overridden and the volume is encrypted with the KMS key specified for Amazon EBS encryption by default. This parameter maps 1:1 with the `Encrypted` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

Type: Boolean

Required: No

filesystemType

The Linux filesystem type for the volume. For volumes created from a snapshot, you must specify the same filesystem type that the volume was using when the snapshot was created. If there is a filesystem type mismatch, the task will fail to start.

The available filesystem types are `ext3`, `ext4`, and `xfs`. If no value is specified, the `xfs` filesystem type is used by default.

Type: String

Valid Values: `ext3` | `ext4` | `xfs` | `ntfs`

Required: No

iops

The number of I/O operations per second (IOPS). For `gp3`, `io1`, and `io2` volumes, this represents the number of IOPS that are provisioned for the volume. For `gp2` volumes, this represents the baseline performance of the volume and the rate at which the volume accumulates I/O credits for bursting.

The following are the supported values for each volume type.

- `gp3`: 3,000 - 16,000 IOPS
- `io1`: 100 - 64,000 IOPS
- `io2`: 100 - 256,000 IOPS

This parameter is required for `io1` and `io2` volume types. The default for `gp3` volumes is 3,000 IOPS. This parameter is not supported for `st1`, `sc1`, or standard volume types.

This parameter maps 1:1 with the `Iops` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

Type: Integer

Required: No

kmsKeyId

The Amazon Resource Name (ARN) identifier of the AWS Key Management Service key to use for Amazon EBS encryption. When a key is specified using this parameter, it overrides Amazon EBS default encryption or any KMS key that you specified for cluster-level managed storage encryption. This parameter maps 1:1 with the `KmsKeyId` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*. For more information about encrypting Amazon EBS volumes attached to a task, see [Encrypt data stored in Amazon EBS volumes attached to Amazon ECS tasks](#).

Important

AWS authenticates the AWS Key Management Service key asynchronously. Therefore, if you specify an ID, alias, or ARN that is invalid, the action can appear to complete, but eventually fails.

Type: String

Required: No

sizeInGiB

The size of the volume in GiB. You must specify either a volume size or a snapshot ID. If you specify a snapshot ID, the snapshot size is used for the volume size by default. You can optionally specify a volume size greater than or equal to the snapshot size. This parameter maps 1:1 with the `Size` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

The following are the supported volume size values for each volume type.

- gp2 and gp3: 1-16,384
- io1 and io2: 4-16,384
- st1 and sc1: 125-16,384
- standard: 1-1,024

Type: Integer

Required: No

snapshotId

The snapshot that Amazon ECS uses to create the volume. You must specify either a snapshot ID or a volume size. This parameter maps 1:1 with the `SnapshotId` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

Type: String

Required: No

tagSpecifications

The tags to apply to the volume. Amazon ECS applies service-managed tags by default. This parameter maps 1:1 with the `TagSpecifications.N` parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

Type: Array of [EBSTagSpecification](#) objects

Required: No

terminationPolicy

The termination policy for the volume when the task exits. This provides a way to control whether Amazon ECS terminates the Amazon EBS volume when the task stops.

Type: [TaskManagedEBSVolumeTerminationPolicy](#) object

Required: No

throughput

The throughput to provision for a volume, in MiB/s, with a maximum of 1,000 MiB/s. This parameter maps 1:1 with the Throughput parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*.

Important

This parameter is only supported for the gp3 volume type.

Type: Integer

Required: No

volumeInitializationRate

The rate, in MiB/s, at which data is fetched from a snapshot of an existing Amazon EBS volume to create a new volume for attachment to the task. This property can be specified only if you specify a snapshotId. For more information, see [Initialize Amazon EBS volumes](#) in the *Amazon EBS User Guide*.

Type: Integer

Required: No

volumeType

The volume type. This parameter maps 1:1 with the VolumeType parameter of the [CreateVolume API](#) in the *Amazon EC2 API Reference*. For more information, see [Amazon EBS volume types](#) in the *Amazon EC2 User Guide*.

The following are the supported volume types.

- General Purpose SSD: gp2|gp3

- Provisioned IOPS SSD: `io1|io2`
- Throughput Optimized HDD: `st1`
- Cold HDD: `sc1`
- Magnetic: `standard`

 Note

The magnetic volume type is not supported on Fargate.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

TaskManagedEBSVolumeTerminationPolicy

The termination policy for the Amazon EBS volume when the task exits. For more information, see [Amazon ECS volume termination policy](#).

Contents

deleteOnTermination

Indicates whether the volume should be deleted on when the task stops. If a value of `true` is specified,

Amazon ECS deletes the Amazon EBS volume on your behalf when the task goes into the `STOPPED` state. If no value is specified, the default value is `true` is used. When set to `false`, Amazon ECS leaves the volume in your account.

Type: Boolean

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

TaskOverride

The overrides that are associated with a task.

Contents

containerOverrides

One or more container overrides that are sent to a task.

Type: Array of [ContainerOverride](#) objects

Required: No

cpu

The CPU override for the task.

Type: String

Required: No

ephemeralStorage

The ephemeral storage setting override for the task.

Note

This parameter is only supported for tasks hosted on Fargate that use the following platform versions:

- Linux platform version 1.4.0 or later.
- Windows platform version 1.0.0 or later.

Type: [EphemeralStorage](#) object

Required: No

executionRoleArn

The Amazon Resource Name (ARN) of the task execution role override for the task. For more information, see [Amazon ECS task execution IAM role](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

inferenceAcceleratorOverrides

The Elastic Inference accelerator override for the task.

Type: Array of [InferenceAcceleratorOverride](#) objects

Required: No

memory

The memory override for the task.

Type: String

Required: No

taskRoleArn

The Amazon Resource Name (ARN) of the role that containers in this task can assume. All containers in this task are granted the permissions that are specified in this role. For more information, see [IAM Role for Tasks](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

TaskSet

Information about a set of Amazon ECS tasks in either an AWS CodeDeploy or an EXTERNAL deployment. An Amazon ECS task set includes details such as the desired number of tasks, how many tasks are running, and whether the task set serves production traffic.

Contents

capacityProviderStrategy

The capacity provider strategy that are associated with the task set.

Type: Array of [CapacityProviderStrategyItem](#) objects

Required: No

clusterArn

The Amazon Resource Name (ARN) of the cluster that the service that hosts the task set exists in.

Type: String

Required: No

computedDesiredCount

The computed desired count for the task set. This is calculated by multiplying the service's `desiredCount` by the task set's `scale` percentage. The result is always rounded up. For example, if the computed desired count is 1.2, it rounds up to 2 tasks.

Type: Integer

Required: No

createdAt

The Unix timestamp for the time when the task set was created.

Type: Timestamp

Required: No

externalId

The external ID associated with the task set.

If an AWS CodeDeploy deployment created a task set, the `externalId` parameter contains the AWS CodeDeploy deployment ID.

If a task set is created for an external deployment and is associated with a service discovery registry, the `externalId` parameter contains the `ECS_TASK_SET_EXTERNAL_ID` AWS Cloud Map attribute.

Type: String

Required: No

fargateEphemeralStorage

The Fargate ephemeral storage settings for the task set.

Type: [DeploymentEphemeralStorage](#) object

Required: No

id

The ID of the task set.

Type: String

Required: No

launchType

The launch type the tasks in the task set are using. For more information, see [Amazon ECS launch types](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Valid Values: EC2 | FARGATE | EXTERNAL | MANAGED_INSTANCES

Required: No

loadBalancers

Details on a load balancer that are used with a task set.

Type: Array of [LoadBalancer](#) objects

Required: No

networkConfiguration

The network configuration for the task set.

Type: [NetworkConfiguration](#) object

Required: No

pendingCount

The number of tasks in the task set that are in the PENDING status during a deployment. A task in the PENDING state is preparing to enter the RUNNING state. A task set enters the PENDING status when it launches for the first time or when it's restarted after being in the STOPPED state.

Type: Integer

Required: No

platformFamily

The operating system that your tasks in the set are running on. A platform family is specified only for tasks that use the Fargate launch type.

All tasks in the set must have the same value.

Type: String

Required: No

platformVersion

The AWS Fargate platform version where the tasks in the task set are running. A platform version is only specified for tasks run on AWS Fargate. For more information, see [AWS Fargate platform versions](#) in the *Amazon Elastic Container Service Developer Guide*.

Type: String

Required: No

runningCount

The number of tasks in the task set that are in the RUNNING status during a deployment. A task in the RUNNING state is running and ready for use.

Type: Integer

Required: No

scale

A floating-point percentage of your desired number of tasks to place and keep running in the task set.

Type: [Scale](#) object

Required: No

serviceArn

The Amazon Resource Name (ARN) of the service the task set exists in.

Type: String

Required: No

serviceRegistries

The details for the service discovery registries to assign to this task set. For more information, see [Service discovery](#).

Type: Array of [ServiceRegistry](#) objects

Required: No

stabilityStatus

The stability status. This indicates whether the task set has reached a steady state. If the following conditions are met, the task set are in `STEADY_STATE`:

- The task `runningCount` is equal to the `computedDesiredCount`.
- The `pendingCount` is 0.
- There are no tasks that are running on container instances in the `DRAINING` status.
- All tasks are reporting a healthy status from the load balancers, service discovery, and container health checks.

If any of those conditions aren't met, the stability status returns `STABILIZING`.

Type: String

Valid Values: STEADY_STATE | STABILIZING

Required: No

stabilityStatusAt

The Unix timestamp for the time when the task set stability status was retrieved.

Type: Timestamp

Required: No

startedBy

The tag specified when a task set is started. If an AWS CodeDeploy deployment created the task set, the `startedBy` parameter is `CODE_DEPLOY`. If an external deployment created the task set, the `startedBy` field isn't used.

Type: String

Required: No

status

The status of the task set. The following describes each state.

PRIMARY

The task set is serving production traffic.

ACTIVE

The task set isn't serving production traffic.

DRAINING

The tasks in the task set are being stopped, and their corresponding targets are being deregistered from their target group.

Type: String

Required: No

tags

The metadata that you apply to the task set to help you categorize and organize them. Each tag consists of a key and an optional value. You define both.

The following basic restrictions apply to tags:

- Maximum number of tags per resource - 50
- For each resource, each tag key must be unique, and each tag key can have only one value.
- Maximum key length - 128 Unicode characters in UTF-8
- Maximum value length - 256 Unicode characters in UTF-8
- If your tagging schema is used across multiple services and resources, remember that other services may have restrictions on allowed characters. Generally allowed characters are: letters, numbers, and spaces representable in UTF-8, and the following characters: + - = . _ : / @.
- Tag keys and values are case-sensitive.
- Do not use `aws :`, `AWS :`, or any upper or lowercase combination of such as a prefix for either keys or values as it is reserved for AWS use. You cannot edit or delete tag keys or values with this prefix. Tags with this prefix do not count against your tags per resource limit.

Type: Array of [Tag](#) objects

Array Members: Minimum number of 0 items. Maximum number of 50 items.

Required: No

taskDefinition

The task definition that the task set is using.

Type: String

Required: No

taskSetArn

The Amazon Resource Name (ARN) of the task set.

Type: String

Required: No

updatedAt

The Unix timestamp for the time when the task set was last updated.

Type: Timestamp

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

TaskVolumeConfiguration

Configuration settings for the task volume that was configuredAtLaunch that weren't set during RegisterTaskDef.

Contents

name

The name of the volume. This value must match the volume name from the Volume object in the task definition.

Type: String

Required: Yes

managedEBSVolume

The configuration for the Amazon EBS volume that Amazon ECS creates and manages on your behalf. These settings are used to create each Amazon EBS volume, with one volume created for each task. The Amazon EBS volumes are visible in your account in the Amazon EC2 console once they are created.

Type: [TaskManagedEBSVolumeConfiguration](#) object

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

TimeoutConfiguration

An object that represents the timeout configurations for Service Connect.

Note

If `idleTimeout` is set to a time that is less than `perRequestTimeout`, the connection will close when the `idleTimeout` is reached and not the `perRequestTimeout`.

Contents

`idleTimeoutSeconds`

The amount of time in seconds a connection will stay active while idle. A value of 0 can be set to disable `idleTimeout`.

The `idleTimeout` default for HTTP/HTTP2/GRPC is 5 minutes.

The `idleTimeout` default for TCP is 1 hour.

Type: Integer

Valid Range: Minimum value of 0. Maximum value of 2147483647.

Required: No

`perRequestTimeoutSeconds`

The amount of time waiting for the upstream to respond with a complete response per request. A value of 0 can be set to disable `perRequestTimeout`. `perRequestTimeout` can only be set if Service Connect `appProtocol` isn't TCP. Only `idleTimeout` is allowed for TCP `appProtocol`.

Type: Integer

Valid Range: Minimum value of 0. Maximum value of 2147483647.

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Tmpfs

The container path, mount options, and size of the tmpfs mount.

Contents

containerPath

The absolute file path where the tmpfs volume is to be mounted.

Type: String

Required: Yes

size

The maximum size (in MiB) of the tmpfs volume.

Type: Integer

Required: Yes

mountOptions

The list of tmpfs volume mount options.

Valid values: "defaults" | "ro" | "rw" | "suid" | "nosuid" | "dev" | "nodev" | "exec" | "noexec" | "sync" | "async" | "dirsync" | "remount" | "mand" | "nomand" | "atime" | "noatime" | "diratime" | "nodiratime" | "bind" | "rbind" | "unbindable" | "runbindable" | "private" | "rprivate" | "shared" | "rshared" | "slave" | "rslave" | "relatime" | "norelatime" | "strictatime" | "nostrictatime" | "mode" | "uid" | "gid" | "nr_inodes" | "nr_blocks" | "mpol"

Type: Array of strings

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

TotalLocalStorageGBRequest

The minimum and maximum total local storage in gigabytes (GB) for instance types with local storage. This is useful for workloads that require local storage for temporary data or caching.

Contents

max

The maximum total local storage in GB. Instance types with more local storage are excluded from selection.

Type: Double

Required: No

min

The minimum total local storage in GB. Instance types with less local storage are excluded from selection.

Type: Double

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Ulimit

The `ulimit` settings to pass to the container.

Amazon ECS tasks hosted on AWS Fargate use the default resource limit values set by the operating system with the exception of the `nofile` resource limit parameter which AWS Fargate overrides. The `nofile` resource limit sets a restriction on the number of open files that a container can use. The default `nofile` soft limit is 65535 and the default hard limit is 65535.

You can specify the `ulimit` settings for a container in a task definition.

Contents

hardLimit

The hard limit for the `ulimit` type. The value can be specified in bytes, seconds, or as a count, depending on the type of the `ulimit`.

Type: Integer

Required: Yes

name

The type of the `ulimit`.

Type: String

Valid Values: `core` | `cpu` | `data` | `fsize` | `locks` | `memlock` | `msgqueue` | `nice` | `nofile` | `nproc` | `rss` | `rtprio` | `rttime` | `sigpending` | `stack`

Required: Yes

softLimit

The soft limit for the `ulimit` type. The value can be specified in bytes, seconds, or as a count, depending on the type of the `ulimit`.

Type: Integer

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

UpdateManagedInstancesProviderConfiguration

The updated configuration for a Amazon ECS Managed Instances provider. You can modify the infrastructure role, instance launch template, and tag propagation settings. Changes apply to new instances launched after the update.

Contents

infrastructureRoleArn

The updated Amazon Resource Name (ARN) of the infrastructure role. The new role must have the necessary permissions to manage instances and access required AWS services.

For more information, see [Amazon ECS infrastructure IAM role](#) in the *Amazon ECS Developer Guide*.

Type: String

Required: Yes

instanceLaunchTemplate

The updated launch template configuration. Changes to the launch template affect new instances launched after the update, while existing instances continue to use their original configuration.

Type: [InstanceLaunchTemplateUpdate](#) object

Required: Yes

propagateTags

The updated tag propagation setting. When changed, this affects only new instances launched after the update.

Type: String

Valid Values: CAPACITY_PROVIDER | NONE

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

VCpuCountRangeRequest

The minimum and maximum number of vCPUs for instance type selection. This allows you to specify a range of vCPU counts that meet your workload requirements.

Contents

min

The minimum number of vCPUs. Instance types with fewer vCPUs than this value are excluded from selection.

Type: Integer

Required: Yes

max

The maximum number of vCPUs. Instance types with more vCPUs than this value are excluded from selection.

Type: Integer

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

VersionInfo

The Docker and Amazon ECS container agent version information about a container instance.

Contents

agentHash

The Git commit hash for the Amazon ECS container agent build on the [amazon-ecs-agent](#) GitHub repository.

Type: String

Required: No

agentVersion

The version number of the Amazon ECS container agent.

Type: String

Required: No

dockerVersion

The Docker version that's running on the container instance.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

Volume

The data volume configuration for tasks launched using this task definition. Specifying a volume configuration in a task definition is optional. The volume configuration may contain multiple volumes but only one volume configured at launch is supported. Each volume defined in the volume configuration may only specify a name and one of either `configuredAtLaunch`, `dockerVolumeConfiguration`, `efsVolumeConfiguration`, `fsxWindowsFileServerVolumeConfiguration`, or `host`. If an empty volume configuration is specified, by default Amazon ECS uses a host volume. For more information, see [Using data volumes in tasks](#).

Contents

configuredAtLaunch

Indicates whether the volume should be configured at launch time. This is used to create Amazon EBS volumes for standalone tasks or tasks created as part of a service. Each task definition revision may only have one volume configured at launch in the volume configuration.

To configure a volume at launch time, use this task definition revision and specify a `volumeConfigurations` object when calling the `CreateService`, `UpdateService`, `RunTask` or `StartTask` APIs.

Type: Boolean

Required: No

dockerVolumeConfiguration

This parameter is specified when you use Docker volumes.

Windows containers only support the use of the `local` driver. To use bind mounts, specify the `host` parameter instead.

Note

Docker volumes aren't supported by tasks run on AWS Fargate.

Type: [DockerVolumeConfiguration](#) object

Required: No

efsVolumeConfiguration

This parameter is specified when you use an Amazon Elastic File System file system for task storage.

Type: [EFSVolumeConfiguration](#) object

Required: No

fsxWindowsFileServerVolumeConfiguration

This parameter is specified when you use Amazon FSx for Windows File Server file system for task storage.

Type: [FSxWindowsFileServerVolumeConfiguration](#) object

Required: No

host

This parameter is specified when you use bind mount host volumes. The contents of the host parameter determine whether your bind mount host volume persists on the host container instance and where it's stored. If the host parameter is empty, then the Docker daemon assigns a host path for your data volume. However, the data isn't guaranteed to persist after the containers that are associated with it stop running.

Windows containers can mount whole directories on the same drive as `$env:ProgramData`. Windows containers can't mount directories on a different drive, and mount point can't be across drives. For example, you can mount `C:\my\path:C:\my\path` and `D:\:D:\`, but not `D:\my\path:C:\my\path` or `D:\:C:\my\path`.

Type: [HostVolumeProperties](#) object

Required: No

name

The name of the volume. Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed.

When using a volume configured at launch, the name is required and must also be specified as the volume name in the `ServiceVolumeConfiguration` or `TaskVolumeConfiguration` parameter when creating your service or standalone task.

For all other types of volumes, this name is referenced in the `sourceVolume` parameter of the `mountPoints` object in the container definition.

When a volume is using the `efsVolumeConfiguration`, the name is required.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

VolumeFrom

Details on a data volume from another container in the same task definition.

Contents

readOnly

If this value is `true`, the container has read-only access to the volume. If this value is `false`, then the container can write to the volume. The default value is `false`.

Type: Boolean

Required: No

sourceContainer

The name of another container within the same task definition to mount volumes from.

Type: String

Required: No

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)
- [AWS SDK for Ruby V3](#)

VpcLatticeConfiguration

The VPC Lattice configuration for your service that holds the information for the target group(s) Amazon ECS tasks will be registered to.

Contents

portName

The name of the port mapping to register in the VPC Lattice target group. This is the name of the `portMapping` you defined in your task definition.

Type: String

Required: Yes

roleArn

The ARN of the IAM role to associate with this VPC Lattice configuration. This is the Amazon ECS infrastructure IAM role that is used to manage your VPC Lattice infrastructure.

Type: String

Required: Yes

targetGroupArn

The full Amazon Resource Name (ARN) of the target group or groups associated with the VPC Lattice configuration that the Amazon ECS tasks will be registered to.

Type: String

Required: Yes

See Also

For more information about using this API in one of the language-specific AWS SDKs, see the following:

- [AWS SDK for C++](#)
- [AWS SDK for Java V2](#)

- [AWS SDK for Ruby V3](#)

Request throttling for the Amazon ECS API

Amazon Elastic Container Service throttles all API requests for each AWS account on a per-Region basis. We do this to ensure consistent performance and fair usage of the service for all Amazon ECS customers. Throttling ensures that calls to the Amazon ECS API do not exceed the maximum allowed API request quotas for both Amazon ECS and the other AWS services that it integrates with. API calls are subject to the request quotas whether they originate from:

- A third-party application
- A command line tool
- The Amazon ECS console

If you exceed an API throttling quota, you get the `ThrottlingException` error code.

```
An error occurred (ThrottlingException) when calling the DescribeClusters operation
(reached max retries: 4): Rate exceeded.
com.amazonaws.services.ecs.model.AmazonECSEException: Rate exceeded (Service: AmazonECS;
Status Code: 400; Error
Code: ThrottlingException; Request ID: 5ed90669-e454-464d-9b2f-6523bc86f537; Proxy:
null)
```

How throttling is applied

Amazon ECS uses the [token bucket algorithm](#) to implement API throttling. With this algorithm, your account has a *bucket* that holds a specific number of *tokens*. The number of tokens in the bucket represents your throttling quota at any given second.

Amazon ECS examines the rate of API request submissions for all Amazon ECS APIs in your account, per Region, and applies two types of API throttling quotas: *sustained* and *burst*. The sustained rate is the average number of API requests allowed per second over time for an operation. The burst rate is the maximum number of API requests allowed in any one second. With burst, you can periodically make a higher number of API requests than the sustained rate. Following which, Amazon ECS throttles subsequent API requests until the rate of API requests allowed over time stabilizes to the sustained rate. In the token bucket algorithm, the *bucket maximum capacity* signifies the burst rate and the *bucket refill rate* is the sustained rate. We will use these terms to provide you an illustration of Amazon ECS API request throttling in the following example.

You are throttled on the number of API requests you make and each request removes one token from the token bucket. For example, the bucket size for *Cluster read actions*, such as the `DescribeClusters` API, is 50 tokens, so you can make up to 50 `DescribeClusters` requests in one second. If you exceed 50 requests in a second, you are throttled and the remaining requests within that second fail.

Buckets automatically refill at a set rate. If the bucket is below its maximum capacity, a set number of tokens is added back to it every second until it reaches its maximum capacity. If the bucket is full when refill tokens arrive, they are discarded. The bucket cannot hold more than its maximum number of tokens. For example, the bucket size for *Cluster read actions*, such as the `DescribeClusters` API, is 50 tokens, and the refill rate is 20 tokens per second. If you make 50 `DescribeClusters` API requests in a second, the bucket is immediately reduced to zero tokens. The bucket is then refilled by 20 tokens every second, until it reaches its maximum capacity of 50 tokens. This means that the previously empty bucket reaches its maximum capacity after 2.5 seconds.

You do not need to wait for the bucket to be completely full before you can make API requests. You can use tokens as they are added to the bucket. If you immediately use the refill tokens, the bucket does not reach its maximum capacity. For example, the bucket size for *Cluster read actions*, such as the `DescribeClusters` API, is 50 tokens, and the refill rate is 20 tokens per second. If you deplete the bucket by making 50 API requests in a second, you can continue to make 20 API requests per second. The bucket can refill to the maximum capacity only if you make fewer than 20 API requests per second.

Request Token Bucket Sizes and Refill Rates

For request rate limiting purposes, API actions are grouped into categories. All API actions in a category share the same token bucket. For instance, `DescribeClusters` and `ListClusters` APIs share the *Cluster read actions* bucket, for which capacity is 50 and refill rate is 20. This means that the cumulative number of API requests for all *Cluster read actions* is throttled by the same burst rate quota of 50 API requests. Thus, you can make 25 `DescribeClusters` and 25 `ListClusters` API requests in one second, or 30 `DescribeClusters` and 20 `ListClusters`, or 50 `DescribeClusters` and 0 `ListClusters`, or 0 `DescribeClusters` and 50 `ListClusters`, but you cannot make 50 `DescribeClusters` and 50 `ListClusters` requests at the same time. Sustained rate is similarly applied cumulatively to all API requests within a bucket.

The following table shows the bucket capacity (or burst) and refill rate (or sustained) for all AWS Regions. All API action categories enforce rate quotas for each AWS account on a per-Region basis.

API action category	Actions	Bucket maximum capacity (or Burst rate)	Bucket refill rate (or Sustained rate)
Cluster modify actions	- CreateCluster - DeleteCluster - PutClusterCapacity Providers - UpdateCluster - UpdateClusterSettings	20	1
Cluster read actions	- DescribeClusters - ListClusters	50	20
Task definition modify actions	- DeregisterTaskDefinition - RegisterTaskDefinition	20	1
Task definition read actions	- DescribeTaskDefinition - ListTaskDefinitions - ListTaskDefinitionFamilies	50	20
Task definition deletion actions	DeleteTaskDefinitions	5	1
Capacity provider modify actions	- CreateCapacityProvider - DeleteCapacityProvider - UpdateCapacityProvider	10	1
Capacity provider read actions	DescribeCapacityProviders	50	20
Tag modify actions	- TagResource - UntagResource	20	10

API action category	Actions	Bucket maximum capacity (or Burst rate)	Bucket refill rate (or Sustained rate)
Tag read actions	ListTagsForResource	50	20
Setting modify actions	- DeleteAccountSetting - PutAccountSetting - PutAccountSettingDefault	10	1
Setting read actions	ListAccountSettings	50	20
Cluster resource modify actions	- DeleteAttributes - DeregisterContainerInstance - ExecuteCommand - PutAttributes - RunTask¹ - StartTask - StopTask - UpdateContainerAgent - UpdateContainerInstancesStates	100	40
Cluster resource read actions	- DescribeContainerInstances - DescribeTasks - ListAttributes - ListContainerInstances - ListTasks	100	20

API action category	Actions	Bucket maximum capacity (or Burst rate)	Bucket refill rate (or Sustained rate)
Agent modify actions	- RegisterContainerInstance - SubmitAttachmentStateChanges - SubmitContainerStateChange - SubmitTaskStateChange	200	120
Service modify actions	- CreateService - DeleteService - UpdateService	50	5
Service read actions	- DescribeServices - ListServices	100	20
Service deployment actions	- DescribeServiceDeployments - ListServiceDeployments	50	20
Service revision actions	DescribeServiceRevisions	50	20
Task protection actions	- UpdateTaskProtection - GetTaskProtection	200	80
Cluster service resource read actions	ListServicesByNamespace	10	1

¹ AWS Fargate additionally throttles Amazon ECS RunTask API to the rates listed [here](#) in the *Amazon ECS Developer Guide*.

Adjusting API throttling quotas

You can request an increase for API throttling quotas for your AWS account. To request a quota adjustment, contact the [AWS Support Center](#).

Handling API throttling

You can implement an error retry and exponential back-off strategy to avoid the impact of throttling errors on your workloads. If you use AWS SDK, the automatic retry logic is already built-in and configurable. You can refer to the following resources for more details:

- [Error retries and exponential backoff in AWS](#) in the AWS General Reference Guide
- [Exponential backoff and jitter](#) blog post
- [Timeouts, retries, and backoff with jitter](#) article in the Amazon Builder's Library

Ensuring idempotency

When you perform a mutating operation, you might see an exception because of timeouts or server issues occurring after the resources are mutated. This can make it difficult to determine whether the mutation occurred, and could lead to multiple retries. However, if the original operation and the subsequent retries actually performed the mutations you might have applied stacking changes or created more resources than you intended to. Idempotency ensures that an operation mutates resources no more than one time. With an idempotent request, if the original request mutated successfully, any subsequent retries complete successfully without performing any further mutation.

Topics

- [Idempotency in Amazon ECS](#)
- [Idempotency for RunTask](#)
- [Examples](#)
- [Retry recommendations for idempotent requests](#)

Idempotency in Amazon ECS

The following API actions optionally support idempotency using a *client token*. The corresponding AWS CLI commands also support idempotency using a client token. A client token is a unique, case-sensitive string. To make an idempotent API request using one of these actions, specify a client token in the request. You should not reuse the same client token for other API requests. If you retry a request that completed successfully using the same client token and the same parameters, the retry succeeds without performing any further actions.

Idempotent using a client token

- `CreateService`

The client token can be up to 36 ASCII characters in the range of 33-126 (inclusive).

- `CreateTaskSet`

The client token can be up to 36 ASCII characters in the range of 33-126 (inclusive).

- `RunTask`

The client token can be up to 64 ASCII characters in the range of 33-126 (inclusive).

Types of idempotency

- **cluster** – Requests with the same token in the same cluster are idempotent. For example, ClientToken A can only be used as a request parameter one time for RunTask requests in Cluster X. RunTask requests to other clusters are considered a separate request. Therefore, you can use ClientToken A for a RunTask request for cluster Y.

Idempotency for RunTask

The RunTask API supports idempotency using a client token. A client token is a unique string that you specify when you make an API request. If you retry an API request with the same client token and the same request parameters after it has completed successfully, the result of the original request is returned. If you retry a successful request using the same client token, but one or more of the parameters are different, other than the Region or Availability Zone, the retry fails with a `ConflictException`. If you do not specify your own client token, the AWS SDK and AWS Command Line Interface automatically generate a client token for the request to ensure that it is idempotent. A client token can be any string that includes up to 64 ASCII characters in the range of 33-126 (inclusive).

The time to live (TTL) for the RunTask client token is 24 hours. You should not reuse the same client token for different requests. The client token maximum TTL is valid for whichever of the following two values is lower:

- 24 hours
- The lifetime of the resource plus one hour

The lifetime of a resource is the timestamp at which the task was created to the timestamp at which the last status (`lastStatus`) transitioned to `STOPPED`. When you use RunTask to launch more than one task, the lifetime of the resource equals the lifetime of the last task that transitioned to `STOPPED`.

RunTask retry rules and responses

When you retry a request because you received a 5xx exception, the retried successful response normally includes all of the information that the original request would have returned. Tasks that have been stopped for under an hour only include the task ARN, last status, and desired status.

The following is an example snippet of the response from a retry when there is one running task, one stopped task, and one task that failed to launch.

```
{
  "failures": [
    {
      "arn": "arn:aws:ecs:us-east-1:123456789012:container/4df26bb4-f057-467b-a079-961675296e64",
      "reason": "RESOURCE:MEMORY"
    }
  ],
  "tasks": [
    {
      "desiredStatus": "RUNNING",
      "taskArn": "arn:aws:ecs:us-east-1:123456789012:task/default/fdf2c302-468c-4e55-b884-5331d816e7fb",
      ...
    },
    {
      "taskArn": "arn:aws:ecs:us-east-1:123456789012:task/default/fdf2c302-468c-4e55-b884-5331d819999",
      "lastStatus": "STOPPED",
      ...
    }
  ]
}
```

Failures that are over an hour old only include the number of failed tasks.

Examples

AWS CLI command examples

To make an AWS CLI command idempotent, add the `--client-token` option.

Example: create-service

The following [create-service](#) command uses idempotency as it includes a client token.

```
aws ecs create-service \  
  --cluster MyCluster \  
  --service MyService \  
  --task-definition MyTaskDefinition:2 \  
  --desired-count 2 \  
  --launch-type FARGATE \  
  --platform-version LATEST \  
  --network-configuration  
  "awsvpcConfiguration={subnets=["subnet-12344321"],securityGroups=["sg-12344321"],assignPublicIP=DISABLED} \  
  \  
  --client-token 550e8400-e29b-41d4-a716-44665544
```

Example: create-task-set

The following [create-task-set](#) command uses idempotency as it includes a client token.

```
aws ecs create-task-set \  
  --cluster MyCluster \  
  --service MyService \  
  --task-definition MyTaskDefinition:2 \  
  --network-configuration  
  "awsvpcConfiguration={subnets=["subnet-12344321"],securityGroups=["sg-12344321"]}" \  
  --client-token 550e8400-e29b-41d4-a716-44665544
```

Example: run-task

The following [run-task](#) command uses idempotency as it includes a client token.

```
aws ecs run-task \  
  --cluster MyCluster \  
  --task-definition MyTaskDefinition:2 \  
  --client-token 550e8400-e29b-41d4-a716-446655440000
```

API request examples

To make an API request idempotent, add the `clientToken` parameter.

Example: CreateService

The following [CreateService](#) API request uses idempotency as it includes a client token.


```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 87
X-Amz-Target: AmazonEC2ContainerServiceV20141113.CreateService
X-Amz-Date: 20150429T170125Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
```

```
{
  "serviceName": "MyService",
  "taskDefinition": "MyTaskDefinition:2",
  "desiredCount": 10,
  "capacityProviderStrategy": [
    {
      "base": "number",
      "capacityProvider": "FARGATE",
      "weight": 1
    }
  ],
  "capacityProviderStrategy": [
    {
      "base": "number",
      "capacityProvider": "FARGATE_SPOT",
      "weight": 1
    }
  ],
  "clientToken": "550e8400-e29b-41d4-a716-44665544"
}
```

Example: CreateTaskSet

The following [CreateTaskSet](#) API request uses idempotency as it includes a client token.

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 87
X-Amz-Target: AmazonEC2ContainerServiceV20141113.CreateTaskSet
X-Amz-Date: 20150429T170125Z
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS
```

```
{
  "serviceName": "MyService",
  "taskDefinition": "mytask:1",
  "desiredCount": 1,
  "capacityProviderStrategy": [
    {
      "base": "number",
      "capacityProvider": "FARGATE",
      "weight": 1
    }
  ],
  "capacityProviderStrategy": [
    {
      "base": "number",
      "capacityProvider": "FARGATE_SPOT",
      "weight": 1
    }
  ],
  "clientToken": "550e8400-e29b-41d4-a716-44665544"
}
```

Example: RunTask

The following [RunTask](#) API request uses idempotency as it includes a client token.

```
POST / HTTP/1.1
Host: ecs.us-east-1.amazonaws.com
Accept-Encoding: identity
Content-Length: 45
X-Amz-Target: AmazonEC2ContainerServiceV20141113.RunTask
X-Amz-Date: 20161121T215740Z
User-Agent: aws-cli/1.11.13 Python/2.7.12 Darwin/16.1.0 botocore/1.4.66
Content-Type: application/x-amz-json-1.1
Authorization: AUTHPARAMS

{
  "count": 1,
  "taskDefinition": "mytask:1",
  "clientToken": "550e8400-e29b-41d4-a716-446655440000"
}
```

Retry recommendations for idempotent requests

The following table shows some common responses that you might get for idempotent API requests, and provides retry recommendations.

Response	Recommendation	Comments
200 (OK)	Do not retry	The original request completed successfully. Any subsequent retries return successfully.
400-series response codes (client errors)	Do not retry	There is a problem with the request, from among the following: • It includes a parameter or parameter combination that is not valid. • It uses an action or resource for which you do not have permissions. • It uses a resource that is in the process of changing states. If the request involves a resource that is in the process of changing states, retrying the request could possibly succeed.
500-series response codes (server errors)	Retry	The error is caused by an AWS server-side issue and is generally transient. Repeat the request with an appropriate back-off strategy.

Common Parameters

The following list contains the parameters that all actions use for signing Signature Version 4 requests with a query string. Any action-specific parameters are listed in the topic for that action. For more information about Signature Version 4, see [Signing AWS API requests](#) in the *IAM User Guide*.

Action

The action to be performed.

Type: string

Required: Yes

Version

The API version that the request is written for, expressed in the format YYYY-MM-DD.

Type: string

Required: Yes

X-Amz-Algorithm

The hash algorithm that you used to create the request signature.

Condition: Specify this parameter when you include authentication information in a query string instead of in the HTTP authorization header.

Type: string

Valid Values: AWS4-HMAC-SHA256

Required: Conditional

X-Amz-Credential

The credential scope value, which is a string that includes your access key, the date, the region you are targeting, the service you are requesting, and a termination string ("aws4_request"). The value is expressed in the following format: *access_key/YYYYMMDD/region/service/aws4_request*.

For more information, see [Create a signed AWS API request](#) in the *IAM User Guide*.

Condition: Specify this parameter when you include authentication information in a query string instead of in the HTTP authorization header.

Type: string

Required: Conditional

X-Amz-Date

The date that is used to create the signature. The format must be ISO 8601 basic format (YYYYMMDD'T'HHMMSS'Z'). For example, the following date time is a valid X-Amz-Date value: 20120325T120000Z.

Condition: X-Amz-Date is optional for all requests; it can be used to override the date used for signing requests. If the Date header is specified in the ISO 8601 basic format, X-Amz-Date is not required. When X-Amz-Date is used, it always overrides the value of the Date header. For more information, see [Elements of an AWS API request signature](#) in the *IAM User Guide*.

Type: string

Required: Conditional

X-Amz-Security-Token

The temporary security token that was obtained through a call to AWS Security Token Service (AWS STS). For a list of services that support temporary security credentials from AWS STS, see [AWS services that work with IAM](#) in the *IAM User Guide*.

Condition: If you're using temporary security credentials from AWS STS, you must include the security token.

Type: string

Required: Conditional

X-Amz-Signature

Specifies the hex-encoded signature that was calculated from the string to sign and the derived signing key.

Condition: Specify this parameter when you include authentication information in a query string instead of in the HTTP authorization header.

Type: string

Required: Conditional

X-Amz-SignedHeaders

Specifies all the HTTP headers that were included as part of the canonical request. For more information about specifying signed headers, see [Create a signed AWS API request](#) in the *IAM User Guide*.

Condition: Specify this parameter when you include authentication information in a query string instead of in the HTTP authorization header.

Type: string

Required: Conditional

Common Errors

This section lists the errors common to the API actions of all AWS services. For errors specific to an API action for this service, see the topic for that API action.

AccessDeniedException

You do not have sufficient access to perform this action.

HTTP Status Code: 400

IncompleteSignature

The request signature does not conform to AWS standards.

HTTP Status Code: 400

InternalFailure

The request processing has failed because of an unknown error, exception or failure.

HTTP Status Code: 500

InvalidAction

The action or operation requested is invalid. Verify that the action is typed correctly.

HTTP Status Code: 400

InvalidClientTokenId

The X.509 certificate or AWS access key ID provided does not exist in our records.

HTTP Status Code: 403

NotAuthorized

You do not have permission to perform this action.

HTTP Status Code: 400

OptInRequired

The AWS access key ID needs a subscription for the service.

HTTP Status Code: 403

RequestExpired

The request reached the service more than 15 minutes after the date stamp on the request or more than 15 minutes after the request expiration date (such as for pre-signed URLs), or the date stamp on the request is more than 15 minutes in the future.

HTTP Status Code: 400

ServiceUnavailable

The request has failed due to a temporary failure of the server.

HTTP Status Code: 503

ThrottlingException

The request was denied due to request throttling.

HTTP Status Code: 400

ValidationError

The input fails to satisfy the constraints specified by an AWS service.

HTTP Status Code: 400